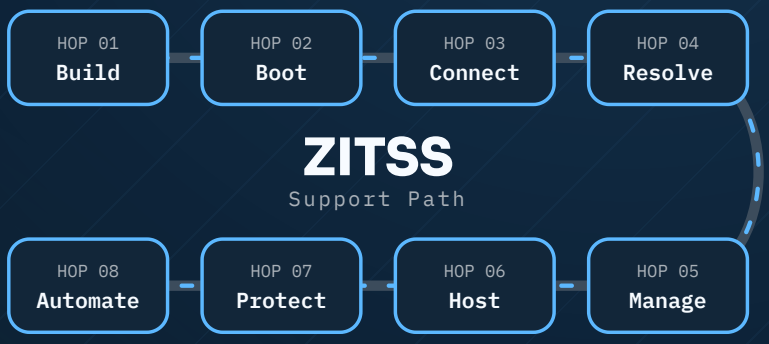
ZITSS

ZyeemZia™ IT Support Specialization

Fix it fast. Fix it right. Keep people working.



COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 38 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A job-ready, hands-on IT support track for the people every organisation calls first: hardware and PC repair, Windows, Linux and macOS, networking, a disciplined troubleshooting method, customer service and ticketing, system administration, cloud and SaaS, security basics, PowerShell and Bash automation, documentation and career skills.

01
Learn
Work through 38 lessons. Each ends with a knowledge check.

02
Practise
Fill in the templates, answer the scenarios and case studies, review the glossary.

03
Certify
Take the 40-question official exam at zitss.zyeemzia.com, graded on the ZyeemZia server. Pass mark 80%.

04
Get certified
Download your certificate online. Anyone can verify its credential ID at zitss.zyeemzia.com/#verify. 90%+ and the capstone earn Distinction.

Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Week 1	m1 Build	Identify parts, build or strip down a PC	☐
Week 2	m2 Boot	Install Windows and Linux in virtual machines	☐
Week 3	m3 Connect	Subnet practice, DNS and DHCP lab	☐
Week 4	m4 Resolve	Work ten practice tickets end to end	☐
Week 5	m5 Manage	Users, groups, permissions, backup restore	☐
Week 6	m6 Host	Virtual server, cloud tenant, SaaS admin	☐
Week 7	m7 Protect	MFA, endpoint and phishing drills	☐
Week 8	m8 Automate	Two scripts, KB articles, capstone, exam	☐

Learner name .....

Start date ..... Target exam date .....

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5. Building, Upgrading and Repairing a PC · 30 min

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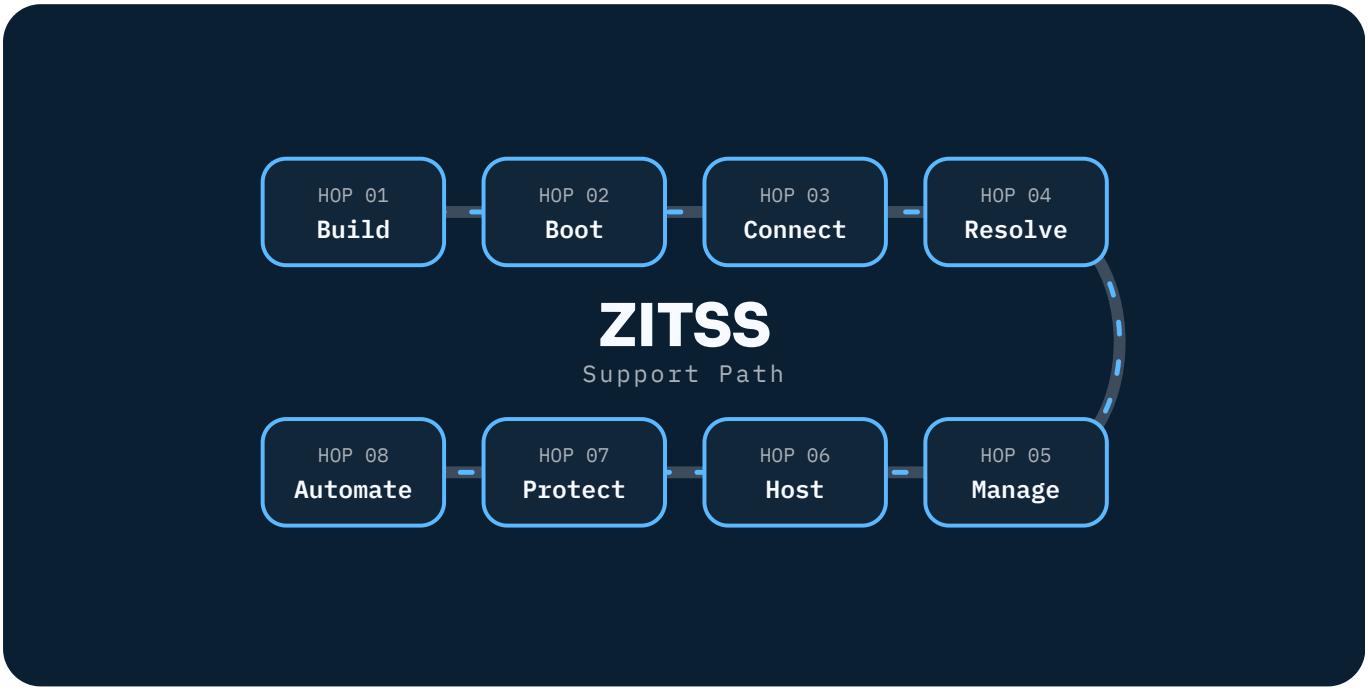
+ Glossary · Capstone guide · Practice exam & answers · Certificate · Notes

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THE FRAMEWORK

The ZITSS Support Path

Eight stops on the road from the repair bench to a trusted support professional: build the hardware, boot the operating system, connect it to the network, resolve problems for real people, manage users and systems, host services in the server room and the cloud, protect everything, and automate the work so you can keep growing.



01 Build
Components, storage, power, peripherals and safe repair: the first stop on the Support Path

02 Boot
Install, configure and support the three main desktop platforms and mobile devices

03 Connect
TCP/IP, addressing, DNS, DHCP, Wi-Fi and the tools that prove where a network fault lives

04 Resolve
A disciplined method, excellent communication and service management that keeps people working

05 Manage
Users, groups, directories, policy, backups and patching: keeping many systems consistent and safe

06 Host
Servers, virtualisation, storage, cloud and SaaS administration, monitoring and remote access

07 Protect
Identity, MFA, malware, phishing, encryption and incident handling for the first line of defence

08 Automate
PowerShell and Bash automation, documentation, asset management and the skills to grow your career

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01

MODULE 1 · BUILD

Hardware and PC Repair

Components, storage, power, peripherals and safe repair: the first stop on the Support Path

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Describe the IT support role and organise your learning around the eight-stage ZITSS Support Path
- Identify and choose compatible CPUs, motherboards, memory, storage and power supplies
- Diagnose display, printer, mobile, power and overheating problems
- Build, upgrade and repair a PC safely and record the work

LESSONS

1. The IT Support Role and How Computers Work · 22 min
2. CPU, Motherboard and Memory · 26 min
3. Storage, Power and Cooling · 26 min
4. Peripherals, Displays, Printers and Mobile Devices · 24 min
5. Building, Upgrading and Repairing a PC · 30 min

MODULE 1 · BUILD · LESSON 1 · 22 MIN

The IT Support Role and How Computers Work

What support specialists do, the Support Path, and the four jobs every computer performs

- Describe the IT support role, its tiers and the skills employers expect in 2026
- Explain the input, processing, storage and output model of a computer
- Convert between bits, bytes and the decimal and binary storage units
- Use the eight stages of the ZITSS Support Path to organise your learning

When a laptop will not boot, a printer jams before a board meeting or a new employee cannot sign in, someone has to fix it quickly and calmly. That person is the **IT support specialist**. In 2026 the job is broader than ever: you look after physical devices, three major operating systems, networks, cloud services and accounts, and you are often the first person to spot a security problem. Above all, you keep other people productive.

The ZITSS Support Path: eight stages

1 Build
Know the hardware: components, assembly, upgrades, safe repair.

2 Boot
Install, configure and support Windows, Linux and macOS.

3 Connect
Get devices onto networks with TCP/IP, DNS, DHCP and Wi-Fi.

4 Resolve
Troubleshoot methodically, serve users well and manage tickets.

5 Manage
Administer users, groups, policies, backups and patches.

6 Host
Run infrastructure services, virtual machines, cloud and SaaS.

7 Protect
Apply security basics: identity, MFA, malware, phishing, encryption.

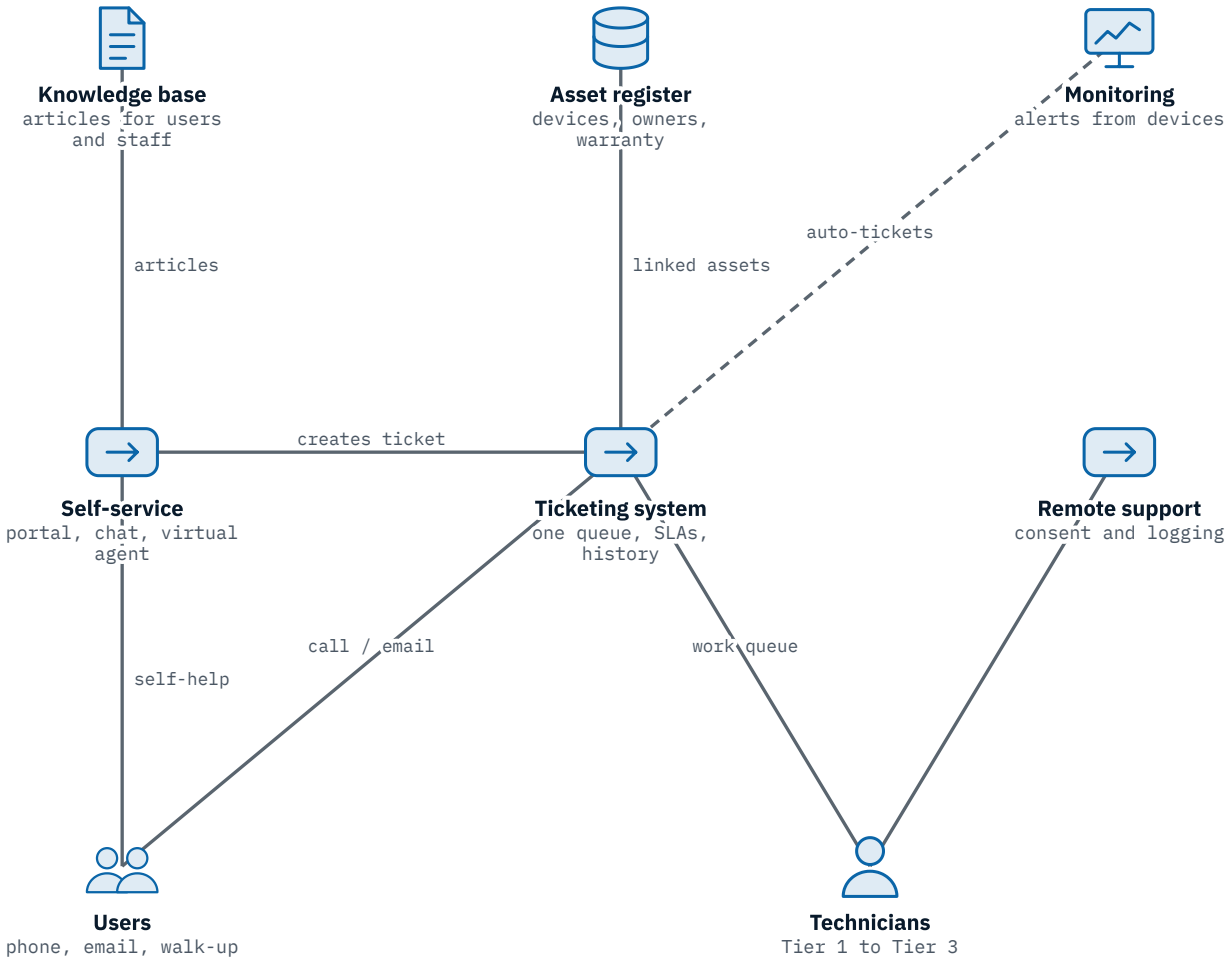
8 Automate
Script repetitive work, document everything and grow your career.

Typical support tiers

TIER	MAIN WORK	EXAMPLE	ESCALATES TO
Tier 0 (self-service)	Knowledge base, portal, self-service password reset, virtual agent (chatbot)	User resets own password with MFA through a portal	Tier 1 if it fails
Tier 1 (service desk)	Logging tickets, first diagnosis, known fixes	Mapping a shared drive, clearing a print queue	Tier 2
Tier 2 (desktop / technical)	Deeper troubleshooting, hardware swaps, remote fixes	Reimaging a laptop, fixing a VPN profile	Tier 3
Tier 3 (specialist / engineering)	Servers, network design, vendor escalation	Fixing a failed domain controller	Vendor or supplier

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Help-desk tooling in a typical organisation



Every channel ends in one ticketing system, which links to knowledge, asset records and monitoring. Notice that self-service and monitoring alerts can create tickets without anyone picking up a phone.

INSIGHT · TECHNICAL SKILL IS ONLY HALF THE JOB

Hiring managers consistently rank **communication, patience and reliability** alongside technical knowledge. A specialist who explains clearly, sets expectations and follows up will be trusted with harder problems faster than a brilliant technician who disappears.

The four jobs every computer performs

01 Input
Keyboard, mouse, touch, camera, microphone, network data arriving.

→

02 Processing
The CPU (and sometimes GPU or NPU) executes instructions on the data.

→

03 Storage
RAM holds working data; SSDs and HDDs keep data when power is off.

→

04 Output
Display, speakers, printer, or data sent over the network.

Storage units you will use daily

UNIT	DECIMAL (DRIVE MAKERS)	BINARY (OPERATING SYSTEMS)	TYPICAL USE
Bit (b)	0 or 1	0 or 1	Network speeds: Mbps, Gbps

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UNIT	DECIMAL (DRIVE MAKERS)	BINARY (OPERATING SYSTEMS)	TYPICAL USE
Byte (B)	8 bits	8 bits	One character of text
Kilobyte	1,000 bytes (kB)	1,024 bytes (KiB)	Small text files
Megabyte	1,000,000 bytes (MB)	1,048,576 bytes (MiB)	Photos, documents
Gigabyte	10^9 bytes (GB)	2^30 bytes (GiB)	RAM, apps, videos
Terabyte	10^12 bytes (TB)	2^40 bytes (TiB)	Drives and backups

Two conversions that answer common tickets

MEASURE	FORMULA	WHAT IT TELLS YOU
Download time	`time (s) = file size (bits) / link speed (bits per second)`	A 2 GB file is about 16,000 megabits. On a 100 Mbps link it takes roughly 160 seconds, plus overhead.
Reported drive size	`size in GiB = size in bytes / 1,073,741,824`	A '1 TB' drive holds 10^12 bytes, which the OS reports as about 931 GiB. Nothing is missing.

EXAMPLE · THE CASE OF THE 'MISSING' STORAGE

A user in a Dhaka buying house complains that a new 512 GB laptop shows only 476 GB. The difference is units (512 x 10^9 bytes is about 476.8 GiB), plus a recovery partition. A two-sentence explanation closes the ticket and builds trust.

Hard skills and soft skills employers test

Hard skills

- Hardware diagnosis and replacement
- Windows, Linux and macOS administration
- Networking basics: IP, DNS, DHCP, Wi-Fi
- Ticketing systems and remote support tools
- Security hygiene and basic scripting

Soft skills

- Listening and asking good questions
- Explaining without jargon
- Staying calm with stressed users
- Writing clear notes and articles
- Owning a problem until it is closed

SCENARIO · DECIDE · YOUR FIRST MORNING ON THE SERVICE DESK

On your first day, three things arrive at once: the CEO's assistant cannot print the board pack due in 20 minutes, a user asks how to change their desktop wallpaper, and an automated alert says one backup job failed overnight.

- (A) Change the wallpaper first because it is the quickest win
- (B) Ignore the backup alert because nobody has complained about it
- (C) Log all three, start on the board-pack printing, and note the backup alert for follow-up within the hour
- (D) Escalate everything to Tier 2 because you are new

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Think about the last time someone fixed a technology problem for you. What did they do, apart from the technical fix, that made the experience good or bad?

Key terms

IT support specialist
A professional who installs, maintains and troubleshoots devices, software, networks and accounts so that people can work without interruption.

Service desk
The single point of contact between users and IT, where requests and incidents are received, logged and managed.

Tier (support level)
A layer of support with increasing skill and access; work moves up tiers through escalation.

Escalation
Passing a ticket to a more specialised person or team when it exceeds your skill, access or authority.

Bit
The smallest unit of digital data, holding a value of 0 or 1; network speeds are measured in bits per second.

Byte
A group of eight bits; storage and file sizes are measured in bytes.

Gibibyte (GiB)
2^30 bytes (1,073,741,824 bytes); operating systems often show this value while labelling it GB.

Key takeaways

- Support exists to keep people productive; technical skill and communication count equally.
- Work flows from self-service up through tiers, and escalation passes evidence, not just the problem.
- One ticketing system ties channels, knowledge, assets and monitoring together.
- Drive makers count in decimal (GB), operating systems in binary (GiB): 1 TB shows as about 931 GiB.
- Network speeds are in bits, file sizes in bytes: multiply bytes by 8 before dividing by link speed.

KNOWLEDGE CHECK

1. A user's new '1 TB' drive shows as 931 GB in the operating system. What is the most likely explanation?
   - (A) The OS reports binary units while the drive maker uses decimal units
   - (B) The drive is faulty and must be replaced
   - (C) Malware has hidden part of the drive
   - (D) The drive was formatted with an old file system that wastes a lot of space
2. Which tier usually handles password resets that a self-service portal could not complete?
   - (A) Tier 3
   - (B) Tier 0
   - (C) The hardware vendor
   - (D) Tier 1
3. In the input-processing-storage-output model, where does RAM belong?
   - (A) Input
   - (B) Output
   - (C) Processing only
   - (D) Storage (working memory)

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4. Roughly how long does a 1 GB file take to download over a 100 Mbps link, ignoring overhead?

   (A) About 80 seconds
   (B) About 8 seconds
   (C) About 10 seconds
   (D) About 800 seconds

5. Which behaviour best shows the soft skills employers look for in support staff?

   (A) Explaining the fix in plain language and confirming the user is working again
   (B) Using precise technical terms so users learn them
   (C) Closing tickets quickly without contacting the user
   (D) Escalating every ticket that involves an unhappy or demanding user straight away

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · BUILD · LESSON 2 · 26 MIN

CPU, Motherboard and Memory

The core trio that decides how fast and how stable a computer is

- Identify motherboard form factors, sockets, slots and connectors
- Compare CPU characteristics: cores, threads, clock speed, cache and integrated graphics
- Choose compatible RAM by type, speed, capacity and channel configuration
- Read hardware details from the command line on Windows, Linux and macOS

The **motherboard** is the backbone that every other part plugs into. The **CPU** does the general-purpose thinking, and **RAM** is the fast, temporary workspace it uses. Most performance and stability problems you will meet on desktops and laptops trace back to one of these three: a CPU running too hot, a mismatched memory module or a failing board component.

What sits on a typical desktop motherboard (top to bottom)

CPU and cooler	`Socket with locking mechanism` `Heatsink and fan or liquid cooler` `Thermal paste between them`
Memory	`DIMM slots (desktops) or SO-DIMM / soldered RAM (laptops)` `Dual-channel pairs, often colour-coded`
Expansion	`PCIe x16 slot for graphics` `PCIe x1 / x4 slots for network or capture cards` `M.2 slots for NVMe SSDs`
Chipset and firmware	`Chipset links CPU to slower devices` `UEFI firmware chip and CMOS battery (CR2032)`
Connectors	`24-pin ATX and 8-pin CPU power` `SATA ports, front-panel headers, USB headers, fan headers` `Rear I/O: USB, video, audio, Ethernet`

Common form factors

FORM FACTOR	APPROX. SIZE	TYPICAL USE	SUPPORT NOTE
ATX	305 x 244 mm	Standard desktops and workstations	Most slots; easiest to work on
Micro-ATX	244 x 244 mm	Office desktops	Fewer expansion slots
Mini-ITX	170 x 170 mm	Small form factor PCs, kiosks	One PCIe slot; tight cooling
Laptop / proprietary	Varies	Notebooks and all-in-ones	Parts often soldered; use service manuals

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CPU characteristics that matter to users

Cores and threads

- More cores help multitasking and heavy apps
- Threads let one core run two instruction streams
- Office PCs: 4-8 cores is ample in 2026

Clock and cache

- Clock speed (GHz) sets cycles per second
- Boost clocks rise briefly when cool
- Larger cache reduces trips to RAM

Extras

- Integrated graphics avoid a separate GPU
- NPU accelerates on-device AI features
- Virtualisation extensions must be enabled in firmware for VMs

WATCH OUT · SOCKET COMPATIBILITY IS NOT OPTIONAL

A CPU only fits a motherboard with the matching **socket and chipset support**, and sometimes needs a firmware update first. Always check the board maker's CPU support list before ordering a replacement or upgrade.

Memory generations at a glance

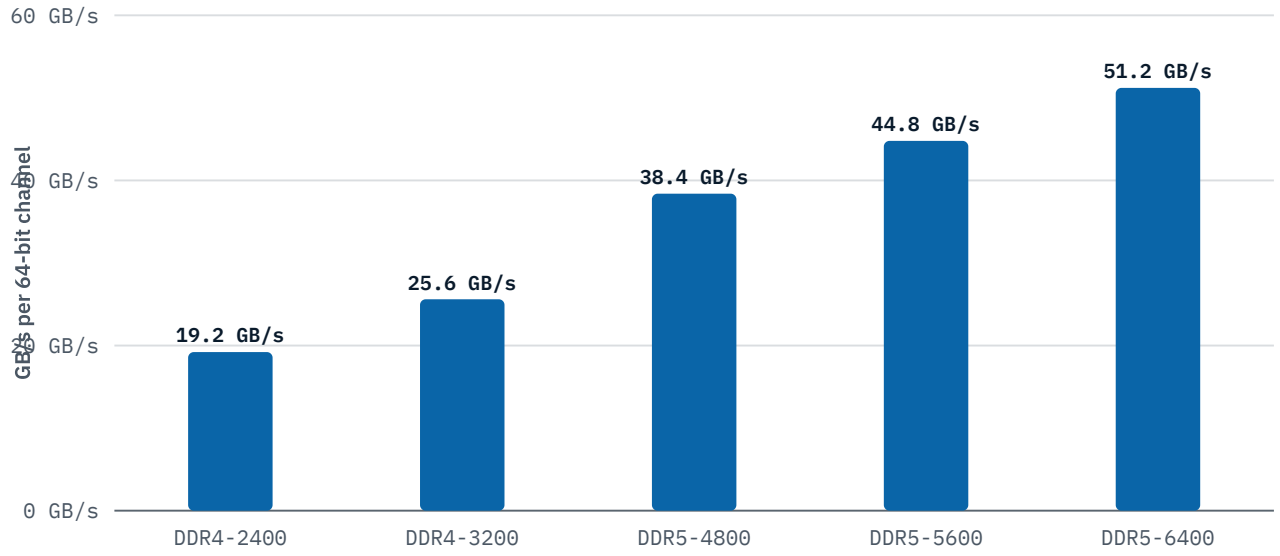
TYPE	WHERE	NOTES
DDR4 DIMM	Desktops from roughly 2015-2022	288 pins, 1.2 V; not interchangeable with DDR5
DDR5 DIMM	Current desktops	288 pins but notch in a different place; on-module power management
SO-DIMM	Laptops, mini PCs	Shorter module; DDR4 and DDR5 versions differ
LPDDR (soldered)	Thin laptops, tablets, phones	Cannot be upgraded; buy enough at purchase
LPCAMM2 / CAMM2	Some newer thin laptops and workstations	Compact, replaceable module for low-power memory; check the service manual
ECC memory	Servers, workstations	Detects and corrects single-bit errors; needs board support

Memory maths

MEASURE	FORMULA	WHAT IT TELLS YOU
Peak bandwidth per channel	`bandwidth (MB/s) = transfers per second (MT/s) x 8 bytes`	DDR5-5600 gives about 5,600 x 8 = 44,800 MB/s (about 44.8 GB/s) per 64-bit channel.
Dual-channel bandwidth	`total = per-channel bandwidth x number of populated channels`	Two matched modules in the correct slots roughly double peak bandwidth compared with one module.

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Peak bandwidth of one memory channel



Each bar is MT/s x 8 bytes, so the numbers are exact theoretical peaks; real throughput is lower. Notice that filling a second channel doubles these figures, which is why matched pairs matter.

Adding RAM to a desktop safely

1. **Confirm the specification** Check type (DDR4 or DDR5), maximum capacity, supported speeds and slot count in the board manual or with a system information tool.
2. **Prepare** Shut down, switch off the power supply, unplug and hold the power button for a few seconds to drain residual charge. Wear an ESD strap.
3. **Install in the right slots** Use the slots the manual recommends for dual channel (often A2 and B2). Align the notch and press firmly until the latches click.
4. **Verify** Boot, enter firmware setup or the OS, confirm the full capacity and speed are detected, then run a memory test if the system was unstable.

Reading CPU and memory details

```Windows PowerShell
Get-CimInstance Win32_Processor | Select-Object Name, NumberOfCores, NumberOfLogicalProcessors,
MaxClockSpeed
Get-CimInstance Win32_PhysicalMemory | Select-Object BankLabel, Capacity, Speed, Manufacturer
Get-CimInstance Win32_BaseBoard | Select-Object Manufacturer, Product, SerialNumber
# Capacity is in bytes: divide by 1GB to read it in GiB
(Get-CimInstance Win32_ComputerSystem).TotalPhysicalMemory / 1GB
```

These read-only queries are safe to run on any user's machine and are ideal for asset records and upgrade planning.

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The same information on Linux and macOS

Linux host

```
lscpu | grep -E 'Model name|^CPU\(s\)|Thread|MHz'
free -h
sudo dmidecode --type memory | grep -E 'Size|Speed|Type:'
# macOS equivalent in Terminal:
# system_profiler SPHardwareDataType SPMemoryDataType
```

SCENARIO · DECIDE · UPGRADE REQUEST FROM ACCOUNTS

An accountant in a Chattogram shipping office has a desktop with one 8 GB DDR4 module and two free slots. Spreadsheets with large pivot tables are slow and the system uses 95% of memory. The manager approves a small budget.

- (A) Buy one 16 GB DDR5 module because it is newer and faster
- (B) Add a matching 8 GB DDR4 module in the paired slot to reach 16 GB in dual channel
- (C) Replace the CPU with a faster model
- (D) Increase the page file size and close the ticket

Circle your choice. Answer and feedback: module answer key.

Key terms

Motherboard
The main circuit board that connects the CPU, memory, storage, expansion cards and power, and holds the firmware.

Chipset
Motherboard logic that connects the CPU to slower peripherals and determines features such as USB, SATA and PCIe lanes.

Cache
Very fast memory built into the CPU that holds recently used data to avoid slower trips to RAM.

ECC memory
Error-correcting code RAM that detects and fixes single-bit errors, used in servers and workstations.

CPU socket
The mounting interface on a motherboard that accepts a specific family of processors.

Core
An independent processing unit inside a CPU that can execute its own instruction stream.

Dual channel
A memory configuration where matched modules in paired slots are accessed in parallel to increase bandwidth.

Key takeaways

- Socket, chipset support and firmware version decide whether a CPU fits a board.
- DDR4 and DDR5 are keyed differently and never interchangeable; check the manual before buying.
- Matched modules in the recommended slots enable dual channel and double peak bandwidth.
- Soldered LPDDR cannot be upgraded, so size laptop memory correctly at purchase.
- Read-only inventory commands give exact CPU and memory details for records and upgrades.

KNOWLEDGE CHECK

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6. A technician tries to install a DDR5 module in a DDR4 slot. What happens?

   (A) It works at DDR4 speed
   (B) It works, but only in single-channel mode at reduced DDR4 speeds and timings
   (C) It works after a firmware update
   (D) It will not physically seat because the key notch is in a different position

7. Which component is most likely to need a firmware update before a newer CPU will work in an existing board?

   (A) The RAM modules
   (B) The power supply
   (C) The motherboard UEFI
   (D) The SSD

8. What is the approximate peak bandwidth of one DDR4-3200 channel?

   (A) 3.2 GB/s
   (B) 12.8 GB/s
   (C) 51.2 GB/s
   (D) 25.6 GB/s

9. Why should two RAM modules be placed in the slots the manual recommends?

   (A) To reduce power consumption
   (B) To enable dual-channel operation
   (C) To make the CMOS battery last longer
   (D) To allow ECC to work on any board

10. A thin laptop is too slow with 8 GB of RAM. The service manual says memory is LPDDR5. What should you advise?

    (A) Memory cannot be upgraded; consider a replacement device with more RAM
    (B) Add a SO-DIMM module
    (C) Replace the RAM chips with larger ones
    (D) Increase the virtual memory size and enable dual-channel mode in the firmware

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · BUILD · LESSON 3 · 26 MIN

Storage, Power and Cooling

Keeping data safe, power clean and components cool

- Compare HDD, SATA SSD and NVMe SSD performance, interfaces and failure signs
- Size a power supply and recognise power-related faults
- Explain how cooling works and diagnose overheating
- Check drive health and file system status with built-in tools

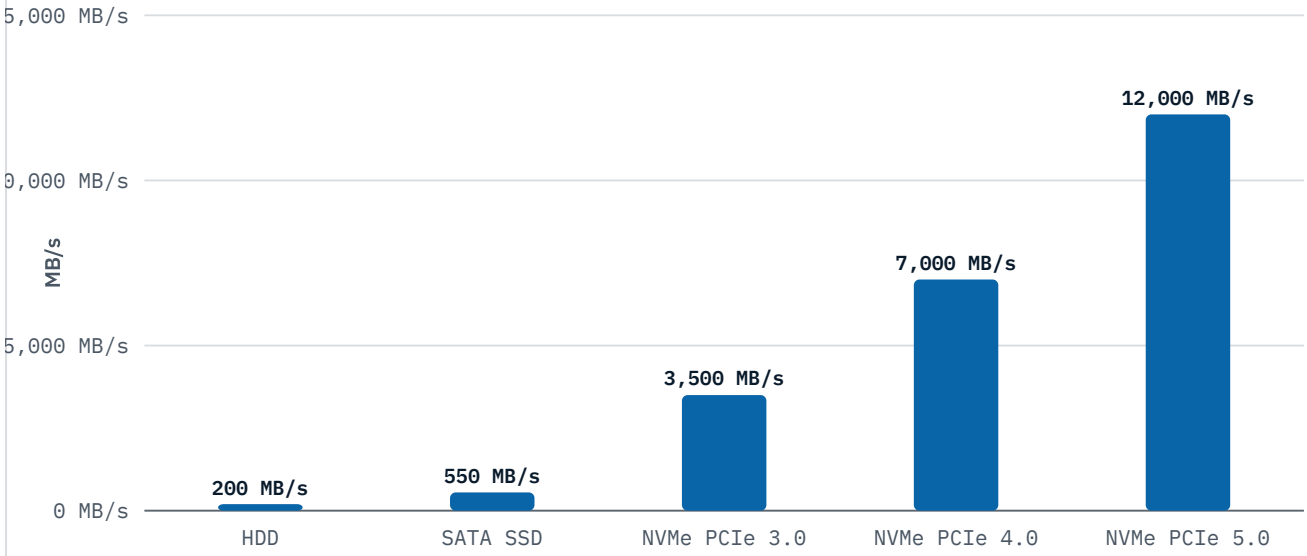
Storage holds everything the user cares about, power keeps everything alive and cooling keeps everything stable. Problems in these three areas often look like software faults: random restarts, freezes, slow boot or corrupted files. A good support specialist checks the physical basics before blaming the operating system.

Storage technologies compared

TYPE	INTERFACE	TYPICAL SEQUENTIAL SPEED	STRENGTHS	WATCH FOR
HDD (spinning disk)	SATA	150-250 MB/s	Cheap per terabyte, good for archives	Clicking, slow response, reallocated sectors
SATA SSD	SATA	Up to about 550 MB/s	Big upgrade over HDD, fits older PCs	Wear on cheap drives, sudden failure
NVMe SSD (M.2)	PCIe	About 3,000-14,000 MB/s (PCIe 3.0 to 5.0)	Fastest boot and app load	Heat throttling; check M.2 key and PCIe support
External / USB	USB 3.x / USB4	Varies by drive and port	Portable backups and transfers	Unplugged without ejecting; loss and theft

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Typical sequential read speed by drive type



Illustrative figures for good drives of each type; real results depend on the model, slot and workload. For users the biggest felt jump is from any HDD to any SSD, because access time drops from milliseconds to microseconds.

PRO TIP · CHECK THE M.2 SLOT BEFORE YOU BUY

M.2 is a **shape**, not a speed. An M.2 slot may support SATA, NVMe or both, and sizes such as 2242, 2260 and 2280 describe width and length in millimetres. Read the board or laptop manual before ordering.

Checking disk health and file systems

Windows PowerShell

```
Get-PhysicalDisk | Select-Object FriendlyName, MediaType, HealthStatus, Size
Get-Volume | Select-Object DriveLetter, FileSystem, SizeRemaining, Size
# Scan a volume for file system errors without taking it offline
Repair-Volume -DriveLetter C -Scan
# Classic equivalent (read-only check):
chkdsk C:
```

Run read-only scans first. Repairs that lock a volume should be scheduled with the user, after a backup.

Disk health on Linux

Linux host

```
lsblk -o NAME,SIZE,TYPE,MOUNTPOINT,MODEL
df -h
sudo smartctl -H /dev/sda
sudo smartctl -a /dev/nvme0 | grep -iE 'percentage used|media errors|temperature'
```


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Power supply sizing

MEASURE	FORMULA	WHAT IT TELLS YOU
Estimated load	`load (W) = CPU TDP + GPU board power + about 50-100 W for board, drives and fans`	A 65 W CPU with integrated graphics in an office PC usually draws well under 200 W.
Recommended PSU rating	`PSU rating = estimated peak load x 1.3 to 1.5`	Headroom keeps the PSU in its efficient range and handles brief power spikes from modern GPUs.
UPS runtime (rough)	`runtime (h) = battery Wh x inverter efficiency / load (W)`	A 600 Wh UPS at 90% efficiency running a 270 W load lasts about 2 hours in theory; real runtimes are lower as batteries age.

Power problems and symptoms

SYMPTOM	POSSIBLE CAUSE	FIRST CHECK
No lights, no fans	Dead outlet, switched-off PSU, failed PSU	Outlet, PSU switch, known-good cable
Fans spin, no display	RAM, CPU power cable, GPU power	8-pin CPU cable, reseat RAM, POST codes
Random restarts under load	Undersized or ageing PSU, overheating	Event logs, temperatures, PSU rating
Restarts during storms or load-shedding	Unstable mains power	UPS or AVR; surge protection

EXAMPLE · LOAD-SHEDDING AND CORRUPTED FILES

A small clinic in Sylhet saw repeated database corruption on its reception PC. The cause was power cuts during writes. A line-interactive UPS with a USB link that triggers a clean shutdown solved the problem, and the clinic stopped losing appointment records.

Cooling options

Air cooling

- Heatsink plus fan; simple and reliable
- Needs dust cleaning every 6-12 months
- Most office PCs and laptops

Liquid cooling (AIO)

- Pump and radiator move heat away
- Quieter under heavy load
- Pump failure can overheat quickly

Case airflow

- Front intake, rear and top exhaust
- Cable management keeps air paths clear
- Positive pressure reduces dust entry

Diagnosing overheating

1. **Confirm the symptom** Ask when it happens: under load, after a period of use or at random. Check fan noise and the heat of the chassis.
2. **Measure** Read CPU and GPU temperatures with firmware setup or a monitoring utility. Many CPUs throttle around 95-100 degrees C.
3. **Clean and inspect** Power off, then blow dust out with compressed air while holding fans still. Check that every fan spins.

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4. **Refresh thermal paste** On older systems, clean the old paste with isopropyl alcohol and apply a small new amount before refitting the cooler.
5. **Retest** Run a sustained load and confirm temperatures and stability are now normal.

SCENARIO · DECIDE · LAPTOP THAT SWITCHES OFF ON ITS OWN

A field officer's laptop shuts down without warning after about 20 minutes of video calls. It is two years old, very hot near the left vent and the fan is loud. Event logs show 'unexpected shutdown' with no blue screen.

(A) Clean the vents and fan, check temperatures and, if needed, replace the thermal paste
(B) Reinstall the operating system
(C) Replace the battery
(D) Lower the screen brightness

Circle your choice. Answer and feedback: module answer key.

Key terms

HDD
Hard disk drive: magnetic storage with spinning platters and moving heads; cheap per terabyte but slower and shock-sensitive.

SSD
Solid-state drive: flash storage with no moving parts, much faster and more durable than an HDD.

NVMe
A storage protocol that runs SSDs over PCIe lanes for very high speed and low latency.

S.M.A.R.T.
Self-monitoring data reported by drives, such as reallocated sectors, wear and temperature, used to predict failure.

PSU
Power supply unit: converts mains AC into the DC voltages a computer needs.

UPS
Uninterruptible power supply: a battery unit that keeps equipment running briefly during outages and can trigger safe shutdown.

Thermal throttling
Automatic reduction of processor speed to protect it when it gets too hot.

Key takeaways

- Physical faults in storage, power or cooling often look like software problems; check them first.
- M.2 is a shape: confirm SATA or NVMe support and the module length before ordering.
- Size a PSU at about 1.3-1.5 times the estimated peak load, and use a UPS where mains power is unstable.
- Overheating follows a pattern: measure, clean, refresh paste, then retest under load.
- Make sure a current backup exists before repairing any volume.

KNOWLEDGE CHECK

11. A user's PC has an M.2 SATA SSD. They buy a faster NVMe drive but it is not detected. What is the most likely cause?

(A) The drive is too large
(B) The PSU is too weak
(C) The M.2 slot only supports SATA
(D) The drive needs to be formatted first

12. An office PC uses a 65 W CPU, integrated graphics and one SSD. Which PSU rating is sensible?

(A) 300-400 W
(B) 150 W
(C) 1,200 W
(D) Exactly 65 W

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13. Which symptom most strongly suggests a failing hard disk drive?

(A) Clicking noises and growing reallocated sector counts

(B) The screen flickers

(C) Wi-Fi drops every few minutes and reconnects on its own without warning

(D) Keyboard keys repeat

14. What should you do FIRST before repairing errors on a user's data volume?

(A) Run the repair immediately so the errors do not spread

(B) Reformat the drive

(C) Update the BIOS

(D) Make sure a current backup exists

15. A desktop restarts during frequent power cuts and files become corrupted. What is the best fix?

(A) A bigger power supply unit with higher wattage headroom

(B) A UPS configured to trigger a clean shutdown

(C) More RAM

(D) A faster SSD

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · BUILD · LESSON 4 · 24 MIN

Peripherals, Displays, Printers and Mobile Devices

Everything users plug in, print with or carry around

- Identify common ports, cables and connectors and what they carry
- Troubleshoot display, audio and input problems
- Explain printer types, drivers and print queues and fix common faults
- Support laptops, phones and tablets including mobile device basics

Peripherals generate a surprising share of tickets: a projector that will not show the slides, a headset with no microphone, a label printer printing blank pages. These problems are rarely difficult, but they are visible and urgent because they usually happen in meetings, at counters or in front of customers.

Ports and connectors you must recognise

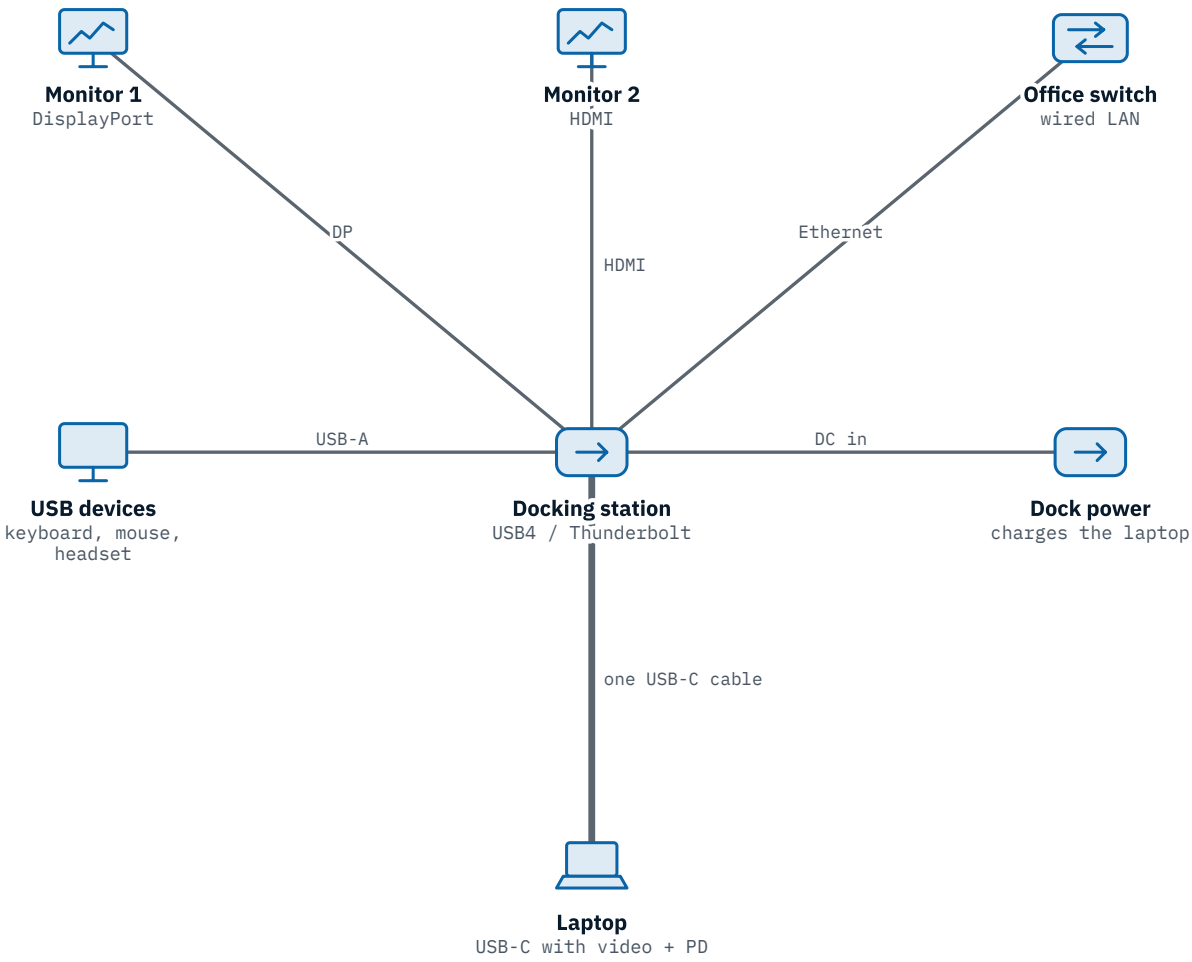
CONNECTOR	CARRIES	NOTES
USB-C	Data, video, power (depends on port)	Same shape, different abilities: look for USB4, Thunderbolt or DisplayPort marks and the charging wattage
USB-A	Data and limited power	Still common for keyboards, mice, drives
HDMI	Digital video and audio	TVs, projectors, monitors
DisplayPort	Digital video and audio	Monitors; supports daisy-chaining on some screens
RJ45	Ethernet	Wired network; may also carry PoE power
3.5 mm jack	Analogue audio	Headsets; combined mic and audio on many laptops

WATCH OUT · NOT EVERY USB-C PORT IS EQUAL

A USB-C port might offer only USB 2.0 data and slow charging, or USB4 or Thunderbolt with 40-80 Gbps, several screens and up to 240 W of charging with a suitable cable. When a docking station or monitor 'does not work', check what that specific port and cable support before replacing hardware.

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A typical USB-C desk setup



One cable carries video, data, network and charging. If one screen or charging fails, check what the laptop port and cable support (video mode, bandwidth, wattage) before blaming the dock.

Display troubleshooting in order

1. **Power and input** Is the monitor or projector on, and set to the right input source?
2. **Cable and port** Reseat both ends, try a known-good cable and another port.
3. **OS display settings** Use the display shortcut (for example Windows key + P on Windows) to choose duplicate or extend, and check resolution and refresh rate.
4. **Driver** Update or roll back the graphics driver if the issue began after a change.
5. **Hardware** Test the screen with another computer to isolate monitor versus PC.

Printer technologies

Laser

- Toner fused by heat; fast and cheap per page
- Common faults: smudging (fuser), streaks (drum)
- Office workgroups

Inkjet

- Liquid ink sprayed through nozzles
- Faults: clogged heads, dried ink
- Photos and low-volume colour

Thermal

- Heat-sensitive paper or ribbon
- Faults: wrong media, dirty print head
- Receipts, labels, wristbands in hospitals

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What happens when a user clicks Print

01 Application — Sends the document to the OS print system. →

02 Driver — Converts it into a language the printer understands. →

03 Spooler / queue — Holds jobs and sends them in order. →

04 Network or USB — Delivers the data to the printer. →

05 Printer — Processes, prints and reports status or errors.

Clearing a stuck print queue

```
Windows PowerShell

Get-Printer | Select-Object Name, PrinterStatus, PortName
Get-PrintJob -PrinterName 'HR-Laser'
# Remove only the stuck job, not everyone's work
Remove-PrintJob -PrinterName 'HR-Laser' -ID 14
# If the spooler itself is hung, restart the service
Restart-Service -Name Spooler
```

Run from an elevated session. Tell users before restarting the spooler on a shared print server, because queued jobs may need to be resent.

Mobile device support essentials

- **Battery**: swelling is a safety hazard; stop using the device and replace the battery.
- **Charging**: test another cable and charger; clean lint from the port gently.
- **Connectivity**: toggle airplane mode, forget and rejoin Wi-Fi, check mobile data limits.
- **Email and apps**: confirm account sign-in, MFA and that the device is enrolled in management if required.
- **Loss or theft**: report immediately so the device can be located, locked or wiped remotely.

EXAMPLE · THERMAL LABEL PRINTERS IN A GARMENT FACTORY

A Gazipur garment factory's packing line stopped because barcode labels were printing faint. The printer had been switched from direct-thermal labels to stock that needs a ribbon. Matching the media setting to the label type fixed it in five minutes; the fix went into the knowledge base for the night shift.

SCENARIO · DECIDE · PROJECTOR BEFORE A SEMINAR

A guest lecturer at a university in Rajshahi connects a laptop to the hall projector with HDMI. The projector shows 'No signal'. The seminar starts in five minutes.

(A) Check the projector input source, then use the display shortcut to duplicate the screen

(B) Replace the projector lamp

(C) Reinstall the laptop's graphics driver

(D) Ask the lecturer to email the slides to the IT office

Circle your choice. Answer and feedback: module answer key.

Key terms

USB-C
A reversible connector whose capabilities (data speed, video, charging power) depend on the port and cable behind it.

DisplayPort
A digital display interface for monitors that carries video and audio and can drive multiple screens.

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Print spooler
The operating system service that queues print jobs and sends them to printers.

Fuser
The heated unit in a laser printer that bonds toner to paper; failure causes smudging.

Remote wipe
A management action that erases a lost or stolen device to protect organisational data.

Printer driver
Software that translates documents into a language a specific printer understands.

Direct thermal printing
Printing by heating special paper so it darkens, without ink or ribbon.

Key takeaways

- USB-C is a connector, not a capability: data speed, video and charging depend on the port and cable.
- Work through display faults in order: power and input, cable and port, OS settings, driver, hardware.
- Remove a single stuck print job before restarting the spooler or reinstalling printers.
- Swollen batteries are a safety hazard: stop use and replace.
- Report lost or stolen devices at once so they can be locked or wiped.

KNOWLEDGE CHECK

16. Laser printer output smears when touched. Which part is most likely failing?

(A) The drum
(B) The fuser
(C) The toner chip
(D) The network card

17. A monitor connected via USB-C shows nothing, but a USB-C phone charger works in that port. What should you check?

(A) Whether the monitor's brightness is turned fully down
(B) The keyboard driver
(C) Whether the port supports video output
(D) The Wi-Fi password

18. Several users' jobs are stuck behind one corrupted job. What is the best first action?

(A) Remove the single stuck job from the queue
(B) Reinstall the printer on every PC
(C) Replace the printer
(D) Disable the print spooler permanently and use USB printers

19. A phone's battery case is visibly swollen. What should you advise?

(A) Keep using it until it stops charging
(B) Press the case flat
(C) Charge it fully overnight to recalibrate the battery, then check it again
(D) Stop using it, keep it away from heat and arrange a battery replacement

20. What should be done FIRST when an employee reports a company phone stolen?

(A) Wait 24 hours in case it is found or handed in at the office reception
(B) Order a new phone
(C) Lock or wipe it through device management and secure the account
(D) Change the Wi-Fi password

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21. A laptop that needs 100 W shows a 'slow charger' warning only when connected through a new dock. Monitors and network work. What is the most likely cause?

(A) The second monitor is running at a refresh rate that is far too high

(B) The dock's Ethernet port is connected to a slow switch

(C) The cable or dock delivers less power than the laptop needs

(D) The battery must be fully drained once before docking works

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 1 · BUILD · LESSON 5 · 30 MIN

Building, Upgrading and Repairing a PC

A safe, repeatable process from parts on a bench to a working system

- Work safely with electricity, ESD and batteries
- Assemble a PC in the correct order and verify it with POST
- Use a structured approach to diagnose hardware faults and choose repair or replace
- Record builds and repairs so others can support the device later

Building or repairing a PC is a core support skill even in a cloud-first world. Office desktops, lab machines, kiosks and point-of-sale terminals still need RAM, SSD and power-supply swaps. A methodical builder works safely, checks compatibility before opening the box and documents every change.

Bench safety checklist

☐ Power off, switch off the PSU and unplug the mains cable before opening the case.

☐ Hold the power button for five seconds to discharge residual power.

☐ Wear an antistatic wrist strap clipped to bare metal on the chassis or an ESD mat.

☐ Keep parts in antistatic bags until you install them.

☐ Never open a power supply or monitor: they hold dangerous charge in capacitors.

☐ Lift heavy equipment such as UPS units and servers with your legs, or with help.

☐ Dispose of batteries, toner and e-waste through approved recyclers, not normal rubbish.

☐ Remove watches and rings, and keep drinks off the bench.

Assembly order for a desktop PC

1. **Prepare the motherboard outside the case** Install the CPU (align the triangle marker), apply thermal paste if the cooler has none, fit the cooler, then the RAM and M.2 SSD.
2. **Prepare the case** Fit the I/O shield if separate and check standoffs match the board's mounting holes.
3. **Mount the board** Lower it in at an angle into the I/O shield and screw it to the standoffs without overtightening.
4. **Install the PSU and power cables** Connect the 24-pin ATX and 8-pin CPU power, then SATA or PCIe power as needed.
5. **Add expansion cards and drives** Seat the GPU in the top PCIe x16 slot and secure it; connect SATA drives.
6. **Front panel and fans** Connect power switch, LEDs, USB and audio headers and fan cables using the manual's diagram.
7. **First boot** Before closing the case, power on, enter UEFI setup, confirm CPU, RAM and drives are detected and temperatures are normal.

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The boot sequence when you press power

01 Power good
The PSU stabilises voltages and signals the board.

→

02 POST
Firmware tests CPU, RAM and video; failures produce beep or LED codes.

→

03 UEFI settings
Firmware reads boot order and Secure Boot settings.

→

04 Boot loader
Loads from the boot drive's EFI partition.

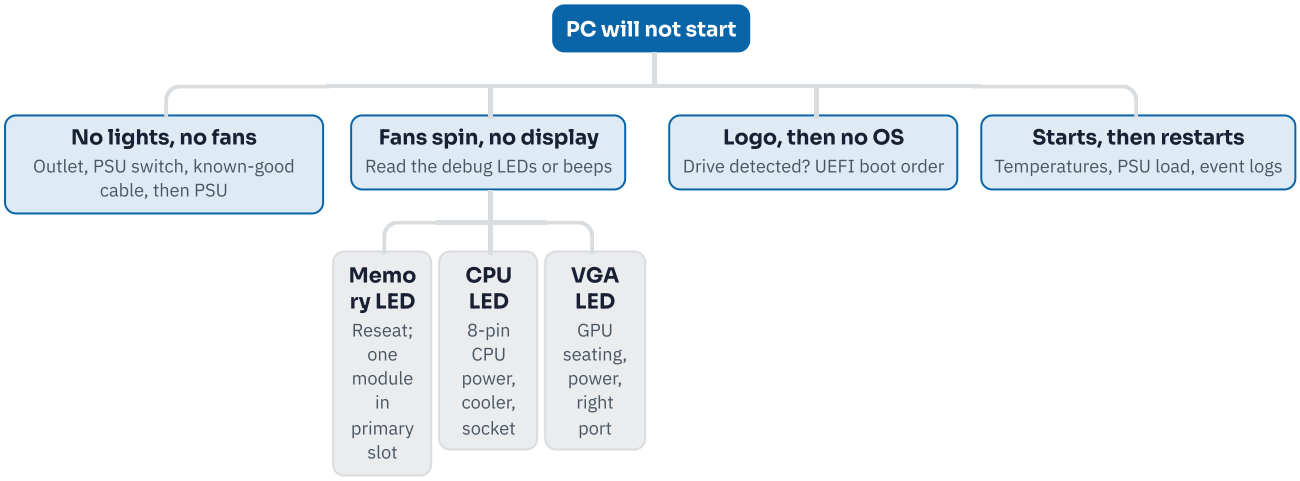
→

05 OS kernel
The operating system starts drivers and services.

Reading POST failures

SYMPTOM OR CODE	LIKELY AREA	ACTION
Memory LED / repeating beeps	RAM	Reseat modules, try one module in the primary slot
CPU LED	CPU or CPU power	Check 8-pin CPU power, socket pins, cooler pressure
VGA LED / no display	Graphics	Reseat GPU, connect monitor to the right port, check GPU power
Boot device LED	Storage or boot order	Check drive detection and UEFI boot order
Clock reset at every start	CMOS battery	Replace the CR2032 coin cell

When a PC will not start: a symptom-first decision tree



Start from what the user actually sees, then let the board's LEDs or beep codes narrow it down. Each leaf is one safe test to run before any part is replaced.

PRO TIP · MINIMUM-CONFIGURATION TESTING
When a system will not POST, strip it to the minimum: board, CPU with cooler, one RAM module and PSU, with on-board video. If it starts, add parts back one at a time. The part that brings the fault back is the culprit.

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Repair or replace?

Repair when

- The part is cheap and standard (RAM, SSD, PSU, fan)
- The device is within its planned lifecycle
- Downtime is acceptable and parts are available
- Data or software setup is complex to rebuild

Replace when

- Repair costs exceed about 50% of a new device
- The device cannot run a supported OS (for example no TPM 2.0 for Windows 11) or is near end of life
- Parts are soldered or proprietary and hard to get
- Failures keep repeating on the same model

CASE STUDY · THE LAB OF TWENTY SLOW PCS

A private university in Dhaka has a computer lab of 20 desktops bought in 2020: quad-core CPUs, 8 GB DDR4 (one module), 1 TB HDDs. Students complain that the PCs take four minutes to boot and freeze in programming tools. The budget is limited and the PCs must last two more years. The CPUs are still supported by current Windows and Linux releases.

1. What two upgrades would give the biggest improvement per taka spent?

2. How would you roll the work out without closing the lab for a week?

3. What would you record for each PC?

Model answer: module answer key.

TEMPLATE **PC build and repair record**

Record every build or repair so the next technician knows exactly what was changed and why.

ASSET TAG	DATE	PROBLEM OR PURPOSE	PARTS INSTALLED OR REPLACED	TESTS PERFORMED	RESULT	TECHNICIAN
LAB-PC-07	*2026-03-12*	*Slow boot, freezing*	*500 GB NVMe SSD (SN 4F9...), +8 GB DDR4*	*POST, memory test 1 pass, boot time 18 s*	*Resolved*	*N. Akter*

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SCENARIO · DECIDE · NO DISPLAY AFTER A RAM UPGRADE

You add a second RAM module to a desktop. It powers on, fans spin, but there is no display and the motherboard memory LED stays lit.

(A) Replace the graphics card

(B) Reinstall the operating system

(C) Replace the power supply

(D) Reseat both modules firmly, then test the new module alone in the primary slot

Circle your choice. Answer and feedback: module answer key.

REFLECT

Which repair task in this module do you feel least confident about? Write down how you will practise it safely, for example on an old donated PC or in a supervised lab.

Key terms

ESD
Electrostatic discharge: a sudden flow of static electricity that can damage components without visible signs.

UEFI
Unified Extensible Firmware Interface: modern firmware that initialises hardware, offers setup screens and starts the boot loader.

Standoff
A small spacer that holds the motherboard away from the metal case to prevent short circuits.

Minimum configuration
Testing with only the essential parts installed to isolate a hardware fault.

POST
Power-on self-test: firmware checks of core hardware that run before the operating system loads.

CMOS battery
A coin cell on the motherboard that keeps firmware settings and the real-time clock when the PC is unplugged.

Thermal paste
A compound that fills microscopic gaps between a CPU and its cooler to improve heat transfer.

Key takeaways

- Safety first: power off, unplug, discharge, use ESD protection and never open a PSU.
- Assemble in a fixed order and test POST before closing the case.
- No POST? Strip to minimum configuration and add parts back one at a time.
- Repair cheap standard parts; replace when costs pass about half a new device or OS support ends.
- Record every build and repair against the asset tag.

KNOWLEDGE CHECK

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22. A PC loses its date and time every time it is unplugged. What should you replace?

(A) The PSU
(B) The CMOS battery
(C) The SSD
(D) The RAM

23. Why should you never open a power supply unit during repair?

(A) It voids the operating system licence
(B) It contains toner
(C) It is sealed at the factory, and opening it voids the PSU's warranty
(D) Capacitors inside can hold a dangerous charge even when unplugged

24. A new build will not POST. What is the best diagnostic approach?

(A) Replace all the parts at once with new ones from a different supplier
(B) Install the OS from USB
(C) Update the SSD firmware
(D) Strip to minimum configuration and add parts back one at a time

25. Repairing a five-year-old laptop's motherboard would cost 70% of a new model, and its OS is near end of support. What should you recommend?

(A) Repair it because repair is always greener
(B) Keep using it without repair
(C) Upgrade its RAM
(D) Replace it and recover the data

26. Which is the correct place to clip an antistatic wrist strap?

(A) The plastic front panel
(B) The CPU heatsink fins, because they are the closest metal part
(C) The user's desk lamp
(D) Bare metal on the chassis or a grounded ESD mat

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · BUILD

Answer key

Knowledge checks

1. A Drive makers count 10^12 bytes, while the OS divides by 2^30 per GiB, giving about 931. Nothing is faulty, hidden or misformatted.

2. D Tier 1 service desk staff handle known, routine fixes such as resets when self-service fails. Tier 3 and vendors deal with complex engineering problems.

3. D RAM stores the data and programs the CPU is working on right now, so it belongs to storage, although it is volatile. It does not take input or produce output itself.

4. A 1 GB is 8,000 megabits; 8,000 / 100 = 80 seconds. Forgetting to multiply bytes by 8 gives the wrong answer of 10 seconds.

5. A Plain explanations and confirming the result show communication and ownership. Jargon confuses users, silent closure skips confirmation and escalating unhappy users avoids responsibility.

6. D DDR4 and DDR5 modules have the notch in different places, so they cannot be inserted into the wrong slot. There is no backwards compatibility mode or firmware fix.

7. C Boards often need updated UEFI firmware to recognise newer CPUs on the same socket. RAM, PSU and SSD firmware do not control CPU support.

8. D 3,200 MT/s x 8 bytes = 25,600 MB/s, about 25.6 GB/s. 51.2 GB/s would be two channels.

9. B Recommended slot pairs enable dual channel and better bandwidth. Slot choice does not affect battery life, power much, or ECC support.

10. A LPDDR is soldered to the board, so it cannot be upgraded in practice. The right advice is to buy enough memory at purchase or replace the device.

11. C Some M.2 slots support only SATA. An NVMe drive in a SATA-only slot is not detected. Size, PSU capacity and formatting do not stop detection by firmware.

12. A Total load is roughly 65 W + 75 W, so a quality 300-400 W unit gives ample headroom and efficiency. 150 W is tight; 1,200 W is wasteful; 65 W ignores other parts.

13. A Clicking and reallocated sectors indicate mechanical and surface failure. The other symptoms point to display, network and input faults.

14. D Repairs can make things worse on a failing disk, so protect the data first. Formatting destroys data and firmware is unrelated.

15. B A UPS bridges short outages and shuts the system down cleanly during long ones. A bigger PSU, more RAM or a faster SSD do nothing when mains power disappears.

16. B The fuser bonds toner with heat; if it fails, toner smears. Drum faults usually cause repeated marks, and toner chip or network faults stop printing.

17. C USB-C ports may provide charging without video. The port's DisplayPort Alt Mode or USB4 support decides whether video works.

18. A Removing the corrupted job releases the queue with minimal disruption. Reinstalling, replacing or disabling the spooler are excessive.

19. D Swollen lithium batteries can catch fire, so use should stop and the battery be replaced. Pressing or charging it increases the risk.

20. C Speed protects data: locking or wiping and securing accounts limits exposure. Waiting increases risk, and the Wi-Fi password is not the main issue.

21. C Charging power depends on the dock's rating, how much it keeps for itself and the cable's rating. Monitor refresh, network speed and battery draining do not limit the charging wattage.

22. B The CMOS coin cell keeps the clock and settings without mains power. A PSU, SSD or RAM fault would show other symptoms.

23. D PSU capacitors can store lethal charge. Faulty PSUs are replaced as a unit, not repaired by support staff.

24. D Minimum configuration isolates the faulty component systematically. Swapping everything is wasteful, and the OS and SSD firmware are irrelevant before POST succeeds.

25. D High repair cost and approaching end of support make replacement the better value. RAM does not fix a failed board.

26. D The strap must connect you to the same ground potential as the equipment, via bare chassis metal or an ESD mat. Plastic and unrelated objects do not ground you.

Scenarios & case studies

> **Scenario · Your first morning on the service desk**
> **C** is best. Correct. You capture everything, work on the item with the most urgent business impact and make sure the silent risk is not forgotten.

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Scenario · Upgrade request from accounts
B is best. Correct. It fixes the memory pressure, enables dual-channel bandwidth and fits the budget.

Scenario · Laptop that switches off on its own
A is best. Correct. Heat-related shutdowns under load with a loud fan and a hot vent are classic symptoms of blocked cooling.

Scenario · Projector before a seminar
A is best. Correct. 'No signal' means the projector is on but not receiving video. Wrong input or the laptop not sending output are the quickest likely causes.

Case study · The lab of twenty slow PCs
Upgrades: replace each HDD with a 500 GB SATA or NVMe SSD (boot time falls from minutes to seconds) and add a second 8 GB module for 16 GB in dual channel. **Rollout:** build and test one PC as a reference, capture a standard image, then upgrade four or five PCs per evening, deploying the image over the network. **Records:** asset tag, serial, parts installed with serials and warranty dates, image version, date, technician and test results (POST, memory test, boot time).

Scenario · No display after a RAM upgrade
D is best. Correct. A partly seated or incompatible module is the most likely cause given the timing and the memory LED.

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02

MODULE 2 · BOOT

Operating Systems: Windows, Linux and macOS

Install, configure and support the three main desktop platforms and mobile devices

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain OS components, partitioning, file systems and the UEFI boot process
- Use Windows tools and PowerShell to diagnose and repair common problems
- Work confidently in a Linux shell with permissions, packages, services and logs
- Support macOS and mobile devices including encryption, backup and MDM
- Deploy standard builds and manage software and licences

LESSONS

1. Operating System Fundamentals and the Boot Process · 24 min
2. Windows Essentials for Support · 30 min
3. Linux Essentials · 32 min
4. macOS and Mobile Operating Systems · 26 min
5. Installing, Imaging and Managing Software · 28 min

MODULE 2 · BOOT · LESSON 1 · 24 MIN

Operating System Fundamentals and the Boot Process

What an OS does, how disks are laid out and how a computer starts

- Explain the main jobs of an operating system: kernel, drivers, processes, memory and files
- Compare GPT and MBR partitioning and common file systems
- Describe the UEFI boot process and Secure Boot
- Explain OS lifecycles and why unsupported versions are a risk

An **operating system** (OS) sits between hardware and the apps people use. It decides which program gets the CPU, hands out memory, talks to devices through drivers and organises data into files. Windows, Linux and macOS look different, but they solve the same problems, so the concepts you learn here transfer across all three.

Layers of a modern operating system (top first)

Applications	`Browsers, office suites, line-of-business apps`
User interface	`Desktop shell and windows` `Command-line shells: PowerShell, Bash, zsh`
System services	`Networking, printing, updates, logging, security`
Kernel	`Process scheduling` `Memory management` `File systems` `Security enforcement`
Drivers and firmware	`Device drivers` `UEFI firmware` `Hardware`

Core OS jobs and what breaks when they fail

JOB	WHAT IT DOES	TYPICAL SYMPTOM WHEN IT FAILS
Process management	Runs programs and shares CPU time	One app hogs CPU; system feels frozen
Memory management	Allocates RAM and uses disk as overflow	Heavy paging, 'out of memory' errors
File systems	Organises data into files and folders with permissions	Corrupt files, 'access denied' errors
Device drivers	Let the OS use hardware	Device missing, crashes after a driver update
Security	Accounts, permissions, isolation	Users can see or change what they should not

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Partition styles

GPT (GUID partition table)

- Used with UEFI on modern systems
- Supports disks far larger than 2 TiB
- Up to 128 partitions by default on Windows
- Keeps a backup header at the end of the disk

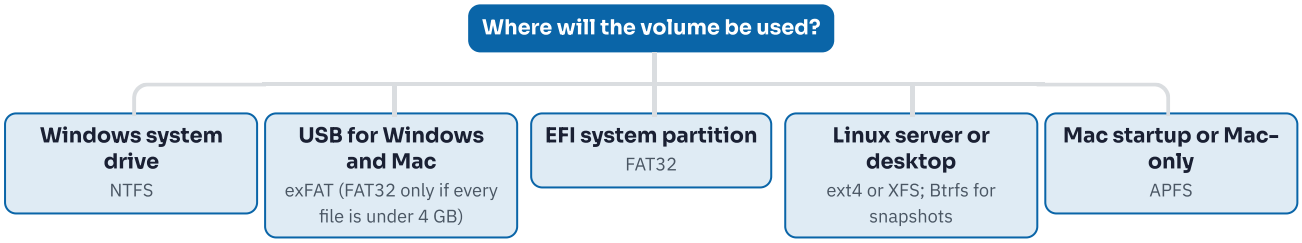
MBR (master boot record)

- Legacy BIOS-era scheme
- Limited to 2 TiB per disk
- Four primary partitions
- Still seen on old PCs and small USB sticks

File systems you will meet

FILE SYSTEM	MAIN OS	NOTES
NTFS	Windows	Permissions, encryption, journaling; default for Windows system drives
exFAT	Cross-platform	Large files on USB drives; readable by Windows, macOS and most Linux
FAT32	Cross-platform	Wide compatibility; 4 GB maximum file size; used for EFI system partitions
ext4 / XFS / Btrfs	Linux	Journaling Linux file systems; Btrfs adds snapshots
APFS	macOS	Snapshots, encryption, space sharing between volumes

Choosing a file system



Start from where the data must be read and written, not from habit. Most 'file too large' and 'read-only on the Mac' tickets come from picking the wrong branch.

How a UEFI system boots

01 Firmware
POST, then reads boot entries stored in firmware NVRAM.

→

02 EFI system partition
Small FAT32 partition holding boot loaders.

→

03 Secure Boot check
Firmware verifies the boot loader's signature.

→

04 Boot loader
Loads the kernel and initial drivers.

→

05 Kernel and services
Mounts file systems, starts services and shows the sign-in screen.

DEFINITION · SECURE BOOT

A UEFI feature that only allows boot loaders and drivers signed by trusted keys to run. It blocks many boot-level rootkits. If you must boot an unsigned tool, turn Secure Boot back on afterwards and record the change.

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WATCH OUT · END OF SUPPORT MEANS END OF SECURITY FIXES

When a vendor stops supporting an OS version, new vulnerabilities are never patched. Windows 10 reached end of support on 14 October 2025, so PCs that cannot move to Windows 11 need a dated plan: replacement, paid Extended Security Updates (ESU) as a short bridge, or another supported OS. Keep a list of OS versions in use with their end-of-support dates, and plan upgrades a year ahead.

Desktop Windows support milestones

JUL 2015
Windows 10 released
Becomes the main business desktop for a decade.

OCT 2021
Windows 11 released
Needs TPM 2.0, UEFI with Secure Boot capability and a supported 64-bit CPU.

2024-2025
Windows 11 24H2 and 25H2
Each yearly version gets 24 months of support (Home, Pro) or 36 months (Enterprise, Education).

14 OCT 2025
Windows 10 end of support
Free security updates stop; paid ESU is the only bridge.

OCT 2026
Consumer ESU ends
Business ESU can be renewed yearly, at rising cost.

OCT 2028
Last year of business ESU
No Windows 10 fixes after this; check the vendor's current timetable.

Plan replacements against these dates, not against the day a PC breaks. ESU buys time to migrate; it is not a long-term strategy.

32-bit and 64-bit

64-bit OS

- Uses far more than 4 GB of RAM
- Required for modern apps and security features
- Runs most older 32-bit apps

32-bit OS

- Limited to about 4 GB of addressable memory
- Largely obsolete on PCs
- Cannot run 64-bit apps or drivers

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SCENARIO · DECIDE · A USB STICK THAT WILL NOT TAKE A BIG FILE

A user tries to copy a 6 GB video onto a new USB stick to show at a partner's office, which uses Macs. The copy fails with 'file too large', although the stick has 30 GB free.

(A) Reformat the stick as NTFS

(B) Compress the file into a ZIP

(C) Buy a bigger USB stick

(D) Reformat the stick as exFAT after backing up its contents

Circle your choice. Answer and feedback: module answer key.

Key terms

Operating system
Software that manages hardware and provides services such as processes, memory, files and security for applications.

Device driver
Software that lets the operating system communicate with a specific piece of hardware.

File system
The structure an OS uses to store, name and protect files on a volume, such as NTFS, ext4 or APFS.

End of support
The date after which a vendor no longer releases security updates or fixes for a product version.

Kernel
The core of the OS that runs with full privileges and controls CPU, memory and devices.

GPT
GUID partition table: the modern disk partitioning scheme used with UEFI, supporting very large disks.

EFI system partition
A small FAT32 partition containing boot loaders used by UEFI firmware.

Key takeaways

- The kernel, drivers and services do the same jobs on every OS, so concepts transfer.
- Modern PCs use UEFI with GPT disks and Secure Boot; MBR is limited to 2 TiB.
- Pick the file system by where it will be used: NTFS, exFAT, ext4/XFS or APFS.
- Windows 10 support ended on 14 October 2025: track every OS's end-of-support date and plan ahead.
- An unsupported OS still runs, but every new vulnerability stays open.

KNOWLEDGE CHECK

1. Which file system should you choose for a USB drive shared between Windows and macOS that must hold 8 GB files?

   (A) exFAT
   (B) FAT32
   (C) ext4
   (D) APFS

2. A 4 TB disk is initialised as MBR. How much of it can be used?

   (A) About 2 TiB
   (B) All 4 TB
   (C) Exactly 3 TB
   (D) Only 512 GB

3. What does Secure Boot primarily protect against?

   (A) Phishing emails that contain malicious links or attachments
   (B) Unsigned or tampered boot loaders and rootkits
   (C) Weak passwords
   (D) Overheating

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4. Why is running an OS past its end-of-support date risky?

   (A) It will stop booting
   (B) It cannot connect to Wi-Fi
   (C) Its licence becomes illegal to use, and it stops booting after a grace period
   (D) New vulnerabilities will not be patched

5. Which component translates OS requests into instructions a specific network card understands?

   (A) The device driver
   (B) The kernel scheduler
   (C) The file system
   (D) The boot loader

6. A finance PC still runs Windows 10 in late 2026, and the organisation never bought ESU. Which statement is accurate?

   (A) It will stop booting after the end-of-support date unless it is reinstalled
   (B) It gets no security fixes, so it should be upgraded, replaced or isolated
   (C) Its licence becomes invalid and it can no longer be activated or used legally
   (D) It keeps receiving security patches, but new apps can no longer be installed

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MODULE 2 · BOOT · LESSON 2 · 30 MIN

Windows Essentials for Support

The tools and commands you will use on Windows every day

- Navigate Settings, Task Manager, Event Viewer, Services and Device Manager confidently
- Use PowerShell for system information, processes, services and logs
- Repair system files and use safe mode and recovery options
- Explain user account control and edition differences relevant to business

Windows is still the most common desktop OS in offices, banks, hospitals and universities, so it dominates support queues. You do not need to memorise every menu. You need a reliable toolkit: where to look for evidence, how to stop or restart things safely and how to repair the system without losing user data.

The Windows support toolkit

TOOL	OPEN WITH	USE IT FOR
Settings	Win + I	Accounts, updates, display, apps, privacy
Task Manager	Ctrl + Shift + Esc	CPU, memory, disk use; end hung apps; startup apps
Event Viewer	eventvwr.msc	System and application errors with timestamps
Services	services.msc	Start, stop and set startup type of background services
Device Manager	devmgmt.msc	Driver status, roll back or update drivers
Disk Management	diskmgmt.msc	Partitions, drive letters, new disks
Computer Management	compmgmt.msc	Local users and groups, shares, all of the above in one console

Business-relevant Windows editions

Home

- Consumer use
- Cannot join a corporate domain
- Limited management features

Pro / Business

- Joins directory domains and cloud identity
- Group policy and drive encryption
- Remote Desktop host

Enterprise / Education

- Advanced security and management
- Licensed through volume agreements
- Used by large organisations and universities

Is this PC ready for Windows 11?

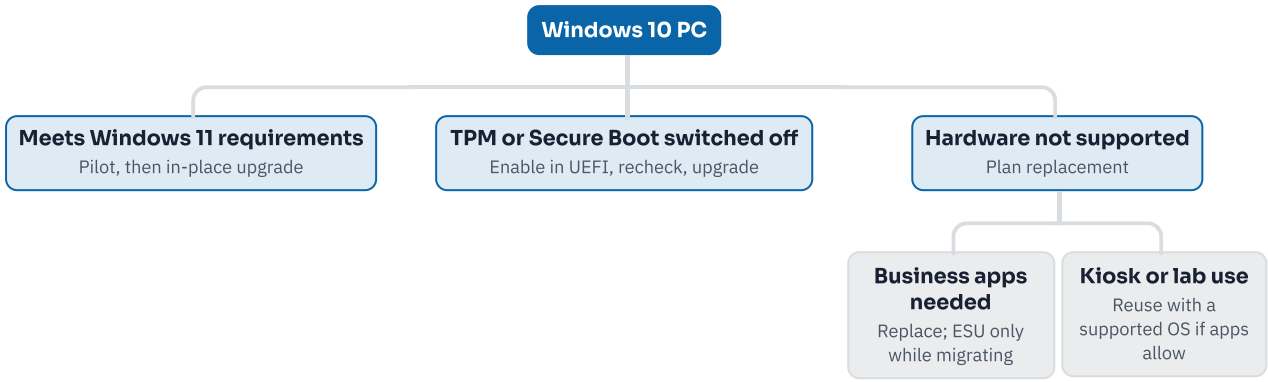
```
Windows PowerShell (elevated)

Get-Tpm | Select-Object TpmPresent, TpmReady, TpmEnabled
(Get-CimInstance -Namespace root/cimv2/security/microsofttpm -ClassName Win32_Tpm).SpecVersion   # must start with 2.0
Confirm-SecureBootUEFI          # True = on; an error usually means legacy BIOS mode
Get-ComputerInfo -Property OsName, OsVersion, BiosFirmwareType
Get-CimInstance Win32_Processor | Select-Object Name   # compare with the supported CPU list
```

A PC that reports no TPM often only needs firmware TPM (fTPM or PTT) and Secure Boot switched on in UEFI setup. Record the change on the asset.

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A Windows 10 PC after end of support



Check the requirements before condemning a PC: many 'unsupported' machines only need a firmware setting. Never leave an unsupported OS on the network without a dated plan.

Everyday diagnostics in PowerShell

```
Windows PowerShell

Get-ComputerInfo -Property OsName, OsVersion, CsName, OsLastBootUpTime
Get-Process | Sort-Object CPU -Descending | Select-Object -First 5 Name, CPU, WS
Get-Service -Name Spooler, wuauserv | Select-Object Name, Status, StartType
Restart-Service -Name Spooler
Get-WinEvent -LogName System -MaxEvents 20 | Where-Object LevelDisplayName -eq 'Error'
Get-CimInstance Win32_StartupCommand | Select-Object Name, Command
```

Everything here is read-only except Restart-Service, which needs an elevated (Run as administrator) session.

> **INSIGHT · EVENT VIEWER TURNS GUESSES INTO EVIDENCE**
> Look at the System and Application logs around the time of the problem. Note the **source** and **event ID**, then search your knowledge base. An unexpected shutdown, a disk warning and a driver crash look identical to a user, but very different in the logs.

Repairing system files

```
Windows PowerShell

# Run in an elevated session. Repair the component store first, then system files
DISM /Online /Cleanup-Image /ScanHealth
DISM /Online /Cleanup-Image /RestoreHealth
sfc /scannow
# Review results, then restart if repairs were made
Restart-Computer -Confirm
```

Recovery options from least to most disruptive

1. **Restart and undo the last change** Uninstall the latest update or app, or roll back a driver in Device Manager.
2. **Safe mode** Boot with minimal drivers and services to remove a problem driver or malware.
3. **System Restore** Return system files and settings to a restore point without touching personal files, if enabled.
4. **Startup repair** Use the Windows Recovery Environment to fix boot files.
5. **Reset this PC** Reinstall Windows keeping or removing files; back up first.

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6. **Reimage** Deploy the organisation's standard image, then restore user data from backup or cloud.

DEFINITION · USER ACCOUNT CONTROL (UAC)

UAC runs even administrators with standard rights until a task needs elevation, then asks for consent or credentials. It limits the damage a mistake or malicious program can do. Do not disable it to 'save time'.

WATCH OUT · THE REGISTRY IS NOT A PLAYGROUND

The registry holds Windows and app settings. Edit it only when a trusted, documented fix requires it, export the key first and never run 'registry cleaner' tools. A wrong change can stop the PC from booting.

SCENARIO · DECIDE · SLOW PC AFTER AN UPDATE

A loan officer at a microfinance office in Bogura says her PC has been slow since yesterday. Task Manager shows 100% disk activity from an indexing service and an update installing in the background. The PC has an HDD.

(A) Disable the update service permanently

(B) Explain that the update is finishing, let it complete, check again after a restart and log the HDD for an SSD upgrade

(C) Reset the PC to factory settings

(D) Run a registry cleaner

Circle your choice. Answer and feedback: module answer key.

Key terms

Task Manager
A Windows tool to view and end processes, monitor performance and manage startup apps.

Windows service
A background program that starts with the system and runs without user interaction, such as the print spooler.

DISM
Deployment Image Servicing and Management: repairs the Windows component store and images.

User Account Control
A Windows feature that requires consent or credentials before tasks run with administrative rights.

Event Viewer
The Windows console that displays system, security and application logs with event IDs and timestamps.

SFC
System File Checker: scans and restores protected Windows system files.

Safe mode
A diagnostic start-up mode that loads only essential drivers and services.

Key takeaways

- Event Viewer turns guesses into evidence: note the source, event ID and time.
- Repair system files with DISM first, then sfc.
- Use the least disruptive recovery option that fixes the fault, from rollback up to reimage.
- Windows 11 needs TPM 2.0, Secure Boot-capable UEFI and a supported CPU; check before you upgrade or replace.
- Keep UAC on and edit the registry only with a documented fix and a backup of the key.

KNOWLEDGE CHECK

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7. Which tool shows the timestamp and event ID of an unexpected shutdown?
   - (A) Event Viewer
   - (B) Device Manager
   - (C) Disk Management
   - (D) Settings > Personalisation
8. What is the recommended order when repairing corrupted Windows system files?
   - (A) sfc then DISM
   - (B) chkdsk then reset the PC
   - (C) DISM RestoreHealth then sfc /scannow
   - (D) Registry cleaner then sfc
9. A new graphics driver causes crashes on start-up. Which is the least disruptive fix?
   - (A) Reset this PC
   - (B) Reimage the laptop
   - (C) Boot to safe mode and roll back the driver
   - (D) Replace the graphics card with a model from a different manufacturer
10. Which Windows edition cannot join a corporate directory domain?
    - (A) Pro
    - (B) Enterprise
    - (C) Education
    - (D) Home
11. Why should UAC stay enabled for administrators?
    - (A) It speeds up the system
    - (B) It is required for Wi-Fi and for network drives to connect at sign-in
    - (C) It encrypts the disk
    - (D) It stops admin rights being used silently by mistakes or malware
12. A 2020 desktop fails the Windows 11 check only because 'TPM 2.0 not detected'. Its CPU is on the supported list. What should you try first?
    - (A) Buy and fit a plug-in TPM module from an online marketplace first
    - (B) Replace the motherboard and CPU with a newer supported generation
    - (C) Keep Windows 10, since the PC still works and the data is safe
    - (D) Turn on firmware TPM and Secure Boot in UEFI, then recheck

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MODULE 2 · BOOT · LESSON 3 · 32 MIN

Linux Essentials

Distributions, the shell, files, permissions, packages and services

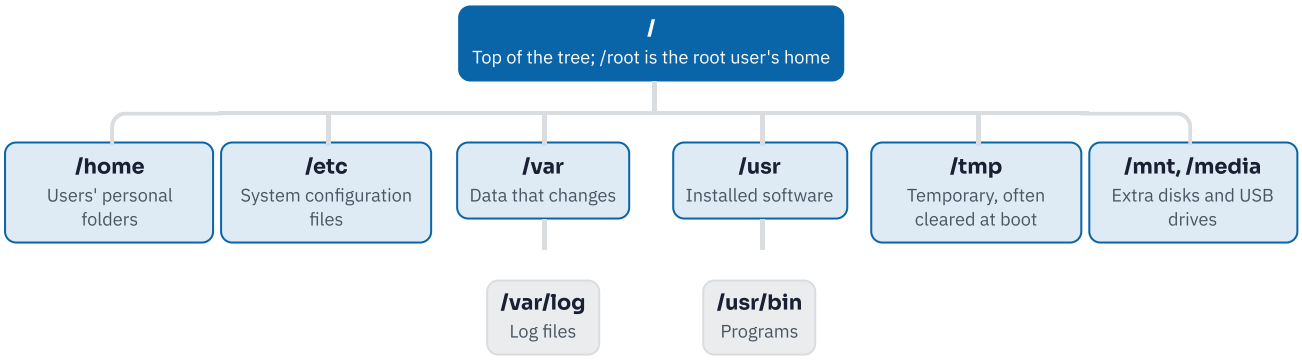
- Navigate the Linux file hierarchy and use core shell commands safely
- Read and change file permissions and ownership
- Install and update software with package managers
- Manage services and read logs with systemd tools

Linux runs most of the world's servers, cloud workloads, network appliances and many developer laptops. Support specialists meet it on web servers, lab machines, kiosks and cloud instances. The good news: a few dozen commands cover most support work, and they are the same across distributions.

Distribution families

FAMILY	EXAMPLES	PACKAGE TOOL	COMMON IN
Debian-based	Debian, Ubuntu, Linux Mint	apt	Desktops, cloud servers, education
Red Hat-based	RHEL, Rocky Linux, AlmaLinux, Fedora	dnf	Enterprise servers
SUSE	openSUSE, SLES	zypper	Enterprise, Europe
Others	Arch, Alpine	pacman, apk	Enthusiasts, containers

Key Linux directories



Everything hangs from a single root, even extra disks, which are mounted into the tree. When a ticket mentions settings, look in /etc; when it mentions errors, look in /var/log or the journal.

Moving around and inspecting files

Linux host

```
pwd                       # where am I?
ls -lah /var/log          # long listing incl. hidden files and sizes
cd ~/Documents
cat /etc/os-release       # distribution and version
tail -n 50 /var/log/syslog   # if rsyslog is installed; otherwise: journalctl -n 50
grep -i error /var/log/syslog | tail -n 20
du -sh ~/* | sort -h      # which folders use the most space
```


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WATCH OUT · RESPECT SUDO AND RM

`sudo` runs a command as root, with no safety net. `rm -r` deletes without a recycle bin. Read every command twice before pressing Enter, never paste commands you do not understand from the internet and prefer moving files to a backup folder over deleting them.

Reading permissions: -rwxr-x---

Owner (rwx)

- Read, write and execute
- Usually the user who created the file

Group (r-x)

- Read and execute, no write
- Members of the file's group

Others (---)

- No access at all
- Everyone else on the system

Numeric (octal) permissions

MEASURE	FORMULA	WHAT IT TELLS YOU
Permission values	`read = 4, write = 2, execute = 1`	Add the values for each of owner, group and others.
Example 750	`owner 7 (4+2+1), group 5 (4+1), others 0`	Owner full control, group can read and run, others nothing; same as rwxr-x---.
Example 640	`owner 6 (4+2), group 4, others 0`	Typical for a private config file the service group must read.

Changing permissions and ownership

Linux host

```
ls -l report.sh
-rw-r--r-- 1 rahim staff 1204 Mar 3 10:15 report.sh
chmod 750 report.sh                  # owner rwx, group r-x, others none
sudo chown rahim:finance report.sh   # owner rahim, group finance
ls -l report.sh
-rwxr-x--- 1 rahim finance 1204 Mar 3 10:15 report.sh
```

Packages and services

Linux host

```
sudo apt update && sudo apt upgrade     # Debian/Ubuntu
sudo dnf upgrade                         # RHEL family
sudo apt install cups                    # install a package
systemctl status cups                    # is the service running?
sudo systemctl restart cups
sudo systemctl enable --now ssh          # start now and at every boot
journalctl -u cups --since '1 hour ago'  # logs for one service
```

PRO TIP · NEVER USE CHMOD 777 AS A FIX

Granting everyone read, write and execute 'makes the error go away' but opens the file to every user and process. Find out which user or group really needs access and grant just that.

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SCENARIO · DECIDE · WEB APP CANNOT WRITE UPLOADS

A small web application on a Linux server in a Dhaka software house cannot save uploaded files. The log says 'Permission denied' on /srv/app/uploads. The app runs as user `www-data`. The folder is owned by `deploy:deploy` with mode 755.

(A) Change the group of the uploads folder to www-data and set mode 775 on that folder only

(B) Run chmod -R 777 /srv/app

(C) Run the web app as root

(D) Reinstall the web server package

Circle your choice. Answer and feedback: module answer key.

Key terms

Distribution
A packaged Linux operating system combining the kernel, tools, package manager and defaults, such as Debian or Rocky Linux.

root
The Linux superuser account with unrestricted access to the system.

chmod
The command that changes read, write and execute permissions on files and directories.

systemd
The init and service manager on most modern Linux distributions, controlled with systemctl and logged with journalctl.

Shell
A command-line interpreter, such as Bash or zsh, that runs commands and scripts.

sudo
A command that lets an authorised user run a single command with elevated privileges, with logging.

Package manager
A tool such as apt or dnf that installs, updates and removes software from trusted repositories.

Key takeaways

- A few dozen commands cover most Linux support work on any distribution.
- Read permissions as owner, group, others; in numbers read = 4, write = 2, execute = 1.
- Grant only the user or group that needs access; chmod 777 is never a fix.
- Use apt or dnf for packages, systemctl for services and journalctl for logs.
- Treat sudo and rm with respect: read every command twice.

KNOWLEDGE CHECK

13. What numeric mode gives the owner full access, the group read and execute, and others nothing?

(A) 777 (B) 755
(C) 640 (D) 750

14. Which command shows recent log entries for only the ssh service on a systemd system?

(A) cat /etc/ssh (B) systemctl enable ssh
(C) ls /var/ssh (D) journalctl -u ssh

15. Where are most system-wide configuration files stored?

(A) /home (B) /etc
(C) /tmp (D) /usr/bin

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16. A user on Rocky Linux needs to apply all updates. Which command is correct?

(A) sudo apt upgrade
(B) sudo dnf upgrade
(C) sudo zypper update
(D) sudo pacman -Syu

17. Why is chmod 777 a poor fix for a permission error?

(A) It is slower
(B) It only works on directories, so files inside them keep the old permissions
(C) It grants every user and process full access, creating a security risk
(D) It deletes the file's owner

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MODULE 2 · BOOT · LESSON 4 · 26 MIN

macOS and Mobile Operating Systems

Supporting Macs, iPhones, iPads and Android devices in the workplace

- Use macOS support tools: System Settings, Activity Monitor, Console, Disk Utility and Terminal
- Explain FileVault, Time Machine and macOS recovery
- Compare iOS/iPadOS and Android management and support basics
- Describe how mobile device management enrols and protects devices

Macs are common among designers, executives, developers and universities, and almost every employee now carries a smartphone with work email. Support specialists are expected to help on all of them. macOS is UNIX-based, so many Linux commands work in its Terminal, but its management tools and recovery steps are its own.

macOS support tools

TOOL	FIND IT	USE IT FOR
System Settings	Apple menu	Users, network, privacy and security, updates
Activity Monitor	Applications > Utilities	CPU, memory, energy, disk and network per process
Console	Applications > Utilities	System logs and crash reports
Disk Utility	Applications > Utilities	First Aid repairs, formatting, APFS volumes
Terminal	Applications > Utilities	Shell commands (zsh by default)
Keychain Access	Applications > Utilities	Stored passwords and certificates

Useful macOS Terminal commands

macOS Terminal

```
sw_vers                                 # macOS version and build
system_profiler SPHardwareDataType      # model, chip, memory, serial
diskutil list                           # disks and APFS volumes
softwareupdate --list                   # available updates
networksetup -listallhardwareports      # network interfaces
fdesetup status                         # is FileVault on?
top -o cpu -l 1 | head -n 20            # top processes by CPU
```

Protecting a Mac

FileVault

- Full-disk encryption of the startup volume
- Recovery key must be escrowed by IT or management
- Protects data if the Mac is lost

Time Machine

- Automatic hourly backups to an external or network disk
- Restore single files or a whole Mac
- Should be paired with off-site or cloud copies

Gatekeeper and updates

- Blocks unsigned or unnotarised apps by default
- Urgent security fixes can arrive between full updates
- Keep automatic updates on

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macOS recovery path

1. **Restart into Recovery** Hold the power button on Apple silicon Macs until options appear (or Command-R on older Intel Macs).
2. **Run Disk Utility First Aid** Check and repair the startup volume.
3. **Restore from Time Machine** If data or settings are damaged and a backup exists.
4. **Reinstall macOS** Reinstalls the OS while keeping user data on the volume.
5. **Erase and reinstall** Last resort after confirming a backup, for example before reassigning a Mac to a new employee.

Mobile platforms

iOS / iPadOS

- Single vendor controls hardware and updates
- Apps from the official store; strong sandboxing
- Supervision and automated enrolment for company-owned devices
- Updates available to all supported devices at once

Android

- Many manufacturers and versions
- Update speed varies by maker and carrier
- Work profile separates company apps from personal ones
- Check monthly security patch level in settings

How mobile device management (MDM) works

01 Enrol
The device registers with the MDM service, automatically or by the user.

→

02 Apply profile
Wi-Fi, email, VPN and passcode rules are pushed.

→

03 Deploy apps
Approved apps install silently or from a company portal.

→

04 Monitor compliance
OS version, encryption and jailbreak status are checked.

→

05 Act
Lock, locate or wipe company data when lost or when the employee leaves.

INSIGHT · PERSONAL PHONES AND COMPANY DATA

For bring-your-own-device (BYOD), a work profile or app-level protection lets IT wipe **only company data** when someone leaves, without touching personal photos. This is easier to explain to staff and respects privacy.

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How much to manage a mobile device

SENSITIVITY OF THE DATA ON IT →

Work profile + app protection

- Access only from approved apps
- Selective wipe of company data

Full management (supervised)

- Encryption, passcode, compliance checks
- Lock, locate and full wipe

Light touch

- Web or app access with MFA
- No device enrolment

Standard baseline

- Automated enrolment
- Standard apps and update rules

COMPANY OWNERSHIP OF THE DEVICE →

Control grows with data sensitivity and with company ownership. Personal phones that hold sensitive data get a work profile, not full device control, which respects privacy and still protects the data.

SCENARIO · DECIDE · MAC USER LOCKED OUT AFTER A PASSWORD CHANGE

A marketing manager changed her network password on a Windows PC. Now her Mac keeps asking for her old password to unlock the keychain, and Wi-Fi and email prompts appear repeatedly.

(A) Erase the Mac and reinstall macOS

(B) Disable FileVault

(C) Update the login keychain password to the new account password, or create a new default keychain if the old password is unknown

(D) Tell her to stop using the Mac until the next update

Circle your choice. Answer and feedback: module answer key.

Key terms

FileVault
macOS full-disk encryption for the startup volume, unlocked with the user's password or a recovery key.

APFS
Apple File System: the macOS file system with snapshots, encryption and space sharing.

MDM
Mobile device management: a service that enrols, configures, monitors and secures phones, tablets and computers.

BYOD
Bring your own device: a policy that lets employees use personal devices for work under agreed controls.

Time Machine
The built-in macOS backup feature that keeps hourly, daily and weekly versions of files.

Keychain
The macOS store for passwords, certificates and keys, unlocked by the user's login password.

Work profile
A separated, managed container on a mobile device for company apps and data.

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Key takeaways

- macOS has its own tools: Activity Monitor, Console, Disk Utility and Terminal (zsh).
- FileVault recovery keys must be escrowed before anyone needs them.
- Follow the recovery path from First Aid to erase-and-reinstall, backing up first.
- MDM enrols, configures, monitors and can lock or wipe devices.
- For BYOD, separate and wipe only company data.

KNOWLEDGE CHECK

18. Which macOS command confirms whether disk encryption is enabled?

(A) fdesetup status
(B) sw_vers
(C) diskutil list
(D) softwareupdate --list

19. A departing employee used a personal Android phone with a work profile. What should IT do?

(A) Remove the work profile and company data only
(B) Factory-reset the whole phone
(C) Ask the employee to keep using company email until the contract fully ends
(D) Do nothing

20. What should be done before using 'erase and reinstall' on a Mac?

(A) Disable Gatekeeper
(B) Confirm a current backup and the FileVault recovery key status
(C) Install a third-party antivirus product and run a full scan of the disk
(D) Change the Wi-Fi channel

21. Why do Android security patch levels vary between devices of the same OS version?

(A) Android has no updates
(B) Users cannot see the patch level, so devices never report which updates they have
(C) Manufacturers and carriers deliver updates on their own schedules
(D) Patches only apply to tablets

22. Which MDM action best protects a lost company-owned tablet holding patient records?

(A) Change its wallpaper
(B) Remote lock and, if not recovered, remote wipe
(C) Disable Bluetooth
(D) Reduce screen brightness and disable notifications on the lock screen

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · BOOT · LESSON 5 · 28 MIN

Installing, Imaging and Managing Software

Clean installs, images, drivers, packages and licences

- Choose between clean install, in-place upgrade and imaging
- Create bootable installers and use virtual machines for testing
- Install, update and remove software with package managers on each OS
- Track licences and follow a standard build

Every new starter needs a working machine on day one, and every broken machine needs to return to a known-good state fast. Organisations solve both with **standard builds**: a documented OS, settings and app list, deployed the same way every time. Your job is to install cleanly, keep software current and avoid unlicensed or unknown software.

Ways to deploy an operating system

Clean install

- Wipe and install fresh
- Removes accumulated problems
- Needs backup and reinstall of apps

In-place upgrade

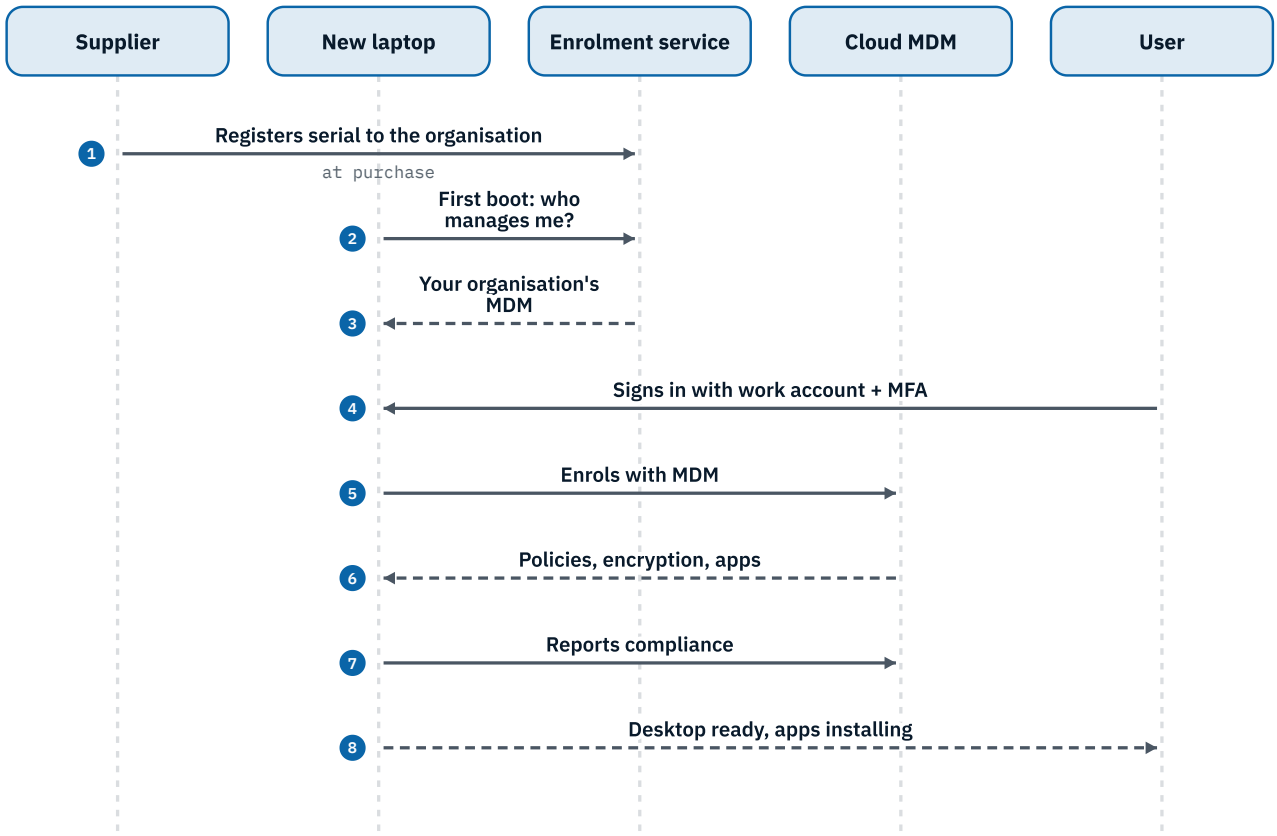
- Moves to a newer version keeping apps and data
- Faster for users
- Can carry old problems forward

Image or zero-touch provisioning

- Standard image, or a device registered to the organisation that sets itself up over the internet
- Consistent and fast at scale
- Images and profiles must be kept current

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Zero-touch provisioning of a company laptop



The laptop never visits an IT bench: ownership is registered at purchase, and its first internet connection pulls the organisation's settings. Notice that the user's MFA sign-in is what ties the device to a person.

Clean install checklist

1. **Back up** Copy user data and note installed apps, printers and licences.
2. **Prepare media** Download the official ISO and write it to a USB stick with a trusted tool; verify the checksum.
3. **Set firmware** Enable UEFI mode and Secure Boot; set the USB stick as a one-time boot device.
4. **Install** Delete old partitions if wiping, let the installer create GPT partitions and install.
5. **Drivers and updates** Install chipset, network and graphics drivers, then all OS updates.
6. **Apps and settings** Install the standard app set, join the domain or cloud identity, apply policies.
7. **Restore and verify** Restore user data, test sign-in, printing, network drives and email with the user.

Verifying a download before you trust it

```
Windows PowerShell

Get-FileHash .\installer.iso -Algorithm SHA256
# Compare the output with the checksum published on the official download page
# Linux equivalent:
# sha256sum installer.iso
```


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Package managers by platform

PLATFORM	TOOL	INSTALL EXAMPLE	UPDATE ALL
Windows	winget	winget install --id Mozilla.Firefox -e	winget upgrade --all
Debian / Ubuntu	apt	sudo apt install vlc	sudo apt update && sudo apt upgrade
RHEL family	dnf	sudo dnf install vlc	sudo dnf upgrade
macOS	App Store, MDM or a community package manager	Install through the company portal	softwareupdate --install --all (OS updates)

Finding and removing software

Windows PowerShell

```
winget list --name zoom
winget upgrade
winget uninstall --id Zoom.Zoom -e
# Linux:
# apt list --installed | grep -i zoom
# sudo apt remove zoom
```

PRO TIP · TEST IN A VIRTUAL MACHINE FIRST

A hypervisor on your own PC lets you install an OS or app in a **virtual machine**, take a snapshot, test and roll back in seconds. Test new images, risky updates and unfamiliar software there before touching a user's laptop.

Software licensing in plain terms

- **Per user or per device** subscriptions are common for productivity and security tools.
- **Volume licences** cover many installations under one agreement.
- **Open-source** licences allow free use but have conditions, especially on redistribution.
- **Freeware for personal use** is often not licensed for business use.
- Unlicensed software brings legal, financial and security risk; keep records of every licence.

CASE STUDY · NEW STARTERS AT A MOBILE FINANCIAL SERVICES COMPANY

A fast-growing MFS provider in Dhaka hires 25 customer-care agents each month. IT currently installs Windows and 14 apps by hand on each laptop, taking about four hours per machine. Settings differ between machines, and two agents found that the softphone app was missing on their first day.

1. What deployment approach would you propose?
2. How would you keep the standard build current?
3. How would you prove each laptop is ready before handover?

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Model answer: module answer key.

TEMPLATE

Standard workstation build sheet

Define exactly what a standard machine contains so every build is identical and auditable.

ITEM	VERSION OR SETTING	INSTALL METHOD	LICENCE OR SOURCE	VERIFIED BY
Operating system	*Windows Pro, latest feature update*	*Image v2026.04*	*Volume agreement*	*Build test*
Softphone client	*Approved current version*	*winget script*	*Per-user subscription*	*Test call to extension 100*

SCENARIO · DECIDE · A USER WANTS TO INSTALL A FREE TOOL

A researcher at a Chattogram university asks you to install a free PDF converter she found online. It is 'free for personal use' and not on the approved software list.

- (A) Install it because it is free
- (B) Give her admin rights so she can install it herself
- (C) Explain the policy, suggest an approved tool that does the same job, or raise a software request for review
- (D) Refuse without explanation

Circle your choice. Answer and feedback: module answer key.

Key terms

Clean install
Installing an OS on a wiped disk so no previous settings, apps or problems remain.

Disk image
A complete copy of a configured system used to deploy identical builds to many machines.

In-place upgrade
Upgrading an OS to a newer version while keeping installed apps, settings and data.

Checksum (hash)
A value calculated from a file's contents, used to confirm the file is genuine and undamaged.

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Virtual machine
A software-based computer running on a hypervisor, isolated from the host and other VMs.

Software licence
A legal agreement setting how software may be installed and used, such as per user, per device or volume.

Snapshot
A saved point-in-time state of a VM or volume that can be restored quickly.

Key takeaways

- Standard builds make every machine predictable and every rebuild fast.
- Zero-touch provisioning ships devices straight to users and configures them over the internet.
- Verify downloads with a published hash before you trust them.
- Package managers (winget, apt, dnf) install and update software repeatably.
- Test images and major updates in a VM with a snapshot first, and keep a record of every licence.

KNOWLEDGE CHECK

23. What is the main purpose of comparing a SHA-256 hash after downloading an ISO?

(A) To speed up installation
(B) To confirm the file is genuine and not corrupted
(C) To activate the licence key that is embedded inside the installer image
(D) To compress the file

24. Which command upgrades all packages managed by winget on Windows?

(A) apt upgrade
(B) winget list
(C) winget upgrade --all
(D) sfc /scannow

25. Which deployment method gives the most consistent results across 100 laptops?

(A) A tested standard image or automated provisioning
(B) Manual installs by different technicians
(C) In-place upgrades of whatever operating system each laptop already has installed
(D) Letting users install their own apps

26. Why test a major update in a virtual machine with a snapshot first?

(A) You can test safely and roll back instantly if it breaks
(B) VMs never need updates
(C) Updates are smaller in VMs
(D) It removes the need for backups because the snapshot keeps every version

27. A tool is labelled 'free for personal use'. What is the risk of installing it on company PCs?

(A) It may breach the licence for business use
(B) None, free means free
(C) It will not run on business editions of Windows or macOS at all
(D) It deletes other licences

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · BOOT

Answer key

Knowledge checks

1. A exFAT handles large files and is supported read-write by both systems. FAT32 has a 4 GB limit; ext4 and APFS are not natively writable on the other OS.

2. A MBR addresses up to 2 TiB. GPT is required to use the full disk.

3. B Secure Boot verifies signatures of boot components, blocking boot-level malware. It does not address phishing, passwords or heat.

4. D Unsupported systems keep working but receive no security fixes, leaving known holes open.

5. A Drivers are hardware-specific translators. The scheduler shares CPU time, the file system stores data and the boot loader starts the kernel.

6. B After end of support the OS keeps working but receives no security fixes, so it needs an upgrade, replacement or isolation plan. It does not stop booting, its licence stays valid and patches do not continue.

7. A Event Viewer records system events with IDs and times. The other tools manage devices, disks and appearance.

8. C DISM repairs the component store that sfc uses as its source, so run DISM first, then sfc.

9. C Safe mode loads basic drivers so you can roll back the faulty one. Resetting or reimaging is more disruptive; hardware is not at fault.

10. D Home lacks domain join. Pro, Enterprise and Education support it.

11. D UAC requires consent for elevation, reducing silent misuse. It does not affect speed, Wi-Fi or encryption.

12. D Most PCs from that era have a firmware TPM that is simply switched off. Buying parts or replacing the board is unnecessary, and staying on Windows 10 leaves it unpatched.

13. D 7 = rwx, 5 = r-x, 0 = none, so 750. 755 lets others read and execute; 777 opens everything; 640 removes execute.

14. D journalctl -u filters the journal by unit. The other commands read config, enable start-up or list a non-standard path.

15. B /etc holds configuration; /home has user files, /tmp temporary data and /usr/bin programs.

16. B Rocky is in the Red Hat family, which uses dnf. The others belong to Debian, SUSE and Arch families.

17. C 777 removes all access control. The proper fix grants only the needed user or group.

18. A fdesetup status reports FileVault state. sw_vers shows the version, diskutil lists disks and softwareupdate lists updates.

19. A Removing the work profile protects company data and leaves personal data untouched.

20. B Erasing destroys data, so a verified backup is essential. The other actions do not protect data.

21. C The maker and carrier control rollout, so patch levels differ. The patch level is visible in settings.

22. B Locking and wiping prevent data access. The other actions do not protect data.

23. B A matching hash proves integrity. It does not install, activate or compress anything.

24. C winget upgrade --all updates all apps it manages. apt is Linux, winget list only lists and sfc repairs system files.

25. A Automation from one tested definition produces identical builds. Manual and user-driven methods drift.

26. A Snapshots allow instant rollback during testing. VMs still need updates and backups still matter.

27. A Personal-use licences typically exclude commercial use, creating legal risk.

Scenarios & case studies

Scenario · A USB stick that will not take a big file
D is best. Correct. exFAT supports large files and is read and written by Windows and macOS.

Scenario · Slow PC after an update
B is best. Correct. You confirm the evidence, avoid harmful shortcuts and fix the underlying weakness: a slow disk.

Scenario · Web app cannot write uploads
A is best. Correct. The app's user gains write access through the group, nothing else changes and other users still cannot write.

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Scenario · Mac user locked out after a password change
C is best. Correct. The keychain is still locked with the old password; aligning it stops the prompts.

Case study · New starters at a mobile financial services company
Approach: define a documented standard build and use automated provisioning (an image or cloud-based enrolment plus a package manager script) so a laptop is ready in under an hour with identical settings. **Keeping it current:** one owner reviews the build monthly, updates app versions and drivers, tests in a VM and a pilot laptop, and records the version number. **Readiness:** a handover checklist run on every laptop: sign-in works, softphone registers, VPN connects, printing tested, encryption on, endpoint protection reporting, user confirms.

Scenario · A user wants to install a free tool
C is best. Correct. You meet the need without legal or security risk and keep the process transparent.

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03

MODULE 3 · CONNECT

Networking Fundamentals

TCP/IP, addressing, DNS, DHCP, Wi-Fi and the tools that prove where a network fault lives

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain how data moves across networks using the OSI and TCP/IP models, TCP, UDP and ports
- Calculate IPv4 subnets and recognise private, APIPA and IPv6 addresses
- Explain and troubleshoot DNS and DHCP
- Set up and secure small office Wi-Fi and routers
- Use command-line tools and a layered method to locate network faults

LESSONS

1. How Networks Work: Models, TCP/IP and Ports · 26 min
2. IP Addressing and Subnetting · 32 min
3. DNS and DHCP · 28 min
4. Wi-Fi and Small Office Networks · 26 min
5. Network Troubleshooting Tools · 30 min

MODULE 3 · CONNECT · LESSON 1 · 26 MIN

How Networks Work: Models, TCP/IP and Ports

Devices, layers, protocols and the port numbers you will see every day

- Identify common network devices and network types
- Use the OSI and TCP/IP models to describe where a problem lives
- Compare TCP and UDP and recognise well-known port numbers
- Explain encapsulation in simple terms

Almost every support ticket touches the network: email, cloud apps, printers, file shares and phones all depend on it. You do not need to be a network engineer, but you must understand how data gets from a laptop to a server and back, so you can tell whether a problem is the user's device, the local network, the internet link or the remote service.

Network devices and what they do

DEVICE	WORKS AT	JOB
Switch	Layer 2	Connects devices in a LAN and forwards frames by MAC address
Router	Layer 3	Moves packets between networks using IP addresses; the default gateway
Wireless access point	Layer 2	Connects Wi-Fi clients to the wired LAN
Firewall	Layers 3-7	Allows or blocks traffic according to rules
Modem / ONT	Layer 1-2	Connects to the provider's DSL, cable or fibre line
SOHO router	Several	Router, switch, Wi-Fi, firewall, DHCP and NAT in one box

Network types

LAN

- One building or floor
- High speed, owned by the organisation
- Ethernet and Wi-Fi

WAN

- Connects distant sites
- Provider links, VPNs or SD-WAN
- Slower and costlier than a LAN

Internet and cloud

- Public networks reaching SaaS and cloud
- Reached through a firewall and NAT
- Performance depends on the provider

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OSI model with support examples (top first)

7 Application	`HTTP, DNS, SMTP` `'The website shows an error'`
6 Presentation	`Encoding, encryption (TLS)` `'Certificate warning'`
5 Session	`Opening and keeping sessions` `'I keep getting logged out of the app'`
4 Transport	`TCP and UDP ports` `'Port 443 is blocked'`
3 Network	`IP addressing and routing` `'Wrong gateway' or 'no route'`
2 Data link	`MAC addresses, switches, Wi-Fi association` `'Port in the wrong VLAN'`
1 Physical	`Cables, signals, radio` `'Cable unplugged' or 'weak Wi-Fi signal'`

INSIGHT · THE TCP/IP MODEL IS THE PRACTICAL ONE

Day to day, engineers use the simpler four-layer TCP/IP model: **link, internet, transport, application**. The OSI numbers are still useful shorthand: 'a Layer 1 problem' means cables or radio, 'Layer 3' means addressing or routing.

Encapsulation: sending an email

01 Data — The mail app prepares the message. →

02 Segment — TCP adds ports and sequence numbers. →

03 Packet — IP adds source and destination addresses. →

04 Frame — Ethernet or Wi-Fi adds MAC addresses. →

05 Signal — The network card sends bits over cable or radio.

TCP versus UDP

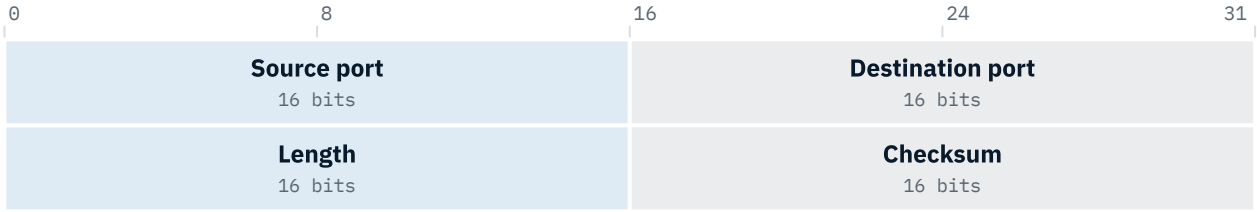
TCP

- Connection-oriented with a three-way handshake
- Reliable: retransmits lost data, keeps order
- Web, email, file transfer, remote desktop

UDP

- Connectionless, no handshake
- Fast, no retransmission
- DNS queries, voice and video calls, DHCP, streaming

The UDP header: just 8 bytes



UDP adds only ports, a length and a checksum: no sequence numbers and no acknowledgements. That is why it suits DNS and live calls, and why a lost UDP packet is simply gone.

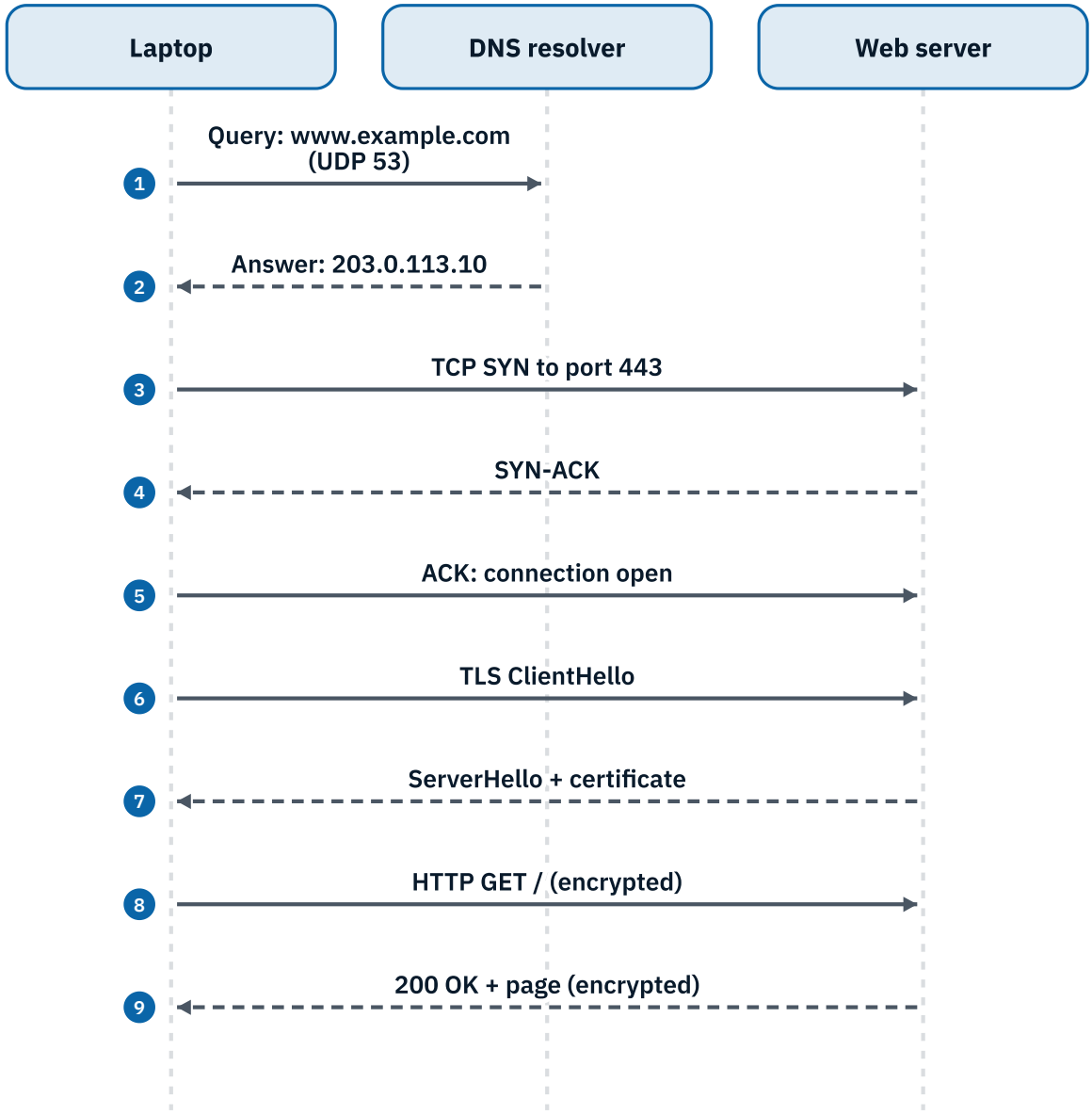
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Ports every support specialist should know

PORT	PROTOCOL	USED FOR
22	TCP	SSH and secure file transfer (SFTP)
25 / 587	TCP	SMTP mail relay / mail submission
53	UDP and TCP	DNS
67 / 68	UDP	DHCP server / client
123	UDP	NTP time synchronisation
80 / 443	TCP (443 also UDP for QUIC)	HTTP / HTTPS
445	TCP	SMB file sharing
3389	TCP	Remote Desktop
993	TCP	IMAP over TLS

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What happens when a user opens a secure website



One click uses name resolution over UDP, a TCP handshake and then TLS before any web content moves; each is a separate place a fault can hide. Browsers using HTTP/3 replace the TCP step with QUIC over UDP 443.

SCENARIO · DECIDE · WEBSITE LOADS, VIDEO CALLS FAIL

Staff at a Khulna NGO office can browse websites normally, but every video call drops after a few seconds with poor audio. A new firewall was installed last week.

- (A) Replace all the network cables
- (B) Reinstall the video call app on every PC
- (C) Check whether the new firewall is blocking or throttling the UDP ports the calling service needs
- (D) Increase the DHCP lease time

Circle your choice. Answer and feedback: module answer key.

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Key terms

LAN
Local area network: a network covering a limited area such as an office, owned and managed by the organisation.

Switch
A Layer 2 device that connects devices in a LAN and forwards frames using MAC addresses.

TCP
Transmission Control Protocol: connection-oriented, reliable transport that resends lost data.

Port number
A number that identifies a specific service or application on a host, such as 443 for HTTPS.

Router
A Layer 3 device that forwards packets between different IP networks.

OSI model
A seven-layer reference model that describes network functions from physical signals to applications.

UDP
User Datagram Protocol: connectionless, low-overhead transport used for DNS, voice and video.

Key takeaways

- Switches forward frames by MAC address inside a LAN; routers forward packets between networks by IP.
- Use OSI numbers as shorthand: Layer 1 is cables and radio, Layer 3 is addressing and routing.
- Data is wrapped layer by layer: segment (ports), packet (IP), frame (MAC), signal.
- TCP is reliable and ordered; UDP is fast and suits DNS, DHCP and live calls.
- Know the everyday ports: 22, 53, 67/68, 123, 443, 445 and 3389.

KNOWLEDGE CHECK

1. Which device should be checked first if a whole office can reach each other's PCs but not the internet?
   - (A) Each user's network card and its driver, one PC at a time across the office
   - (B) The router or firewall acting as the default gateway
   - (C) The printers
   - (D) The keyboard cables
2. Which port must be reachable for users to connect with Remote Desktop?
   - (A) 22
   - (B) 3389
   - (C) 443
   - (D) 53
3. A user sees a certificate warning on a website. Roughly which OSI layer is involved?
   - (A) Layer 1
   - (B) Layer 2
   - (C) Layer 6 / application layer security
   - (D) Layer 3
4. Why do voice and video calls usually use UDP?
   - (A) UDP guarantees delivery
   - (B) UDP encrypts voice traffic by default, while TCP sends it in clear text
   - (C) Late, resent audio is useless, so low delay matters more
   - (D) TCP cannot carry audio
5. What is added to data at the transport layer?
   - (A) Port numbers
   - (B) MAC addresses
   - (C) IP addresses
   - (D) Radio signals

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · CONNECT · LESSON 2 · 32 MIN

IP Addressing and Subnetting

IPv4, masks, gateways, private ranges and a first look at IPv6

- Read an IPv4 address with its subnet mask or CIDR prefix
- Calculate network, broadcast and usable host range for common prefixes
- Recognise private, loopback and APIPA addresses and what they mean in a ticket
- Read basic IPv6 addresses

Every device on a network needs an IP address, a subnet mask and a default gateway to talk beyond its own segment, plus DNS servers to find names. Most tickets about 'no internet' are solved by checking these four values. Subnetting tells you which addresses are local and which must go through the router.

The four settings every host needs

SETTING	EXAMPLE	WHAT IT DOES
IP address	192.168.10.25	Identifies this host on the network
Subnet mask / prefix	255.255.255.0 or /24	Splits the address into network and host parts
Default gateway	192.168.10.1	Router used for anything outside the local subnet
DNS server	192.168.10.10	Translates names into IP addresses

Special IPv4 ranges

RANGE	MEANING	WHAT IT TELLS YOU IN A TICKET
10.0.0.0/8	Private	Normal inside organisations
172.16.0.0/12	Private	Normal inside organisations
192.168.0.0/16	Private	Typical home and small office
127.0.0.0/8	Loopback	The host talking to itself
169.254.0.0/16	APIPA / link-local	The host asked for DHCP and got no answer
100.64.0.0/10	Carrier-grade NAT	Provider is sharing public addresses

Subnet maths

MEASURE	FORMULA	WHAT IT TELLS YOU
Addresses in a subnet	`addresses = 2^(32 - prefix)`	A /24 has 2^8 = 256 addresses; a /26 has 2^6 = 64.
Usable hosts	`hosts = 2^(32 - prefix) - 2`	Subtract the network and broadcast addresses: a /26 gives 62 usable hosts.
Block size	`block = 256 - last non-255 mask octet`	Mask 255.255.255.192 gives block 64, so subnets start at .0, .64, .128, .192.

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Common prefixes

PREFIX	MASK	ADDRESSES	USABLE HOSTS
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/30	255.255.255.252	4	2

Worked example: 192.168.10.77/26

1. **Find the block size** /26 means mask 255.255.255.192, so the block is 256 - 192 = 64.
2. **Find the subnet** Multiples of 64 are 0, 64, 128, 192. 77 falls in the block starting at 64, so the network is 192.168.10.64.
3. **Find the broadcast** The next block starts at 128, so the broadcast is 192.168.10.127.
4. **Usable range** 192.168.10.65 to 192.168.10.126, which is 62 hosts.
5. **Check the gateway** A gateway of 192.168.10.1 would be outside this subnet, so this host could not reach it.

PRACTICE Practise with the subnet calculator

Enter 192.168.10.77/26 and confirm the network (.64), broadcast (.127) and 62 usable hosts. Then try 10.20.5.200/27 and 172.16.40.9/28 and work out the range on paper before checking.

ADDRESS / PREFIX	NETWORK	MASK	FIRST HOST	LAST HOST	BROADCAST	HOSTS
`192.168.10.77/26`						
`10.45.130.9/20`						
`172.20.5.200/27`						
`10.0.0.129/25`						
`192.168.1.5/30`						
`172.16.99.1/22`						

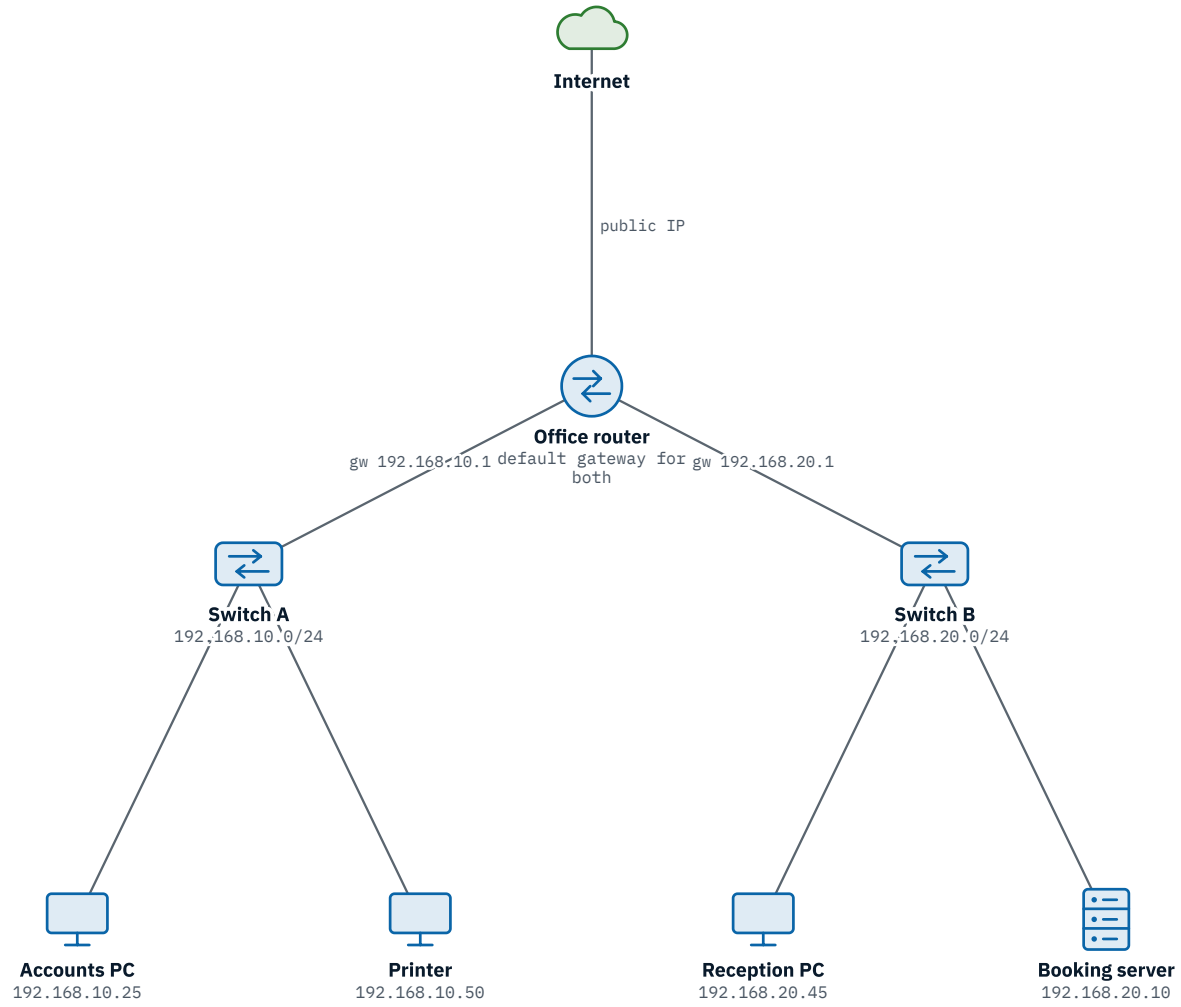
Answers: use the subnet calculator on the course website, or check with the answer key at the end of the book.

EXAMPLE · THE WRONG GATEWAY

A new receptionist's PC at a Chattogram hotel could print locally but not reach the booking website. Its static settings were 192.168.20.45/24 with gateway 192.168.2.1, a typing error. Correcting the gateway to 192.168.20.1 fixed it instantly. Local traffic worked because it never needed the router.

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Two subnets joined by a router



Hosts reach anything in their own subnet directly, but everything else goes to the gateway, which must sit inside that same subnet. The receptionist's mistyped gateway, 192.168.2.1, exists in neither subnet, so only local traffic worked.

Checking IP settings

```Windows PowerShell
ipconfig /all
Get-NetIPConfiguration
# Linux:
# ip -br addr
# ip route show default
# macOS:
# ifconfig en0 | grep 'inet '
```

IPv4 and IPv6

IPv4

- 32-bit, dotted decimal: 192.168.1.10
- About 4.3 billion addresses; relies on NAT
- DHCP for automatic addressing

IPv6

- 128-bit, hexadecimal: 2001:db8:10::25
- Huge address space; no NAT needed
- SLAAC or DHCPv6; fe80:: link-local always present

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PRO TIP · SHORTENING IPV6 ADDRESSES

Leading zeros in each group can be dropped, and one run of all-zero groups can be replaced by `::` . So 2001:0db8:0000:0000:0000:0000:0000:0025 becomes **2001:db8::25**.

SCENARIO · DECIDE · A 169.254 ADDRESS

A user in a Dhaka garment factory office says 'no internet'. `ipconfig` shows IPv4 address 169.254.33.18 with no default gateway. Other PCs on the same switch work.

- (A) Set a static address of 169.254.33.1 as the gateway
- (B) Check the cable and switch port, then release and renew the DHCP lease
- (C) Reinstall the web browser
- (D) Change the DNS server to a public resolver

Circle your choice. Answer and feedback: module answer key.

Key terms

IP address
A logical address that identifies a host interface on an IP network.

CIDR prefix
Notation such as /24 that states how many leading bits form the network portion of an address.

Broadcast address
The last address in an IPv4 subnet, used to reach every host on that subnet.

IPv6
The 128-bit successor to IPv4 with a vastly larger address space, written in hexadecimal.

Subnet mask
A value that separates the network part of an IPv4 address from the host part, such as 255.255.255.0.

Default gateway
The router address a host uses to reach destinations outside its local subnet.

APIPA
Automatic Private IP Addressing: a 169.254.x.x address a host gives itself when DHCP does not respond.

Key takeaways

- Check four values on every 'no internet' ticket: IP address, mask, gateway and DNS.
- Addresses = 2^(32 - prefix); usable hosts = that number minus 2.
- Use the block size (256 minus the mask octet) to find network and broadcast addresses fast.
- 169.254.x.x means DHCP never answered; 127.x is loopback.
- IPv6 is 128-bit hexadecimal; every interface also has an fe80:: link-local address.

KNOWLEDGE CHECK

6. How many usable host addresses are in a /27 subnet?
   - (A) 32
   - (B) 62
   - (C) 30
   - (D) 14
7. What is the network address of 10.1.1.200/26?
   - (A) 10.1.1.192
   - (B) 10.1.1.128
   - (C) 10.1.1.200
   - (D) 10.1.1.0

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8. A PC has address 169.254.10.5. What does this most likely mean?

   (A) It could not obtain an address from DHCP
   (B) It has a public address
   (C) It is using IPv6
   (D) It is a loopback address that the PC uses to talk to itself

9. Which is the correct compressed form of 2001:0db8:0000:0000:0000:0000:0000:0001?

   (A) 2001:db8::1
   (B) 2001:db8:0:1
   (C) 2001::db8::1
   (D) 21:db8::1

10. Host 192.168.5.20/24 has gateway 192.168.6.1. What will happen?

    (A) Local traffic works; traffic to other networks fails
    (B) Everything works
    (C) No traffic works at all, because the mask and gateway conflict completely
    (D) Only DNS works

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · CONNECT · LESSON 3 · 28 MIN

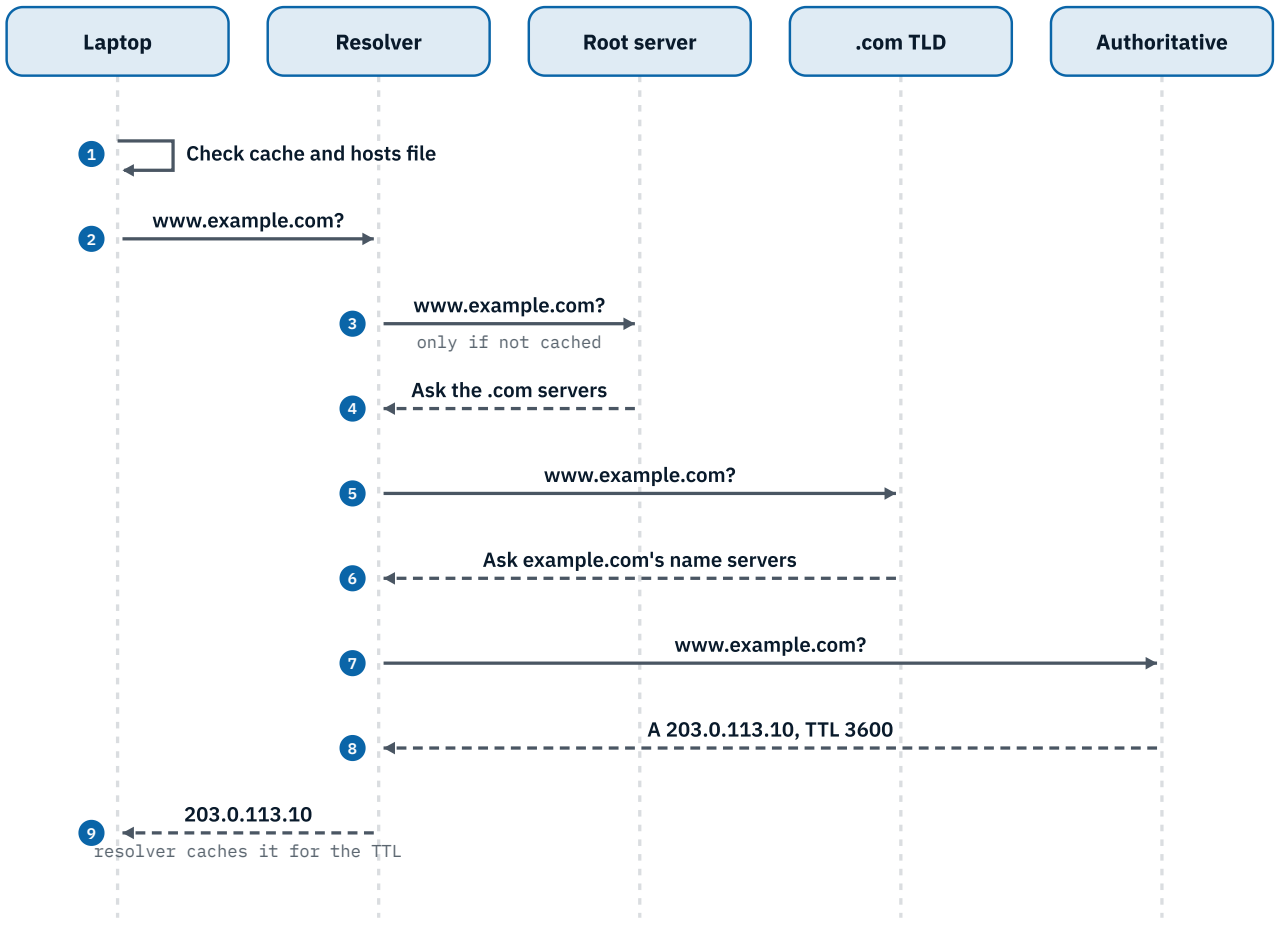
DNS and DHCP

The two services behind most 'the network is down' tickets

- Explain how DNS resolves a name and identify common record types
- Describe the DHCP DORA process, scopes, leases and reservations
- Use command-line tools to test and clear DNS and renew DHCP leases
- Diagnose typical DNS and DHCP failures

People remember names, computers use addresses. **DNS** translates names such as `intranet.company.local` into IP addresses. **DHCP** hands out IP settings automatically. When either fails, users see 'the internet is down' even though cables, switches and routers are fine. That makes these two services the most valuable to understand deeply.

How a name is resolved



The laptop asks one question and gets one answer; the resolver does the walking. Caching at every step is why a changed record can take up to one TTL to be seen everywhere.

DNS record types

RECORD	PURPOSE	EXAMPLE
A	Name to IPv4 address	www -> 203.0.113.10

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RECORD	PURPOSE	EXAMPLE
AAAA	Name to IPv6 address	www -> 2001:db8::10
CNAME	Alias to another name	mail -> mail.provider.example
MX	Mail servers for a domain	company.com -> mx1.provider.example
TXT	Text, used for SPF, DKIM and domain verification	v=spf1 include:...
PTR	Reverse: address to name	203.0.113.10 -> www.company.com
NS	Authoritative name servers	company.com -> ns1.host.example

Testing DNS

Windows PowerShell

```
Resolve-DnsName intranet.company.local
Resolve-DnsName company.com -Type MX
nslookup www.example.com 1.1.1.1      # ask a specific server
ipconfig /displaydns | Select-String example
ipconfig /flushdns                    # clear stale cache entries
```

Testing DNS on Linux and macOS

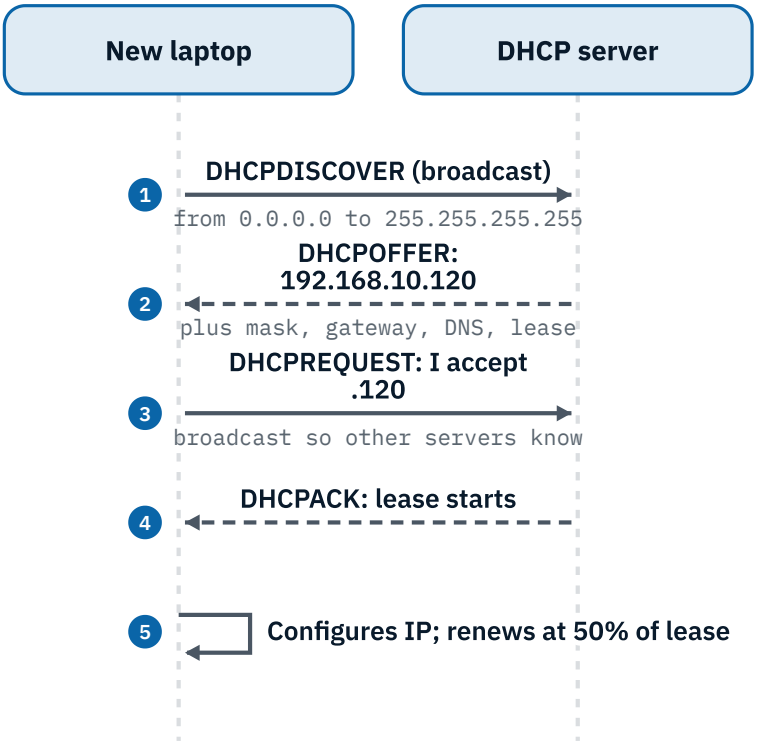
Linux host

```
dig www.example.com +short
dig company.com MX
dig @9.9.9.9 www.example.com          # query a specific resolver
resolvectl status                     # which DNS servers are in use (systemd)
# macOS: scutil --dns | head -n 20
```

> **DEFINITION · TTL (TIME TO LIVE)**
> Each DNS answer carries a TTL in seconds telling resolvers how long to cache it. A long TTL means changes take longer to reach everyone. Before a planned server move, lower the TTL a day ahead.

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DHCP: the DORA exchange



Discover and Request are broadcasts because the client has no address yet, which is why routers need a DHCP relay to serve other subnets. If no Offer arrives, the client falls back to a 169.254 address.

DHCP terms in practice

ITEM	MEANING	EXAMPLE
Scope	The pool of addresses offered on a subnet	192.168.10.100-192.168.10.199
Exclusion	Addresses inside the range the server must not give out	Printers' static block
Reservation	A fixed address always given to one MAC address	Label printer always gets .50
Lease	How long a client may use the address	8 hours for guests, 8 days for staff
Options	Extra settings handed out	Gateway, DNS servers, domain name
Relay	Forwards DHCP requests from other subnets to the server	Router relays VLAN 20 to the server

Renewing a DHCP lease

Windows PowerShell

```
ipconfig /release
ipconfig /renew
ipconfig /all | Select-String 'DHCP Server|Lease'
# Linux (NetworkManager): nmcli connection down 'Wired 1'; nmcli connection up 'Wired 1'
# macOS: sudo ipconfig set en0 DHCP
```


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WATCH OUT · SCOPE EXHAUSTION

If a scope runs out of addresses, new devices get APIPA addresses and cannot connect. Guest Wi-Fi with long leases is a common cause. Shorten guest leases, enlarge the scope or split the network.

SCENARIO · DECIDE · ONLY NAMES FAIL

After an office move in Sylhet, users can reach `142.250.x.x` addresses by IP but no websites load by name. `ipconfig /all` shows DNS server 192.168.1.10, the old office's server, which no longer exists.

- (A) Fix the DNS server option in the new DHCP scope and have clients renew their leases
- (B) Replace the users' network cards
- (C) Add hosts-file entries on each PC for popular sites
- (D) Reboot the internet router

Circle your choice. Answer and feedback: module answer key.

Key terms

DNS
Domain Name System: the distributed service that translates names into IP addresses and other records.

MX record
A DNS record that names the mail servers responsible for a domain.

DHCP
Dynamic Host Configuration Protocol: automatically assigns IP addresses and network settings to clients.

DHCP relay
A function on a router that forwards DHCP broadcasts from one subnet to a DHCP server on another.

A record
A DNS record that maps a hostname to an IPv4 address.

TTL
Time to live: how long, in seconds, a DNS answer may be cached.

DHCP reservation
A setting that always gives the same IP address to a specific device, identified by its MAC address.

Key takeaways

- If IP addresses work but names fail, suspect DNS first.
- A, AAAA, CNAME, MX, TXT, PTR and NS records each answer a different question.
- TTL controls caching: lower it before a planned change and flush client caches after.
- DHCP follows Discover, Offer, Request, Acknowledge; reservations keep printers on fixed addresses.
- Fix wrong settings at the DHCP scope so every client gets the correction at renewal.

KNOWLEDGE CHECK

11. What is the correct order of the DHCP exchange?

- (A) Request, Offer, Discover, Acknowledge
- (B) Offer, Discover, Acknowledge, Request
- (C) Discover, Request, Offer, Acknowledge
- (D) Discover, Offer, Request, Acknowledge

12. Which record tells other mail servers where to deliver email for a domain?

- (A) A
- (B) MX
- (C) CNAME
- (D) PTR

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13. A user can open a site by IP address but not by name. What is the most likely fault area?

(A) Cabling
(B) DNS
(C) DHCP lease time
(D) Printer driver

14. A network printer keeps changing address and users lose it. What is the best fix?

(A) Create a DHCP reservation for the printer's MAC address
(B) Shorten the lease time so the printer renews its address more often
(C) Turn off DHCP
(D) Use APIPA

15. After moving a website to a new server, some users still reach the old one. What explains this?

(A) DHCP leases
(B) A wrong subnet mask on the clients that still reach the old server
(C) MX record errors
(D) Cached DNS answers that have not reached their TTL expiry

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · CONNECT · LESSON 4 · 26 MIN

Wi-Fi and Small Office Networks

Standards, bands, security and setting up a small office or home office

- Compare Wi-Fi standards, bands and channel choices
- Configure secure Wi-Fi with WPA3 and separate guest access
- Explain NAT and port forwarding on a small office router
- Troubleshoot weak signal, interference and roaming problems

Wi-Fi is how most people connect in 2026, and it is the network they complain about most. Radio is shared, invisible and affected by walls, crowds and neighbours. A support specialist who understands bands, channels and security can fix many 'slow internet' tickets without touching the internet link.

Wi-Fi generations

NAME	STANDARD	BANDS	NOTES
Wi-Fi 4	802.11 n	2.4 and 5 GHz	Older devices; still common in printers and IoT
Wi-Fi 5	802.11 ac	5 GHz	Widely deployed in offices
Wi-Fi 6 / 6E	802.11 ax	2.4, 5 and 6 GHz (6E)	Better in crowded spaces; 6E adds clean 6 GHz channels and requires WPA3 there
Wi-Fi 7	802.11 be	2.4, 5 and 6 GHz	Up to 320 MHz channels in 6 GHz and multi-link operation (MLO) across bands
Wi-Fi 8 (in development)	802.11 bn	2.4, 5 and 6 GHz	Aims at reliability rather than peak speed; not yet a purchasing factor

Choosing a band

2.4 GHz

- Longest range, passes walls better
- Only three non-overlapping channels: 1, 6, 11
- Crowded with neighbours, Bluetooth and microwaves

5 GHz

- Many more channels, faster
- Shorter range
- Best default for laptops and phones

6 GHz

- Very clean spectrum, widest channels
- Shortest range; needs Wi-Fi 6E or 7 devices
- Requires WPA3

Wi-Fi security options

OPTION	STATUS IN 2026	USE
Open	No encryption	Avoid; use a captive portal with enhanced open if guests need easy access
WEP / WPA (TKIP)	Broken	Never use
WPA2-Personal (AES)	Acceptable	Legacy devices that cannot do WPA3

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OPTION	STATUS IN 2026	USE
WPA3-Personal (SAE)	Recommended	Small offices with a strong passphrase
WPA2/WPA3-Enterprise (802.1X)	Best for organisations	Each user signs in with their own credentials or certificate

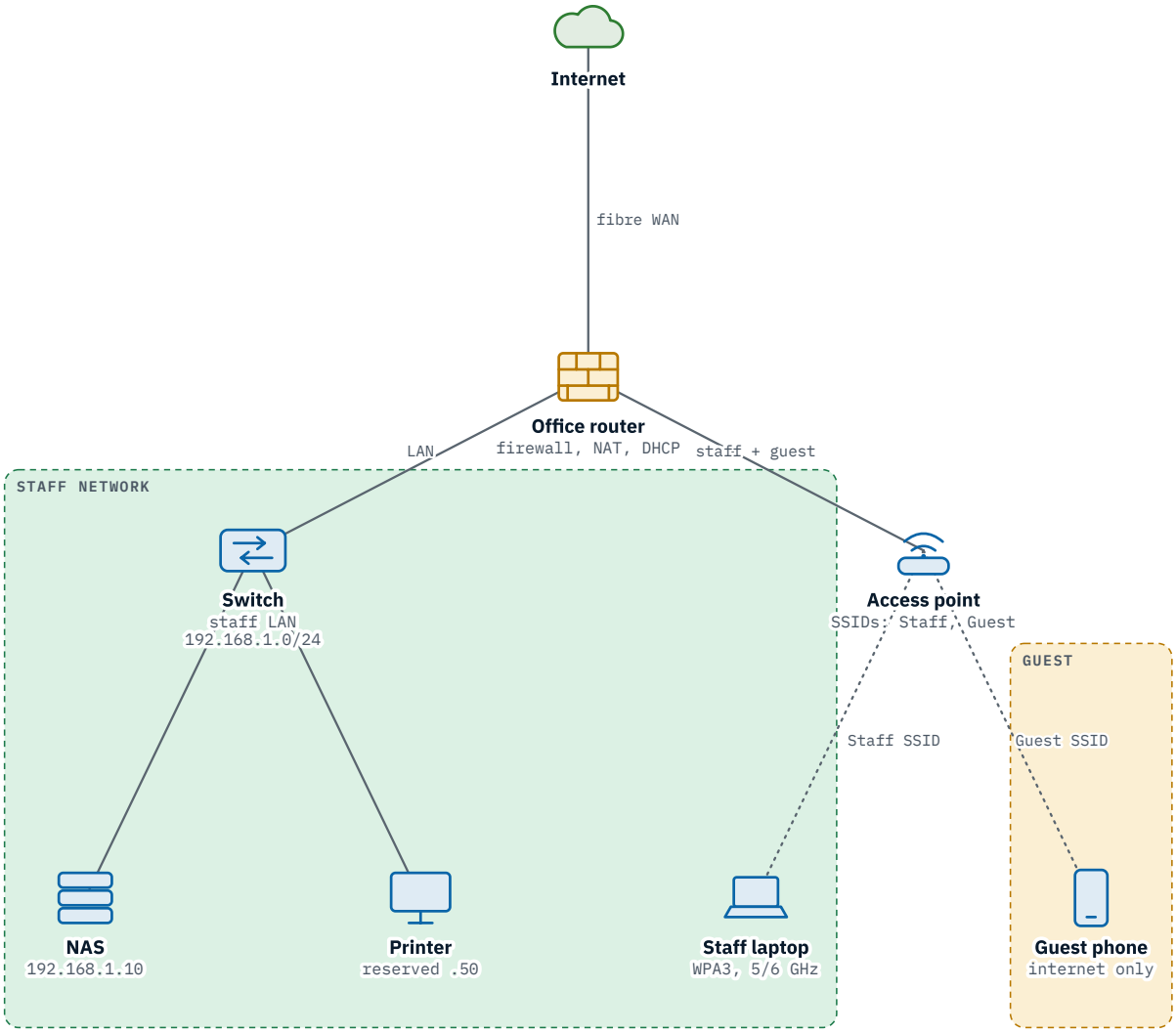
> **PRO TIP · SEPARATE GUESTS FROM STAFF**
> Put guests on their own SSID and network segment that can reach only the internet. A visitor's infected laptop should never be able to see your file server or printers.

Setting up a small office router securely

1. **Change the admin credentials** Replace the default admin password and disable remote admin from the internet.
2. **Update the firmware** Install the latest firmware and turn on automatic updates if available.
3. **Configure Wi-Fi** WPA3 (or WPA2/WPA3 mixed), a long passphrase and a neutral SSID that does not reveal the company name.
4. **Create guest Wi-Fi** Isolated from the internal LAN, with a separate passphrase changed regularly.
5. **Set DHCP and DNS** Size the scope for growth and reserve addresses for printers.
6. **Disable risky features** Turn off WPS and UPnP unless there is a specific need.
7. **Document** Record SSIDs, admin access method, firmware version and passphrase location in the password manager.

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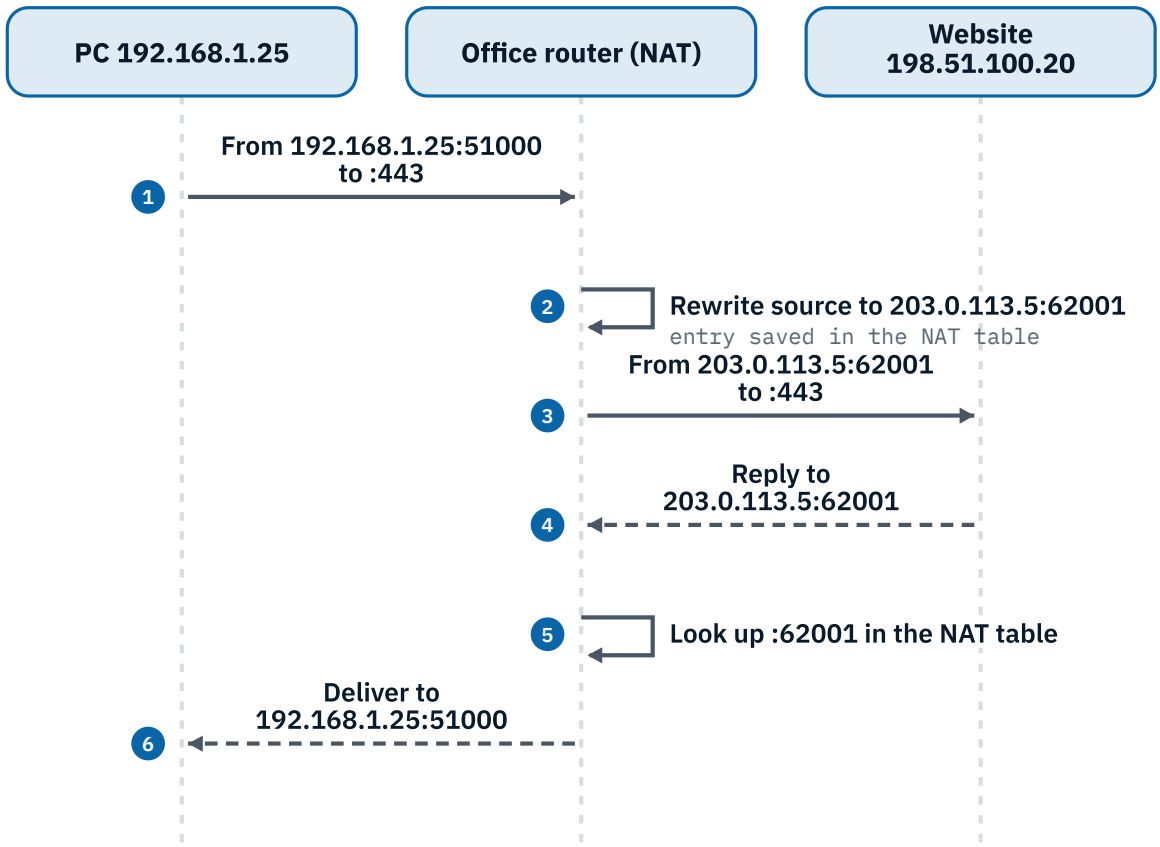
A secure small office network



One router and one access point carry two separate networks. Staff devices share the LAN with the NAS and printer, while guests land in their own segment with a path to the internet only.

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NAT on a small office router



Many inside devices share one public address because the router tracks each conversation by port. Unsolicited inbound traffic has no table entry, which is why reaching an inside device needs port forwarding, and why that is risky.

WATCH OUT · PORT FORWARDING OPENS A DOOR

Port forwarding exposes an internal device to the internet. Never forward Remote Desktop (3389) or camera admin pages directly. Use a VPN or a cloud-managed remote access service instead.

Fixing 'slow Wi-Fi' tickets

- Check signal strength (RSSI): above about -67 dBm is good for calls; below -75 dBm expect trouble.
- Move clients to 5 or 6 GHz where possible.
- On 2.4 GHz use only channels 1, 6 or 11 and avoid overlapping neighbours.
- Add access points rather than turning up power; louder APs do not make phones louder.
- Test wired speed at the same time to separate Wi-Fi from internet problems.

Checking Wi-Fi from the command line

```
Windows PowerShell

netsh wlan show interfaces          # SSID, band, channel, signal %, rates
netsh wlan show networks mode=bssid # nearby networks and channels
# Linux (NetworkManager):
# nmcli dev wifi list
# macOS: hold Option and click the Wi-Fi icon for RSSI, channel and noise
```


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SCENARIO · DECIDE · MEETING ROOM CALLS KEEP DROPPING

In a Dhaka office, video calls drop in the large meeting room only. `netsh wlan show interfaces` shows signal 38%, channel 6 on 2.4 GHz. The nearest access point is two rooms away behind a concrete wall.

(A) Increase the access point's transmit power to maximum

(B) Change the SSID name

(C) Ask users to switch off video

(D) Add an access point in or near the meeting room and steer clients to 5 GHz

Circle your choice. Answer and feedback: module answer key.

Key terms

SSID
Service set identifier: the network name broadcast by Wi-Fi access points.

802.1X
A port and Wi-Fi authentication framework where each user or device signs in individually, usually via a RADIUS server.

RSSI
Received signal strength indicator, usually in dBm; values closer to zero mean a stronger signal.

Port forwarding
A router rule that sends inbound traffic on a public port to a specific internal device.

WPA3
The current Wi-Fi security standard, using SAE for stronger protection of passphrase-based networks.

Channel
A specific frequency range within a Wi-Fi band; overlapping channels interfere with each other.

NAT
Network address translation: rewriting private source addresses to a public address so many devices share one connection.

Key takeaways

- Prefer 5 or 6 GHz for staff devices; on 2.4 GHz use only channels 1, 6 and 11.
- Use WPA3 (or 802.1X for organisations); WEP and TKIP are broken.
- Isolate guests on their own SSID and segment with internet-only access.
- Harden every small office router: new admin password, current firmware, WPS and UPnP off.
- Never port-forward RDP or camera pages; use a VPN or secure remote-access service.

KNOWLEDGE CHECK

16. Which 2.4 GHz channels do not overlap?

(A) 1, 2, 3
(B) 3, 7, 13
(C) 1, 6, 11
(D) 6, 12, 18

17. Which Wi-Fi security option should a small office choose for staff today?

(A) WEP
(B) WPA3-Personal or enterprise 802.1X
(C) WPA-TKIP
(D) Open with a hidden SSID

18. A signal reading of -82 dBm is:

(A) Excellent
(B) Weak, likely to cause problems
(C) Good for calls
(D) Too strong

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19. A manager wants to reach an office CCTV recorder from home. What is the safest approach?

(A) Forward the recorder's web port to the internet so he can log in directly

(B) Disable the router firewall

(C) Use a VPN or a secure cloud remote-access service

(D) Use the default admin password

20. Why should guest Wi-Fi be isolated from the internal LAN?

(A) So guest devices cannot reach internal servers and devices

(B) To make it faster

(C) Because guests use IPv6

(D) To save DHCP addresses and give staff devices more of the wireless bandwidth

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · CONNECT · LESSON 5 · 30 MIN

Network Troubleshooting Tools

Ping, traceroute, port tests, sockets and a layered method

- Apply a bottom-up method to network tickets
- Use ping, traceroute, port tests and socket tools on Windows, Linux and macOS
- Interpret common outputs to decide where a fault lies
- Document network evidence clearly in a ticket

Network tickets feel vague: 'the internet is slow', 'the system is down'. Good support specialists turn vague reports into precise facts using a handful of tools and a fixed order of checks. Each test either passes, so you move up a layer, or fails, so you know where to dig.

The five-check method

1. **Link** Is the cable connected or Wi-Fi associated? Check link lights, `ipconfig` or `ip link`.
2. **Address** Does the host have a valid IP, mask, gateway and DNS? Look for 169.254 or wrong subnets.
3. **Gateway** Can it ping the default gateway? If not, the problem is local.
4. **Beyond** Can it ping a known external IP? If yes, routing and the internet link are fine.
5. **Name and service** Does DNS resolve, and is the service's port reachable? Test the actual application last.

The core toolkit on Windows

Windows PowerShell

```
ping 192.168.1.1 -n 4
Test-Connection 8.8.8.8 -Count 4
tracert www.example.com
Test-NetConnection mail.company.com -Port 587
Get-NetTCPConnection -State Established | Select-Object -First 10
arp -a                              # IP-to-MAC mappings learned on this subnet
```

The same checks on Linux

Linux host

```
ping -c 4 192.168.1.1
traceroute www.example.com          # or tracepath if traceroute is not installed
nc -zv mail.company.com 587         # is the port open?
ss -tulpn                           # listening ports and owning processes
ip neigh                            # neighbour (ARP) table
curl -I https://intranet.company.local   # HTTP status and headers
```

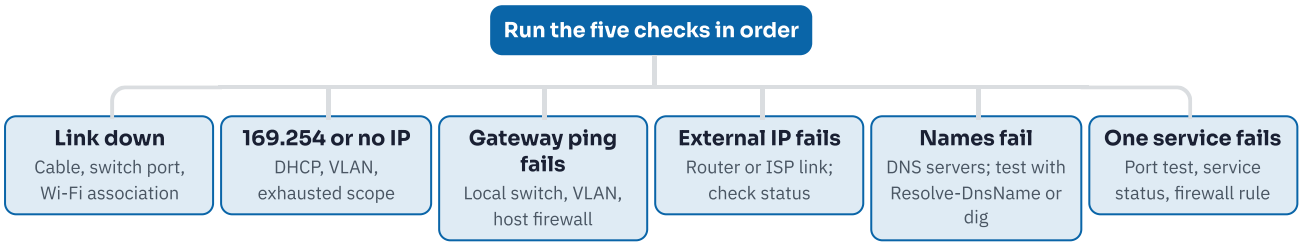
Reading common results

RESULT	MEANING	NEXT STEP
Request timed out to the gateway	Local link, VLAN or firewall issue	Check cable, Wi-Fi, port and host firewall
Ping to 8.8.8.8 works, names fail	DNS problem	Check DNS servers and test with Resolve-DnsName or dig

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RESULT	MEANING	NEXT STEP
Traceroute stops at hop 3	Path breaks after that router	Report the hop to the network team or provider
TcpTestSucceeded : False	Port blocked or service not listening	Check firewall rules and service status
High, variable ping times	Congestion or weak Wi-Fi	Test wired, check utilisation and signal

'The internet is down': where is the fault?



Stop at the first check that fails: that branch tells you where the fault lives and who to involve. Everything to its left has already been proven to work.

INSIGHT · PING CAN LIE

Many servers and firewalls block ICMP, so a failed ping does not always mean a host is down. Confirm with a port test to the service the user actually needs, such as HTTPS on 443.

Bandwidth and latency checks

MEASURE	FORMULA	WHAT IT TELLS YOU
Throughput	`Mbps = (bytes transferred x 8) / seconds / 1,000,000`	Transferring 500 MB in 50 s is 4,000 megabits / 50 = 80 Mbps.
Packet loss	`loss % = (sent - received) / sent x 100`	Sending 100 pings and receiving 96 is 4% loss, enough to ruin voice calls.
Round-trip time	`RTT = time for request plus reply`	Local LAN under 1-2 ms; same-country cloud 10-50 ms; overseas 150 ms or more.

Physical-layer tools

- **Cable tester** confirms each pair is wired and continuous.
- **Tone generator and probe** traces an unlabelled cable to its patch-panel port.
- **Loopback plug** tests a network port without another device.
- **Wi-Fi analyser app** shows channels, neighbours and signal strength.
- **Link lights** on switches and NICs give an instant yes or no.

TEMPLATE **Network troubleshooting log**

Capture evidence so the next person, or the provider, can act without repeating your tests.

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TIME	TEST	FROM (HOST)	TO (TARGET)	RESULT	CONCLUSION
10:05	*ping gateway*	*ACC-PC-12*	*192.168.10.1*	*4/4 replies, 1 ms*	*Local link OK*
10:07	*Test-NetConnection port 443*	*ACC-PC-12*	*erp.company.com*	*TcpTestSucceeded False*	*Port blocked or service down; check firewall*

CASE STUDY · THE BRANCH THAT CANNOT REACH THE ERP

A bank branch in Cumilla reports that no one can open the head-office ERP system since 9:00, although internet browsing works. Branch PCs get correct DHCP addresses, the gateway replies to ping, and `Resolve-DnsName erp.bank.local` returns 10.50.1.20. `Test-NetConnection 10.50.1.20 -Port 443` fails, and `tracert 10.50.1.20` stops at the branch router.

1. Which layers have you proven are working?

2. Where is the fault most likely?

3. What do you write in the escalation to the network team?

Model answer: module answer key.

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SCENARIO · DECIDE · ONLY ONE PC IS SLOW

One accountant says 'the internet is slow'. Colleagues on the same floor are fine. Her PC shows 100 Mbps link speed while others show 1 Gbps, and the cable is coiled under a chair wheel.

(A) Call the internet provider
(B) Reinstall the network driver
(C) Replace the damaged patch cable and confirm the link returns to 1 Gbps
(D) Change her DNS server

Circle your choice. Answer and feedback: module answer key.

Key terms

Ping
A tool that sends ICMP echo requests to test reachability and round-trip time.

Test-NetConnection
A PowerShell cmdlet that tests ping, TCP port reachability and routes to a target.

Latency
The delay for data to travel across a network, usually measured as round-trip time in milliseconds.

Cable tester
A device that checks network cable pairs for continuity, wiring faults and sometimes length.

Traceroute
A tool that lists each router hop on the path to a destination, showing where a path breaks or slows.

ss
A Linux utility that lists open sockets and listening ports, replacing older netstat.

Packet loss
The percentage of packets that fail to reach their destination; small amounts damage voice and video.

Key takeaways

- Turn vague reports into facts with the same five checks every time.
- A failed ping is not proof a host is down; test the real service port.
- Traceroute shows where a path breaks, which tells you who to escalate to.
- Loss above about 1-2% or high, variable latency ruins voice and video.
- Write down every test and result so nobody repeats them.

KNOWLEDGE CHECK

21. A user can ping 8.8.8.8 but cannot browse to any site by name. What is the next test?

(A) Test DNS resolution with Resolve-DnsName or dig
(B) Replace the cable
(C) Check the printer
(D) Renew the Wi-Fi password and reconnect the laptop to the wireless network

22. A server blocks ICMP. How can you confirm its web service is reachable?

(A) ping -t
(B) Test-NetConnection server -Port 443
(C) arp -a
(D) ipconfig /flushdns

23. Which Linux command shows which process is listening on port 8080?

(A) ip route
(B) dig
(C) ss -tulpn
(D) chmod

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24. Out of 200 pings, 190 replies arrive. What is the packet loss?

(A) 10%
(B) 5%
(C) 2%
(D) 19%

25. Traceroute to a head-office server stops at the first hop. What does this suggest?

(A) DNS failure
(B) The server's disk is full
(C) The path breaks at or just beyond the local router
(D) The user's password expired, so the router refuses to forward the traffic

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · CONNECT

Answer key

Knowledge checks

1. B Local traffic works, so the LAN is fine; the path out through the gateway or internet link is suspect.

2. B Remote Desktop uses TCP 3389. 22 is SSH, 443 HTTPS and 53 DNS.

3. C TLS and certificates are usually placed at the presentation layer, near the application. Lower layers deal with cables, frames and addressing.

4. C Real-time media prefers speed; retransmitting old audio does not help. UDP itself has no encryption or delivery guarantee.

5. A TCP or UDP headers add ports. IP adds addresses at Layer 3 and MACs at Layer 2.

6. C 2^5 = 32 addresses minus network and broadcast gives 30.

7. A Block size 64: 0, 64, 128, 192. 200 falls in the block starting at 192.

8. A 169.254.x.x is APIPA, used when DHCP fails.

9. A Drop leading zeros and replace the single run of zero groups with ::. '::' can appear only once.

10. A The gateway is outside the host's subnet, so the host cannot reach it; same-subnet traffic still works.

11. D DORA: Discover, Offer, Request, Acknowledge.

12. B MX lists mail servers. A maps names to IPv4, CNAME is an alias and PTR is reverse lookup.

13. B IP connectivity works, so name resolution is the problem.

14. A A reservation keeps the address constant while still managed centrally.

15. D Resolvers keep cached answers until the TTL expires; flushing caches or waiting resolves it.

16. C Channels 1, 6 and 11 are the standard non-overlapping set in 2.4 GHz.

17. B WPA3 or 802.1X are current best practice. WEP and TKIP are broken and hiding an SSID is not security.

18. B Values below about -75 dBm are weak. Closer to zero is stronger.

19. C A VPN or secure service avoids exposing the recorder to the internet. The other options create serious risks.

20. A Isolation prevents untrusted devices from reaching internal resources.

21. A IP connectivity beyond the gateway works, so name resolution is the next layer to test.

22. B A TCP port test checks the actual service even when ping is blocked.

23. C ss -tulpn lists listening sockets with process names.

24. B (200 - 190) / 200 x 100 = 5%.

25. C Traceroute shows where forwarding stops, pointing to the local router or the next link.

Scenarios & case studies

Scenario · Website loads, video calls fail
C is best. Correct. Calls depend heavily on UDP, while web browsing uses mostly TCP 443. A recent firewall change is the obvious suspect.

Scenario · A 169.254 address
B is best. Correct. APIPA means the PC never heard back from DHCP. A physical or port issue on this one PC, or a DHCP failure, are the likely causes; renewing tests the fix.

Scenario · Only names fail
A is best. Correct. The scope is still handing out the old DNS server. Fixing the option at the source fixes every client.

Scenario · Meeting room calls keep dropping
D is best. Correct. Weak signal through concrete on a crowded band is the root cause; closer coverage and a cleaner band fix it.

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Case study · The branch that cannot reach the ERP

Working: link, addressing, gateway and DNS all pass, and the internet path works. **Likely fault:** the route or VPN tunnel from the branch router to head office (traceroute stops at the branch router), or a firewall rule change. **Escalation note:** time the fault started, scope (whole branch), the exact tests and outputs (DNS answer 10.50.1.20, port 443 test failed, traceroute stops at hop 1), what still works (internet), and a request to check the branch-to-HQ tunnel and recent firewall changes. This avoids anyone repeating basic tests.

Scenario · Only one PC is slow

C is best. Correct. A crushed cable often drops to 100 Mbps or produces errors. The single affected PC and the evidence point there.

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04

MODULE 4 · RESOLVE

Troubleshooting, Customer Service and Ticketing

A disciplined method, excellent communication and service management that keeps people working

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Apply a seven-step troubleshooting method and isolation techniques to any IT problem
- Find and remove root causes so incidents do not recur
- Communicate with users clearly, calmly and professionally
- Categorise, prioritise and document tickets within SLAs

LESSONS

1. A Troubleshooting Method That Works · 26 min
2. Diagnostic Techniques and Root Cause · 28 min
3. Customer Service and Communication · 24 min
4. Ticketing and IT Service Management · 30 min

MODULE 4 · RESOLVE · LESSON 1 · 26 MIN

A Troubleshooting Method That Works

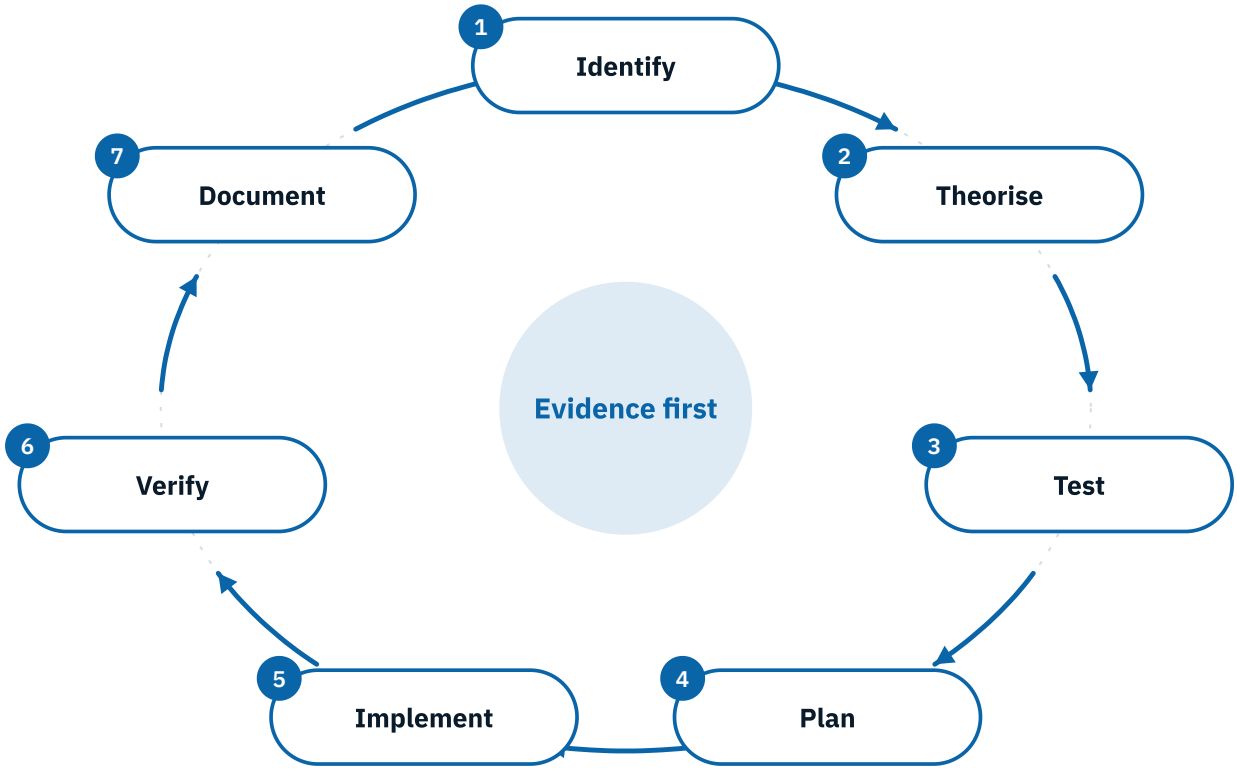
Seven steps that turn guesswork into evidence

- Apply a seven-step troubleshooting method to any IT problem
- Ask questions that define the scope, timing and recent changes
- Form and test theories one at a time, in order of likelihood and ease
- Verify the fix with the user and document the outcome

Experienced support staff are not faster because they know every answer. They are faster because they follow the same method every time: they gather facts before acting, change one thing at a time and confirm the result. Guessing feels quick but often creates new problems, especially on systems other people depend on.

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The ZITSS seven-step troubleshooting method



1. **Identify** — Gather facts: symptoms, error messages, scope, timing and recent changes. Reproduce if you can. Back up data before risky work.
2. **Theorise** — List probable causes, starting with the most likely and easiest to test. Question the obvious.
3. **Test** — Test one theory at a time. If it is disproved, move to the next or escalate.
4. **Plan** — Decide how to fix it, considering impact, approvals, change windows and rollback.
5. **Implement** — Apply the fix, or escalate with your evidence.
6. **Verify** — Confirm full functionality with the user and add preventive measures.
7. **Document** — Record symptoms, cause, actions and outcome in the ticket and knowledge base.

The method is a loop: a disproved theory sends you back to Theorise, and what you document feeds the next Identify. Never skip Verify or Document to save time.

Questions that define the problem

QUESTION	WHY IT MATTERS
What exactly happens? Can you show me or send the error text?	Exact messages narrow the cause quickly
When did it start? Did it ever work?	Points to a change or a new set-up issue
What changed recently: updates, new software, moves, passwords?	Most incidents follow a change

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QUESTION	WHY IT MATTERS
Who else is affected?	One user, one team or everyone changes where you look
Is it constant or intermittent? Any pattern by time or place?	Patterns reveal heat, load, schedules or Wi-Fi coverage
What have you already tried?	Avoids repeating tests and reveals side effects

INSIGHT · SCOPE IS YOUR FASTEST CLUE

If **one user** is affected, look at their device, account or location. If **one team** is affected, look at what they share: a printer, a group, a switch. If **everyone** is affected, look at shared services such as DNS, the internet link or a cloud service, and declare an incident early.

Good theories versus weak theories

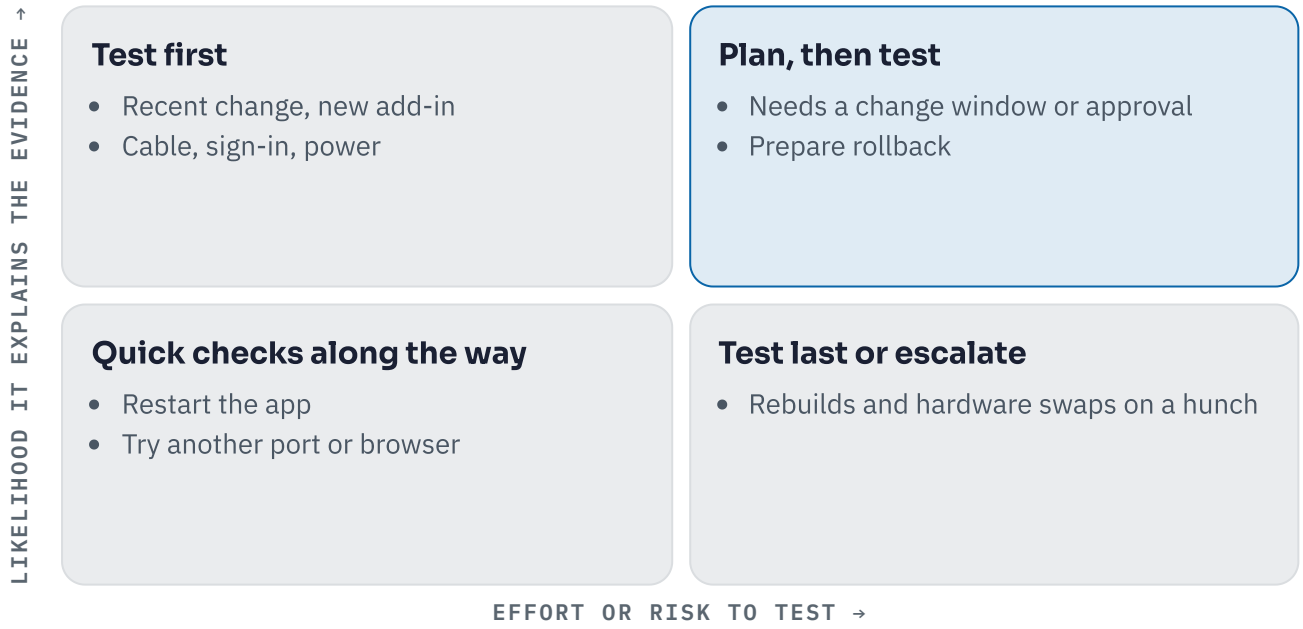
Good theories

- Explain all the symptoms
- Fit the scope and timing
- Can be tested quickly and safely
- Start simple: power, cables, sign-in, recent change

Weak theories

- Explain only part of the evidence
- Ignore who is affected
- Require major changes to test
- Come from 'last time it was...' without checking

Which theory to test first



Rank theories by how likely they are and how cheap and safe they are to test. Likely, cheap tests come first; expensive, unlikely ones come last or go to a specialist.

WATCH OUT · CHANGE ONE THING AT A TIME

If you change three settings and the problem disappears, you do not know which change fixed it, and you may have introduced two new problems. Change one variable, test, and undo it if it did not help.

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Escalating well

01 Recognise the limit
Your skill, access or authority is exceeded, or the SLA is at risk.

→

02 Package evidence
Symptoms, scope, tests done and results, timestamps, screenshots.

→

03 Choose the right team
Network, server, application owner or vendor.

→

04 Tell the user
Who has it now and when they will hear next.

→

05 Follow up
Stay the user's contact until it is resolved.

Before you close any ticket

- ☐ The user confirms the problem is solved in their own words.
- ☐ Related functions still work (printing, email, shared drives).
- ☐ Root cause is recorded, or marked unknown with evidence.
- ☐ Any temporary workarounds or test changes are removed.
- ☐ Preventive action is noted, such as a KB article or a problem record.
- ☐ Time spent and category are recorded accurately.

SCENARIO · DECIDE · EMAIL APP WILL NOT OPEN FOR ONE USER

A sales manager says email will not open on his laptop. You notice a new add-in was installed this morning. Colleagues' email works. He has a client meeting in an hour.

- (A) Reinstall the whole office suite immediately
- (B) Reset his password
- (C) Escalate to the email server team
- (D) Start the mail app in safe mode or with add-ins disabled to test the new add-in theory

Circle your choice. Answer and feedback: module answer key.

Key terms

Troubleshooting methodology
A repeatable sequence of steps for finding and fixing problems based on evidence rather than guesswork.

Theory of probable cause
A testable explanation of a problem that fits the symptoms, scope and timing.

Rollback
Reversing a change to return a system to its previous working state.

Root cause
The underlying reason a problem occurred, which, if fixed, prevents it recurring.

Scope (of an incident)
How many users, devices, locations or services are affected by a problem.

Reproduce
Make a problem happen again on purpose to observe it and test fixes.

Verification
Confirming with the user and through testing that the fix works and nothing else broke.

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Key takeaways

- Gather facts first: symptoms, scope, timing and recent changes.
- Scope is the fastest clue: one user, one team or everyone points to different places.
- Test the most likely, cheapest theory first, and change one thing at a time.
- Escalate with evidence and stay the user's contact until it is fixed.
- Close only after the user confirms and the root cause and fix are documented.

KNOWLEDGE CHECK

1. What should you do FIRST when a user reports a problem?

   (A) Gather information about symptoms, scope and recent changes
   (B) Restart their computer
   (C) Escalate to Tier 2
   (D) Reinstall the application they were using when the problem started

2. You changed a setting and it did not fix the problem. What should you do?

   (A) Leave it and try more changes
   (B) Undo it before testing the next theory
   (C) Close the ticket
   (D) Change several more settings at once to save time on the ticket

3. Every user in the building suddenly loses access to a cloud app. Where should you look first?

   (A) One user's browser cache and saved cookies, starting with the first caller
   (B) Shared services: internet link, DNS or the provider's status
   (C) Each user's keyboard
   (D) The printer queue

4. Which step comes right after implementing a fix?

   (A) Document the change in the knowledge base and close the ticket
   (B) Theorise
   (C) Verify full functionality with the user
   (D) Identify

5. Why is 'What changed recently?' such a valuable question?

   (A) Users enjoy answering it
   (B) Most incidents are triggered by a recent change
   (C) It replaces testing
   (D) It is required by law in most service management standards and audits

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · RESOLVE · LESSON 2 · 28 MIN

Diagnostic Techniques and Root Cause

Isolate faults, read the evidence and stop problems coming back

- Use isolation techniques: half-split, known-good swap and minimum configuration
- Collect evidence from logs, performance tools and the user's screen
- Use remote support tools safely and with consent
- Find root causes with the five whys and a cause-and-effect diagram

Once you know what the problem is, you need to find where it lives. Diagnostic techniques are ways of shrinking the search area quickly. Root-cause techniques then make sure that, once fixed, the same problem does not come back next week for fifty more users.

Isolation techniques

TECHNIQUE	HOW IT WORKS	EXAMPLE
Half-split	Test the middle of a chain; discard the half that works	Ping the gateway: works, so the fault is beyond it
Known-good swap	Replace a suspect part with one known to work	Swap the monitor cable with a tested one
Move the problem	Move the device, user or part to see if the fault follows	Try the user's laptop at another desk
Compare with a working system	Find differences between good and bad	Compare driver versions on two identical PCs
Minimum configuration	Remove everything non-essential, then add back	Clean boot with start-up apps disabled

Collecting evidence on Windows

Windows PowerShell

```
Get-WinEvent -FilterHashtable @{LogName='System'; Level=1,2; StartTime=(Get-Date).AddHours(-24)} |
  Select-Object TimeCreated, Id, ProviderName, Message -First 15
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 5
Get-CimInstance Win32_OperatingSystem | Select-Object LastBootUpTime, FreePhysicalMemory
Get-Volume C | Select-Object SizeRemaining, Size
```

Level 1 is Critical and Level 2 is Error. Recent updates and the last boot time are quick ways to correlate a fault with a change.

Collecting evidence on Linux

Linux host

```
journalctl -p err -b                # errors since the last boot
uptime                              # load averages and time since boot
free -h && df -h
top -b -n 1 | head -n 15            # snapshot of busiest processes
grep -i 'out of memory' /var/log/kern.log   # Debian/Ubuntu kernel log
```


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PRO TIP · ASK FOR THE SCREEN, NOT A DESCRIPTION

A screenshot or a short screen recording of the error is worth ten minutes of questions. Teach users the screenshot shortcut on their platform and ask for one in your first reply.

Remote support done right

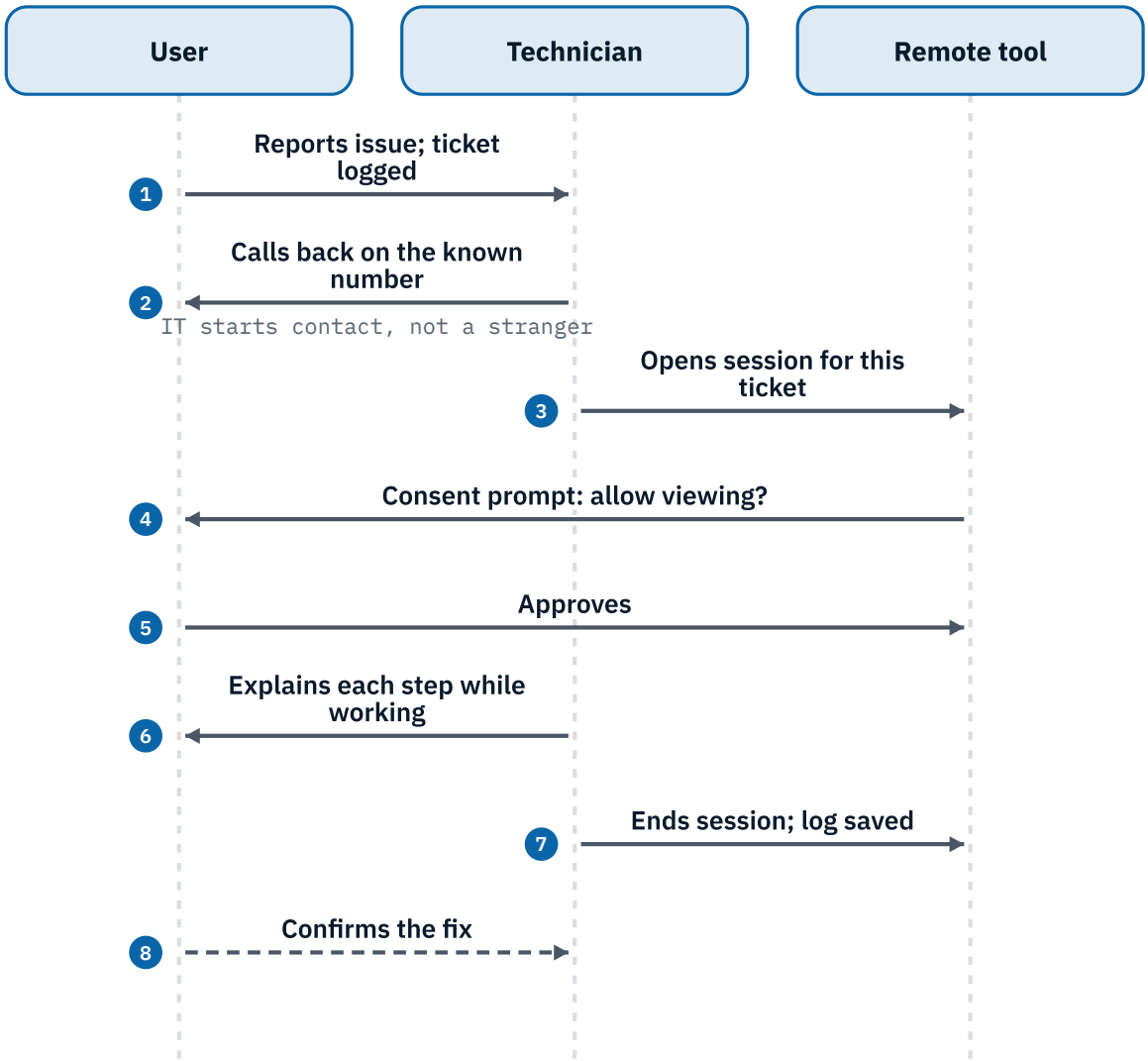
Do

- Ask permission before connecting and say what you will do
- Use the organisation's approved, logged remote tool
- Close sensitive windows or ask the user to
- Disconnect and confirm when finished

Do not

- Use personal or free remote tools with no logging
- Browse the user's personal files
- Leave unattended access enabled without a reason
- Accept a 'remote support' call you did not initiate (common scam)

A legitimate remote support session



Real support starts from a ticket, is initiated by IT through an approved, logged tool and needs the user's consent each time. A caller who reverses this order and asks the user to install something is running the classic support scam.

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Root-cause techniques

5W
Five whys
Ask 'why?' repeatedly until you reach a cause you can act on, usually a process or design gap.

FB
Fishbone
Group possible causes under people, process, technology, environment and suppliers.

TL
Timeline
Lay out events with timestamps to see what happened before the failure.

CA
Change analysis
List every change in the period before the problem and test each.

Five whys: lab PCs out of disk space

1. **Why did lab PCs stop logging users in?** The system drives were full.
2. **Why were the drives full?** Hundreds of user profiles were stored locally.
3. **Why were profiles kept?** Each student who logs in creates a profile and nothing removes old ones.
4. **Why does nothing remove them?** The profile clean-up policy was never set when the lab was rebuilt.
5. **Why was it missed?** The lab build checklist did not include it. Fix: add the policy and update the checklist.

CASE STUDY · MONDAY MORNING PRINTING FAILURES

Every Monday, about 30 users at a pharmaceutical company in Savar cannot print for the first hour. By 10:00 it works. Restarting the print server 'fixes' it. Logs show the print server installs updates and reboots at 03:00 on Sunday, and its print spooler service is set to 'Manual' start. Two months ago it was set to 'Automatic'.

1. Which diagnostic technique revealed the pattern?

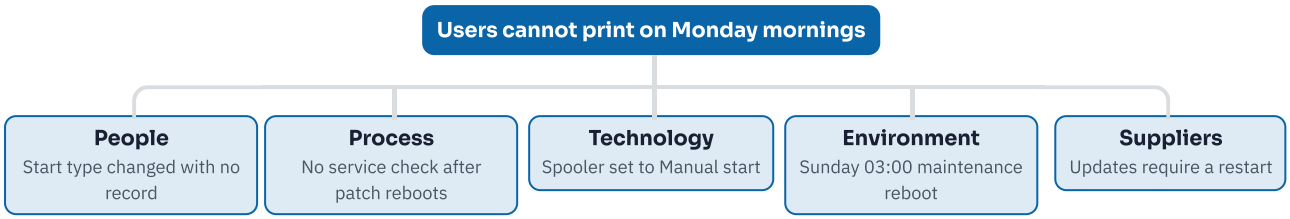
2. What is the root cause?

3. What permanent fixes and preventive actions would you propose?

Model answer: module answer key.

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Fishbone, drawn as a tree: Monday printing failures



Grouping candidate causes by category stops a team fixating on technology alone. The Manual start type is the technical fault, but the missing post-reboot check is what let it recur every week.

SCENARIO · DECIDE · INTERMITTENT FREEZES ON ONE MODEL

Twelve identical laptops in a finance team freeze for several seconds a few times a day. Twelve laptops of a different model are fine. All run the same image.

- (A) Compare drivers and firmware between the two models and check the affected model's known issues
- (B) Reimage all twelve laptops
- (C) Replace the users' keyboards
- (D) Ask the users to restart more often

Circle your choice. Answer and feedback: module answer key.

Key terms

Half-split
A diagnostic technique that tests the midpoint of a system or path to eliminate half of the possible fault locations.

Clean boot
Starting an OS with non-essential start-up programs and services disabled to isolate software conflicts.

Fishbone diagram
A cause-and-effect diagram that groups possible causes into categories such as people, process and technology.

Problem record
A record that tracks the investigation of the underlying cause of one or more incidents.

Known-good component
A part or device verified to work, used to isolate faults by substitution.

Five whys
A root-cause technique that asks 'why?' repeatedly until a fixable underlying cause is found.

Remote support tool
Approved software that lets a technician view or control a user's device over the network with consent and logging.

Key takeaways

- Shrink the search area with half-split, known-good swaps, moving the problem and comparison.
- Collect evidence from logs, recent updates and boot times before changing anything.
- Remote sessions need an approved, logged tool, the user's consent and a ticket.
- Unsolicited 'support' calls asking users to install tools are scams: refuse and report.
- Root-cause tools (five whys, fishbone, timeline, change analysis) target process gaps, not people.

KNOWLEDGE CHECK

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6. A fault follows a laptop when it is moved to another desk and another network port. What does this suggest?
   - (A) The original network port is faulty
   - (B) The building's DNS server is failing
   - (C) The switch needs replacing
   - (D) The fault is in the laptop itself
7. Which technique would you use to rule out half of a long network path at once?
   - (A) Half-split testing
   - (B) Five whys
   - (C) Fishbone
   - (D) Clean install
8. A caller claiming to be from a well-known software company's support team asks a user to install a remote tool. What should the user do?
   - (A) Install it if the caller sounds professional
   - (B) Hang up and report it to IT
   - (C) Give them the password only
   - (D) Restart the PC
9. What is the main purpose of the five whys?
   - (A) To assign blame
   - (B) To speed up password resets by asking the user five security questions
   - (C) To list all users affected
   - (D) To reach an actionable underlying cause
10. Which Linux command shows error-level journal entries since the last boot?
    - (A) df -h
    - (B) journalctl -p err -b
    - (C) uptime
    - (D) ls /var

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · RESOLVE · LESSON 3 · 24 MIN

Customer Service and Communication

Listening, explaining and calming people down while you fix things

- Use active listening and empathy to understand users' needs
- Explain technical issues and fixes in plain language
- Handle frustrated or difficult users professionally
- Write clear updates by email and chat and set realistic expectations

For the user, the IT problem is never really about the printer or the password. It is about a deadline, a patient, a customer or a manager who is waiting. Great support specialists solve the technical problem **and** the human one. The way you communicate decides whether users trust IT or work around it.

The LEAP model for every interaction

L

Listen

Let the user explain without interruption, then summarise back what you heard.

E

Empathise

Acknowledge the impact: 'I understand the report is due at noon.'

A

Act

Explain the next step, do it and keep them informed.

P

Promise and follow up

Commit to a time for the next update and keep that promise.

Words that help and words that hurt

Say

- 'Let me check that for you.'
- 'Here is what I am going to do next.'
- 'I will update you by 3 pm, even if it is not fixed yet.'
- 'Thanks for the screenshot, that helps a lot.'

Avoid

- 'That is impossible.'
- 'You must have done something.'
- 'It works on my machine.'
- Acronyms and jargon such as 'your DHCP lease expired'

Choosing the right channel

CHANNEL	BEST FOR	TIPS
Phone or video	Urgent, complex or emotional issues	Confirm details in writing afterwards
Chat	Quick questions, guided fixes	Short messages, one step at a time
Email or ticket portal	Updates, non-urgent requests, records	Clear subject, summary first, next step and time
In person	Hardware, VIP or vulnerable users, training	Respect the workspace and privacy

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Picking a channel for the moment

COMPLEXITY OR EMOTION →

Scheduled call or visit
- Training, a new device
- A sensitive conversation

Phone or video now
- Outage before a deadline
- An upset or vulnerable user

Portal, email or KB
- Requests and how-to questions
- Routine status updates

Chat
- Quick guided fix
- One step at a time

URGENCY →

Match the channel to the urgency and the emotional weight of the issue. Whatever channel you use, confirm the outcome in writing in the ticket.

Handling a frustrated user

1. **Stay calm and do not take it personally** Their frustration is with the situation, not with you.
2. **Let them vent briefly** Interrupting escalates emotion; listen for facts inside the complaint.
3. **Acknowledge and apologise for the impact** 'I am sorry this has disrupted your morning.' You are not admitting fault.
4. **Move to action** Explain exactly what you will do now and when they will hear next.
5. **Set boundaries if needed** If abuse continues, say you want to help, and involve your team lead according to policy.

EXAMPLE · TRANSLATING JARGON

Instead of 'Your profile is corrupted, so I will rebuild it', say: 'Your laptop's saved settings for your account are damaged. I will create a fresh set. Your files are safe in the company cloud storage, but you will need to sign in to a few apps again.' The user knows what will happen and what to expect.

TEMPLATE Status update message

A consistent structure for updates by email, chat or ticket so users always know where things stand.

SECTION	WHAT TO WRITE	EXAMPLE
Summary	*One line on the issue and current state*	*Your shared-drive access is still blocked; we have found the cause.*
Next step and time	*What happens next and when you will update*	*The server team is correcting permissions; I will update you by 2:30 pm.*

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Professional conduct

- Be on time for appointments and tell users if you are delayed.
- Respect confidentiality: never discuss what you saw on someone's screen.
- Ask before touching a user's device, files or desk.
- Avoid personal calls and distractions while helping a user.
- Be inclusive: adapt to language, accessibility needs and seniority without condescension.
- Own your mistakes quickly and explain how you will fix them.

SCENARIO · DECIDE · THE ANGRY HEAD OF DEPARTMENT

The head of the Accounts department at a Chattogram port company calls, angry, because his team's shared folder has been inaccessible for an hour and month-end reports are due. He says IT is 'useless'. You already know the server team is restoring the folder.

(A) Explain that it is the server team's fault, not yours

(B) Acknowledge the impact, explain the restore in plain terms, give a realistic time and offer a workaround such as access to yesterday's copy

(C) Tell him to calm down before you will help

(D) Promise it will be fixed in five minutes to end the call

Circle your choice. Answer and feedback: module answer key.

REFLECT

Recall a time you were frustrated with a service provider. Which of their phrases or actions made it worse, and which helped? Write two phrases you will use, and two you will avoid, in your own support work.

Key terms

Active listening
Fully concentrating on the speaker, then summarising back what was said to confirm understanding.

Expectation setting
Telling users clearly what will happen next and when, so they can plan.

Workaround
A temporary way for the user to keep working while the real fix is in progress.

Confidentiality
Keeping private any information you see or hear while supporting users.

Empathy
Recognising and acknowledging how a problem affects the other person.

Jargon
Technical language that the listener may not understand; it should be translated into plain terms.

Follow-up
Contacting the user after a fix or at an agreed time to update them or confirm the fix held.

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Key takeaways

- Solve the human problem as well as the technical one: Listen, Empathise, Act, Promise.
- Translate jargon into what the user will see and what they need to do.
- Choose the channel by urgency and emotion, and confirm in writing.
- Keep every promised update time, even when there is no fix yet.
- Respect confidentiality: what you see on a user's screen stays private.

KNOWLEDGE CHECK

11. A user is very upset about a lost file. What should you do first?

(A) Listen, acknowledge the impact and gather the facts
(B) Explain why it was their fault
(C) Escalate to management immediately
(D) Tell them deleted files cannot be recovered so they can start again quickly

12. Which update is the most useful to a user?

(A) 'Working on it.'
(B) 'Your DHCP option 6 is wrong, so the scope options need to be reconfigured.'
(C) 'Cause found; vendor patch installing; I will confirm by 4 pm.'
(D) 'Should be fine soon.'

13. When is phone or video better than email?

(A) For routine updates
(B) For sending large log files and screenshots to the vendor's support team
(C) For password sharing
(D) For urgent, complex or emotional issues

14. You promised an update by 2 pm but have no progress. What should you do?

(A) Wait until you have news
(B) Close the ticket
(C) Send an update at 2 pm with progress so far and a next update time
(D) Ask the user to call you when they want an update, so you are not interrupted

15. While fixing a laptop you notice a colleague's salary file open on the screen. What should you do?

(A) Mention it to the team
(B) Copy it for safekeeping
(C) Ignore it and keep the information confidential
(D) Report the salary amount to HR in case it is higher than it should be

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · RESOLVE · LESSON 4 · 30 MIN

Ticketing and IT Service Management

Incidents, requests, priorities, SLAs and knowledge

- Distinguish incidents, service requests, problems and changes
- Prioritise tickets using impact and urgency
- Work within SLAs and track key support metrics
- Write high-quality ticket notes and knowledge-base articles

A **ticketing system** is the memory of the support team. It makes sure nothing is forgotten, shows managers where time goes and lets anyone pick up a ticket where someone else left off. Good service management practices, described in widely used frameworks, give everyone the same language for what kind of work a ticket is and how urgent it is.

Four kinds of work

TYPE	DEFINITION	EXAMPLE	GOAL
Incident	Unplanned interruption or reduced quality of a service	Email is down for the sales team	Restore service fast
Service request	A routine, pre-approved request	New laptop, access to a shared folder	Fulfil reliably
Problem	The unknown cause of one or more incidents	Why does the VPN drop every afternoon?	Find and remove the root cause
Change	Adding, modifying or removing something that could affect services	Upgrading the firewall firmware	Change safely with approval

Prioritising tickets

MEASURE	FORMULA	WHAT IT TELLS YOU
Priority	`priority = impact x urgency (looked up in a matrix)`	Impact is how many people or how much business is affected; urgency is how quickly it hurts.
SLA compliance	`compliance % = tickets resolved within target / total tickets x 100`	If 470 of 500 tickets met target, compliance is 94%.
First contact resolution	`FCR % = tickets solved at first contact / total tickets x 100`	A high FCR means users get answers without being passed around.

A simple priority matrix

IMPACT \ URGENCY	HIGH URGENCY	MEDIUM URGENCY	LOW URGENCY
High impact (many users or critical service)	P1 Critical	P2 High	P3 Medium
Medium impact (a team or important user)	P2 High	P3 Medium	P4 Low
Low impact (one user, workaround exists)	P3 Medium	P4 Low	P4 Low

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Impact and urgency at a glance

IMPACT ↑

P3 Medium: plan it soon
- A team affected, workaround exists

P1 or P2: act now
- Clinic registration down, patients queuing

P4 Low
- Request for a new mouse

P3 Medium: quick fix
- One user, meeting in 10 minutes

URGENCY →

Seniority and volume of complaints are not on either axis. Set priority from impact and urgency, then let the SLA targets follow from the priority.

Example SLA targets

PRIORITY	RESPOND WITHIN	RESOLVE WITHIN	EXAMPLE
P1 Critical	15 minutes	4 hours	Hospital patient system down
P2 High	30 minutes	8 business hours	One department cannot print payroll
P3 Medium	4 business hours	3 business days	Single user's app crashes, workaround exists
P4 Low	1 business day	10 business days	Request a new mouse

Ticket lifecycle

01 **Logged** — Captured from phone, portal, email, chat or monitoring. →

02 **Categorised and prioritised** — Type, category and priority set. →

03 **Assigned** — Owned by a person or team. →

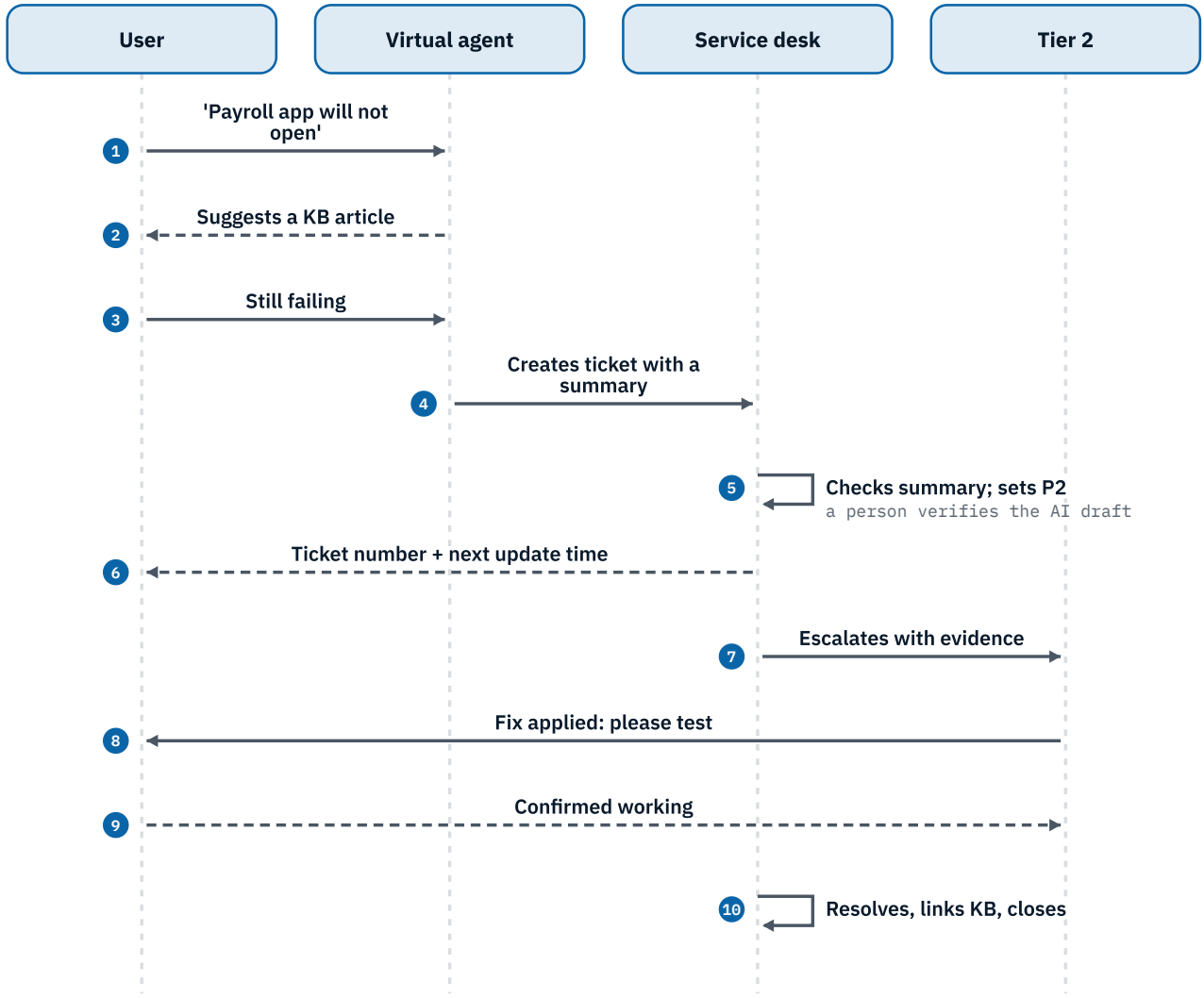
04 **In progress** — Diagnosed, updated, escalated if needed. →

05 **Resolved** — Fix applied and user confirms. →

06 **Closed** — Notes complete; automatically closes after a set time.

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A ticket from first contact to closure



A virtual agent can deflect simple issues and draft the ticket, but a person checks its summary and priority before work starts. Notice that the user hears something at every hand-off.

WATCH OUT · AI ASSISTANTS NEED HUMAN VERIFICATION

Many service desks now use AI assistants to suggest articles, summarise long tickets and draft replies. Treat every suggestion as a draft: check facts and commands before anyone runs them, never paste passwords or personal data into unapproved tools, and never let an assistant approve identity checks, access or MFA changes. The technician who sends the reply owns it.

PRO TIP · WRITE NOTES FOR A STRANGER

Write every ticket note as if a colleague who has never seen the ticket will pick it up at 2 am. Include what you tested, the results, what you changed and what is next. 'Fixed' is not a note.

TEMPLATE Ticket note

A standard structure for work notes so every ticket can be picked up by anyone.

FIELD	CONTENT	EXAMPLE
Symptom	*What the user sees, exact errors*	*Mail app shows 'Cannot start the mail client' since 09:10*

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FIELD	CONTENT	EXAMPLE
Tests and results	*What you checked and found*	*Safe mode opens; disabling CRM add-in fixes start-up*
Action and outcome	*Change made, user confirmation, next step*	*Add-in disabled, user confirmed at 09:40; problem record raised for add-in update*

What makes a good knowledge-base article

- A title written the way users describe the problem, such as 'Cannot connect to office Wi-Fi'.
- Who it applies to: OS, device type, location.
- Short numbered steps with screenshots.
- Separate sections for users and for technicians.
- An owner and a review date so it stays accurate.

CASE STUDY · A SERVICE DESK DROWNING IN TICKETS

A university IT service desk in Dhaka receives about 1,200 tickets a month with four staff. Users complain that tickets 'disappear'. Analysis shows 35% are password resets, 15% are 'how do I connect to Wi-Fi' questions, many tickets have no category and P1 incidents are logged by email like everything else.

1. Which changes would reduce ticket volume most?
2. How would you fix the 'disappearing' tickets?
3. How should P1 incidents be handled differently?

Model answer: module answer key.

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SCENARIO · DECIDE · TWO TICKETS AT ONCE

At 10:00 you have two new tickets: (A) the managing director's second monitor flickers; he can still work on his laptop screen. (B) The outpatient registration system at a clinic is down for all 12 reception staff and patients are queuing.

- (A) Do A first because the requester is the most senior person
- (B) Work on both at the same time, switching every five minutes
- (C) Do B first as a P1 incident and send A a quick acknowledgement with a time
- (D) Escalate both to Tier 3

Circle your choice. Answer and feedback: module answer key.

Key terms

Incident
An unplanned interruption to a service or a reduction in its quality.

Problem
The underlying cause, often unknown at first, of one or more incidents.

SLA
Service level agreement: agreed targets for responding to and resolving tickets.

Knowledge base
A searchable library of articles that explain fixes and procedures for users and support staff.

Service request
A formal request from a user for something routine, such as access, information or equipment.

Priority
The relative importance of a ticket, usually derived from impact and urgency.

First contact resolution
The share of tickets resolved during the user's first contact with the service desk.

Key takeaways

- Classify work correctly: incident, service request, problem or change.
- Priority comes from impact and urgency, never from seniority alone.
- SLA targets, FCR and compliance turn service quality into numbers you can improve.
- Write ticket notes a stranger could act on at 2 am.
- AI assistants can draft and summarise, but a person verifies before anything is sent or changed.

KNOWLEDGE CHECK

16. A user asks for access to the marketing shared folder. What type of ticket is this?
    - (A) Service request
    - (B) Incident
    - (C) Problem
    - (D) Major change
17. Investigating why the VPN drops every afternoon for many users is best tracked as:
    - (A) A service request
    - (B) A change
    - (C) A problem
    - (D) A purchase order
18. A service desk resolved 900 of 1,000 tickets within SLA. What is its SLA compliance?
    - (A) 90%
    - (B) 9%
    - (C) 100%
    - (D) 10%

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19. A single user's app crashes but a web version works. What priority fits best?

(A) P3 or P4
(B) P1 Critical
(C) P2 High
(D) No ticket needed

20. Which ticket note is most useful to a colleague?

(A) 'Fixed.'
(B) 'Looked at it.'
(C) 'User happy.'
(D) 'Printer offline: spooler stopped after update; restarted service, test page OK, user confirmed 11:20.'

21. An AI assistant drafts a reply telling a user to make a registry change and switch off a security tool. What should the technician do?

(A) Send it, since the assistant was trained on the vendor's documentation
(B) Check and test the steps, remove unsafe ones, then send it
(C) Ask the assistant whether its own answer is correct, then send it
(D) Forward it to the user and ask them to judge whether it looks safe

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 4 · RESOLVE

Answer key

Knowledge checks

1. A Identifying the problem comes before theories or fixes. Restarts and reinstalls without facts are guesses.

2. B Undo failed changes so you do not create new problems and can attribute any later fix correctly.

3. B Wide scope points to shared components, not individual devices.

4. C Verification confirms the fix works before documentation and closure.

5. B Recent changes are the most common trigger, so they point to likely causes quickly; you still test them.

6. D If the fault moves with the device, the device is the cause, not the location.

7. A Half-split tests the midpoint to eliminate half the path.

8. B Unsolicited support calls are a common scam. Users should refuse and report.

9. D The aim is to find a cause you can fix to prevent recurrence, not to blame people.

10. B journalctl -p err filters by priority and -b limits to the current boot.

11. A Listening and empathy calm the situation and reveal the facts you need.

12. C It states progress, the next step and a time, in plain language.

13. D Live conversation suits urgency and emotion; written channels suit records. Passwords should never be shared.

14. C Keeping the promise maintains trust even when there is no fix yet.

15. C Support staff must keep anything they see confidential.

16. A Access to an existing resource is a routine service request.

17. C Finding the underlying cause of recurring incidents is problem management.

18. A 900 / 1,000 x 100 = 90%.

19. A Low impact with a workaround is medium or low priority.

20. D It states symptom, cause, action and confirmation.

21. B The technician owns every reply: verify the steps and remove anything unsafe, such as disabling protection. The assistant cannot check itself, and users cannot judge technical safety.

Scenarios & case studies

Scenario · Email app will not open for one user
D is best. Correct. It tests the most likely, recent change quickly and safely, with minimal disruption.

Case study · Monday morning printing failures
Technique: timeline and change analysis: failures follow the Sunday reboot every week, and the spooler start type was changed two months ago. **Root cause:** after the weekly reboot the spooler does not start automatically; it only starts when an administrator restarts the server or someone starts the service manually. **Fix:** set the spooler to Automatic (with recovery options to restart on failure), add a service health check after maintenance reboots, raise a problem record noting who changed the start type and why, and add 'verify key services after patch reboot' to the patching checklist.

Scenario · Intermittent freezes on one model
A is best. Correct. The fault follows the hardware model, so model-specific drivers or firmware are the most likely cause.

Scenario · The angry head of department
B is best. Correct. Empathy, clear facts, a time commitment and a practical workaround address both the emotion and the business need.

Case study · A service desk drowning in tickets
Reduce volume: introduce self-service password reset with MFA (removing up to a third of tickets) and publish Wi-Fi set-up articles linked from the portal and the login page. **Disappearing tickets:** make category and priority mandatory, assign every ticket to an owner, send automatic acknowledgements with ticket numbers and use queue dashboards showing ageing tickets. **P1 handling:** a phone line or monitoring alert for critical incidents, immediate assignment, a named incident lead, regular status updates to affected users and a short review afterwards.

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Scenario · Two tickets at once

C is best. Correct. B has high impact and high urgency. A has a workaround and low impact, so it waits, with a polite update.

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05

MODULE 5 · MANAGE

System Administration

Users, groups, directories, policy, backups and patching: keeping many systems consistent and safe

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Manage users, groups and permissions using least privilege and joiner-mover-leaver processes
- Explain directory services, domains, OUs and cloud identity, and perform common account tasks
- Apply and troubleshoot group policy and configuration management
- Design, run and test backups against agreed RPO and RTO targets
- Run a ring-based patch process with emergency handling and rollback

LESSONS

1. Users, Groups and Permissions · 28 min
2. Directory Services and Identity · 30 min
3. Group Policy and Configuration Management · 26 min
4. Backup and Recovery · 30 min
5. Patch and Update Management · 26 min

MODULE 5 · MANAGE · LESSON 1 · 28 MIN

Users, Groups and Permissions

Giving the right people the right access, and nothing more

- Create and manage local user accounts and groups on Windows and Linux
- Explain role-based access and why permissions are assigned to groups
- Combine share and file-system permissions correctly
- Run joiner, mover and leaver processes safely

Most security incidents and many support tickets come down to access: someone cannot open what they need, or someone can open what they should not. System administration starts with **identity**: every person has their own account, accounts belong to groups that match roles, and permissions are granted to groups, never to individuals one by one.

Account types

Standard user

- Everyday work: email, documents, apps
- Cannot install system software or change security settings
- The default for every employee

Administrator

- Full control of a device or system
- Separate admin account used only for admin tasks
- Never used for email or browsing

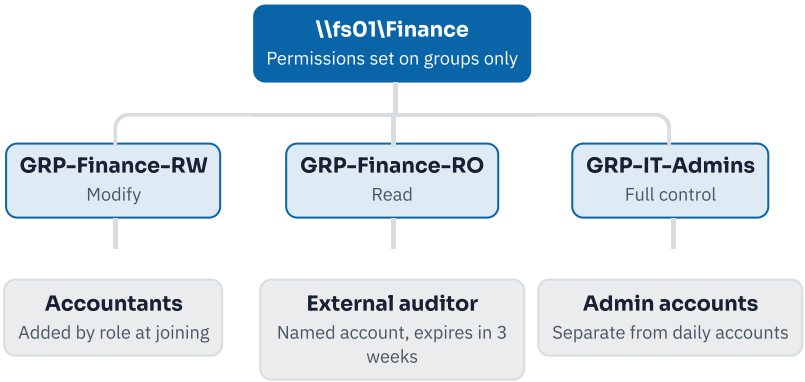
Service account

- Runs an application or scheduled task
- Long random password or managed credential
- No interactive sign-in; documented owner

INSIGHT · ASSIGN TO GROUPS, NOT TO PEOPLE

If ten accountants each get individual permissions on a folder, you will spend the year fixing exceptions. Create a group such as **GRP-Finance-RW**, give the group access and add or remove people from it. A new hire then needs one change, not ten.

Group-based access to a finance folder



Permissions are granted once, to groups; people join or leave groups as their role changes. A new accountant needs one change, and the auditor's access ends automatically with her account.

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Local users and groups on Windows

Windows PowerShell

```
Get-LocalUser | Select-Object Name, Enabled, LastLogon
$pw = Read-Host -AsSecureString 'Temporary password'
New-LocalUser -Name 'kiosk01' -Password $pw -FullName 'Lobby kiosk' -Description 'Reception kiosk'
Add-LocalGroupMember -Group 'Users' -Member 'kiosk01'
Get-LocalGroupMember -Group 'Administrators'
Disable-LocalUser -Name 'olduser'
```

Read-Host -AsSecureString keeps the password out of the screen and command history. Review the Administrators group regularly: it should contain very few members.

Users and groups on Linux

Linux host

```
sudo useradd -m -s /bin/bash -c 'Nusrat Jahan' nusrat
sudo passwd nusrat                  # set an initial password
sudo groupadd finance
sudo usermod -aG finance nusrat     # -a appends; without it other groups are removed
id nusrat                           # show user and group membership
sudo usermod -L olduser             # lock an account
```

Share permissions and file-system permissions

Share permissions

- Apply only when accessed over the network
- Simple: read, change, full control
- Common practice: give authenticated users broad share access, control detail with file-system permissions

File-system (NTFS) permissions

- Apply locally and over the network
- Detailed: read, write, modify, full control, special
- Inherited from parent folders unless inheritance is broken

Effective result

- When both apply, the most restrictive wins
- An explicit deny overrides allows
- Test with the 'effective access' view

Working out effective access

MEASURE	FORMULA	WHAT IT TELLS YOU
Over the network	`effective = most restrictive of (combined share permission, combined NTFS permission)`	Share 'Change' plus NTFS 'Read' gives Read.
Within one permission type	`combined = union of all allows from user and groups, minus any denies`	Member of Sales-Read and Sales-Modify gets Modify unless a deny applies.

Joiner, mover, leaver

01 Joiner — HR request triggers account, groups, licences and device for the role. →

02 Mover — Add new role groups and remove old ones on the day of the move. →

03 Leaver — Disable sign-in at the agreed time, revoke sessions and MFA, recover devices. →

04 Retain — Hand mailbox and files to the manager per policy. →

05 Delete — Remove the account after the retention period.

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WATCH OUT · PRIVILEGE CREEP

People who move between teams often keep old access 'just in case'. After a few years they can see far more than any single role needs. Removing old groups when someone moves, and reviewing access every quarter, stops this creep.

TEMPLATE **Access request record**

Capture who asked for access, who approved it and when it should be reviewed.

USER	RESOURCE	ACCESS LEVEL	GROUP USED	APPROVED BY	REVIEW DATE
T. Rahman	*\\fs01\Finance*	*Modify*	*GRP-Finance-RW*	*Head of Finance*	*2026-09-30*

SCENARIO · DECIDE · ACCESS FOR A TEMPORARY AUDITOR

An external auditor will work at a Dhaka bank's head office for three weeks and needs to read the Finance folder. The finance manager asks you to 'just give her the same access as me'.

(A) Create a named account with an expiry date and add it to a read-only finance group, with the manager's written approval

(B) Copy the finance manager's account permissions to the auditor

(C) Share the finance manager's password with the auditor

(D) Give the auditor a domain admin account so nothing blocks her

Circle your choice. Answer and feedback: module answer key.

Key terms

User account
An identity that lets one person sign in and be held accountable for actions on a system.

Role-based access control
Granting permissions according to job roles, implemented through groups.

NTFS permissions
File-system permissions on Windows volumes that control read, write, modify and full control access.

Privilege creep
The gradual build-up of unnecessary access as people change roles without old access being removed.

Group
A collection of accounts that receives permissions as a unit, usually matching a role or team.

Service account
A non-human account used by an application or service, with a documented owner and strong credentials.

Effective permissions
The access a user actually has after combining all user, group, share and deny entries.

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Key takeaways

- Every person gets their own account; admins also get a separate admin account.
- Grant permissions to role groups, never to individuals one by one.
- Over the network, the most restrictive of share and NTFS permissions applies, and an explicit deny wins.
- Run joiner, mover and leaver steps on the day they happen to stop privilege creep.
- Temporary access gets a named account, approval and an expiry date.

KNOWLEDGE CHECK

1. A user has share permission 'Full control' and NTFS permission 'Read' on a folder. What can they do over the network?

   (A) Full control
   (B) Modify
   (C) Nothing
   (D) Read

2. Why should `usermod` be run with `-aG` when adding a Linux user to a group?

   (A) -a makes the user an administrator of the group so they can add other members
   (B) -a sets a password
   (C) -a appends the group; without it existing supplementary groups are replaced
   (D) -a locks the account

3. A team of twelve needs access to a project folder. What is the best practice?

   (A) Add each user to the folder's permissions individually so access is precise
   (B) Give everyone full control
   (C) Share one account among the team
   (D) Create a group, grant it access and add the users to the group

4. An employee leaves today at 17:00. When should their account be disabled?

   (A) Next month
   (B) At or immediately after 17:00 as agreed with HR
   (C) When their manager remembers
   (D) Never, in case they return or need to collect personal files later

5. What is privilege creep?

   (A) Accumulation of unneeded access as people change roles
   (B) A virus that elevates privileges
   (C) Slow sign-in times
   (D) Too many administrators logging in at once and slowing the domain controllers

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · MANAGE · LESSON 2 · 30 MIN

Directory Services and Identity

Domains, organisational units, LDAP, Kerberos and cloud identity

- Explain what a directory service stores and why organisations use one
- Describe domains, domain controllers, organisational units and groups
- Compare on-premises directories, cloud identity and hybrid models
- Perform common account tasks: reset, unlock and find users

Once an organisation has more than a handful of people, managing accounts on each computer becomes impossible. A **directory service** keeps one central database of users, groups, computers and policies. People sign in anywhere with the same account, and IT can manage thousands of devices from one place. The most widely deployed example on-premises is Active Directory; cloud identity platforms now play the same role for SaaS apps.

A typical on-premises directory (top first)

Forest	`Security boundary containing one or more domains`
Domain	`company.local` `Shared user database and policies`
Organisational units (OUs)	`Dhaka, Chattogram, Laptops, Servers` `Used to delegate admin and apply policy`
Objects	`Users, groups, computers, printers, service accounts`
Domain controllers	`Servers holding a replicated copy of the directory` `Answer sign-ins and LDAP queries`

Directory protocols and services

NAME	WHAT IT DOES	PORT
LDAP	Query and modify directory entries	389 (636 for LDAPS)
Kerberos	Ticket-based authentication for domain sign-in	88
DNS	Clients find domain controllers through DNS records	53
SAML / OpenID Connect	Single sign-on to web and SaaS apps	443
SCIM	Automatic provisioning of users into SaaS apps	443

INSIGHT · DNS IS THE HEARTBEAT OF A DOMAIN

Domain-joined PCs locate domain controllers through DNS. If a PC uses a public DNS server instead of the internal one, sign-ins, policies and file shares fail in confusing ways. One of the first checks on a 'cannot log on to the domain' ticket is the PC's DNS setting.

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Identity models

On-premises directory

- Domain controllers in the office or data centre
- Strong control of PCs and file servers
- Needs VPN or line of sight for some functions

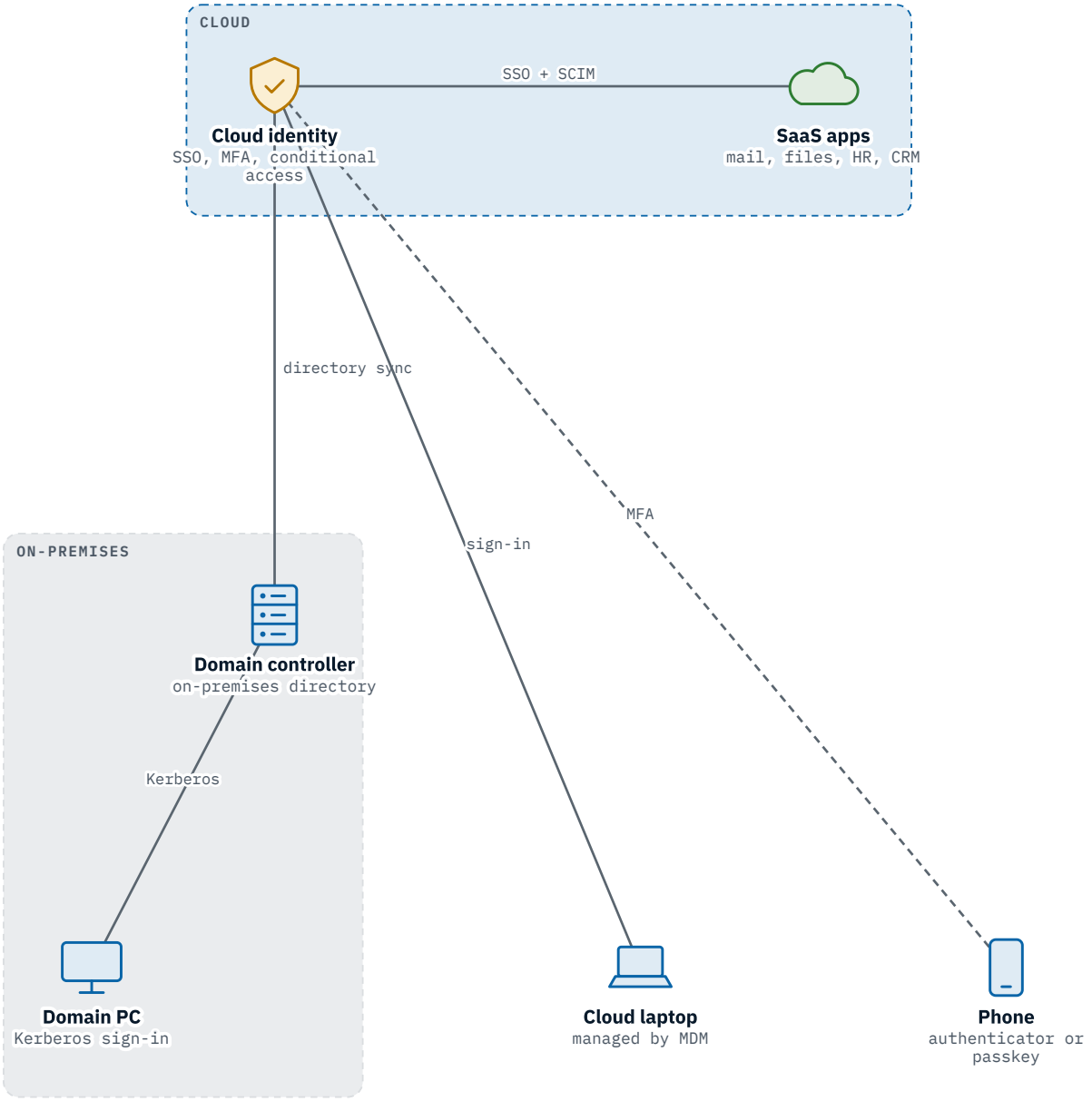
Cloud identity

- Identity provider hosted as a service
- SSO, MFA and passkeys for SaaS apps
- Devices joined and managed over the internet

Hybrid

- On-premises accounts synchronised to the cloud
- One password for local and cloud resources
- Very common in 2026; many newer organisations start cloud-only

Hybrid identity in one picture



Accounts are created once on-premises and synchronised to the cloud identity provider, which signs users in to every SaaS app with SSO and MFA. A cloud-only organisation simply has no on-premises zone.

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Common account tasks with the directory module

```
Windows PowerShell

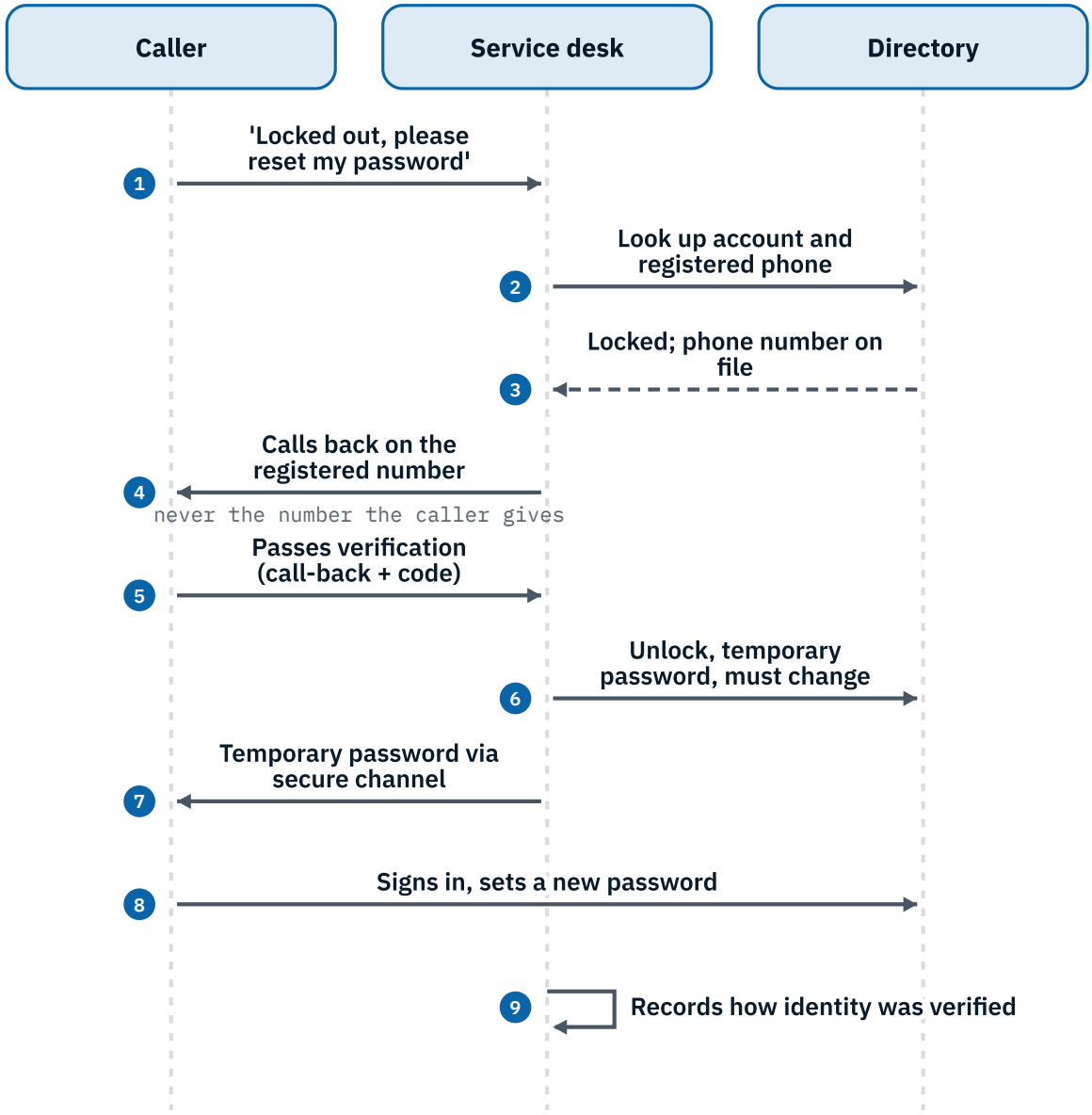
# Requires the directory management tools (RSAT) and suitable delegated rights
Get-ADUser -Identity sabbir -Properties LockedOut, LastLogonDate, PasswordExpired
Unlock-ADAccount -Identity sabbir
Set-ADAccountPassword -Identity sabbir -Reset -NewPassword (Read-Host -AsSecureString 'New
password')
Set-ADUser -Identity sabbir -ChangePasswordAtLogon $true
Get-ADGroupMember -Identity 'GRP-Finance-RW' | Select-Object Name
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -UsersOnly
```

Safe password reset procedure

1. **Verify identity** Use the organisation's method: call-back to a known number, manager confirmation or identity questions from HR records. Never accept 'I am the CEO, hurry'.
2. **Check the account** Is it locked, expired or disabled? A lockout may point to a stale password on a phone.
3. **Reset** Set a temporary password through a secure channel and require change at next sign-in.
4. **Review MFA** If compromise is suspected, revoke sessions and re-register MFA.
5. **Record** Log who requested, how identity was verified and what was done.

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Password reset with identity verification



The desk, not the caller, decides how identity is proven, using contact details already on record. The temporary password works once, so IT never knows the user's real password.

WATCH OUT · REPEATED LOCKOUTS

When an account locks again minutes after an unlock, a device is usually trying an old password: a phone's mail app, a mapped drive with saved credentials or a scheduled task. Find and update the source rather than unlocking all day. Repeated lockouts from unknown locations may be an attack.

CASE STUDY · LOCKOUTS AFTER A PASSWORD CHANGE

A senior engineer at a power plant near Ashuganj changes his password on Monday. From Tuesday his account locks every 20 to 30 minutes. Directory logs show failed sign-ins coming from a hostname 'ENG-TAB-04', a shared engineering tablet, and from his phone's mail client.

1. What is causing the lockouts?

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2. What steps fix it without weakening security?

3. What would you add to the knowledge base?

Model answer: module answer key.

SCENARIO · DECIDE · DOMAIN SIGN-IN FAILS AT A BRANCH

New PCs at a branch office in Rangpur cannot join the domain. The error says the domain controller cannot be contacted. Pinging the head-office domain controller by IP works. The PCs are using the internet provider's DNS servers.

(A) Add the domain controller's IP to each PC's hosts file

(B) Point the branch DHCP scope's DNS option to the internal DNS servers and renew

(C) Disable the Windows firewall on the domain controller

(D) Rebuild the domain controller

Circle your choice. Answer and feedback: module answer key.

Key terms

Directory service
A central database of users, groups, computers and other objects used for authentication, authorisation and management.

Organisational unit (OU)
A container in a directory used to organise objects, delegate administration and apply policies.

Kerberos
A ticket-based authentication protocol used for secure sign-in within domains.

Account lockout
A security control that disables sign-in temporarily after too many failed password attempts.

Domain controller
A server that holds a copy of the directory and authenticates domain sign-ins.

LDAP
Lightweight Directory Access Protocol: the standard protocol for querying and updating directory data.

Single sign-on (SSO)
Signing in once to access many applications without re-entering credentials.

Key takeaways

- A directory holds users, groups, computers and policy in one place; domain PCs find it through DNS.
- Hybrid identity syncs on-premises accounts to a cloud identity provider for SSO and MFA.
- Verify identity with details already on record before any reset or MFA change.
- Repeated lockouts usually mean a device still holds an old password.
- Record who asked, how identity was verified and what was changed.

KNOWLEDGE CHECK

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6. Which service do domain-joined PCs rely on to locate domain controllers?

   (A) DHCP
   (B) DNS
   (C) SMTP
   (D) NTP only

7. What is the main purpose of an organisational unit?

   (A) To store email
   (B) To encrypt disks
   (C) To organise objects for delegation and policy
   (D) To replace security groups when granting access to folders and printers

8. A user's account locks repeatedly after a password change. What is the most likely cause?

   (A) A failing hard drive that corrupts the stored password hash
   (B) Expired licence
   (C) Wrong time zone
   (D) A device still using the old saved password

9. Before resetting a caller's password, what must you do?

   (A) Reset immediately to save time
   (B) Email the new password to their manager so it can be passed on securely
   (C) Disable MFA
   (D) Verify the caller's identity using the approved method

10. Which protocol is commonly used for single sign-on to SaaS apps?

    (A) SMB
    (B) DHCP
    (C) SNMP
    (D) SAML or OpenID Connect

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · MANAGE · LESSON 3 · 26 MIN

Group Policy and Configuration Management

Setting rules once and applying them to thousands of devices

- Explain how group policy objects are linked, inherited and processed
- Troubleshoot policy application with gpupdate and gpresult
- Compare directory policy with cloud device management and Linux configuration tools
- Design a simple, testable policy structure

Configuring 500 PCs by hand guarantees 500 slightly different PCs. **Group policy** and modern **configuration management** let you define settings once, such as password rules, mapped drives, screen locks, update schedules and software, and apply them automatically to the right users and devices. Consistency makes support easier and security stronger.

Policy processing order on a domain-joined Windows PC

01 Local
Policy set on the PC itself.

→

02 Site
Policies linked to the network site.

→

03 Domain
Policies linked to the whole domain.

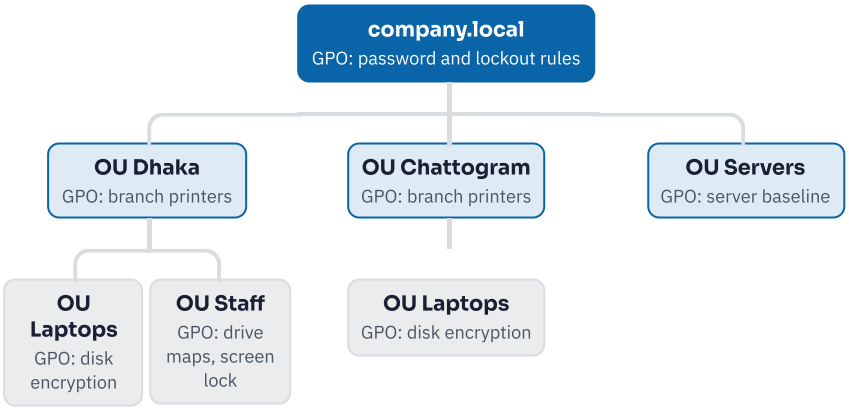
→

04 OU
Policies linked to OUs, parent first then child.

→

05 Result
Later policies win when settings conflict, so the closest OU usually wins.

Where policies are linked in a typical domain



An object receives every policy linked on the path from the domain down to its own OU, and the closest link wins conflicts. Moving an object to another OU silently changes its policy set.

Computer settings and user settings

Computer configuration

- Applies at start-up to the device
- Firewall, update schedule, encryption, local admin rules
- Follows the computer object's OU

User configuration

- Applies at sign-in to the person
- Mapped drives, desktop restrictions, browser settings
- Follows the user object's OU

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DEFINITION · ENFORCED AND BLOCKED INHERITANCE

An OU can **block inheritance** so higher-level policies do not apply, and a higher-level link can be **enforced** so it applies even through a block. Use both sparingly: they make troubleshooting much harder.

Checking which policies applied

Windows PowerShell

```
gpupdate /force                                  # re-apply policies now
gpresult /r                                      # summary of applied policies
gpresult /h C:\Temp\gp-report.html               # detailed HTML report
Get-WinEvent -LogName 'Microsoft-Windows-GroupPolicy/Operational' -MaxEvents 10
```

Run gpresult as the affected user for user settings, and in an elevated session to see computer settings.

Common reasons a policy does not apply

- The user or computer is in a different OU than you thought.
- Security filtering excludes the user or computer.
- A WMI or item-level filter evaluates to false.
- A later policy overrides the setting.
- The PC cannot reach a domain controller (often DNS or VPN).
- The setting needs a restart or sign-out to take effect.

Configuration management beyond the domain

Cloud device management (MDM)

- Configuration profiles pushed over the internet
- Works for laptops that never visit the office
- Manages Windows, macOS, iOS and Android
- Compliance results can gate sign-in (conditional access)

Linux configuration tools

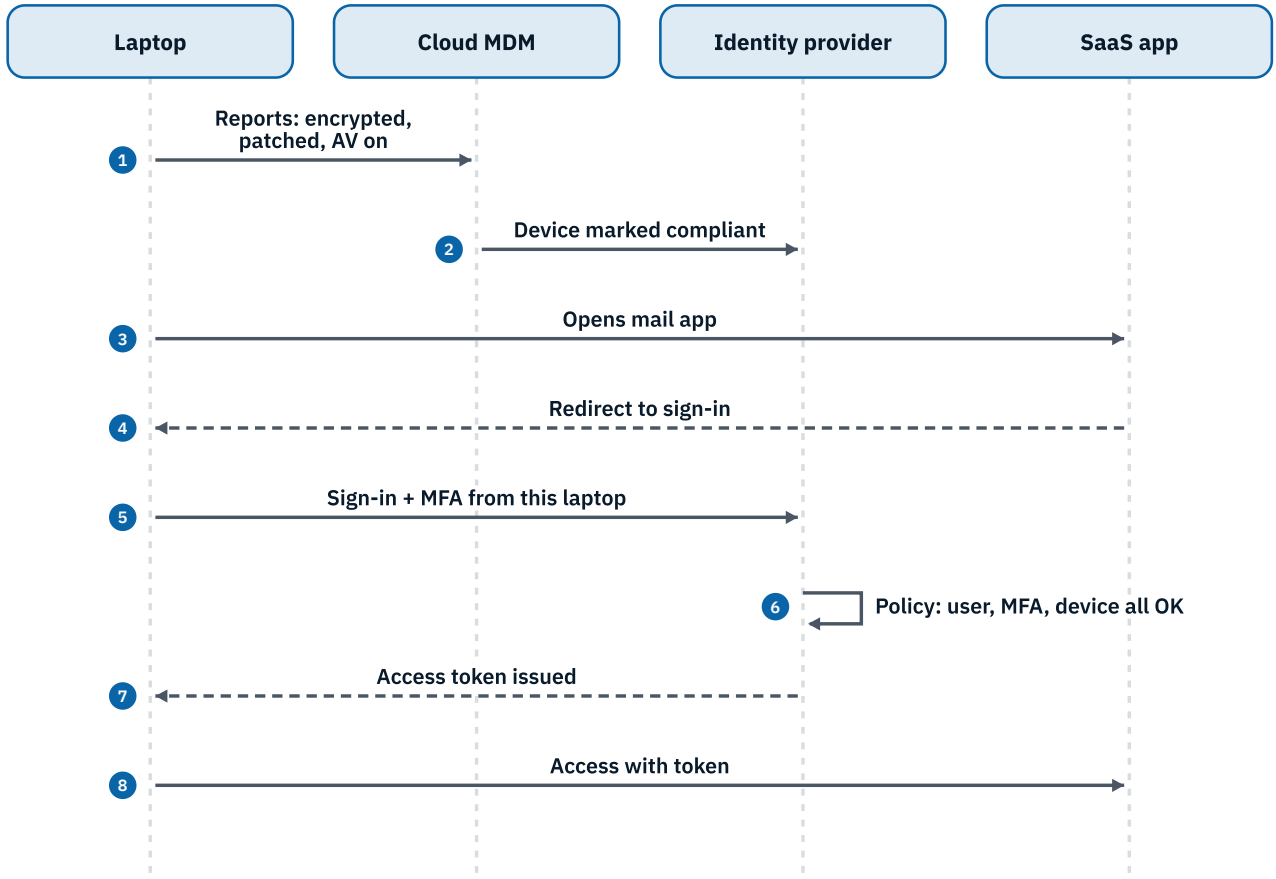
- Declarative tools describe the desired state
- Run on schedule and fix drift automatically
- Files stored in version control

Shared principles

- Define once, apply many
- Test on a pilot group first
- Document every policy's purpose and owner

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Conditional access: device health gates sign-in



The same password and MFA from an unmanaged or out-of-date device can be blocked or limited. When a user is 'blocked for no reason', check the device's compliance status first.

Rolling out a new policy safely

1. **Define** Write the purpose, the setting, who it applies to and the owner.
2. **Build** Create a new policy object rather than editing a large existing one.
3. **Pilot** Link it to a pilot OU or group of IT and friendly users for a week.
4. **Communicate** Tell affected users what will change and when.
5. **Deploy** Link to production OUs in stages.
6. **Verify and document** Check with gpresult or compliance reports and record the change.

TEMPLATE Policy register

Keep track of every policy, what it does and who owns it.

POLICY NAME	PURPOSE	LINKED TO	KEY SETTINGS	OWNER	LAST REVIEWED
SEC-ScreenLock-10min	*Lock idle screens*	*All Workstations OU*	*Lock after 10 minutes, require password*	*Security lead*	*2026-02-15*

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POLICY NAME	PURPOSE	LINKED TO	KEY SETTINGS	OWNER	LAST REVIEWED

SCENARIO · DECIDE · MAPPED DRIVE MISSING FOR ONE TEAM

The Procurement team at a Chattogram shipping company no longer gets its S: drive mapped at sign-in. Other teams' drives work. Yesterday, user accounts were moved into a new 'Procurement' OU during a clean-up.

(A) Map the drive manually on every PC

(B) Restart the file server

(C) Check that the drive-mapping policy is linked to the new OU, or that its targeting includes the moved users, then run gpupdate and test

(D) Give the team local administrator rights

Circle your choice. Answer and feedback: module answer key.

Key terms

Group policy object (GPO)
A collection of settings linked to sites, domains or OUs to configure users and computers centrally.

Security filtering
Limiting a policy so it applies only to specified users, computers or groups.

Configuration drift
Gradual divergence of systems from their intended configuration over time.

Pilot group
A small set of users or devices that receives a change first to catch problems before wide rollout.

Inheritance
The way policies linked at higher levels apply to objects in lower containers unless blocked.

gpresult
A Windows command that reports which group policies have been applied to a user and computer.

Desired state
A declared target configuration that management tools enforce automatically.

Key takeaways

- Define settings once and apply them automatically: consistency makes support easier.
- On a domain, policies apply local, site, domain, then OU, and the closest OU usually wins.
- Use gpresult to see what actually applied, and check OU, filtering and DNS when it did not.
- Cloud MDM manages devices anywhere, and its compliance results can gate sign-in.
- Pilot every new policy, roll out in stages and record it in a policy register.

KNOWLEDGE CHECK

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11. Which command shows the policies applied to the current user and computer?
    - (A) ipconfig /all
    - (B) sfc /scannow
    - (C) gpresult /r
    - (D) netstat -an
12. Two policies set conflicting screen-lock times: one at the domain and one on the user's OU. Which normally wins?
    - (A) The domain policy
    - (B) Whichever policy object was created first, because older policies win
    - (C) The OU policy, because it is processed later
    - (D) Neither applies
13. What is the safest way to deploy a new restrictive policy?
    - (A) Link it to the whole domain at once
    - (B) Apply it to administrators only
    - (C) Edit an existing large policy without testing so fewer objects need linking
    - (D) Pilot it on a small group, then roll out in stages
14. A laptop that rarely visits the office misses policy updates. What can help most?
    - (A) A longer DHCP lease
    - (B) Cloud-based device management that works over the internet
    - (C) Removing the laptop from the domain without any replacement management at all
    - (D) Disabling the firewall
15. What is configuration drift?
    - (A) Moving a server to a new rack or data centre without updating its documentation
    - (B) Slow network speed
    - (C) Systems gradually differing from the intended configuration
    - (D) A type of malware

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MODULE 5 · MANAGE · LESSON 4 · 30 MIN

Backup and Recovery

If it is not tested, it is not a backup

- Apply the 3-2-1 rule and choose backup types and schedules
- Use RPO and RTO to agree recovery targets with the business
- Protect backups from ransomware with offline or immutable copies
- Run and document a test restore

Hardware fails, people delete things, laptops are stolen and ransomware encrypts whole servers. Backups are the last line of defence. The goal is not to have backups but to be able to **restore** the right data, quickly enough, when it matters. Many organisations only discover their backups do not work on the day they need them.

The 3-2-1-1-0 rule

3

Three copies

The live data plus at least two backups.

2

Two media types

For example disk on site and cloud or tape.

1

One off site

A copy in another location or region to survive fire, flood or theft.

+1

One offline or immutable

A copy ransomware cannot alter or delete.

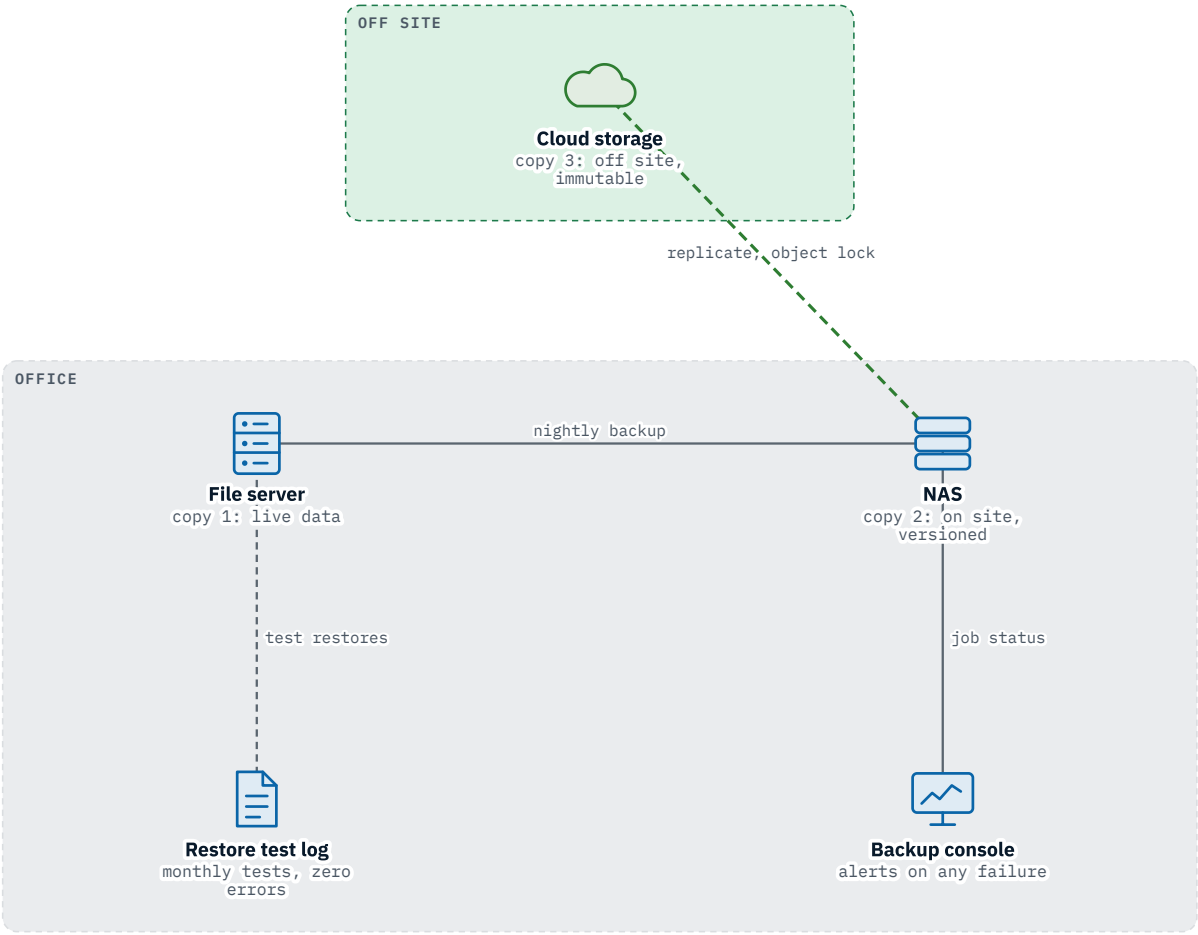
0

Zero errors

Backups verified and restores tested on a schedule.

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3-2-1-1-0 in a small office



Copy 2 is fast to restore on site; copy 3 is off site and immutable, so ransomware on the office network cannot alter it. The restore test log is the zero: proof that restores really work.

Backup types

TYPE	WHAT IT COPIES	BACKUP SPEED	RESTORE NEEDS
Full	Everything selected	Slowest, largest	Just the full backup
Incremental	Changes since the last backup of any type	Fastest, smallest	Last full plus every incremental since
Differential	Changes since the last full	Grows during the week	Last full plus the latest differential
Snapshot	Point-in-time state of a volume or VM	Seconds	Same system; not a backup on its own

Recovery targets

MEASURE	FORMULA	WHAT IT TELLS YOU
RPO	`RPO = maximum acceptable data loss, measured in time`	An RPO of 4 hours means backups or replication must run at least every 4 hours.

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MEASURE	FORMULA	WHAT IT TELLS YOU
RTO	`RTO = maximum acceptable time to restore service`	An RTO of 2 hours means the restore process, not just the copy, must finish within 2 hours.
Retention storage (rough)	`storage = full size + (daily change x days retained)`	A 500 GB server with 10 GB daily change kept 30 days needs about 500 + 300 = 800 GB before compression.

EXAMPLE · RPO IN A HOSPITAL

A hospital in Sylhet agreed an RPO of 15 minutes for its patient records system, because re-entering a morning of clinical notes is unacceptable. Its payroll system has an RPO of 24 hours. Different systems deserve different targets; agree them with the data owners, not alone in IT.

File-level backups from the command line

Windows PowerShell

```
# Mirror a folder to a backup share and log the result
robocopy D:\Shared \\nas01\backup\Shared /MIR /R:2 /W:5 /LOG:C:\Logs\backup.log
# /MIR also deletes files in the destination that were deleted at the source:
# use it only for a mirror, and keep versioned backups elsewhere
```

Linux backup with rsync and tar

Linux host

```
rsync -a --delete /srv/data/ backup@nas01:/backups/srv-data/
tar -czf /backups/etc-$(date +%F).tar.gz /etc
tar -tzf /backups/etc-2026-05-01.tar.gz | head   # list contents to verify
# test restore into a scratch folder, never over live data
mkdir -p /tmp/restore && tar -xzf /backups/etc-2026-05-01.tar.gz -C /tmp/restore
```

WATCH OUT · SYNC IS NOT BACKUP

File-sync services and RAID copy changes instantly, including deletions and encrypted files. They protect against a failed disk, not against mistakes or ransomware. You still need versioned backups with retention.

Running a test restore

1. **Pick a sample** Choose a realistic item: a folder, a mailbox or a whole VM.
2. **Restore to a safe location** An isolated folder or test network, never over production.
3. **Time it** Measure how long it takes and compare with the RTO.
4. **Verify** Open files, check the database starts and ask the data owner to confirm.
5. **Record** Write the date, item, time taken, result and any issues.

TEMPLATE Backup and restore test log

Prove backups work and track restore times against agreed targets.

SYSTEM	BACKUP TYPE AND SCHEDULE	RPO / RTO	LAST TEST DATE	RESTORE TIME	RESULT AND NOTES
File server FS01	*Nightly incremental, weekly full, cloud copy*	*24 h / 8 h*	*2026-04-20*	*1 h 40 min (300 GB)*	*Pass; owner verified Finance folder*

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SYSTEM	BACKUP TYPE AND SCHEDULE	RPO / RTO	LAST TEST DATE	RESTORE TIME	RESULT AND NOTES

SCENARIO · DECIDE · RANSOMWARE IN A SMALL OFFICE

A garment buying house in Dhaka finds its file server encrypted by ransomware on Monday morning. Backups run nightly to a USB disk that is always connected to the server, plus a weekly cloud copy with 30-day versioning.

(A) Restore from the USB disk immediately

(B) Pay the ransom to get files back quickly

(C) Isolate the server, report the incident, then restore from the cloud versioned copy after confirming it predates the infection

(D) Wipe the server and start again without restoring

Circle your choice. Answer and feedback: module answer key.

Key terms

3-2-1 rule
A backup principle: three copies of data, on two types of media, with one copy off site; often extended to 3-2-1-1-0 with one offline or immutable copy and zero restore errors.

Differential backup
A backup of all data changed since the last full backup.

RTO
Recovery time objective: the maximum acceptable time to restore a service after a failure.

Test restore
A planned recovery of backed-up data to prove backups work and measure restore time.

Incremental backup
A backup of only the data changed since the previous backup of any type.

RPO
Recovery point objective: the maximum amount of data loss, measured in time, that is acceptable.

Immutable backup
A backup copy that cannot be changed or deleted until its retention period ends.

Key takeaways

- Backups exist to restore: agree RPO and RTO with data owners, system by system.
- Follow 3-2-1-1-0: three copies, two media, one off site, one immutable or offline, zero errors.
- Sync, RAID and snapshots are not backups on their own.
- A backup that anything on the infected network can write to can be encrypted too.
- Only a timed, verified test restore proves a backup works.

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KNOWLEDGE CHECK

16. A system must lose no more than one hour of data. Which target is this?

(A) RPO of 1 hour
(B) RTO of 1 hour
(C) SLA of 1 hour
(D) Retention of 1 hour

17. Which backup type needs the last full plus every backup since to restore?

(A) Differential
(B) Full
(C) Incremental
(D) Mirror

18. Why is a continuously connected USB backup disk weak protection against ransomware?

(A) USB is too slow
(B) Ransomware can reach and encrypt it like any other drive
(C) USB disks cannot hold backups
(D) It breaks the 3-2-1 rule because a USB disk never counts as one of the three copies

19. Which statement about RAID is correct?

(A) RAID protects against a disk failure but not deletion or ransomware
(B) RAID replaces backups
(C) RAID keeps 30 days of file versions, so deleted files can always be restored from it
(D) RAID stores data off site

20. What proves a backup will work?

(A) The backup job reports 'success' every night in the console for a whole month
(B) The disk has free space
(C) A successful, timed test restore verified by the data owner
(D) The backup is encrypted

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MODULE 5 · MANAGE · LESSON 5 · 26 MIN

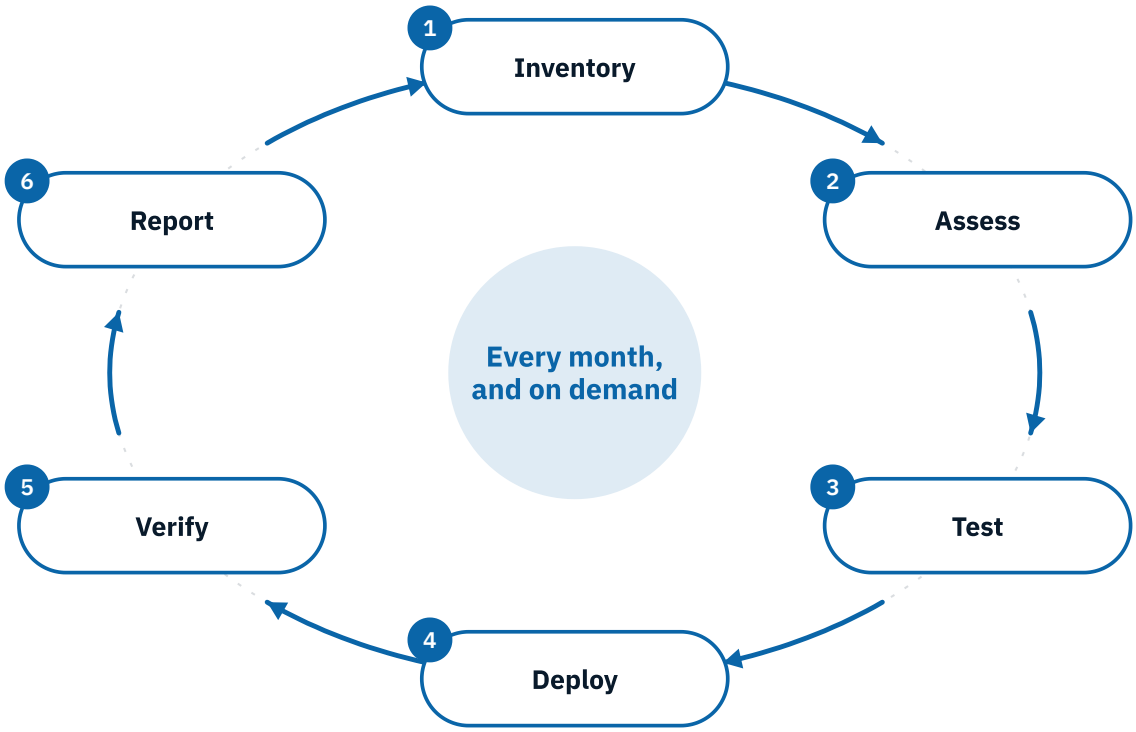
Patch and Update Management

Closing security holes quickly without breaking the business

- Describe a patch management cycle from inventory to reporting
- Use deployment rings, testing and maintenance windows
- Handle emergency patches and failed updates with rollback
- Check and install updates on Windows, Linux and macOS

Most successful attacks use vulnerabilities for which a patch already existed. Patching is not glamorous, but it is one of the most effective security controls a support team owns. The challenge is speed with safety: apply updates quickly, while making sure a bad update does not stop a hospital, a factory or a bank branch.

The patch management cycle



1. **Inventory** — Know every device, OS, app and firmware version.
2. **Assess** — Read advisories; rate severity and exposure.
3. **Test** — Apply to a lab and a pilot ring.
4. **Deploy** — Roll out in rings within maintenance windows.
5. **Verify** — Confirm installation and service health.
6. **Report** — Track compliance and exceptions.

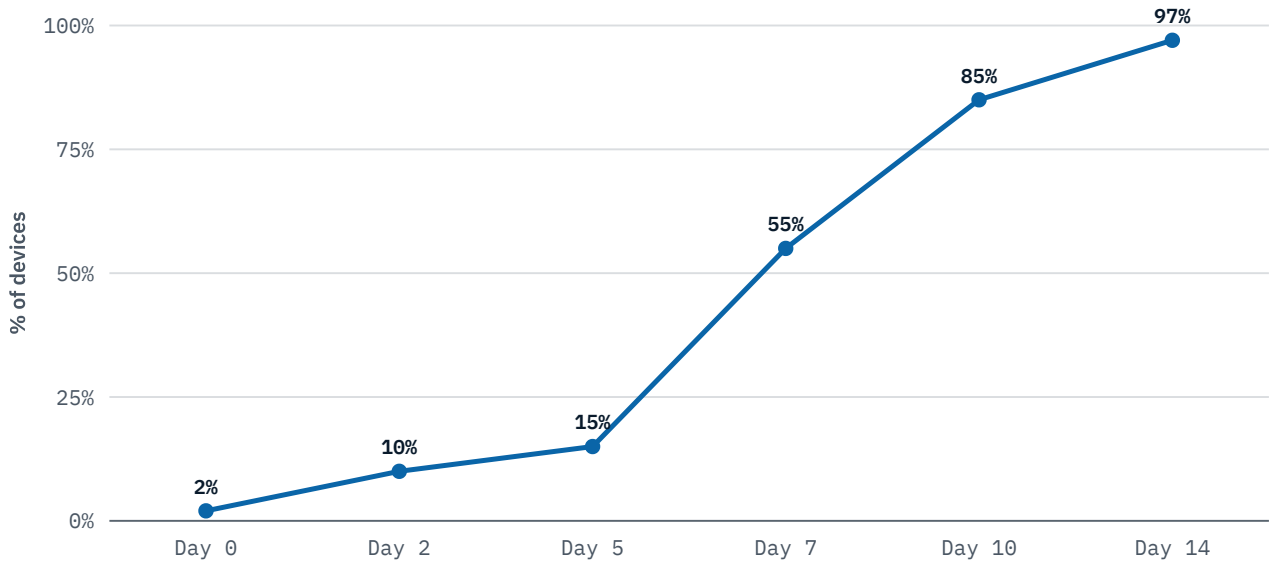
Reporting feeds the next inventory, so the loop never ends. Emergency patches run the same cycle in hours instead of weeks.

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Deployment rings

RING	WHO	WHEN AFTER RELEASE	PURPOSE
Ring 0	IT test machines	Day 0-1	Catch obvious breakage
Ring 1 (pilot)	IT staff and volunteers from each department	Day 2-5	Real-world testing on varied apps
Ring 2 (broad)	Most users	Day 6-14	Main rollout
Ring 3 (critical)	Sensitive systems: clinical, production lines, finance servers	After broad success, in agreed windows	Protect critical operations

Devices updated during a ring rollout (illustrative)



Illustrative numbers. Coverage stays low while the test and pilot rings prove the update, then climbs steeply in the broad ring; the last few percent are critical systems and stragglers that need follow-up.

Measuring patch performance

MEASURE	FORMULA	WHAT IT TELLS YOU
Patch compliance	`compliance % = devices fully patched / devices in scope x 100`	If 1,140 of 1,200 PCs are current, compliance is 95%.
Mean time to patch	`MTTP = sum of (install date - release date) / number of patches`	Track separately for critical patches; many organisations aim for days, not weeks.

WATCH OUT · EMERGENCY PATCHES

When a vulnerability is being actively exploited, normal timelines shrink to hours or days. Follow the emergency change process: quick risk assessment, accelerated pilot, deployment to exposed systems first (internet-facing servers, VPN gateways) and close monitoring. Public catalogues of known exploited vulnerabilities help decide which patches jump the queue.

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Updates on Windows

```
Windows PowerShell

Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 10
# List and install updates with the built-in Windows Update client (UI) or management tools
# Check the last time updates were installed and any pending restart:
Test-Path 'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto
Update\RebootRequired'
winget upgrade --all --silent          # third-party apps managed by winget
```

Third-party apps such as browsers, PDF readers and runtimes are frequent attack targets; patch them as seriously as the OS.

Updates on Linux and macOS

```
Linux host

sudo apt update && apt list --upgradable
sudo apt upgrade
sudo dnf check-update                      # RHEL family: list available updates
sudo dnf upgrade --security                # install security updates only
[ -f /var/run/reboot-required ] && cat /var/run/reboot-required   # Debian/Ubuntu
# macOS: softwareupdate --list ; sudo softwareupdate --install --all --restart
```

Do not forget

- **Firmware and BIOS/UEFI** on PCs, servers, switches and printers.
- **Network and security appliances**: firewalls and VPN gateways are prime targets.
- **Third-party apps**: browsers, document readers, runtimes, conferencing tools.
- **Mobile devices**: enforce minimum OS versions through MDM.
- **Unsupported systems**, such as Windows 10 PCs without ESU: isolate them and plan replacement.

When an update breaks something

1. **Pause the rollout** Stop further rings from receiving the update.
2. **Confirm the link** Check that affected devices share the update and that unaffected ones do not.
3. **Roll back or mitigate** Uninstall the update on affected devices or apply the vendor's workaround.
4. **Communicate** Tell users and the change board what happened and the plan.
5. **Resume with a fix** Deploy the corrected update and record lessons learned.

CASE STUDY · PATCHING A FACTORY FLOOR

A textile factory in Narayanganj runs 40 production PCs that control dyeing machines with software certified only for a specific OS build. Production runs 24 hours a day, six days a week. The office has 150 standard PCs. Last month a critical vulnerability was exploited in the region, and the factory's office PCs were 60 days behind on updates.

1. How would you organise rings for the office and production PCs?

2. How would you protect production PCs that cannot be patched quickly?

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3. What would you report to management each month?

Model answer: module answer key.

SCENARIO · DECIDE · A PATCH BREAKS THE BILLING APP

On Wednesday, Ring 1 pilot users in a telecom company's billing team report that the billing app crashes after this month's OS update. Ring 2 (all staff) is scheduled for Thursday.

(A) Continue to Ring 2 because security patches must not wait

(B) Pause the update for the billing team's devices, roll back on affected pilots, alert the vendor and continue for unaffected groups with monitoring

(C) Cancel all patching for the month

(D) Tell billing users to stop using the app

Circle your choice. Answer and feedback: module answer key.

Key terms

Patch
An update that fixes security vulnerabilities, bugs or adds improvements to software or firmware.

Deployment ring
A group of devices that receives updates at a defined stage of a phased rollout.

Patch compliance
The percentage of devices in scope that have all required updates installed.

Firmware update
An update to the low-level software stored on hardware such as motherboards, SSDs, printers and network devices.

Security advisory
A vendor notice describing a vulnerability, the affected versions, its severity and the fix or workaround.

Maintenance window
An agreed time when changes such as updates and restarts may be made with minimal business impact.

Emergency change
A change, such as an urgent patch, that must be implemented faster than the normal process allows.

Key takeaways

- Patching is one of the most effective controls a support team owns.
- Know your inventory, then test, pilot and deploy in rings.
- Actively exploited flaws on internet-facing systems go through the emergency process within days.
- Patch firmware, appliances and third-party apps, not only the OS.
- When an update breaks something, pause the rollout, roll back the affected devices and communicate.

KNOWLEDGE CHECK

21. Which group should receive a new monthly update first?

(A) Critical production servers
(B) IT test machines and a pilot ring
(C) All users at once
(D) Executives

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22. 1,800 of 2,000 devices are fully patched. What is patch compliance?

(A) 90%
(B) 80%
(C) 95%
(D) 18%

23. Which Linux command on the RHEL family installs only security updates?

(A) sudo apt upgrade
(B) sudo systemctl upgrade
(C) sudo dnf upgrade --security
(D) sudo yum remove

24. A legacy PC cannot be patched because its machine software would break. What is the best mitigation?

(A) Isolate it on a segmented network and plan its replacement
(B) Connect it directly to the internet so the vendor can support it remotely
(C) Disable its antivirus
(D) Give all users admin rights on it

25. What should happen first when an update breaks an important app in the pilot ring?

(A) Push it to everyone anyway
(B) Delete the app
(C) Reimage the pilot PCs and continue the rollout to the next ring as planned
(D) Pause further rollout of that update

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MODULE 5 · MANAGE

Answer key

Knowledge checks

1. D Over the network, the most restrictive of share and NTFS permissions applies, so Read.

2. C Without -a, the user's supplementary groups are replaced by the new list.

3. D Group-based access is easier to manage, audit and change.

4. B Leaver accounts should be disabled at the agreed time to remove access promptly.

5. A Privilege creep is access building up over time without removal.

6. B Service records in DNS point clients to domain controllers.

7. C OUs group objects for administration and policy; they are not security groups.

8. D Stale saved credentials retry the old password and trigger lockout.

9. D Identity verification prevents social-engineering attacks.

10. D SAML and OpenID Connect are standard web SSO protocols.

11. C gpresult reports applied policies.

12. C Later-processed policies win; OU links process after domain links.

13. D Piloting catches problems before they affect everyone.

14. B MDM delivers configuration over the internet without needing the office network.

15. C Drift is divergence from the desired state over time.

16. A Acceptable data loss is the recovery point objective.

17. C Incremental restores need the full and the whole chain of incrementals.

18. B Anything the infected system can write to can be encrypted.

19. A RAID provides redundancy, not versioned, independent copies.

20. C Only a tested restore proves recoverability.

21. B Testing on IT and pilot rings catches problems before wide rollout.

22. A 1,800 / 2,000 x 100 = 90%.

23. C dnf supports a security filter. apt is for Debian and the others are wrong.

24. A Segmentation and controls reduce exposure while replacement is planned.

25. D Pausing prevents wider impact while you investigate.

Scenarios & case studies

Scenario · Access for a temporary auditor
A is best. Correct. Least privilege, accountability through a named account and an automatic end date.

Case study · Lockouts after a password change
Cause: the shared tablet and his phone still hold the old password and keep retrying it, each attempt counting as a failure until the lockout threshold is reached. **Fix:** unlock the account, update or remove the saved credentials on the phone mail app and the tablet (ideally stop using personal credentials on shared devices and use a service or kiosk account), then watch the logs for an hour. **Knowledge base:** an article 'Account keeps locking after a password change' listing common stale-credential sources (mobile mail, mapped drives, saved Wi-Fi, scheduled tasks, shared devices) and how to find the source in sign-in logs.

Scenario · Domain sign-in fails at a branch
B is best. Correct. Domain members need internal DNS to find domain controllers through service records.

Scenario · Mapped drive missing for one team
C is best. Correct. The change that matches the timing is the OU move; the policy probably is not linked or filtered for the new location.

Scenario · Ransomware in a small office
C is best. Correct. Containment first, then restore from a copy the attacker could not reach, checking its date.

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Case study · Patching a factory floor

Rings: office PCs follow IT test, pilot and broad rings within 14 days, with critical patches within days. Production PCs form a separate ring patched only in the weekly shutdown window, after the machine vendor confirms compatibility. **Protection:** segment production PCs on their own network, block internet access, allow only required connections, use application allow-listing and strong endpoint protection, and disable USB storage. **Report:** compliance per ring, mean time to patch critical updates, exceptions with business owner sign-off, and any failed updates and rollbacks.

Scenario · A patch breaks the billing app

B is best. Correct. You contain the impact, keep security moving where it is safe and work on a fix.

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06

MODULE 6 · HOST

IT Infrastructure Services and Cloud

Servers, virtualisation, storage, cloud and SaaS administration, monitoring and remote access

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain and check core infrastructure services such as file, print, DNS, DHCP, time and email
- Use virtualisation, RAID and shared storage appropriately
- Apply cloud service and deployment models with the shared responsibility model and cost awareness
- Administer users, licences, mail flow and sharing in a cloud productivity suite
- Monitor services, calculate availability and provide secure remote access

LESSONS

1. Core Infrastructure Services · 26 min
2. Virtualisation and Storage · 28 min
3. Cloud Computing Fundamentals · 28 min
4. SaaS and Productivity Suite Administration · 28 min
5. Monitoring, Availability and Remote Access · 26 min

MODULE 6 · HOST · LESSON 1 · 26 MIN

Core Infrastructure Services

The servers and services every organisation quietly depends on

- Identify common server roles and what users experience when each fails
- Explain how file, print, DNS, DHCP, time and email services fit together
- Describe server hardware features and good server-room practice
- Check service health on Windows and Linux servers

Behind every working desk is a set of shared services: file shares, printers, name resolution, addressing, time, email and remote access. Support specialists may not build all of these, but they are expected to know what each does, recognise when one has failed and gather the right evidence before escalating to the infrastructure team.

Server roles and failure symptoms

SERVICE	WHAT IT PROVIDES	WHAT USERS NOTICE WHEN IT FAILS
File server / NAS	Shared folders with permissions	'S: drive missing', 'access denied'
Print server	Central queues and drivers	Nobody can print to shared printers
DNS	Name resolution	Everything 'down' though IP works
DHCP	Automatic addressing	New devices get 169.254 addresses
Directory / identity	Sign-in and policy	Cannot log on, policies not applied
NTP (time)	Synchronised clocks	Sign-in failures, certificate errors, wrong log timestamps
Email / messaging	Mail flow and calendars	Messages stuck in outbox or bounced

INSIGHT · TIME MATTERS MORE THAN YOU THINK

Ticket-based sign-in protocols reject requests when clocks differ by more than a few minutes, and certificates appear 'not yet valid'. If several PCs suddenly cannot sign in, check that their clocks and the domain's time source are correct.

Server versus desktop hardware

Server features

- Redundant power supplies and fans
- ECC memory
- Hot-swap drives in RAID
- Out-of-band management (remote console even when the OS is down)
- Rack mounting and higher duty cycles

Desktop features

- Single PSU
- Consumer components
- Designed for 8-10 hours a day
- No remote management controller

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Server room essentials

- Locked room with access logging; no storage of boxes or cleaning supplies.
- Cooling sized for the heat load, with temperature monitoring and alerts.
- UPS for clean shutdown and, where possible, a generator for longer outages.
- Labelled cables and patch panels; a current rack diagram.
- Fire detection and suitable suppression; no water pipes overhead if avoidable.
- Environmental monitoring for heat, humidity and water leaks.

Checking services on a Windows server

Windows PowerShell

```
Get-Service -Name DNS, DHCPServer, Spooler, LanmanServer | Select-Object Name, Status
Get-SmbShare | Select-Object Name, Path, Description
Get-SmbSession | Select-Object ClientComputerName, ClientUserName, NumOpens
w32tm /query /status                    # time source and last sync
Get-Printer | Where-Object PrinterStatus -ne 'Normal'
```

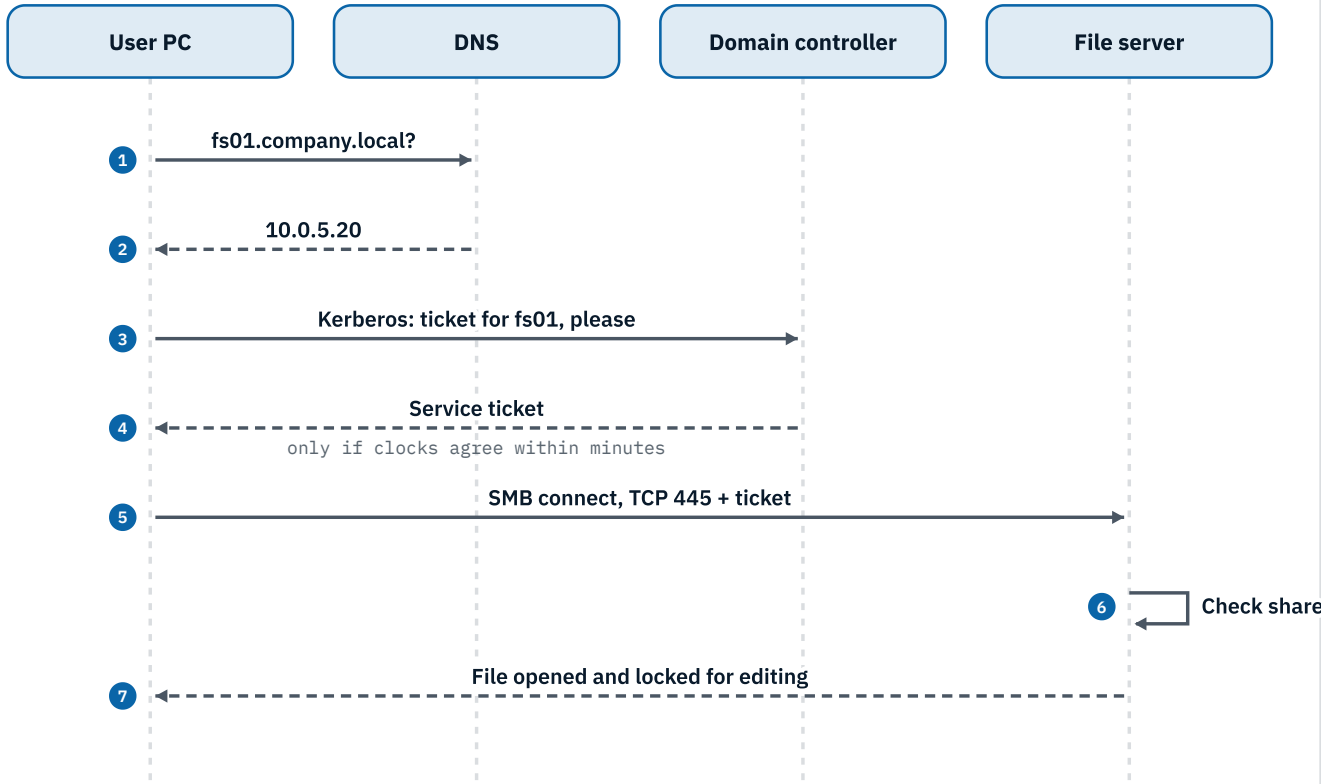
Checking services on a Linux server

Linux host

```
systemctl --failed                      # any failed units?
systemctl status smbd nginx
timedatectl                             # clock and NTP sync status
ss -tlnp | grep -E ':53|:445|:443'
df -h /srv && free -h
```


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What happens when a user opens a shared file



Four services take part before the file appears, so 'the S: drive is broken' could be DNS, time, the directory or permissions. Test them in this order.

SCENARIO · DECIDE · NOBODY CAN PRINT ON THE SECOND FLOOR

At 08:30 every user on the second floor of a Dhaka insurance office reports they cannot print. Users on other floors print normally to their own printers through the same print server. The second-floor printer's display shows 'Ready'.

- (A) Check the second-floor printer's queue on the server for a stuck job and its network connectivity
- (B) Restart the print server immediately
- (C) Reinstall the printer driver on every PC on the floor
- (D) Replace the printer's toner

Circle your choice. Answer and feedback: module answer key.

Key terms

File server
A server that stores shared files and controls access through share and file-system permissions.

NTP
Network Time Protocol: keeps device clocks synchronised with a reliable time source.

Out-of-band management
A dedicated controller that lets administrators manage a server remotely even when its OS is not running.

Print server
A server that hosts shared print queues and drivers for network printers.

SMB
Server Message Block: the protocol used for Windows file and printer sharing, on TCP port 445.

Hot-swap
Replacing a component, such as a drive or PSU, while the system keeps running.

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Server room
A secured, cooled and powered space that houses servers and network equipment.

Key takeaways

- Know what each shared service does and how its failure looks to users.
- Clock drift breaks Kerberos sign-in and certificates: check time sync early.
- Servers add redundancy (PSUs, ECC, RAID) and out-of-band management.
- Use scope to choose between one queue, one server or a shared service.
- Gather service status and logs before escalating to the infrastructure team.

KNOWLEDGE CHECK

1. Several PCs suddenly fail to sign in and show certificate 'not yet valid' warnings. What should you check?
   - (A) Printer queues
   - (B) Monitor cables
   - (C) Clock and time synchronisation
   - (D) Toner levels
2. Which port does SMB file sharing use?
   - (A) UDP 53
   - (B) TCP 445
   - (C) TCP 3389
   - (D) TCP 25
3. Which server feature allows a technician to see the console of a crashed server remotely?
   - (A) Out-of-band management
   - (B) ECC memory
   - (C) Hot-swap fans
   - (D) RAID
4. A whole office cannot reach anything by name, but IP addresses work. Which service is most likely down?
   - (A) DHCP
   - (B) Print
   - (C) NTP
   - (D) DNS
5. Which Linux command lists failed services?
   - (A) systemctl --failed
   - (B) timedatectl
   - (C) df -h
   - (D) ss -tlnp

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · HOST · LESSON 2 · 28 MIN

Virtualisation and Storage

Hypervisors, virtual machines, containers, RAID and shared storage

- Compare type 1 and type 2 hypervisors, VMs and containers
- Choose a RAID level and calculate usable capacity
- Compare direct-attached, NAS and SAN storage
- Use snapshots safely

Few organisations run one service per physical server any more. **Virtualisation** runs many isolated virtual machines on one host, cutting hardware, power and space, and making backup and recovery faster. Storage has evolved too: RAID protects against disk failure and shared storage lets VMs move between hosts.

Hypervisor types

Type 1 (bare metal)

- Installs directly on server hardware
- Used in data centres and server rooms
- High performance and isolation

Type 2 (hosted)

- Runs as an app on Windows, Linux or macOS
- Ideal for labs, testing and training
- Shares resources with the host OS

Virtual machines and containers

Virtual machine

- Full OS with its own kernel
- Strong isolation
- Gigabytes in size; boots in minutes

Container

- Shares the host kernel; packages an app and its libraries
- Lightweight, starts in seconds
- Common for web apps and microservices

RAID levels

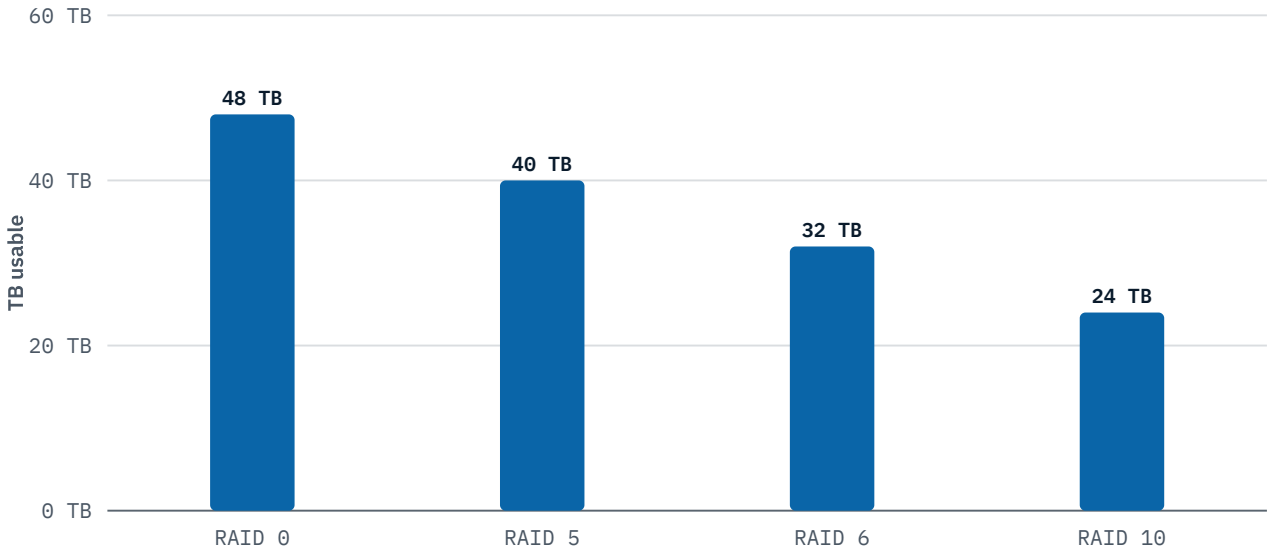
LEVEL	MINIMUM DISKS	SURVIVES	USABLE CAPACITY	TYPICAL USE
RAID 0	2	No disk failure	All disks	Scratch space only; never for important data
RAID 1	2	1 disk failure	One disk's worth	Boot drives, small servers
RAID 5	3	1 disk failure	(n - 1) disks	File servers with smaller disks
RAID 6	4	2 disk failures	(n - 2) disks	Large-disk arrays, archives
RAID 10	4	1 per mirror pair	Half of all disks	Databases, VMs needing speed

RAID capacity

MEASURE	FORMULA	WHAT IT TELLS YOU
RAID 5	`usable = (n - 1) x disk size`	Four 4 TB disks give 3 x 4 = 12 TB usable.
RAID 6	`usable = (n - 2) x disk size`	Six 8 TB disks give 4 x 8 = 32 TB usable.
RAID 10	`usable = (n / 2) x disk size`	Eight 2 TB disks give 4 x 2 = 8 TB usable.

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Usable capacity from six 8 TB disks



Exact usable capacity before formatting. Protection costs capacity: RAID 6 gives up two disks to survive two failures, and RAID 10 gives up half for speed.

WATCH OUT · RAID REBUILDS ARE RISKY

When a disk fails, the array rebuilds onto a replacement by reading every other disk. With large disks this can take a day or more, and a second failure during a RAID 5 rebuild loses the array. Replace failed disks promptly and keep backups current.

Storage options

Direct-attached (DAS)

- Disks inside or cabled to one server
- Simple and fast
- Not shared between hosts

NAS

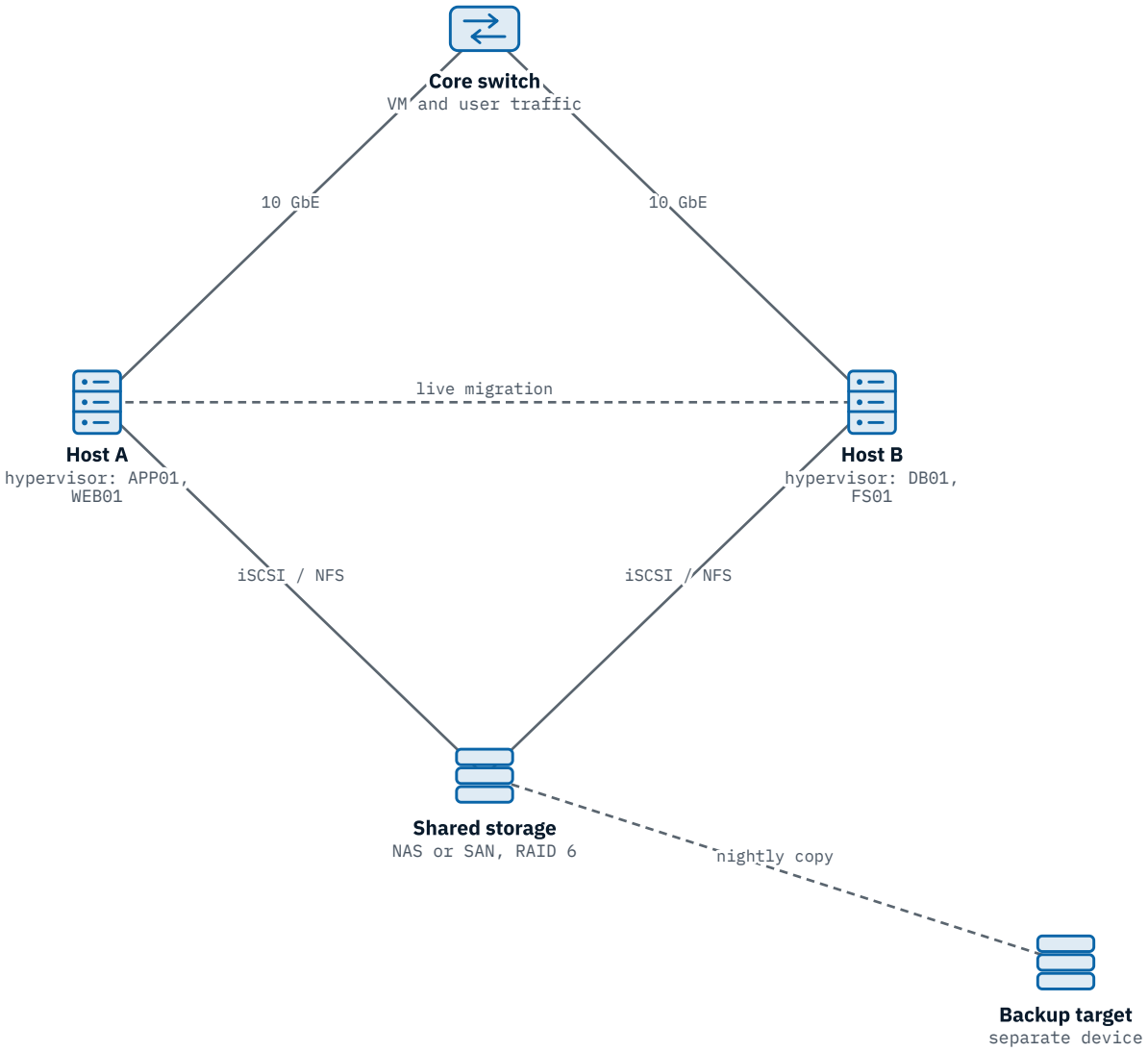
- File-level storage over the network (SMB, NFS)
- Easy shared folders and backups
- Common in small offices

SAN

- Block-level storage over a dedicated network
- Shared by hypervisor clusters
- Higher cost and complexity

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Two virtualisation hosts sharing storage



Because both hosts see the same storage, a VM can move to the other host for maintenance or restart there after a host failure. The shared array then becomes a single point of failure, so it needs RAID, monitoring and a separate backup.

Inspecting virtual machines on a Windows hypervisor host

```Windows PowerShell
Get-VM | Select-Object Name, State, CPUUsage, MemoryAssigned, Uptime
Checkpoint-VM -Name 'APP01' -SnapshotName 'Before-update-2026-05-10'
Get-VMSnapshot -VMName 'APP01'
# After a successful update and a few days of monitoring, remove the snapshot:
Remove-VMSnapshot -VMName 'APP01' -Name 'Before-update-2026-05-10'
```

Other hypervisors offer equivalent commands; the principles are the same everywhere.

PRO TIP · SNAPSHOTS ARE SHORT-TERM UNDO BUTTONS

Take a snapshot before a risky change, verify the change and delete the snapshot within days. Long-lived snapshots grow, slow the VM and are not backups: if the underlying storage fails, they go with it.

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Containers at a glance

```
Linux host

docker ps                          # running containers
docker logs --tail 50 web01        # recent logs from one container
docker restart web01
# Podman and other runtimes use very similar commands
```

SCENARIO · DECIDE · CHOOSING RAID FOR A NEW FILE SERVER

A university department in Rajshahi buys a file server with six 8 TB disks for research data. Data must survive disk failures during long rebuilds, and capacity matters more than raw speed.

- (A) RAID 0 for maximum capacity
- (B) RAID 1 using only two of the disks
- (C) No RAID, because backups exist
- (D) RAID 6, giving 32 TB usable and surviving two disk failures, plus separate backups

Circle your choice. Answer and feedback: module answer key.

Key terms

Hypervisor
Software that creates and runs virtual machines by sharing physical hardware between them.

Container
A lightweight package of an application and its dependencies that shares the host operating system kernel.

NAS
Network-attached storage: a device that provides file-level storage over the network.

Checkpoint (VM snapshot)
A saved point-in-time state of a VM that can be reverted to quickly, intended for short-term use.

Type 1 hypervisor
A hypervisor installed directly on hardware, used for production servers.

RAID
Redundant array of independent disks: combining disks for redundancy, performance or both.

SAN
Storage area network: a dedicated network that provides block-level storage to servers.

Key takeaways

- Type 1 hypervisors run production servers; type 2 suit labs and testing.
- Containers share the host kernel and start in seconds; VMs bring their own OS.
- Choose RAID by what must survive: RAID 6 for large disks, RAID 10 for speed, never RAID 0 for data.
- Shared storage lets VMs move between hosts but must itself be protected and backed up.
- Snapshots are short-term undo buttons, not backups.

KNOWLEDGE CHECK

6. How much usable space do five 4 TB disks provide in RAID 5?

- (A) 20 TB
- (B) 12 TB
- (C) 16 TB
- (D) 10 TB

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7. Which RAID level offers no redundancy?

   (A) RAID 1
   (B) RAID 5
   (C) RAID 0
   (D) RAID 6

8. Which statement about VM snapshots is correct?

   (A) They replace backups
   (B) They are short-term rollback points to remove after use
   (C) They make VMs faster over time
   (D) They protect against storage failure because they are stored as separate files

9. A trainer needs to run three test OS installs on her laptop. Which option suits best?

   (A) A SAN
   (B) RAID 10
   (C) A type 2 hypervisor
   (D) A print server

10. What is the main difference between a container and a VM?

    (A) Containers include a full OS kernel, so each one needs its own licence
    (B) VMs cannot be backed up
    (C) Containers share the host kernel and are lighter
    (D) Containers only run on Windows

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MODULE 6 · HOST · LESSON 3 · 28 MIN

Cloud Computing Fundamentals

Service models, deployment models, shared responsibility and cost

- Compare IaaS, PaaS and SaaS and give examples of each
- Explain public, private and hybrid cloud
- Apply the shared responsibility model to support tasks
- Estimate and control basic cloud costs

Cloud computing means renting computing resources over the internet on demand instead of buying and running everything yourself. For support specialists, cloud changes the job: fewer server rooms, more consoles, identities, permissions and bills. Knowing what the provider looks after, and what you still own, is essential.

Service models (top: you manage least)

SaaS	`Complete application: email, office suite, CRM, HR system` `You manage users, data and settings`
PaaS	`Platform to run your code: managed databases, app hosting` `You manage the app and data`
IaaS	`Virtual machines, networks and storage` `You manage the OS, patches, apps and data`
On-premises	`Your own building, hardware and everything above it` `You manage it all`

Deployment models

Public cloud

- Shared provider infrastructure
- Pay as you go, global regions
- Fast to start

Private cloud

- Dedicated to one organisation
- More control and customisation
- Higher cost and skills needed

Hybrid / multi-cloud

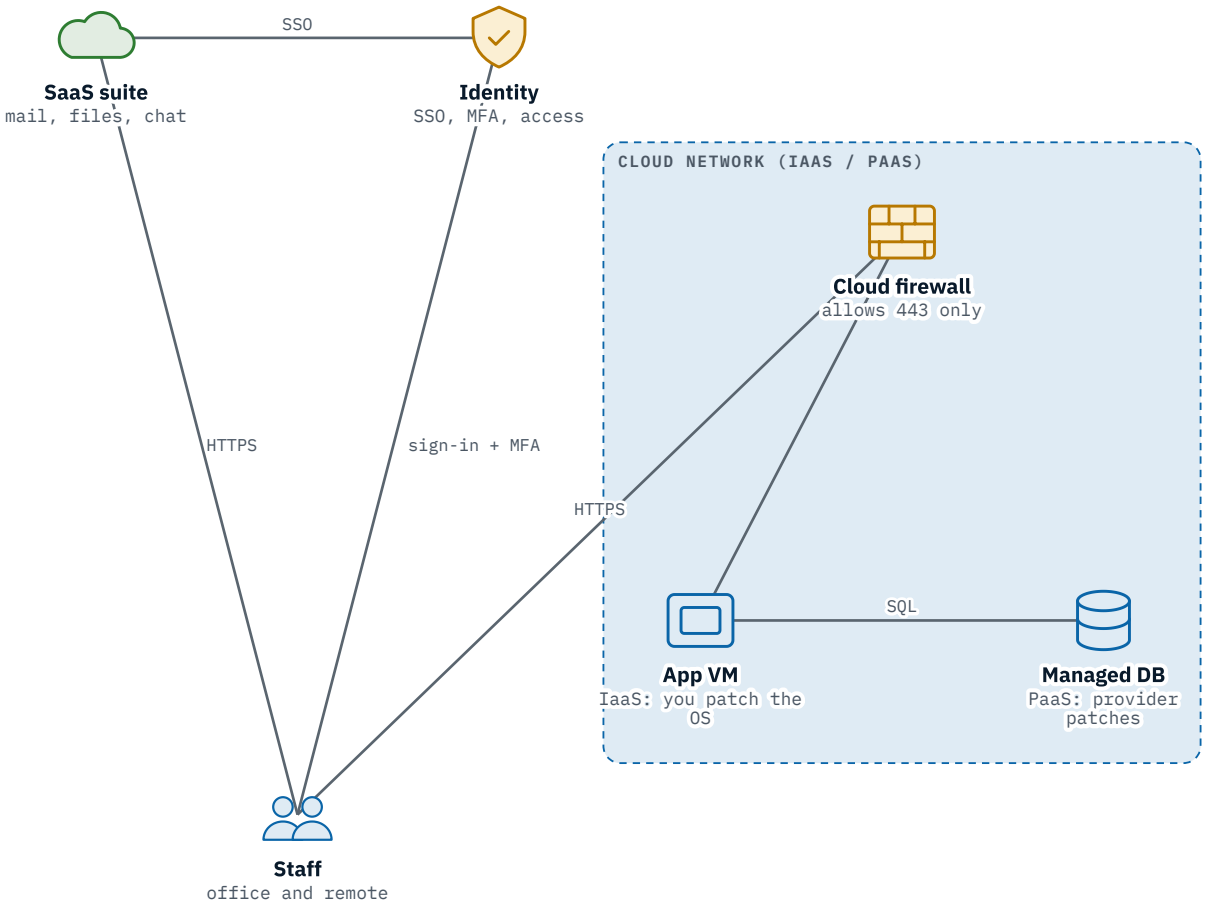
- Mix of on-premises and one or more clouds
- Common in banks, hospitals, government
- Needs consistent identity and security

Shared responsibility

AREA	IAAS	PAAS	SAAS
Physical data centre	Provider	Provider	Provider
Operating system patches	Customer	Provider	Provider
Application configuration	Customer	Customer	Customer (settings)
Identities and access	Customer	Customer	Customer
Data and its classification	Customer	Customer	Customer

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One organisation, three service models



The same organisation consumes SaaS, runs its own VM on IaaS and uses a managed database on PaaS. Who patches what differs per box, but identities, access and data are the customer's in every one.

WATCH OUT · THE PROVIDER WILL NOT SAVE YOU FROM YOURSELF

In every model, **you** own your accounts, MFA, permissions and data. A misconfigured public storage bucket or a shared admin password is the customer's failure, not the provider's. Many cloud breaches are configuration mistakes.

Cloud characteristics that matter to support

- **On-demand self-service**: resources in minutes through a console or API.
- **Elasticity**: scale up for peaks and down afterwards.
- **Regions and availability zones**: choose where data lives and design for failure.
- **Metered service**: you pay for what you use, so idle resources still cost money.
- **Data residency**: some regulators require certain data to stay in-country.

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Basic cloud cost estimates

MEASURE	FORMULA	WHAT IT TELLS YOU
Compute cost	`monthly cost = hourly rate x hours running x number of instances`	A VM at USD 0.10 per hour running all month (730 hours) costs about USD 73.
Savings from scheduling	`saving = hourly rate x hours switched off`	Shutting a test VM down 14 hours a day saves about 14 x 30 x 0.10 = USD 42 a month.
Storage cost	`monthly cost = GB stored x price per GB-month`	2,000 GB at USD 0.02 per GB-month costs USD 40.

Working with a cloud provider's command-line tool (generic)

Linux host

```
# Each provider ships a CLI; commands differ but the pattern is the same
cloud login --tenant company-prod            # sign in (opens browser MFA)
cloud vm list --output table                 # what is running and where
cloud vm stop --name test-web-01             # stop an idle test VM
cloud cost show --last 30d --group-by project
```

This illustrative syntax shows the workflow. Use your provider's real CLI and always confirm which tenant and project you are in before changing anything.

PRO TIP · TAG EVERYTHING

Tags such as owner, cost centre, environment and expiry date let you find who owns a resource, who pays for it and when it can be removed. Untagged resources are the first ones to become forgotten, insecure and expensive.

SCENARIO · DECIDE · A SURPRISE CLOUD BILL

A software company in Dhaka gets a cloud bill three times higher than usual. The cost report shows twelve large VMs created by a developer for a load test five weeks ago, still running, untagged.

(A) Contact the developer to confirm, stop or delete the test VMs, then introduce mandatory tags and budget alerts

(B) Delete all untagged resources immediately

(C) Move everything back on-premises

(D) Ask finance to pay and ignore it

Circle your choice. Answer and feedback: module answer key.

Key terms

IaaS
Infrastructure as a service: renting virtual machines, storage and networks while managing the OS and above yourself.

SaaS
Software as a service: complete applications delivered over the internet and managed by the provider.

Elasticity
The ability to add or remove resources automatically as demand changes.

PaaS
Platform as a service: a managed platform for running applications and databases without managing servers.

Shared responsibility model
The division of security and operational duties between a cloud provider and its customer.

Availability zone
A physically separate data centre location within a cloud region, used to design for failure.

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Resource tag
A key-value label attached to a cloud resource to record owner, cost centre, environment or purpose.

Key takeaways

- SaaS, PaaS and IaaS differ in how much you still manage.
- Whatever the model, accounts, MFA, permissions and data stay your responsibility.
- Many cloud breaches are configuration mistakes, not provider failures.
- Cloud costs accrue every hour a resource runs: schedule, right-size and delete idle resources.
- Tag every resource with owner, cost centre, environment and expiry.

KNOWLEDGE CHECK

11. In IaaS, who is responsible for patching the guest operating system?

(A) The cloud provider
(B) The customer
(C) Nobody
(D) The hardware vendor

12. Which is an example of SaaS?

(A) A virtual machine you install Linux on
(B) A managed database you deploy code to
(C) A hosted email and office suite
(D) Your own server room

13. A VM costs USD 0.20 per hour and runs 24/7 for 30 days. What is the monthly cost?

(A) USD 48
(B) USD 200
(C) USD 72
(D) USD 144

14. Which is always the customer's responsibility, whatever the cloud model?

(A) Physical security of data centres
(B) Managing user access and data
(C) Hypervisor patching
(D) Replacing failed disks

15. Why use resource tags?

(A) To make VMs faster
(B) To encrypt data
(C) To identify owner, cost and purpose and control spending
(D) To avoid using MFA when automation scripts sign in to the cloud console

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MODULE 6 · HOST · LESSON 4 · 28 MIN

SaaS and Productivity Suite Administration

Users, licences, mail, sharing and collaboration in cloud suites

- Perform common admin tasks in a cloud productivity suite
- Manage licences, shared mailboxes, groups and calendars
- Troubleshoot mail flow and sharing problems
- Configure safe external sharing and offboarding

Most organisations now run email, documents, chat and video meetings on a cloud productivity suite. The support specialist becomes a **tenant administrator**: creating users, assigning licences, fixing mail flow, recovering deleted files and controlling what can be shared outside the organisation. The menus differ between providers, but the tasks are almost identical.

Everyday SaaS admin tasks

TASK	WHERE	WATCH OUT FOR
Create user and assign licence	Admin centre or identity directory	Correct licence tier; usage location for compliance
Reset password / MFA	Identity admin	Verify identity first; revoke sessions if compromise suspected
Shared mailbox	Mail admin	Grant access to a group; usually needs no licence of its own
Distribution or team group	Groups admin	Owner assigned; external members only if approved
Restore deleted file or mailbox	Retention / recycle settings	Retention period limits; act quickly
Offboard a leaver	Identity, mail and storage admin	Block sign-in, transfer data, convert mailbox, reclaim licence

Offboarding in a cloud suite

01 Block sign-in — Disable the account and revoke active sessions and tokens. →

02 Protect data — Set mail forwarding or convert to a shared mailbox for the manager. →

03 Transfer files — Move ownership of documents and shared drives. →

04 Remove access — Remove from groups, apps and devices; wipe company data on phones. →

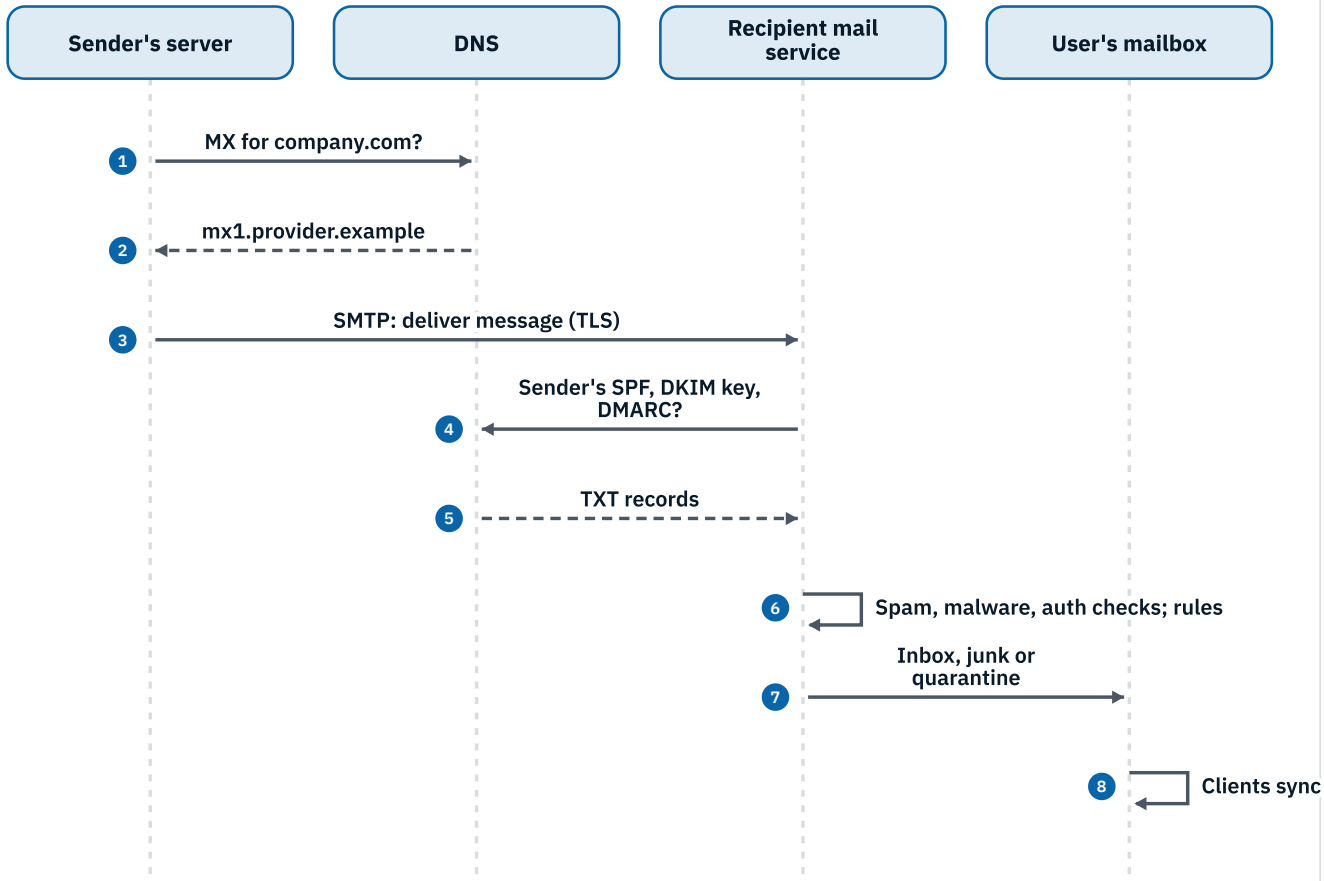
05 Reclaim licence — Free the licence after the retention steps are done.

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Licence management basics

- Licences are assigned per user; unused licences still cost money every month.
- Group-based licensing assigns licences automatically by role.
- Review licence reports monthly for inactive accounts.
- Shared mailboxes and resource calendars often need no paid licence (check limits).
- Plan for growth: new starters cannot work without a licence on day one.

How an email reaches a user



Most 'missing email' tickets are answered at step 6: the message arrived but was filtered, quarantined or redirected by a rule. Message trace shows exactly which step a message reached.

Mail troubleshooting clues

Never arrived

- Check message trace in the admin centre
- Check quarantine and junk
- Check the sender's bounce message

Arrived late

- Look at message header timestamps for each hop
- Large attachments or greylisting
- Sender-side delays

Bounced

- Read the status code and text
- Mailbox full, address wrong, or blocked by policy
- Failed SPF, DKIM or DMARC checks; large mailbox providers have required all three from bulk senders since 2024

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WATCH OUT · AUTO-FORWARDING TO EXTERNAL ADDRESSES

Attackers who compromise a mailbox often create rules to forward mail outside the organisation silently. Block automatic external forwarding by default, alert on new forwarding rules and check rules first when investigating a suspicious account.

Sharing settings

Internal only

- Files shared within the organisation
- Default for sensitive teams such as HR and finance

Specific external people

- Invite named guests who must sign in
- Expiry dates on guest access
- Good for partners and auditors

Anyone with the link

- No sign-in needed
- Easy to leak
- Disable or restrict with expiry and view-only

TEMPLATE SaaS offboarding checklist record

Track each leaver through every offboarding step with evidence.

LEAVER	SIGN-IN BLOCKED (TIME)	SESSIONS REVOKED	MAIL HANDLED	FILES TRANSFERRED TO	DEVICES WIPED	LICENCE RECLAIMED
M. Chowdhury	*2026-06-30 17:05*	*Yes*	*Shared mailbox, manager access*	*Team lead K. Das*	*Phone work profile removed*	*2026-07-30*

SCENARIO · DECIDE · MISSING INVOICES FROM A SUPPLIER

The accounts team at a Chattogram trading company says invoices from one supplier stopped arriving three days ago. Other external mail arrives normally.

(A) Ask the supplier to send invoices by WhatsApp instead

(B) Increase the accounts mailbox size

(C) Reset the accounts team's passwords

(D) Run a message trace for the supplier's domain and check quarantine and the supplier's SPF/DMARC results

Circle your choice. Answer and feedback: module answer key.

Key terms

Tenant
An organisation's dedicated instance of a cloud service, with its own users, settings and data.

Licence assignment
Allocating a paid service plan to a user so they can use specific cloud applications.

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Shared mailbox
A mailbox that several users can access, such as accounts@ or support@, without a separate sign-in.

Quarantine
A holding area for messages blocked as spam, phishing or malware, which admins can review and release.

Retention policy
A rule that keeps or deletes data for a defined period to meet legal or business needs.

Message trace
An admin tool that shows the path and status of an email through the mail service.

Guest access
Permission for an external person to access specific internal resources after signing in.

Key takeaways

- Tenant admins create users, assign licences, fix mail flow and control sharing.
- Offboard in order: block sign-in and revoke sessions first, reclaim the licence last.
- Use message trace, quarantine and SPF/DKIM/DMARC results for missing mail.
- Block automatic external forwarding and watch for new forwarding rules.
- Prefer named guest access with expiry over 'anyone with the link'.

KNOWLEDGE CHECK

16. Which tool shows whether an email from a partner reached the organisation and what happened to it?

(A) Disk Management
(B) Message trace
(C) gpresult
(D) ipconfig

17. What should be done first when offboarding a leaver in a cloud suite?

(A) Delete the account immediately so the licence is freed the same day
(B) Block sign-in and revoke sessions
(C) Reclaim the licence
(D) Email their contacts

18. Why should automatic forwarding to external addresses be blocked by default?

(A) It slows the mail server and fills the outbound queue during busy periods
(B) It breaks calendars
(C) Attackers use it to exfiltrate mail silently
(D) It uses licences

19. A team needs a common support@ inbox that four people can read and answer. What is the best choice?

(A) Create a shared mailbox and grant access to a group
(B) Share one user's password
(C) Forward each message manually
(D) Create four separate accounts named support1 to support4 and forward between them

20. Which sharing setting carries the highest risk of data leakage?

(A) Internal only
(B) Anyone with the link
(C) Specific signed-in guests
(D) Owner only

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MODULE 6 · HOST · LESSON 5 · 26 MIN

Monitoring, Availability and Remote Access

Seeing problems before users do and keeping services reachable

- Choose useful metrics and alerts for servers, networks and services
- Calculate availability and allowed downtime
- Describe redundancy options from power to cloud regions
- Provide secure remote access with VPN and remote desktop gateways

The best support teams hear about problems from their monitoring, not from angry users. Monitoring watches health and performance; availability design adds redundancy so single failures do not stop the business; and secure remote access lets staff and support work from anywhere without opening the network to attackers.

What to monitor

AREA	METRICS	EXAMPLE ALERT
Servers	CPU, memory, disk space, disk latency	Disk C: under 10% free for 15 minutes
Services	Up/down, response time, error rate	Intranet HTTP check fails 3 times in a row
Network	Link status, utilisation, errors, packet loss	WAN link above 90% for 10 minutes
Backups	Job success, duration, last good copy	No successful backup in 26 hours
Security	Failed sign-ins, new admin accounts, disabled protection	Antivirus disabled on any endpoint
Environment	Server room temperature, UPS battery, humidity	Temperature above 27 degrees C

INSIGHT · ALERT ON WHAT NEEDS ACTION

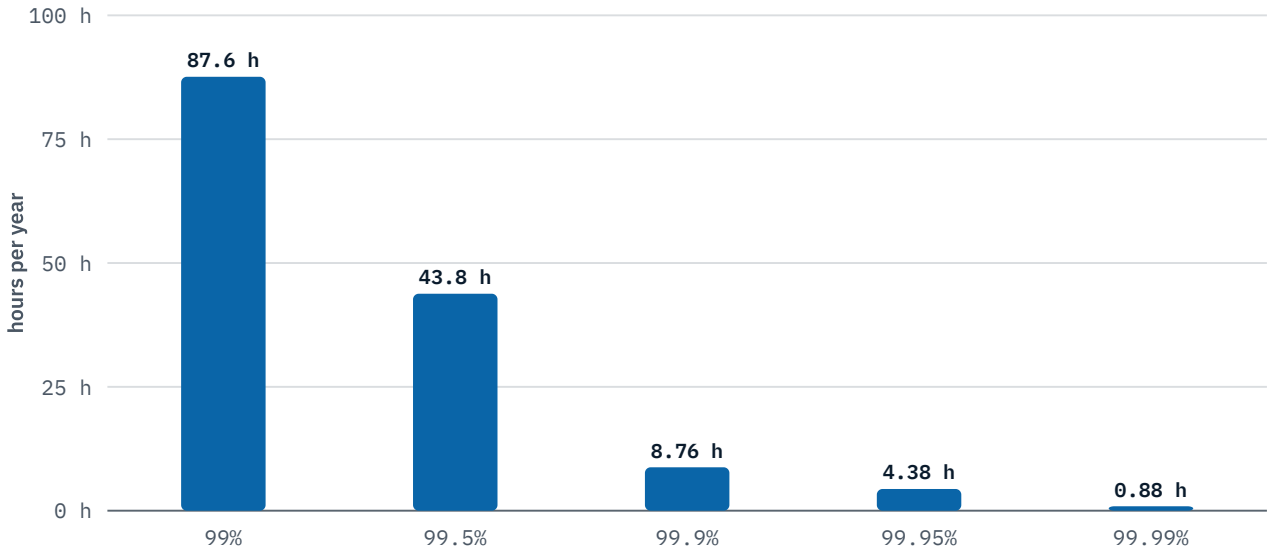
If an alert fires and no one needs to do anything, it is noise. Too many noisy alerts train people to ignore all of them. Each alert should have an owner, a threshold based on real behaviour and a short runbook saying what to do.

Availability maths

MEASURE	FORMULA	WHAT IT TELLS YOU
Availability	`availability % = (total time - downtime) / total time x 100`	A month of 720 hours with 3.6 hours of downtime is 99.5% available.
Allowed downtime per year	`downtime = (1 - availability) x 8,760 hours`	99.9% allows about 8.8 hours a year; 99.99% allows about 53 minutes.
Two redundant components	`combined availability = 1 - (1 - A1) x (1 - A2)`	Two independent 99% internet links give about 99.99% if failover works.

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Downtime allowed per year



Exact values from (1 - availability) x 8,760 hours. Each extra nine cuts allowed downtime tenfold, and usually raises the cost of redundancy sharply.

Redundancy from the ground up (top first)

Sites and regions	`Second site or cloud region` `Disaster recovery plans`
Services	`Clustered or load-balanced servers` `Two DNS and DHCP servers`
Network	`Two internet links from different providers` `Redundant switches and firewall pairs`
Hardware	`RAID, dual PSUs, spare parts on site`
Power	`UPS, generator, dual power feeds`

Remote access options

VPN

- Encrypted tunnel into the internal network
- Use MFA and up-to-date VPN software
- Gives broad network access, so restrict by role

Remote desktop gateway or bastion

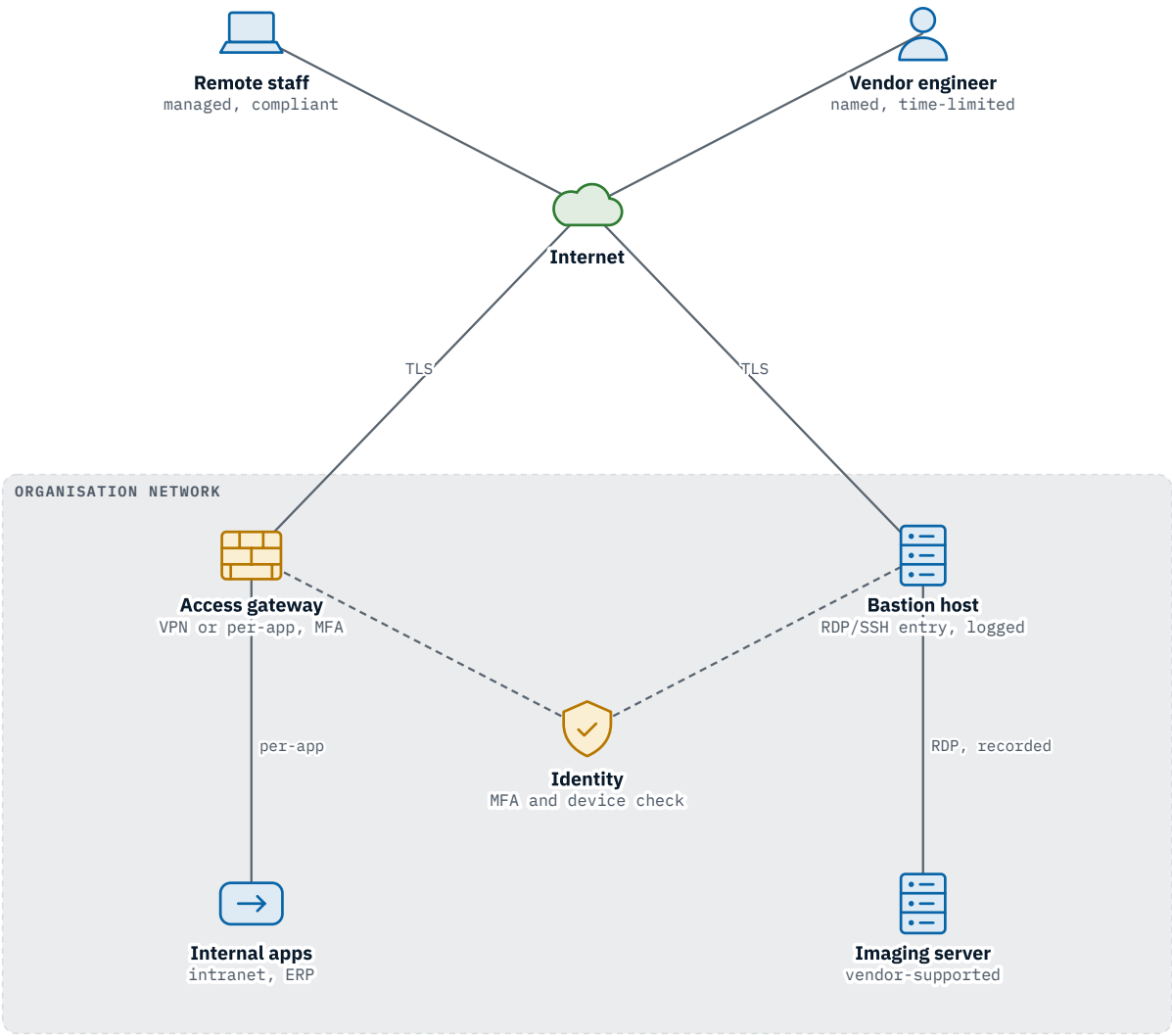
- Single hardened entry point for RDP or SSH
- MFA and logging at the gateway
- Never expose RDP directly to the internet

Zero-trust access

- Per-app access checked by identity and device health
- No broad network access
- Increasingly common in 2026

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Secure remote access without exposed ports



Nothing internal is reachable directly: every path enters through a gateway or bastion that checks identity, MFA and device health and logs the session. The vendor's account works only during the agreed window.

Quick health checks for remote access

Windows PowerShell

```
Test-NetConnection vpn.company.com -Port 443
Get-NetIPConfiguration | Where-Object InterfaceAlias -like '*VPN*'
# Linux / macOS: secure shell through a bastion host
# ssh -J admin@bastion.company.com admin@app01.internal
```

Simple monitoring checks on Linux

Linux host

```
uptime                                  # load averages
df -h | awk '$5+0 > 85'                 # file systems above 85% full
curl -s -o /dev/null -w '%{http_code} %{time_total}\n' https://intranet.company.local
journalctl -p warning --since today | tail -n 20
```


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CASE STUDY · OUTAGE ON PAYDAY

A microfinance institution in Bogura has 60 branches connected to a head-office data centre. On payday, the loan system was unreachable for 3 hours because the single internet link at head office failed. No alert was raised; branches phoned in. The monthly availability target for the loan system is 99.5%.

1. Did the system meet its monthly target (30-day month)?

2. What monitoring would have detected the outage sooner?

3. What redundancy would you recommend?

Model answer: module answer key.

SCENARIO · DECIDE · REMOTE ACCESS FOR A VENDOR

A medical equipment vendor needs to support an imaging server at a Dhaka hospital once a month. They ask for the server's Remote Desktop port to be opened to the internet.

(A) Open TCP 3389 to the internet permanently

(B) Give the vendor a domain admin password by email

(C) Ask the vendor to bring a 4G router and connect it to the server

(D) Provide time-limited access through the hospital's VPN or remote access gateway with MFA, logging and a named vendor account

Circle your choice. Answer and feedback: module answer key.

Key terms

Monitoring
Continuously collecting metrics and checks from systems to detect problems and trends.

Availability
The percentage of time a service is usable as agreed.

Single point of failure
A component whose failure stops the whole service.

Bastion host
A hardened server that acts as the controlled entry point for administrative remote access.

Alert threshold
The value or condition at which monitoring raises an alert, such as disk space below 10%.

Redundancy
Extra components that take over when one fails, removing single points of failure.

VPN
Virtual private network: an encrypted tunnel that gives remote users access to internal resources.

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Key takeaways

- Good monitoring finds problems before users do; every alert needs an owner and a runbook.
- Availability = (total time - downtime) / total time; each extra nine costs far more.
- Remove single points of failure layer by layer, from power to sites.
- Never expose RDP or SSH directly; use a VPN, zero-trust gateway or bastion with MFA.
- Give vendors named, time-limited, logged access.

KNOWLEDGE CHECK

21. Roughly how much downtime per year does 99.9% availability allow?

(A) About 8.8 hours
(B) 53 minutes
(C) 3.65 days
(D) 1 hour

22. Which alert is most useful?

(A) No successful backup in 26 hours for a critical server
(B) CPU above 1% on any server
(C) Every user sign-in
(D) Any change in any printer's page count, reported to every technician

23. Which remote access practice is unsafe?

(A) VPN with MFA
(B) Bastion host with logging
(C) Per-app zero-trust access
(D) RDP exposed directly to the internet

24. Two independent internet links are each 99% available. Approximately what is the combined availability with working failover?

(A) 98%
(B) 99.99%
(C) 99%
(D) 100%

25. A service was down 7.2 hours in a 720-hour month. What was its availability?

(A) 99.9%
(B) 99%
(C) 98%
(D) 99.5%

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MODULE 6 · HOST

Answer key

Knowledge checks

1. C Wrong clocks break ticket-based authentication and certificate validity.

2. B SMB uses TCP 445.

3. A The management controller works independently of the OS.

4. D Name resolution failure with working IP connectivity points to DNS.

5. A systemctl --failed lists failed units.

6. C (5 - 1) x 4 = 16 TB.

7. C RAID 0 stripes without parity or mirroring.

8. B Snapshots are for short-term rollback, not backup.

9. C A hosted hypervisor runs test VMs on a laptop.

10. C Containers share the host kernel; VMs run their own.

11. B In IaaS the customer manages the OS and above.

12. C Complete applications delivered as a service are SaaS.

13. D 0.20 x 24 x 30 = USD 144.

14. B Identity and data stay with the customer.

15. C Tags support ownership, cost tracking and clean-up.

16. B Message trace follows a message through the mail service.

17. B Blocking access comes first; deleting too early loses data.

18. C Forwarding rules are a common data-theft technique after compromise.

19. A A shared mailbox with group access keeps accountability.

20. B Anonymous links can be forwarded to anyone.

21. A 0.001 x 8,760 hours is about 8.76 hours.

22. A It signals a real risk needing action.

23. D Direct RDP exposure is heavily attacked.

24. B 1 - (0.01 x 0.01) = 0.9999.

25. B (720 - 7.2) / 720 = 99%.

Scenarios & case studies

Scenario · Nobody can print on the second floor
A is best. Correct. The scope points to that printer's queue or connection, not to the whole server.

Scenario · Choosing RAID for a new file server
D is best. Correct. With large disks and long rebuilds, double-parity protection is the safer choice, and backups remain essential.

Scenario · A surprise cloud bill
A is best. Correct. You fix the immediate cost safely and prevent repeats with tagging and alerts.

Scenario · Missing invoices from a supplier
D is best. Correct. The trace shows whether messages arrived and why they were filtered; a supplier-side authentication change is a common cause.

Case study · Outage on payday
Target: 30 days = 720 hours; 3 hours down gives (720 - 3) / 720 = 99.58%, so it narrowly met 99.5%, but one more incident would breach it. **Monitoring:** external checks of the loan system's URL from outside the network, WAN link status and utilisation alerts at head office, and alerts routed to on-call phones. **Redundancy:** a second internet link from a different provider with automatic failover, tested quarterly, and possibly a cloud-hosted standby for critical services.

Scenario · Remote access for a vendor
D is best. Correct. Access is controlled, monitored, attributable and only available when needed.

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07

MODULE 7 · PROTECT

Security Basics for IT Support

Identity, MFA, malware, phishing, encryption and incident handling for the first line of defence

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Apply the CIA triad, risk scoring, least privilege and defence in depth to support decisions
- Configure modern password, MFA and privileged-account controls
- Recognise malware and phishing and respond safely
- Support encryption and data protection and play the support role in incidents

LESSONS

1. Security Foundations for Support Staff · 26 min
2. Passwords, MFA and Account Security · 28 min
3. Malware, Phishing and Endpoint Protection · 28 min
4. Encryption, Data Protection and Incident Handling · 28 min

MODULE 7 · PROTECT · LESSON 1 · 26 MIN

Security Foundations for Support Staff

CIA, risk, least privilege and defence in depth in everyday support work

- Explain confidentiality, integrity and availability with support examples
- Describe threats, vulnerabilities and risk, and estimate simple risk scores
- Apply least privilege and defence in depth to support decisions
- Recognise the support team's role in physical and social security

Support specialists hold a lot of power: admin rights, password resets, access to devices and data. That makes them both a strong line of defence and an attractive target. You do not need to be a security engineer, but every decision you make, from granting access to installing software, has a security side.

The CIA triad in support work

C

Confidentiality

Only authorised people see data. Example: locking a screen, restricting a payroll folder, encrypting a laptop.

I

Integrity

Data is accurate and unaltered. Example: checksums on downloads, change control, file versioning.

A

Availability

Systems and data are there when needed. Example: backups, UPS, redundancy, patching to avoid crashes.

Threats, vulnerabilities and risk

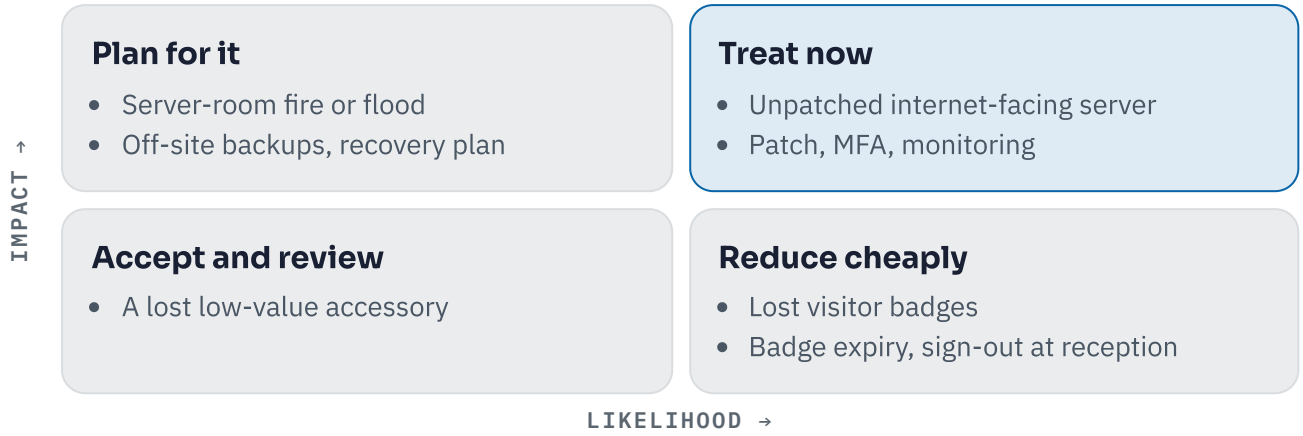
TERM	MEANING	EXAMPLE
Asset	Something of value	Customer database, laptops, reputation
Threat	Something that could cause harm	Ransomware gang, careless insider, flood
Vulnerability	A weakness a threat can exploit	Unpatched VPN, weak password, open server room
Risk	Likelihood that a threat exploits a vulnerability, and the impact	High risk: internet-facing unpatched server
Control	Measure that reduces risk	Patch, MFA, lock, backup, training

Simple risk scoring

MEASURE	FORMULA	WHAT IT TELLS YOU
Risk score	`risk = likelihood (1-5) x impact (1-5)`	An unpatched internet-facing server: likelihood 4 x impact 5 = 20, a top priority.
Residual risk	`residual risk = risk remaining after controls are applied`	After patching and MFA, likelihood drops to 1, giving 1 x 5 = 5.

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Deciding what to do with a risk



Place each risk by likelihood and impact before choosing a control. High-high risks need action now; rare disasters need plans and backups rather than daily effort.

DEFINITION · LEAST PRIVILEGE

Give every account, person and program only the access it needs to do its job, for only as long as it needs it. Least privilege limits the damage from mistakes, malware and stolen credentials.

Defence in depth (outer layer first)

Policies and people	`Acceptable use policy` `Security awareness training`
Physical	`Locked rooms, badges, visitor logs, cable locks`
Network	`Firewalls, segmentation, secure Wi-Fi, VPN`
Identity	`Strong passwords, MFA, least privilege`
Endpoint	`Patching, endpoint protection, encryption`
Data	`Classification, backups, access control`

Physical security habits

- Lock your screen every time you step away (Windows key + L, Ctrl + Cmd + Q on macOS).
- Challenge politely or report people without badges in restricted areas.
- Do not hold secure doors open for strangers (tailgating).
- Keep laptops locked or with you; never leave them visible in vehicles.
- Shred printed documents containing personal or financial data.

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Common social engineering tricks

Pretexting

- Caller pretends to be a manager, auditor or IT
- Creates a believable story to get information

Baiting

- Infected USB sticks left in car parks
- 'Free' downloads carrying malware

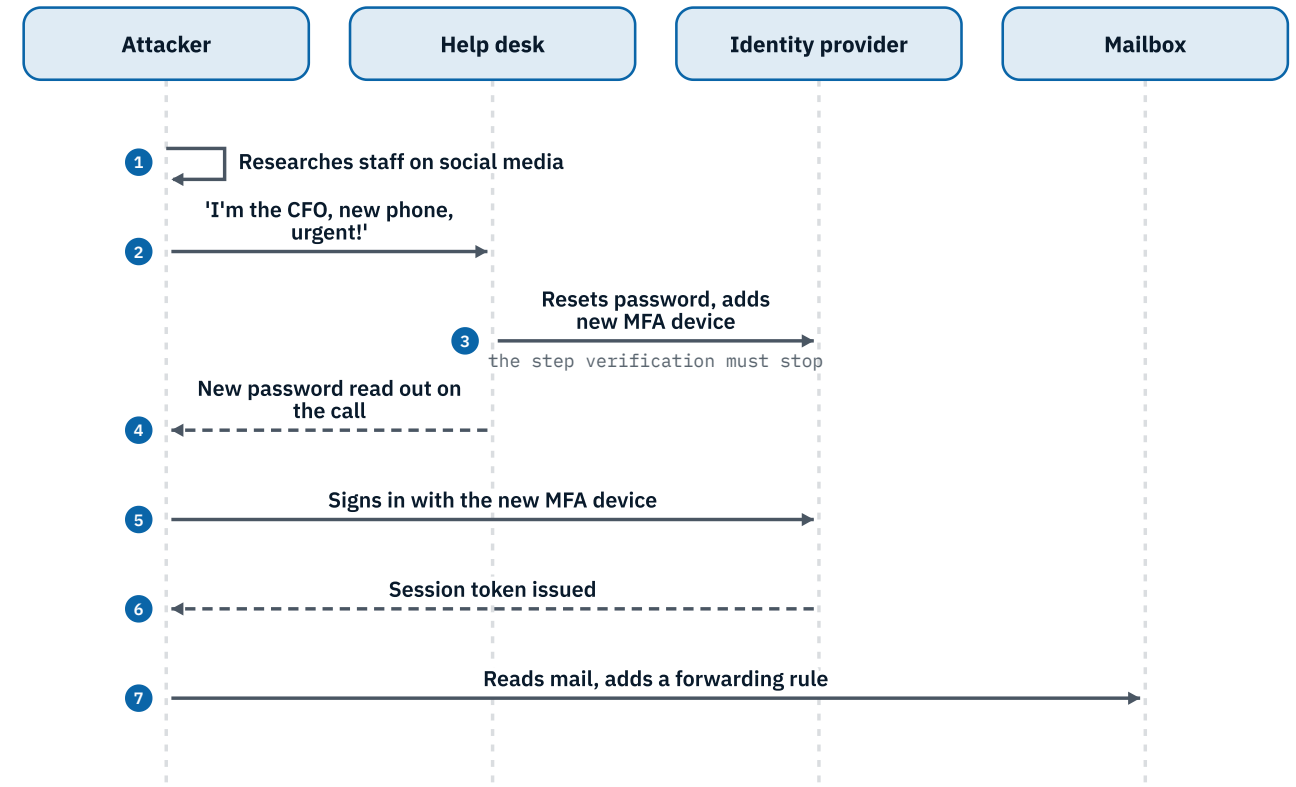
Tailgating

- Following staff through secure doors
- Often carrying boxes to seem legitimate

WATCH OUT · ADMINS ARE PRIME TARGETS

Attackers specifically target help desk staff to reset passwords and add MFA devices for them. Follow identity verification procedures every time, especially when a caller is urgent, senior or angry.

How a help-desk social engineering attack works



No malware is needed: one rushed reset hands over the account. Verifying identity with details already on record at step 3 breaks the whole chain.

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SCENARIO · DECIDE · AN UNKNOWN VISITOR IN THE SERVER ROOM CORRIDOR

You see a man in a contractor's vest near the server room door at a Gazipur factory. He says he is 'here to check the air conditioning' but has no visitor badge and no one told you about a visit.

(A) Let him in because he looks like a contractor

(B) Politely ask him to wait in reception while you confirm the visit with facilities, and do not let him into the room

(C) Confront him aggressively and call the police

(D) Ignore him; security is not your job

Circle your choice. Answer and feedback: module answer key.

Key terms

CIA triad
The three core goals of information security: confidentiality, integrity and availability.

Vulnerability
A weakness in a system, process or person that a threat could exploit.

Least privilege
The principle of granting only the minimum access needed, for the minimum time.

Social engineering
Manipulating people into revealing information or performing actions that weaken security.

Threat
Any circumstance or actor with the potential to cause harm to an asset.

Risk
The combination of the likelihood of a harmful event and its impact.

Defence in depth
Using multiple layers of security controls so that one failure does not lead to compromise.

Key takeaways

- Every support decision has a confidentiality, integrity or availability side.
- Risk = likelihood x impact; treat the highest scores first.
- Least privilege and defence in depth limit the damage of any single failure.
- Help desks are prime social-engineering targets: verify identity every time, especially under pressure.
- Physical security is everyone's job: lock screens, challenge politely, stop tailgating.

KNOWLEDGE CHECK

1. Encrypting a laptop's disk mainly protects which part of the CIA triad?
   (A) Availability
   (B) Confidentiality
   (C) Integrity
   (D) Accountability
2. A risk has likelihood 3 and impact 4. What is its score on a 1-5 x 1-5 scale?
   (A) 7
   (B) 34
   (C) 1
   (D) 12
3. Which action best demonstrates least privilege?
   (A) Making all IT staff domain admins
   (B) Sharing one admin password in the team so that nobody gets locked out of systems
   (C) Disabling UAC
   (D) Using a separate admin account only when admin tasks are needed

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4. Someone follows you through a badge-controlled door without swiping. What is this called?

(A) Phishing
(B) Baiting
(C) Pretexting
(D) Tailgating

5. Why are help desk staff targeted by attackers?

(A) They can reset passwords and change MFA for others
(B) They have slow PCs
(C) They never use email
(D) They work night shifts when security monitoring is usually switched off

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · PROTECT · LESSON 2 · 28 MIN

Passwords, MFA and Account Security

Protecting the identities that unlock everything else

- Set and explain modern password guidance
- Compare MFA methods by strength and usability, including passkeys
- Recognise signs of account compromise and respond correctly
- Separate admin accounts and protect privileged access

Most breaches involve stolen or weak credentials. The good news is that a few well-chosen controls stop the vast majority of account attacks: long unique passwords stored in a password manager, multi-factor authentication everywhere and careful handling of privileged accounts. Support specialists configure these controls and help users live with them.

Password guidance: then and now

Outdated advice

- Short passwords with forced symbols
- Mandatory change every 30-90 days
- Password hints and security questions
- Writing passwords on sticky notes when rules get too hard

Modern practice

- Long passphrases: at least 15 characters when a password is the only factor
- No forced symbol rules or scheduled changes; change on compromise
- Block common and breached passwords
- Password manager for unique passwords; passkeys wherever offered

INSIGHT · WHAT CURRENT GUIDANCE SAYS

The 2025 revision of a widely used digital identity standard (NIST SP 800-63B) tells organisations not to force periodic password changes or composition rules, to screen new passwords against breached lists, and to accept long passphrases, with at least 15 characters when a password is used on its own. Many regulators and auditors now follow the same approach.

Why length beats complexity

MEASURE	FORMULA	WHAT IT TELLS YOU
Possible combinations	`combinations = (characters available) ^ (length)`	8 characters from 95 symbols is about 6.6 x 10^15; 16 lowercase letters is about 4.4 x 10^22, millions of times more.
Entropy (bits)	`entropy = length x log2(characters available)`	A 4-word passphrase chosen at random from 7,776 words has about 51.7 bits; each extra word adds about 12.9 bits.

MFA methods compared

METHOD	STRENGTH	NOTES
SMS or voice code	Weak to moderate	Vulnerable to SIM swap and interception; better than nothing
Authenticator app code (TOTP)	Moderate	Works offline; can still be phished on a fake site
Push approval with number matching	Good	Number matching defeats most 'MFA fatigue' spam

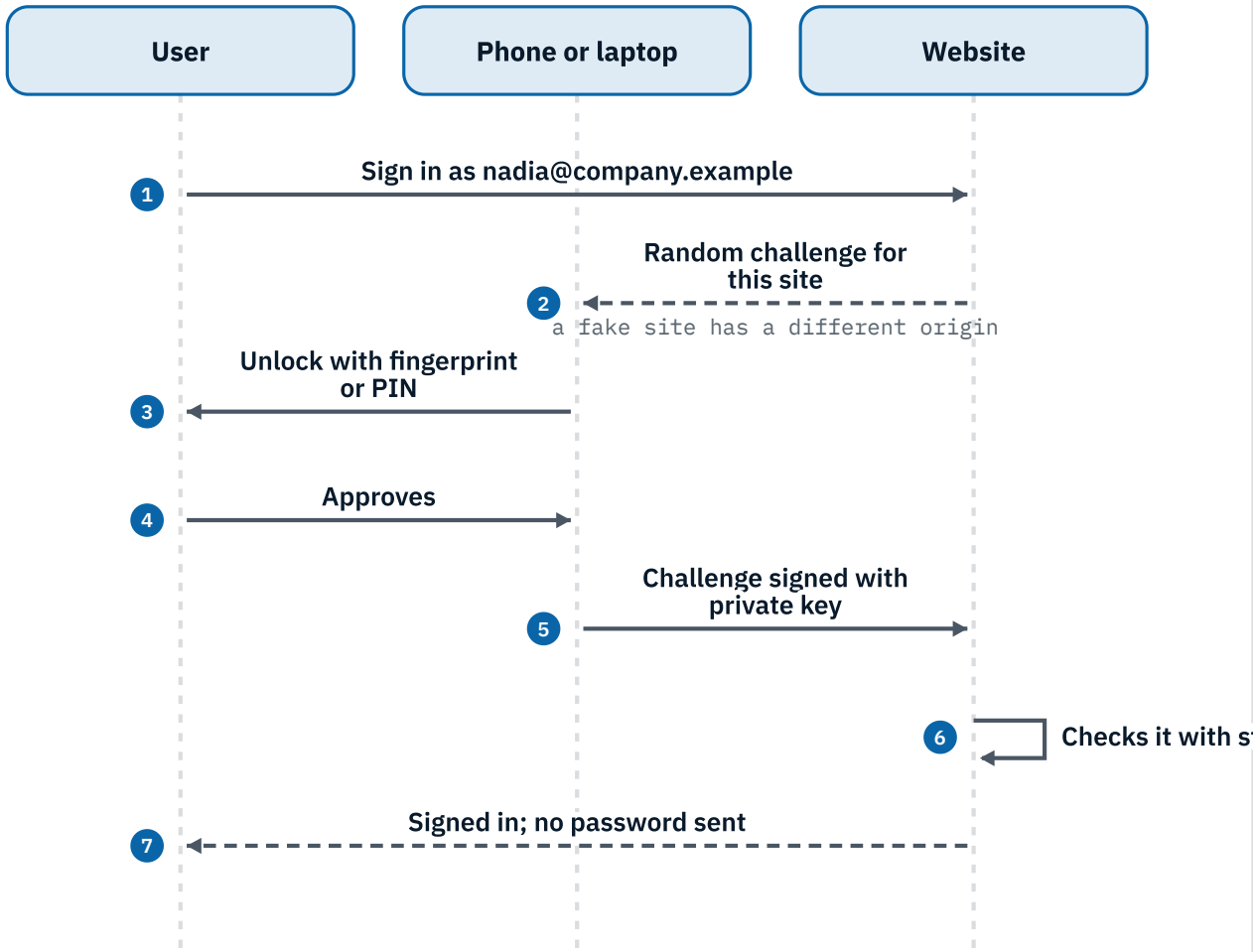
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METHOD	STRENGTH	NOTES
Hardware security key (FIDO2)	Strong	Phishing-resistant; tied to the real site
Passkey (device-bound or synced)	Strong	Phishing-resistant sign-in without a password

INSIGHT · MFA FATIGUE ATTACKS

Attackers with a stolen password send dozens of push prompts at night, hoping the user taps 'Approve' to make them stop. Teach users: **never approve a prompt you did not start**, and report it. Number matching and phishing-resistant methods stop most of these attacks.

How a passkey sign-in works



The private key never leaves the device, and the signature only works for the genuine site, so a phishing page has nothing to steal or replay. If the device is lost, recovery goes through verified identity, never a quick reset.

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Signs an account may be compromised

- Sign-ins from unusual countries or impossible travel between locations.
- MFA prompts the user did not trigger.
- New inbox rules forwarding or deleting mail.
- Password reset emails the user did not request.
- Contacts receiving odd messages or invoices from the user.

Responding to a suspected compromised account

1. **Contain** Reset the password, revoke all sessions and tokens, and block sign-in if needed.
2. **Check MFA** Remove unknown MFA methods and re-register the user's own.
3. **Inspect** Look for malicious inbox rules, forwarding, app consents and recent file sharing.
4. **Report** Raise a security incident and follow the incident process.
5. **Educate** Explain to the user what happened and how to spot it next time, without blame.

Checking local account security settings

Windows PowerShell

```
net accounts                                    # local password and lockout policy
Get-LocalUser | Select-Object Name, Enabled, PasswordLastSet, LastLogon
Get-LocalGroupMember Administrators             # who has local admin?
# Linux: sudo chage -l nusrat  (password ageing for a user)
# Linux: getent group sudo     (who can use sudo on Debian/Ubuntu)
```

Privileged access done well

Separate accounts

- Daily account for email and web
- Admin account only for admin tasks
- Admin accounts never have mailboxes

Limit and log

- Just-in-time elevation that expires
- All admin actions logged
- Quarterly review of admin group membership

Local admin passwords

- Unique, random local admin password per device
- Rotated automatically by a management tool
- Never the same password on every PC

PRO TIP · RECOMMEND A PASSWORD MANAGER

A good password manager generates and stores unique passwords for every site, fills them only on the correct website (which helps against phishing) and can share credentials securely within a team. It is the single most useful tool you can give users.

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SCENARIO · DECIDE · MIDNIGHT MFA PROMPTS

A finance officer at a mobile financial services company calls at 08:00: she received about 30 sign-in approval requests overnight and approved one by mistake around 02:00 to stop them.

(A) Treat it as a compromise: reset her password, revoke sessions, review MFA methods, inbox rules and recent activity, and raise a security incident

(B) Tell her not to worry because she has MFA

(C) Disable MFA so she will not be disturbed again

(D) Ask her to change her password next month

Circle your choice. Answer and feedback: module answer key.

Key terms

Passphrase
A long password made of several words, easier to remember and harder to crack than a short complex one.

Multi-factor authentication
Sign-in that requires two or more types of evidence: something you know, have or are.

Passkey
A cryptographic credential stored on a device that replaces passwords and is bound to the genuine website.

Privileged account
An account with elevated rights, such as administrator or root, that needs extra protection.

Password manager
Software that generates, stores and fills unique strong passwords securely.

Phishing-resistant MFA
Authentication methods such as FIDO2 security keys and passkeys that cannot be replayed on a fake site.

MFA fatigue
An attack that floods a user with approval prompts hoping they will accept one.

Key takeaways

- Length beats complexity: long passphrases plus a password manager.
- Drop forced periodic changes and composition rules; block breached passwords instead.
- Prefer phishing-resistant MFA: passkeys and FIDO2 keys; number matching for push.
- Never approve a prompt you did not start; unexpected prompts mean the password is known.
- Separate admin accounts, just-in-time elevation and unique local admin passwords protect privileged access.

KNOWLEDGE CHECK

6. Which MFA method is phishing-resistant?

(A) SMS code
(B) Authenticator app code
(C) FIDO2 security key or passkey
(D) Email code

7. Which password policy reflects modern practice?

(A) Long passphrases, blocking breached passwords, change on compromise
(B) Change every 30 days
(C) 8-character passwords with forced symbols, numbers and a change every 30 days
(D) Security questions for resets

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8. A user reports MFA prompts they did not trigger. What does this suggest?
   - (A) Their phone is broken
   - (B) MFA is misconfigured and sending duplicate prompts after each sign-in
   - (C) Nothing; ignore them
   - (D) Someone has their password and is trying to sign in
9. Why should admins have separate admin accounts?
   - (A) To get more licences
   - (B) Because admins change roles often and forget which password belongs to which system
   - (C) So email and browsing run without admin rights, limiting damage
   - (D) To avoid MFA
10. Which local admin practice is safest?
    - (A) The same strong password on every PC, so technicians can remember it
    - (B) No password
    - (C) Password written on the PC
    - (D) Unique, automatically rotated passwords per device
11. A small organisation is updating its password policy to current guidance. Which existing rule should it drop?
    - (A) Checking new passwords against lists of breached passwords
    - (B) Forcing every password to change every 60 days
    - (C) Allowing long passphrases that contain spaces
    - (D) Requiring MFA for email and remote access

☐ Lesson complete · Date _____ · Knowledge check score ____ / 6

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MODULE 7 · PROTECT · LESSON 3 · 28 MIN

Malware, Phishing and Endpoint Protection

Recognising, preventing and cleaning up the most common attacks

- Identify common malware types and their symptoms
- Recognise phishing and teach users to report it
- Explain endpoint protection, EDR and application control
- Follow a safe malware response procedure

Phishing emails and malware are where most incidents begin, and support staff are usually the first to hear about them: 'My PC is acting strange', 'I clicked a link', 'My files have weird extensions'. A calm, correct first response limits the damage. A panicked or careless one can spread an infection across the organisation.

Malware types

TYPE	WHAT IT DOES	TYPICAL SIGN
Virus	Infects files and spreads when they run	Modified files, antivirus alerts
Worm	Spreads across networks on its own	Sudden network traffic, many infected hosts
Trojan	Hides inside apparently legitimate software	Unknown processes after installing 'free' tools
Ransomware	Encrypts data and demands payment	Renamed files, ransom notes
Spyware / keylogger	Steals data and keystrokes	Account takeovers, unexpected outbound traffic
Rootkit	Hides deep in the OS or firmware	Security tools disabled or blind
Cryptominer	Uses CPU/GPU to mine currency	High CPU, loud fans, slow PC

Red flags in a phishing message

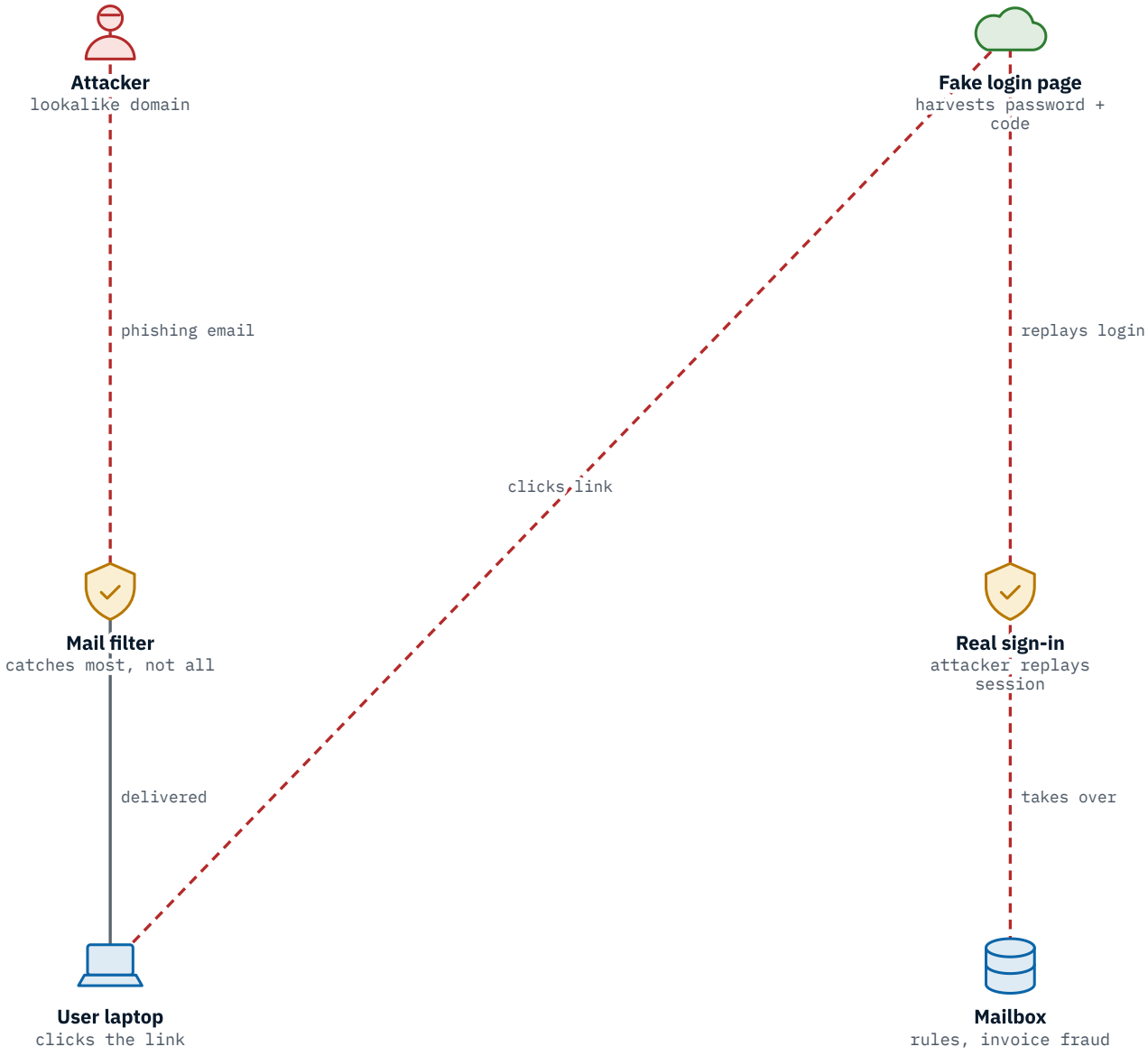
- Urgency or threats: 'Your account will be closed in 2 hours'.
- Sender address that does not match the claimed organisation, or a lookalike domain.
- Links whose real destination (hover to check) differs from the text.
- Unexpected attachments, especially archives, macro documents or HTML files.
- Requests to change bank details, buy gift cards or bypass normal process.
- Generic greetings and unusual language for that sender.
- QR codes or voice messages that rush you to a sign-in page or payment; text and even voices can now be AI-generated.

EXAMPLE · PAYMENT DIVERSION FRAUD

A Chattogram importer received an email that looked like it came from a long-standing supplier asking for 'updated bank details'. The domain had one letter changed. The accounts officer phoned the supplier on the number in their files, not the one in the email, and avoided losing a USD 48,000 payment.

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How a phishing attack unfolds



Each arrow is a place to break the chain: filtering, a user who pauses and reports, phishing-resistant MFA that will not work on the fake page, and alerts on new forwarding rules.

What users should do with a suspicious email

01 Pause
Do not click links, open attachments or reply.

→

02 Report
Use the report-phishing button or forward as instructed.

→

03 If you clicked
Tell IT immediately; do not feel embarrassed.

→

04 If you entered a password
Change it now from a trusted device and tell IT.

→

05 Delete
Remove it once reported.

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Endpoint protection layers

Antivirus / anti-malware

- Signature and behaviour detection
- Real-time scanning and scheduled scans
- Must be on and up to date

EDR

- Records endpoint activity for investigation
- Detects suspicious behaviour chains
- Can isolate a device remotely

Hardening controls

- Application allow-listing
- Blocking macros from the internet
- Host firewall, removal of local admin rights

Checking built-in protection on Windows

```
Windows PowerShell

Get-MpComputerStatus | Select-Object AMServiceEnabled, RealTimeProtectionEnabled,
AntivirusSignatureLastUpdated
Update-MpSignature
Start-MpScan -ScanType QuickScan
Get-MpThreatDetection | Select-Object -First 5 ThreatID, InitialDetectionTime, ActionSuccess
```

These cmdlets work with the operating system's built-in protection. Third-party tools provide their own consoles.

Malware response for support staff

1. **Isolate** Disconnect the device from the network (unplug Ethernet, turn off Wi-Fi) or isolate it through EDR. Do not power it off unless told to; memory evidence may matter.
2. **Report** Raise a security incident and note the time, user, symptoms and what was clicked.
3. **Assess scope** Check whether other devices or accounts show the same signs.
4. **Clean or rebuild** Follow the security team's direction; rebuilding from a known-good image is often safer than cleaning.
5. **Recover** Restore data from clean backups, reset credentials used on the device.
6. **Learn** Record the cause and improve controls or training.

WATCH OUT · DO NOT 'JUST RUN A SCAN AND CLOSE IT'

If ransomware or a remote-access trojan is suspected, a clean scan does not prove the device is safe, and the attacker may already have credentials. Isolate, report and follow the incident process.

SCENARIO · DECIDE · FILES WITH STRANGE EXTENSIONS

A user at a Dhaka architecture firm calls: files on her laptop and on the shared S: drive now end in `.lockd` and there is a text file demanding payment.

(A) Ask her to keep working while you investigate remotely

(B) Delete the ransom note and rename the files back

(C) Restart the laptop to clear the infection

(D) Tell her to disconnect from the network immediately, then report a security incident and alert the file server team

Circle your choice. Answer and feedback: module answer key.

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Key terms

Malware
Malicious software designed to damage, disrupt, steal from or gain unauthorised access to systems.

Trojan
Malware disguised as legitimate software to trick users into installing it.

Spear phishing
Phishing tailored to a specific person or organisation using researched details.

Application allow-listing
A control that lets only approved applications run on a device.

Ransomware
Malware that encrypts data or systems and demands payment for their release.

Phishing
Fraudulent messages that trick people into revealing information, clicking malicious links or opening attachments.

EDR
Endpoint detection and response: tools that monitor endpoint behaviour, detect threats and support investigation and isolation.

Key takeaways

- Most incidents start with phishing or malware, and support hears about them first.
- Teach users to pause, report and tell IT at once if they clicked or typed a password.
- Verify payment or bank-detail changes through a known contact, never the message itself.
- Layer antivirus, EDR and hardening such as allow-listing and macro blocking.
- Suspected ransomware: isolate first, report, then follow the incident process.

KNOWLEDGE CHECK

12. Which malware type spreads across networks without user action?
    - (A) Trojan
    - (B) Worm
    - (C) Keylogger
    - (D) Adware
13. A PC's fans run loudly at 100% CPU even when idle, with an unknown process using the GPU. What is the most likely malware?
    - (A) Ransomware
    - (B) Cryptominer
    - (C) Rootkit
    - (D) Phishing
14. What is the first action when ransomware is suspected on a laptop?
    - (A) Run a full scan and keep working
    - (B) Pay the ransom
    - (C) Delete the encrypted files
    - (D) Disconnect it from the network
15. Which is a strong sign of phishing?
    - (A) An email from a known colleague confirming the time of a meeting you both scheduled
    - (B) A message urging immediate bank detail changes from a lookalike domain
    - (C) A newsletter you subscribed to
    - (D) An internal IT announcement on the intranet
16. What does EDR add beyond traditional antivirus?
    - (A) Faster internet
    - (B) Free licences
    - (C) Automatic backups of every file before any malware can encrypt it
    - (D) Behaviour recording, investigation and remote isolation

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MODULE 7 · PROTECT · LESSON 4 · 28 MIN

Encryption, Data Protection and Incident Handling

Protecting data at rest and in transit, and what to do when things go wrong

- Explain symmetric and asymmetric encryption, hashing and certificates
- Enable and support full-disk encryption and recovery keys
- Classify and handle personal and sensitive data correctly
- Carry out the support role in a security incident

Encryption is how we keep data confidential on lost laptops, across public Wi-Fi and in the cloud. Data protection rules decide how personal data may be collected, stored and shared. And when something does go wrong, a clear incident process turns chaos into a coordinated response. Support specialists touch all three every week.

Encryption building blocks

Symmetric encryption

- One shared key encrypts and decrypts
- Very fast; used for disks and bulk data
- Problem: sharing the key safely

Asymmetric encryption

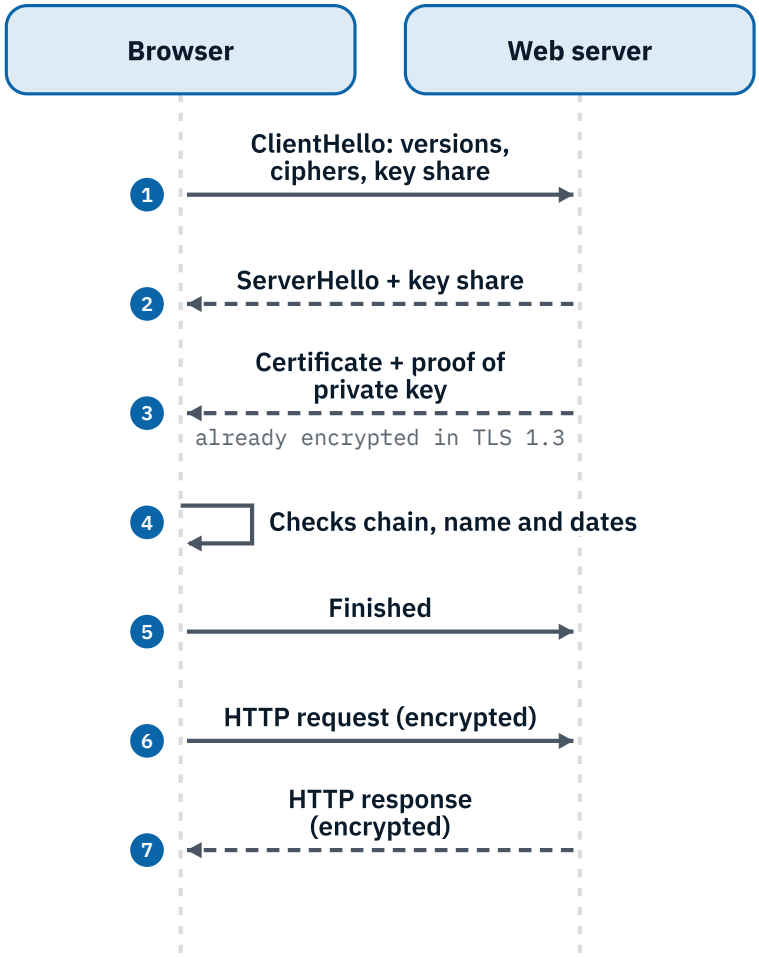
- Public key encrypts or verifies; private key decrypts or signs
- Solves key exchange and enables digital signatures
- Slower; used to set up sessions

Hashing

- One-way fingerprint of data
- Detects changes; stores passwords safely when salted
- Cannot be reversed to the original

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How HTTPS protects a session



Both sides derive the same session key from their key shares, so the key never crosses the network. A certificate warning means step 4 failed: wrong name, expired, untrusted issuer, or someone intercepting.

WATCH OUT · CERTIFICATE WARNINGS ARE NOT A NUISANCE

A browser certificate warning can mean an expired certificate, a wrong name, an untrusted issuer or an attacker intercepting traffic. Do not teach users to click through. Find the cause, especially on internal systems where an expired certificate is common and fixable.

Data at rest and in transit

STATE	EXAMPLE CONTROL	SUPPORT TASKS
At rest on laptops	Full-disk encryption with TPM	Enable, verify status, escrow recovery keys
At rest on servers and cloud	Volume or service-side encryption	Confirm enabled, control key access
On removable media	Encrypted USB drives or blocking USB storage	Issue approved drives, enforce policy
In transit	TLS (HTTPS), VPN, SSH, encrypted email	Fix certificate errors, keep protocols current

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Checking disk encryption

```
Windows PowerShell

manage-bde -status C:
manage-bde -protectors -get C:   # shows the recovery password ID to match against escrow
Get-BitLockerVolume | Select-Object MountPoint, VolumeStatus, ProtectionStatus,
EncryptionPercentage
# Linux (LUKS): sudo cryptsetup status /dev/mapper/cryptroot
# macOS: fdesetup status
```

Recovery keys must be stored in the directory or device management system before a user ever needs one. Never send recovery keys over chat or email without verifying identity.

Data classification example

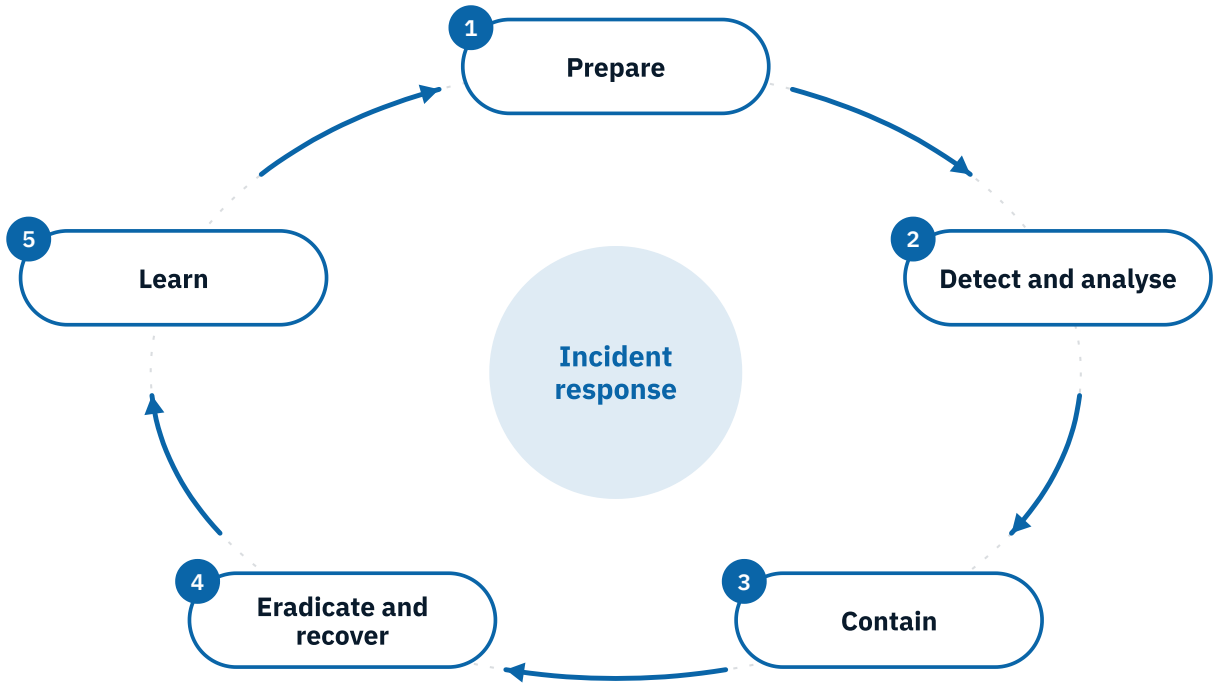
LEVEL	EXAMPLES	HANDLING
Public	Brochures, published reports	No restrictions
Internal	Policies, internal memos	Staff only; not on personal accounts
Confidential	Contracts, salaries, customer lists	Need-to-know access; encrypted when shared externally
Restricted	Patient records, national ID numbers, card data	Strict access, logging, encryption; follow legal requirements

Data protection habits for support staff

- Collect and keep only the personal data needed for the task.
- Never copy user data to personal devices or unapproved cloud storage.
- Wipe or destroy drives before disposal and record it.
- Report suspected data breaches immediately; many laws set short notification deadlines.
- Respect national data protection laws and sector rules for banks, telecoms and health.

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Incident handling phases



1. **Prepare** — Contacts, tools, playbooks and training in place before anything happens.
2. **Detect and analyse** — Recognise and confirm the incident; record facts and times.
3. **Contain** — Isolate devices, disable accounts, block indicators to stop the spread.
4. **Eradicate and recover** — Remove the cause, rebuild, restore from clean backups, monitor.
5. **Learn** — Review what happened and improve controls and processes.

Lessons learned feed back into preparation, so each incident should make the next one smaller. Support staff act mostly in Detect and Contain: notice, report fast and isolate.

TEMPLATE

Security incident first report

Capture the essential facts the security team needs in the first minutes.

TIME NOTICED	REPORTED BY	DEVICE / ACCOUNT	WHAT WAS OBSERVED	ACTIONS ALREADY TAKEN	DATA POSSIBLY INVOLVED
2026-05-14 09:12	*R. Ahmed (Accounts)*	*ACC-LT-22 / rahmed*	*Clicked invoice link, entered password on fake page*	*Password reset 09:20, sessions revoked, laptop isolated*	*Supplier payment records (confidential)*

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TIME NOTICED	REPORTED BY	DEVICE / ACCOUNT	WHAT WAS OBSERVED	ACTIONS ALREADY TAKEN	DATA POSSIBLY INVOLVED

CASE STUDY · A LOST LAPTOP AT THE AIRPORT

A sales director of a pharmaceutical company leaves his laptop in a taxi at Dhaka airport on Friday evening. He reports it on Saturday morning. The laptop holds customer price lists and a cached copy of the CRM. Device management shows it last checked in on Friday at 18:40.

1. What should the support specialist check and do first?

2. How does encryption change the risk?

3. What should be reported, and to whom?

Model answer: module answer key.

SCENARIO · DECIDE · A USER ASKS FOR A DISK ENCRYPTION RECOVERY KEY BY CHAT

A user messages the help desk chat: 'My laptop is asking for a recovery key, please send it here, I am in a hurry.' The chat account name matches a real employee.

(A) Send the key in the chat immediately

(B) Tell them to reinstall Windows

(C) Disable encryption remotely so the key is not needed

(D) Verify the user's identity through the approved method, confirm the device, then provide the key through a secure channel and record it

Circle your choice. Answer and feedback: module answer key.

Key terms

Symmetric encryption
Encryption that uses the same secret key to encrypt and decrypt data.

Asymmetric encryption
Encryption that uses a public and private key pair, enabling secure key exchange and digital signatures.

Hash
A fixed-length, one-way fingerprint of data used to check integrity or store passwords securely.

Digital certificate
A file that binds a public key to an identity, signed by a certificate authority.

Full-disk encryption
Encryption of an entire storage volume so data is unreadable without the key.

Data classification
Labelling data by sensitivity so it receives appropriate protection.

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Security incident
An event that compromises, or threatens, the confidentiality, integrity or availability of information or systems.

Key takeaways

- Symmetric keys encrypt bulk data, asymmetric keys solve key exchange and signatures, and hashes detect change.
- Encrypt data at rest and in transit, and escrow recovery keys before anyone needs them.
- Recovery keys get the same identity checks as a password reset.
- Classify data and handle it by its label; report suspected breaches immediately.
- Incidents run prepare, detect, contain, eradicate and recover, learn.

KNOWLEDGE CHECK

17. Which technique detects whether a file has been altered but cannot be reversed?
    - (A) Hashing
    - (B) Symmetric encryption
    - (C) Compression
    - (D) Asymmetric encryption

18. What makes a stolen, powered-off, fully encrypted laptop low risk?
    - (A) The screen saver
    - (B) The laptop's serial number being recorded in the asset register
    - (C) The Wi-Fi password
    - (D) Data cannot be read without the key

19. In incident handling, which phase includes isolating devices and disabling accounts?
    - (A) Prepare
    - (B) Learn
    - (C) Detect
    - (D) Contain

20. Patient records are best classified as:
    - (A) Public
    - (B) Internal
    - (C) Uncontrolled
    - (D) Restricted

21. Which command shows the encryption status of drive C: on Windows?
    - (A) manage-bde -status C:
    - (B) chkdsk C:
    - (C) ipconfig /all
    - (D) gpupdate /force

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 7 · PROTECT

Answer key

Knowledge checks

1. B Encryption keeps data confidential if the laptop is lost.

2. D 3 x 4 = 12.

3. D Separate, just-in-time admin use limits exposure.

4. D Following through secured doors is tailgating.

5. A Their privileges can be abused to take over accounts.

6. C FIDO2 and passkeys are bound to the genuine site.

7. A Length, breach checks and event-driven change are modern guidance.

8. D Unexpected prompts mean an attacker likely knows the password.

9. C Separating daily and admin use reduces exposure.

10. D Unique rotated passwords stop one stolen password unlocking every device.

11. B Scheduled changes push people towards weak, predictable passwords, so current guidance drops them and changes passwords only on compromise. Breach screening, long passphrases and MFA all stay.

12. B Worms self-propagate across networks.

13. B Heavy unexplained CPU/GPU use suggests cryptomining.

14. D Isolation stops the spread to shared drives.

15. B Urgent payment changes from lookalike domains are classic fraud.

16. D EDR records and responds to suspicious activity.

17. A Hashes are one-way fingerprints.

18. D Encryption keeps data unreadable without the key.

19. D Containment stops the spread.

20. D Health data needs the highest protection.

21. A manage-bde reports drive encryption status.

Scenarios & case studies

Scenario · An unknown visitor in the server room corridor
B is best. Correct. You verify before granting physical access, without confrontation.

Scenario · Midnight MFA prompts
A is best. Correct. The attacker had her password and now likely has a session; containment and investigation are urgent.

Scenario · Files with strange extensions
D is best. Correct. Isolation stops further encryption of shared data; rapid reporting starts containment and recovery.

Case study · A lost laptop at the airport
First: confirm in device management whether full-disk encryption was on and the recovery key escrowed; issue a remote lock and wipe command (it will run when the laptop connects); reset the director's password and revoke sessions; disable any certificates or VPN profiles tied to the device. **Encryption:** if the disk was encrypted and the laptop was locked or off, the data is very likely unreadable, which greatly lowers the risk of a reportable breach. If it was not encrypted, assume the data is exposed. **Reporting:** raise a security incident, inform the security or data protection lead, record the timeline and decisions, and let them decide on any legal or customer notification.

Scenario · A user asks for a disk encryption recovery key by chat
D is best. Correct. Recovery keys unlock all data on a device, so identity and device checks come first.

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08

MODULE 8 · AUTOMATE

Scripting, Documentation and Career Growth

PowerShell and Bash automation, documentation, asset management and the skills to grow your career

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Write safe PowerShell and Bash scripts with variables, conditions, loops and error handling
- Schedule and scale automation for onboarding, inventory and health checks
- Produce runbooks, knowledge articles and asset records that others can rely on
- Plan a career path and present job-ready evidence of your skills

LESSONS

1. Scripting Foundations with PowerShell · 32 min
2. Bash Scripting for Linux and macOS · 30 min
3. Automating Everyday Support Tasks · 30 min
4. Documentation, Knowledge and Asset Management · 26 min
5. Career Skills for IT Support Professionals · 24 min

MODULE 8 · AUTOMATE · LESSON 1 · 32 MIN

Scripting Foundations with PowerShell

Variables, objects, pipelines, conditions and loops for Windows support

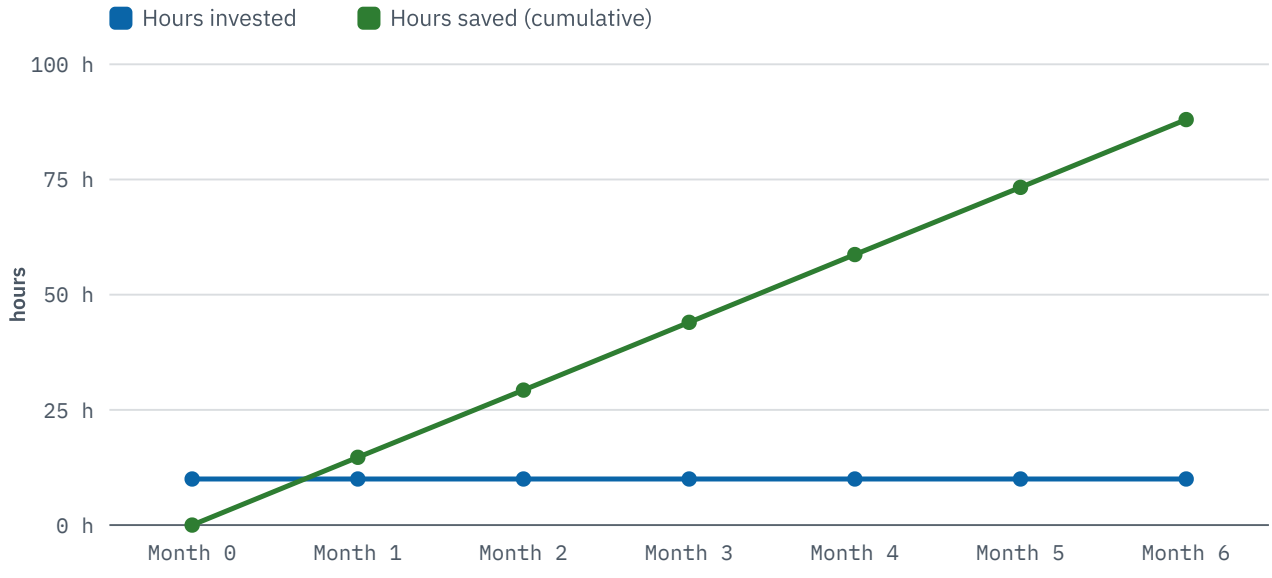
- Explain why support teams automate and what makes a task worth scripting
- Use PowerShell variables, objects, the pipeline, conditions and loops
- Write, run and safely test a small PowerShell script
- Apply execution policy and safe-scripting practices

Every support team has tasks it repeats dozens of times a week: checking disk space, creating accounts, collecting inventory, clearing caches. Scripting turns those tasks into a few seconds of reliable work, frees time for real problems and removes typing mistakes. **PowerShell** is the standard automation shell on Windows: Windows PowerShell 5.1 ships with the OS, and the newer, cross-platform PowerShell 7 also runs on Linux and macOS.

Is it worth automating?

MEASURE	FORMULA	WHAT IT TELLS YOU
Time saved per month	`saved = (manual minutes - automated minutes) x runs per month`	A 12-minute task done 80 times a month, reduced to 1 minute, saves 11 x 80 = 880 minutes (about 14.7 hours).
Payback period	`payback (months) = hours to build and test / hours saved per month`	If the script takes 10 hours to build and test, it pays back in under a month.

When the script pays back



Uses the example above: 10 hours to build and test, about 14.7 hours saved each month. The lines cross within the first month; after that every run is pure gain.

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PowerShell building blocks

CONCEPT	EXAMPLE	MEANING
Cmdlet (Verb-Noun)	Get-Service	A command; verbs include Get, Set, New, Remove, Start, Stop
Variable	$limit = 15	Stores a value; names start with $
Object and property	(Get-Date).DayOfWeek	Output is objects with properties, not plain text
Pipeline	Get-Process | Sort-Object CPU	Passes objects from one cmdlet to the next
Filter	Where-Object { $_.Status -eq 'Stopped' }	$_ is the current object in the pipeline
Comparison	-eq, -ne, -gt, -lt, -like	Operators for conditions
Help	Get-Help Get-Service -Examples	Built-in documentation with examples

Exploring with the pipeline

```
Windows PowerShell
```

```
Get-Command -Verb Get -Noun *Service*
Get-Service | Where-Object { $_.Status -eq 'Stopped' -and $_.StartType -eq 'Automatic' }
Get-Process | Sort-Object WS -Descending | Select-Object -First 5 Name, @{n='MemMB';e=
{[math]::Round($_.WS/1MB)}}
Get-ChildItem C:\Users\*\Downloads -Recurse -File | Measure-Object Length -Sum
Get-Service | Export-Csv C:\Temp\services.csv -NoTypeInformation
```

A first script: Check-DiskSpace.ps1

```
Windows PowerShell
```

```
# Check-DiskSpace.ps1 - warn when any fixed drive is below a free-space threshold
param([int]$Threshold = 15)
$drives = Get-Volume | Where-Object { $_.DriveLetter -and $_.DriveType -eq 'Fixed' -and $_.Size -gt
0 }
foreach ($d in $drives) {
    $pctFree = [math]::Round(($d.SizeRemaining / $d.Size) * 100, 1)
    if ($pctFree -lt $Threshold) {
        Write-Warning "Drive $($d.DriveLetter): only $pctFree% free"
    } else {
        Write-Output "Drive $($d.DriveLetter): $pctFree% free"
    }
}
# Run it:  .\Check-DiskSpace.ps1 -Threshold 20
```

The param block makes the threshold adjustable without editing the script. The script only reads information, so it is safe to run anywhere.

> **DEFINITION · EXECUTION POLICY**
> A Windows setting that controls whether scripts may run, for example only signed scripts or locally created ones. It is a safety guard against accidentally running scripts, not a security boundary. Organisations usually set it centrally; do not weaken it on a user's PC to run something you downloaded.

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Safe scripting habits

- Read every line of a script before running it, especially from the internet.
- Treat AI-generated scripts like forum code: understand and test every line before it runs.
- Use `-WhatIf` on cmdlets that change things to preview the effect.
- Test in a VM or on a test account before production.
- Start with read-only commands; add changes only when the output is correct.
- Never put passwords in scripts; use secure credential stores.
- Add comments and a header: purpose, author, date, version.

Preview before you change

```
Windows PowerShell
```

```
Get-ChildItem C:\Temp\*.log | Where-Object LastWriteTime -lt (Get-Date).AddDays(-30) | Remove-Item
-WhatIf
# Output: What if: Performing the operation "Remove File" on target "C:\Temp\old.log".
# When the list is exactly what you expect, run it again without -WhatIf
```

SCENARIO · DECIDE · A SCRIPT FROM A FORUM

A colleague finds a PowerShell one-liner on a forum that 'speeds up all PCs' and wants to run it on every computer in the office tonight. It downloads and runs a file from an unknown website.

- (A) Run it tonight on all PCs since it is popular
- (B) Decline to run it; if the goal is valid, write or source a reviewed script, test it in a VM and pilot it
- (C) Change execution policy to Unrestricted so it runs smoothly
- (D) Run it on the manager's PC first

Circle your choice. Answer and feedback: module answer key.

Key terms

Script
A text file of commands that runs as a program to automate a task.

Variable
A named storage location for a value, written with a $ prefix in PowerShell and Bash.

Loop
A structure that repeats commands, for example for each item in a list.

Execution policy
A PowerShell setting that controls which scripts are allowed to run on a system.

Cmdlet
A PowerShell command written in Verb-Noun form, such as Get-Service.

Pipeline
Passing the output of one command as input to the next, using the | symbol.

-WhatIf
A PowerShell parameter that shows what a command would do without making changes.

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Key takeaways

- Automate tasks that are frequent, repetitive and error-prone; check the payback.
- PowerShell passes objects through the pipeline; $_ is the current object.
- Preview changes with -WhatIf and start with read-only commands.
- Parameters make scripts reusable without editing them.
- Execution policy prevents accidents but is not a security boundary.

KNOWLEDGE CHECK

1. In a PowerShell pipeline, what does $_ represent?
   - (A) The last error
   - (B) The script's file name
   - (C) The current object being processed
   - (D) An environment variable
2. Which parameter previews the effect of Remove-Item without deleting anything?
   - (A) -Force
   - (B) -Recurse
   - (C) -WhatIf
   - (D) -Verbose
3. A task takes 6 minutes manually and runs 100 times a month. A script reduces it to 1 minute. How much time is saved monthly?
   - (A) 100 minutes
   - (B) 500 minutes
   - (C) 600 minutes
   - (D) 60 minutes
4. Which statement about execution policy is correct?
   - (A) It prevents accidental script runs and is usually set centrally
   - (B) It is a strong security boundary that stops malware from running any script at all
   - (C) It encrypts scripts
   - (D) It is only for Linux
5. Why use a param block in a script?
   - (A) To hide the code
   - (B) To make the script run faster by loading its variables before the main code
   - (C) To pass values, such as a threshold, without editing the script
   - (D) To sign the script

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · AUTOMATE · LESSON 2 · 30 MIN

Bash Scripting for Linux and macOS

Shebangs, variables, tests, loops, exit codes and text tools

- Write and run a Bash script with a shebang, variables and arguments
- Use conditions, loops and exit codes
- Combine text tools such as grep, awk, sort and cut
- Make scripts safer with strict modes, quoting and logging

Bash is the default automation language of Linux servers, cloud instances, network appliances and many developer workstations (macOS uses zsh by default, which runs most Bash scripts with care). Bash scripts glue together small tools that each do one job well, which makes them perfect for health checks, log searches, backups and clean-ups.

Bash building blocks

CONCEPT	EXAMPLE	NOTES
Shebang	#!/usr/bin/env bash	First line; chooses the interpreter
Variable	THRESHOLD=85	No spaces around =; use "$THRESHOLD"
Arguments	$1, $2, $#, $@	First argument, second, count, all
Command substitution	today=$(date +%F)	Captures a command's output
Test	[-f /etc/hosts], ["$a" -gt 5]	File checks and numeric comparisons
Exit code	$?	0 means success; anything else is failure
Make executable	chmod +x script.sh	Then run with ./script.sh

A health-check script: disk-check.sh

Linux host

```
#!/usr/bin/env bash
# disk-check.sh - warn when any file system is above a usage threshold
set -euo pipefail
THRESHOLD="${1:-85}"          # first argument, default 85
df -P -x tmpfs -x devtmpfs | tail -n +2 | while read -r fs size used avail pct mount; do
  usage=$(echo "$pct" | tr -d '%')
  if [ "$usage" -ge "$THRESHOLD" ]; then
    echo "WARNING: $mount is ${usage}% full"
  fi
done
# Run it:  chmod +x disk-check.sh && ./disk-check.sh 90
```

set -euo pipefail stops the script on errors, unset variables and failed pipeline steps, which prevents silent damage.

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Loops and exit codes

Linux host

```
for host in web01 web02 db01; do
  if ping -c 1 -W 2 "$host" > /dev/null 2>&1; then
    echo "$host is up"
  else
    echo "$host is DOWN"
  fi
done
```

Text tools for log analysis

Linux host

```
grep 'Failed password' /var/log/auth.log | wc -l                 # how many failed SSH sign-ins
grep 'Failed password' /var/log/auth.log | awk '{print $(NF-3)}' | sort | uniq -c | sort -rn | head
cut -d: -f1,7 /etc/passwd | grep -v nologin                      # users with login shells
find /var/log -name '*.gz' -mtime +30 -print                     # list old compressed logs first
```

The second line counts failed sign-ins per source address, a quick way to spot brute-force attempts. Always list with -print before you add -delete to a find command.

WATCH OUT · QUOTE YOUR VARIABLES

Unquoted variables split on spaces and expand wildcards. `rm -rf $DIR/` with an empty DIR becomes `rm -rf /` . Always write "$VAR", use `set -u` so unset variables stop the script, and test destructive commands with echo first.

PowerShell and Bash side by side

PowerShell

- Passes objects with properties
- Verb-Noun cmdlets, consistent help
- Strong for Windows, directory and cloud admin

Bash

- Passes text between small tools
- Short commands, huge ecosystem
- Strong for Linux servers, containers and network devices

Scheduling a script with cron

01 Write and test
Run the script by hand until the output is correct.

→

02 Use absolute paths
cron runs with a minimal environment.

→

03 Edit the crontab
crontab -e for the right user.

→

04 Add the schedule
30 6 * * * /usr/local/bin/disk-check.sh 90 >> /var/log/disk-check.log 2>&1

→

05 Check the log
Confirm it ran at the expected time.

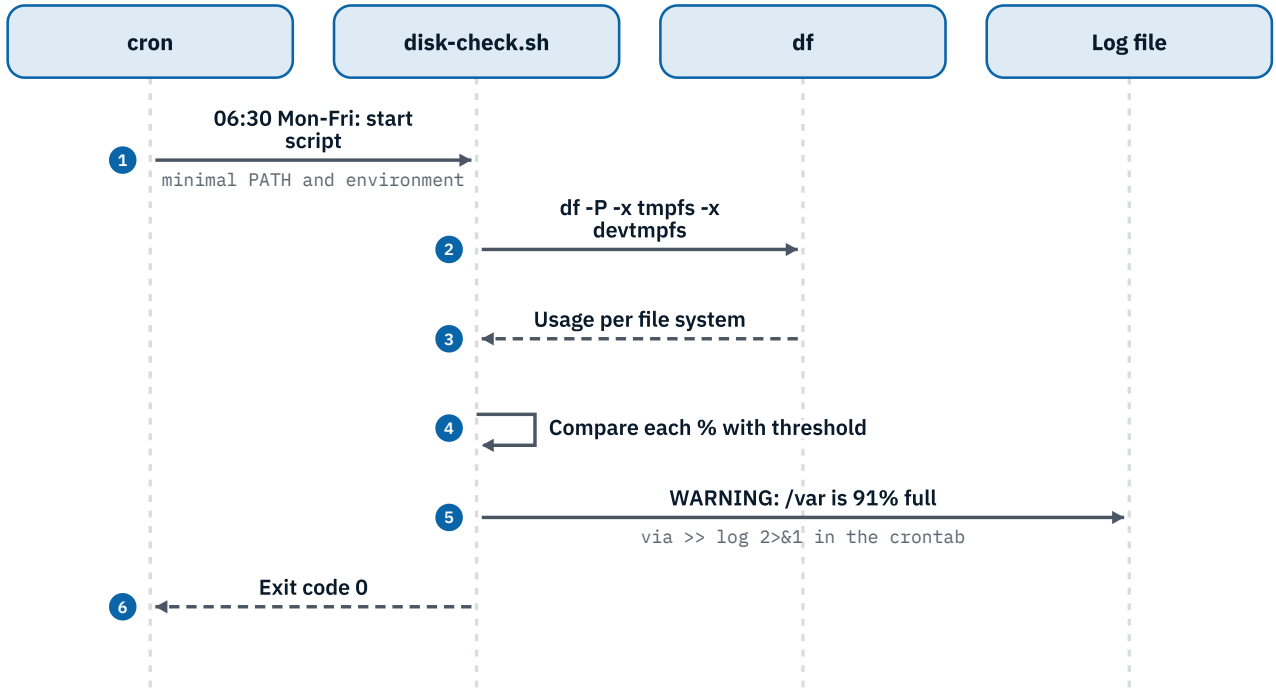
Reading a cron schedule

FIELD	RANGE	EXAMPLE 30 6 * * 1-5
Minute	0-59	30
Hour	0-23	6 (06:30)

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FIELD	RANGE	EXAMPLE 30 6 * * 1-5
Day of month	1-31	* every day
Month	1-12	* every month
Day of week	0-7 (0 and 7 are Sunday)	1-5 Monday to Friday

What happens when cron runs the health check



cron starts the script with a minimal environment, which is why absolute paths matter. Redirecting output to a log file is the only way to see what an unattended run actually did.

SCENARIO · DECIDE · A CLEAN-UP SCRIPT DELETED TOO MUCH

A junior admin wrote a Bash script to delete old files in `$LOGDIR`. On one server the variable was not set and the script removed files from the wrong place before someone stopped it.

(A) Ban scripting for junior staff

(B) Restore from backup, then fix the script with set -u, quoting, a check that the directory exists and a dry-run mode, and review scripts before deployment

(C) Run the script as root so permissions do not block it

(D) Delete the script and do the clean-up manually forever

Circle your choice. Answer and feedback: module answer key.

Key terms

Shebang
The first line of a script, such as #!/usr/bin/env bash, which tells the system which interpreter to use.

Exit code
A number a command returns when it finishes; 0 means success and other values signal errors.

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Command substitution
Capturing the output of a command into a variable, written $(command) in Bash.

grep
A tool that searches text for lines matching a pattern.

set -euo pipefail
Bash options that stop a script on errors, unset variables and failed pipeline commands.

cron
The Linux and macOS scheduler that runs commands at defined times using crontab entries.

awk
A text-processing tool that splits lines into fields and prints or calculates on them.

Key takeaways

- Start scripts with a shebang and set -euo pipefail.
- Always quote variables and check that target directories exist before deleting.
- Exit code 0 means success; test $? or use if to react to failures.
- grep, awk, sort and uniq turn logs into answers quickly.
- Schedule with cron using absolute paths, and log every run.

KNOWLEDGE CHECK

6. What does the cron entry `0 2 * * 0` mean?
   - (A) Every day at 02:00
   - (B) At 00:02 every day
   - (C) Every Sunday at 02:00
   - (D) Every 2 minutes on Sunday

7. What exit code indicates success in Bash?
   - (A) 1
   - (B) 0
   - (C) -1
   - (D) 255

8. Why use `set -u` in a Bash script?
   - (A) To speed it up
   - (B) To run as root
   - (C) To enable coloured output so that warnings stand out in the terminal
   - (D) To stop the script if an unset variable is used

9. Which command counts the lines containing 'error' in app.log?
   - (A) grep -c error app.log
   - (B) cut -d error app.log
   - (C) awk error app.log
   - (D) sort error app.log

10. What should you do before adding -delete to a find command?
    - (A) Run it as root
    - (B) Run it with -print to see which files match
    - (C) Add sudo twice
    - (D) Nothing, because find always asks for confirmation before deleting files

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · AUTOMATE · LESSON 3 · 30 MIN

Automating Everyday Support Tasks

Scheduled jobs, onboarding, inventory, logging and version control

- Schedule scripts on Windows and Linux
- Automate onboarding and inventory collection from structured data
- Make automation safe: idempotence, logging, error handling and least privilege
- Keep scripts in version control and share them with the team

Individual scripts save minutes. Automation that runs on a schedule, reads its input from a file and reports what it did saves whole days and makes results consistent. This lesson turns the scripting basics into dependable tools: scheduled health checks, onboarding from a spreadsheet and automatic hardware inventory.

Scheduling a PowerShell script

Windows PowerShell

```
$action    = New-ScheduledTaskAction -Execute 'powershell.exe' -Argument '-NoProfile -File
C:\Scripts\Check-DiskSpace.ps1 -Threshold 15'
$trigger   = New-ScheduledTaskTrigger -Daily -At 7am
$principal = New-ScheduledTaskPrincipal -UserId 'SYSTEM' -LogonType ServiceAccount -RunLevel
Highest
Register-ScheduledTask -TaskName 'Daily disk check' -Action $action -Trigger $trigger -Principal
$principal
Get-ScheduledTaskInfo -TaskName 'Daily disk check'      # last run time and result
```

Run scheduled tasks under the least-privileged account that can do the job. SYSTEM is used here because the check reads all local volumes.

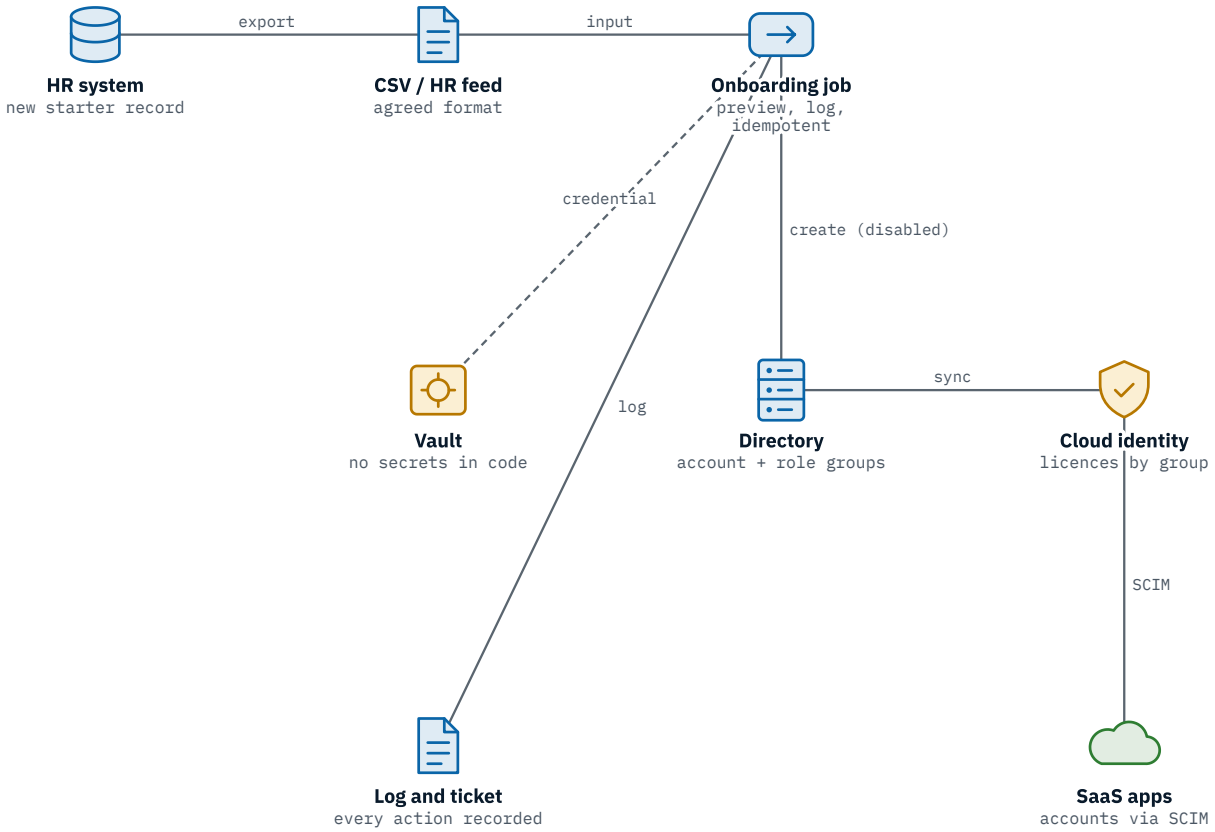
Onboarding from a CSV file (preview first)

Windows PowerShell

```
# newstarters.csv columns: Name,Username,Department,Title
Import-Csv .\newstarters.csv | ForEach-Object {
    New-ADUser -Name $_.Name -SamAccountName $_.Username -Department $_.Department -Title $_.Title
-Enabled $false -WhatIf
}
# Remove -WhatIf when the preview is correct; set passwords and enable accounts in a separate,
logged step
```


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Onboarding automation from HR data to SaaS accounts



HR data drives everything: the job reads an agreed format, takes credentials from a vault, creates disabled accounts in the directory and logs every action. Licences and SaaS accounts then follow from group membership.

Collecting hardware inventory

```Windows PowerShell
$cs   = Get-CimInstance Win32_ComputerSystem
$bios = Get-CimInstance Win32_BIOS
$os   = Get-CimInstance Win32_OperatingSystem
[pscustomobject]@{
    Computer = $env:COMPUTERNAME; Model = $cs.Model; Serial = $bios.SerialNumber
    RAMGB = [math]::Round($cs.TotalPhysicalMemory / 1GB); OS = $os.Caption; Build = $os.BuildNumber
    Collected = (Get-Date -Format 'yyyy-MM-dd')
} | Export-Csv -Path "\\fs01\inventory$\$env:COMPUTERNAME.csv" -NoTypeInformation
```

Linux inventory one-liner set

```Linux host
hostnamectl | grep -E 'Static hostname|Operating System|Kernel'
sudo dmidecode -s system-serial-number
lscpu | grep 'Model name'; free -g | awk '/Mem:/ {print "RAM GB: " $2}'
ip -br addr | grep -v '^lo'
```


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Principles of safe automation

ID

Idempotent

Running the script twice gives the same result as running it once; check before you create or change.

LOG

Logged

Write what was done, when and by which account to a log file or ticket.

ERR

Error-aware

Catch errors, stop safely and report failures instead of carrying on.

LP

Least privilege

Run with the minimum rights; store credentials in a secure vault.

REV

Reviewed

A second person reviews changes; scripts live in version control.

Error handling and logging in PowerShell

Windows PowerShell

```
$log = 'C:\Logs\cleanup.log'
try {
    Remove-Item 'C:\Temp\Cache\*' -Recurse -ErrorAction Stop
    Add-Content $log "$(Get-Date -Format s) OK cache cleared by $env:USERNAME"
} catch {
    Add-Content $log "$(Get-Date -Format s) FAILED $($_.Exception.Message)"
    exit 1
}
```

Version control basics with git

Linux host

```
git init support-scripts && cd support-scripts
git add disk-check.sh Check-DiskSpace.ps1
git commit -m 'Add disk space checks for Linux and Windows'
git log --oneline
git diff                # see what changed before the next commit
```

Version control records who changed what and why, and lets you roll back a bad change. Never commit passwords or keys.

TEMPLATE **Script register**

Keep a catalogue of team scripts so anyone can find, trust and maintain them.

SCRIPT	PURPOSE	RUNS ON / SCHEDULE	ACCOUNT USED	OWNER	LAST REVIEWED	REPOSITORY PATH
Check-DiskSpace.ps1	*Warn when fixed drives fall below 15% free*	*All Windows servers, daily 07:00*	*SYSTEM*	*Service desk lead*	*2026-06-01*	*support-scripts/windows/*

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SCRIPT	PURPOSE	RUNS ON / SCHEDULE	ACCOUNT USED	OWNER	LAST REVIEWED	REPOSITORY PATH

SCENARIO · DECIDE · AUTOMATING NEW-STARTER ACCOUNTS

An NGO in Khulna onboards 30 field staff every quarter. HR sends a spreadsheet, and two technicians spend a full day creating accounts by hand, with frequent typos in names and departments.

(A) Keep doing it by hand but double-check everything

(B) Agree a fixed CSV format with HR, write a reviewed script that previews with -WhatIf, logs every action and assigns standard groups by role

(C) Give HR domain admin rights so they can create accounts themselves

(D) Create one shared account for all field staff

Circle your choice. Answer and feedback: module answer key.

Key terms

Scheduled task
A Windows job that runs a program or script automatically at set times or events.

CSV
Comma-separated values: a simple text format for tabular data used as input and output for scripts.

Version control
A system, such as git, that records changes to files over time with author and reason.

Inventory script
A script that collects hardware and software details from devices for asset records.

Idempotent
Describes an operation that produces the same result whether it runs once or many times.

try/catch
A structure that runs code and handles any errors in a controlled way.

Commit
A saved snapshot of changes in a version control repository, with a message describing it.

Key takeaways

- Scheduled, logged automation saves days, not minutes.
- Drive bulk changes from structured input such as an agreed CSV, and preview first.
- Safe automation is idempotent, logged, error-aware, least-privileged and reviewed.
- Keep credentials in a vault, never in scripts, CSV files or commits.
- Store scripts in version control with a register of owners and schedules.

KNOWLEDGE CHECK

11. What does idempotent mean for an onboarding script?

(A) It runs only once

(B) It runs very fast

(C) Running it again does not create duplicates or extra changes

(D) It needs no testing because it always produces exactly the same output

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12. Where should a script's credentials be kept?

(A) In a secure credential store or vault
(B) In the script as plain text
(C) In the commit message
(D) In the CSV input file, so HR can update passwords with each new starter

13. Which git command records staged changes with a message?

(A) git log
(B) git diff
(C) git commit -m 'message'
(D) git init

14. Why should a scheduled task run with a least-privileged account?

(A) It runs faster
(B) If the script is misused or buggy, the damage is limited
(C) It avoids logging
(D) It is required for scripts that read CSV files from network shares

15. What is the benefit of try/catch with logging?

(A) Errors are hidden
(B) Failures are caught and recorded instead of silently ignored
(C) The script can no longer fail, because every error is automatically retried
(D) It replaces testing

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · AUTOMATE · LESSON 4 · 26 MIN

Documentation, Knowledge and Asset Management

Writing things down so the organisation, not just one person, knows how IT works

- Choose the right type of IT documentation for each need
- Write runbooks and knowledge articles that others can follow
- Run an asset lifecycle from purchase to disposal
- Track licences, warranties and total cost of ownership

If the only person who knows how the VPN is configured goes on leave, the organisation has a problem. Good documentation and asset records turn individual knowledge into organisational knowledge. They speed up troubleshooting, make audits painless and let new staff become useful in days rather than months.

Types of IT documentation

TYPE	PURPOSE	EXAMPLE
Network diagram	Show how things connect	Office LAN, Wi-Fi, firewall, ISP links
Runbook / procedure	Exact steps for a routine or emergency task	Restore the file server from backup
Knowledge article	Help users or technicians fix a known issue	'Printer shows offline' fix
Configuration record	Current settings of a system	DHCP scopes, firewall rules, Wi-Fi SSIDs
Policy / standard	Rules everyone must follow	Password and acceptable use policy
Change log	What changed, when, by whom and why	Firewall rule added for new vendor

Writing documentation people actually use

- Start with the purpose and who the document is for.
- Use numbered steps, one action per step, with expected results.
- Include screenshots or command output where helpful.
- State prerequisites: access, tools, maintenance windows.
- Add an owner and a review date; outdated documents are dangerous.
- Store documents in one searchable place, never on personal drives.

TEMPLATE **Runbook**

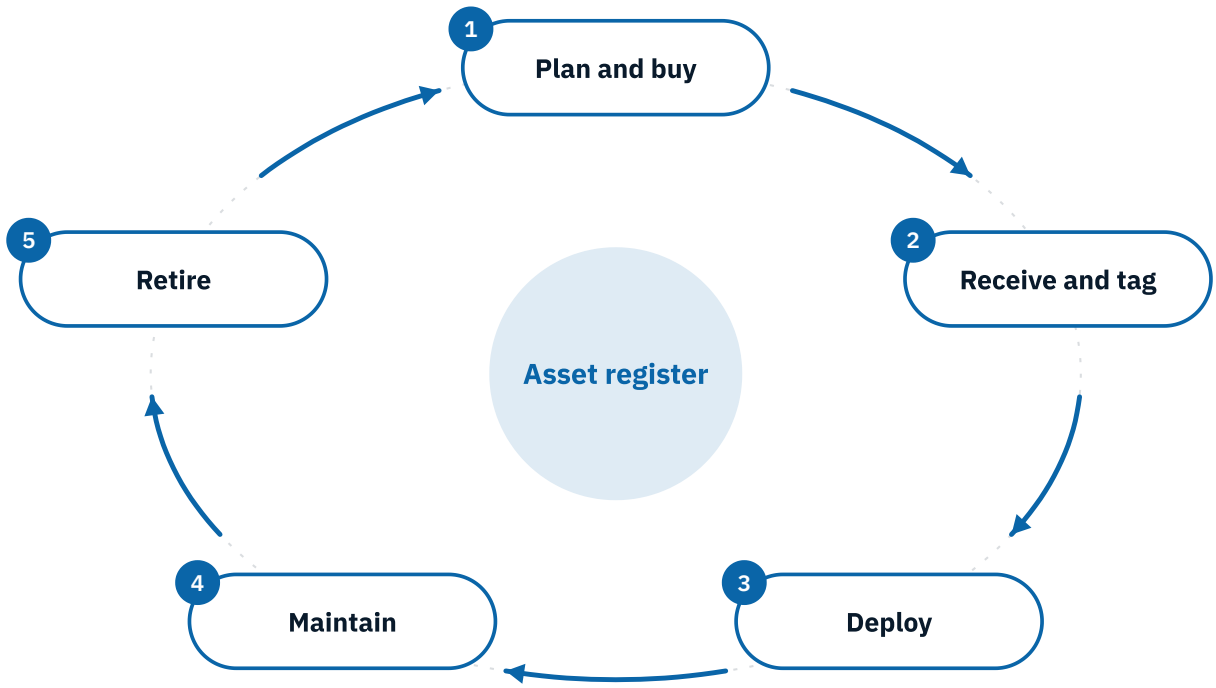
A step-by-step procedure that any trained technician can follow under pressure.

STEP	ACTION	EXPECTED RESULT	IF IT FAILS
1	*Confirm the latest successful backup of FS01 in the backup console*	*Job shows 'Success' within the last 24 hours*	*Escalate to the backup owner before continuing*
2	*Restore the requested folder to D:\Restore\<ticket number>*	*Files appear with original timestamps*	*Check restore job log; retry from previous backup*

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STEP	ACTION	EXPECTED RESULT	IF IT FAILS

The IT asset lifecycle



1. **Plan and buy** — Standard models, approved suppliers, budget.
2. **Receive and tag** — Asset tag, serial number, warranty and owner recorded.
3. **Deploy** — Imaged, enrolled in management, assigned to a user.
4. **Maintain** — Patches, repairs, moves and upgrades recorded.
5. **Retire** — Data wiped, record updated, disposal or reuse.

Retirement feeds the next purchase plan, so lifecycle dates in the register drive the refresh budget. Wiping data at Retire is never optional.

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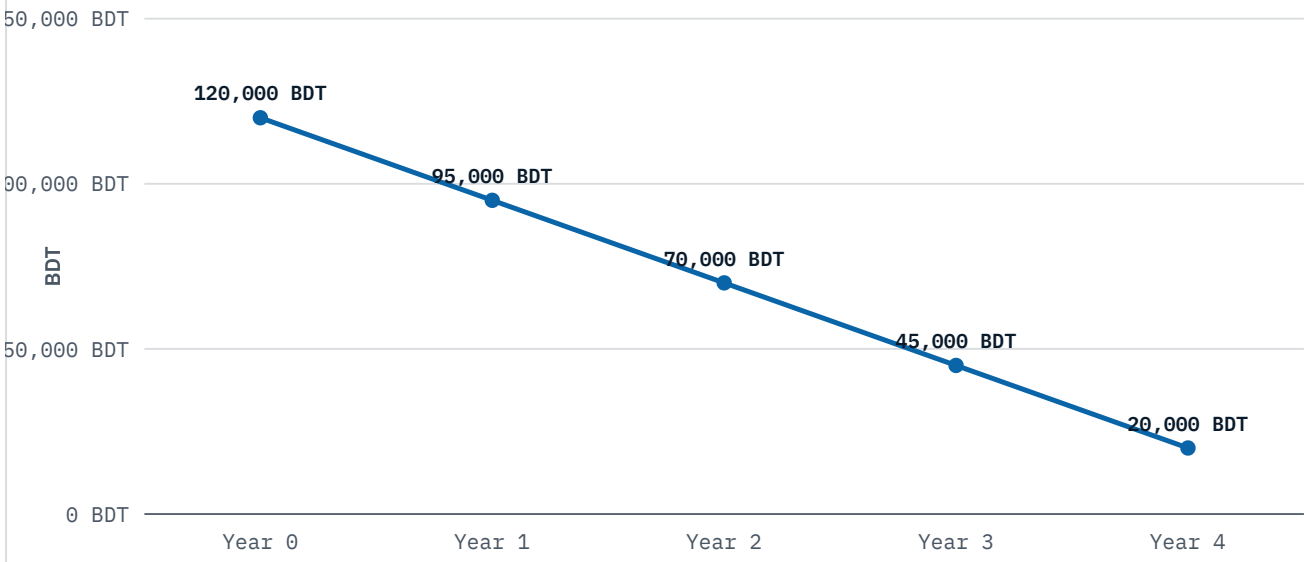
What an asset record should hold

FIELD	EXAMPLE
Asset tag	ZZ-LT-0412
Type, make and model	Laptop, 14-inch business model
Serial number	5CG3281XYZ
Assigned user and location	S. Islam, Dhaka HQ, 4th floor
Purchase date and cost	2025-08-14, BDT 115,000
Warranty end	2028-08-13
Status	In use / in repair / in stock / retired

Lifecycle and cost maths

MEASURE	FORMULA	WHAT IT TELLS YOU
Straight-line depreciation	`annual depreciation = (cost - residual value) / useful life in years`	A BDT 120,000 laptop with BDT 20,000 residual over 4 years depreciates BDT 25,000 a year.
Total cost of ownership	`TCO = purchase + (annual support + licences + energy) x years + disposal`	TCO, not purchase price, is the fair way to compare devices or services.
Refresh planning	`devices due = assets with (purchase date + useful life) within the next 12 months`	Lets you budget replacements a year ahead.

Book value of a BDT 120,000 laptop (straight-line)



Straight-line depreciation removes the same BDT 25,000 each year until the residual value is reached. Book value is an accounting figure: a device can stay useful after it, or become unsafe before it if OS support ends.

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PRO TIP · LINK ASSETS, TICKETS AND USERS

When every ticket is linked to an asset record, patterns jump out: one laptop model with repeated keyboard failures, or a printer generating 40% of all print tickets. That evidence wins budget arguments.

CASE STUDY · THE UNDOCUMENTED NETWORK

A private school in Sylhet hired a new IT officer after the previous one left without notice. There is no network diagram, the Wi-Fi admin password is unknown, licences for the lab software expire 'sometime this year' and 30 laptops have no asset records. Parents complain that the online learning portal is slow.

1. What should the new IT officer document first, and why?

2. How should the asset and licence situation be fixed?

3. How can documentation stay current after this effort?

Model answer: module answer key.

SCENARIO · DECIDE · A LAPTOP RETURNED BY A LEAVER

A leaver returns a laptop. A manager wants to give it straight to a new employee starting tomorrow 'to save time'.

(A) Hand it over as it is

(B) Delete the leaver's desktop files only

(C) Check its condition, back up any required business data, wipe and reimage it, update the asset record with the new user, then issue it

(D) Put it in a cupboard without updating records

Circle your choice. Answer and feedback: module answer key.

Key terms

Runbook
A documented, step-by-step procedure for carrying out a routine or emergency operational task.

Asset tag
A unique label attached to a device and used to identify it in the asset register.

CMDB
Configuration management database: a record of IT components and the relationships between them.

Depreciation
The reduction in an asset's book value over its useful life.

Network diagram
A drawing that shows network devices, connections, addressing and links.

Asset register
A record of an organisation's IT assets with details such as owner, location, cost and status.

Total cost of ownership
The full cost of an asset over its life, including purchase, support, licences, energy and disposal.

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Key takeaways

- Documentation turns one person's knowledge into the organisation's knowledge.
- Write runbooks with numbered steps, expected results and what to do if a step fails.
- Track every asset from purchase to disposal and wipe data before reuse.
- Compare options by total cost of ownership, not purchase price.
- Give every document an owner and a review date.

KNOWLEDGE CHECK

16. Which document gives exact steps to restore a server from backup under pressure?

(A) Policy
(B) Runbook
(C) Network diagram
(D) Change log

17. A BDT 100,000 printer with BDT 10,000 residual value over 3 years depreciates how much each year?

(A) BDT 33,333
(B) BDT 36,667
(C) BDT 10,000
(D) BDT 30,000

18. What must happen before a returned laptop is reissued?

(A) Nothing
(B) Only change the wallpaper
(C) Delete the browser history and the previous user's desktop shortcuts only
(D) Back up needed data, wipe and reimage, update the asset record

19. Why should each document have an owner and review date?

(A) So it is kept accurate and someone is accountable for updates
(B) To make it longer
(C) For printing
(D) To restrict who can read it, so that only the owner sees the latest version

20. Linking tickets to asset records helps most with:

(A) Choosing wallpaper
(B) Speeding up Wi-Fi by matching each laptop to its nearest access point
(C) Spotting models or devices with repeated failures
(D) Encrypting disks

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MODULE 8 · AUTOMATE · LESSON 5 · 24 MIN

Career Skills for IT Support Professionals

From first job to a growing career in infrastructure, cloud or security

- Build a CV, portfolio and home lab that prove job-ready skills
- Answer technical and behavioural interview questions with structure
- Map career paths from IT support to specialist roles
- Create a personal learning plan and act ethically as a trusted professional

IT support is one of the best entry points into technology. It exposes you to every part of IT, from hardware to cloud, and teaches the communication skills specialists often lack. The people who grow fastest treat every ticket as a lesson, document what they learn and keep building skills deliberately.

Proving your skills

CV

- One or two pages, clear headings
- Skills section matched to the job advert
- Achievements with numbers: 'cut onboarding time from 4 h to 1 h with a script'
- Certifications and projects, not just duties

Portfolio

- Home-lab write-ups with screenshots
- Scripts in a public repository (no secrets)
- Sample knowledge-base articles
- Your ZITSS capstone pack

Home lab

- Old PC or laptop with a type 2 hypervisor
- Windows and Linux VMs, a domain, DHCP and DNS
- A free cloud tier for SaaS and IaaS practice
- Break things on purpose and document the fix

STAR: answering behavioural questions

S
Situation
Set the scene briefly: where, when, who.

T
Task
What you were responsible for.

A
Action
What you did, step by step, using 'I' not 'we'.

R
Result
What changed, ideally with numbers, and what you learned.

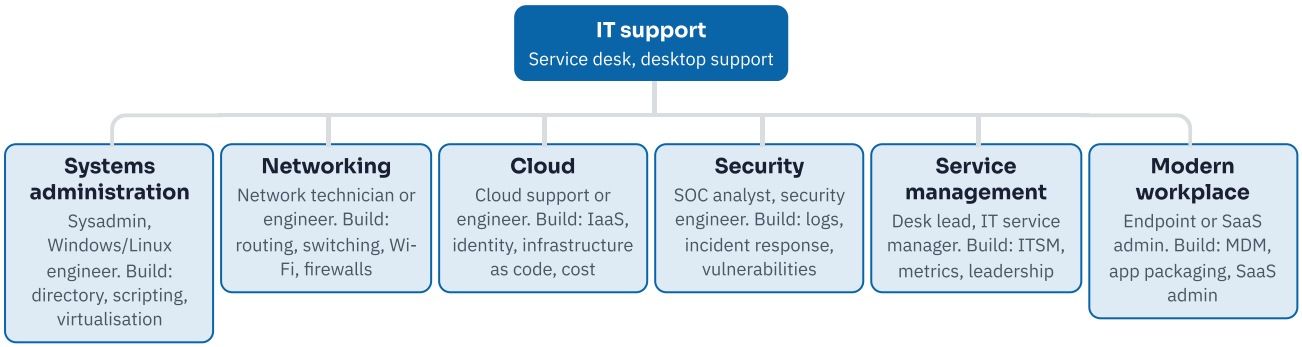
Common interview questions and what they test

QUESTION	WHAT THE INTERVIEWER WANTS TO SEE
'A user cannot reach the internet. Walk me through it.'	A layered method: link, IP, gateway, DNS, service
'Tell me about a time you handled an angry user.'	Empathy, calm, ownership, a positive result (use STAR)
'What is the difference between DNS and DHCP?'	Clear fundamentals explained simply
'How do you prioritise several urgent tickets?'	Impact and urgency, SLA awareness, communication
'What have you automated?'	Initiative and practical scripting

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QUESTION	WHAT THE INTERVIEWER WANTS TO SEE
'What would you do if you did not know the answer?'	Honesty, research skills and escalation judgement

Where IT support can lead



Every branch starts from skills you use daily in support. Pick one direction, then build lab evidence for the skills listed under it.

INSIGHT · CERTIFICATES OPEN DOORS, SKILLS KEEP YOU IN THE ROOM

Vendor and independent certifications help CVs pass screening, but interviewers test whether you can actually troubleshoot and communicate. Pair every certificate with a lab project or work example that shows you can apply it.

Professional ethics for IT support

- Access only what a task requires, even when your rights allow more.
- Keep users' data and conversations confidential.
- Never install pirated software or bypass licensing.
- Use AI tools within policy: no confidential data in unapproved tools, and verify every output.
- Report security issues and mistakes, including your own, quickly.
- Be honest about what you do not know and about timelines.
- Treat every user with the same respect, whatever their role.

TEMPLATE 90-day learning plan

Turn career goals into weekly, measurable learning actions.

GOAL	SKILL OR TOPIC	RESOURCE OR LAB	EVIDENCE OF PROGRESS	TARGET DATE
Move towards a sysadmin role	*PowerShell automation*	*Automate onboarding in the home-lab domain*	*Script in repository with README and demo video*	*2026-09-30*

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GOAL	SKILL OR TOPIC	RESOURCE OR LAB	EVIDENCE OF PROGRESS	TARGET DATE

SCENARIO · DECIDE · YOUR FIRST JOB OFFER DECISION

You have two offers: a service desk role at a large bank in Dhaka with structured training, shift work and rotation into infrastructure teams after a year, or a 'solo IT person' role at a 20-person start-up with a higher salary but no senior colleagues.

(A) Weigh learning, mentoring, progression and your goals; for most newcomers, the structured role with rotation builds stronger foundations

(B) Always pick the higher salary

(C) Accept both and decide later

(D) Reject both and wait for a perfect offer

Circle your choice. Answer and feedback: module answer key.

REFLECT

Look back over the eight stages of the ZITSS Support Path. Which stage is your strongest, and which needs the most work? Write three specific actions for the next 30 days that will strengthen your weakest stage.

Key terms

Home lab
A personal practice environment of physical or virtual machines used to build and test IT skills safely.

STAR method
A structure for answering behavioural interview questions: situation, task, action, result.

Career path
A planned sequence of roles and skills leading towards a long-term professional goal.

Professional ethics
Principles of honest, confidential and responsible conduct expected of trusted IT staff.

Portfolio
A collection of work samples, such as lab write-ups, scripts and articles, that proves practical skills.

Behavioural question
An interview question about how you handled real situations in the past.

Continuous learning
The ongoing habit of updating skills as technology and practices change.

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Key takeaways

- Prove skills with evidence: measurable CV lines, a portfolio and home-lab write-ups.
- Answer behavioural questions with STAR, using 'I' for your own actions.
- Support leads to systems, networking, cloud, security, service management or modern workplace roles.
- Pair every certificate with a project that shows you can apply it.
- Ethics, honesty and confidentiality are what make a support professional trusted.

KNOWLEDGE CHECK

21. In the STAR method, what does the 'A' describe?

(A) The audience
(B) The approval you received
(C) The actions you personally took
(D) The amount of time spent

22. Which CV line is strongest?

(A) Responsible for tickets
(B) Reduced new-starter set-up from 4 hours to 1 hour by scripting onboarding
(C) Helped users with IT
(D) Hard-working, passionate team player with excellent communication skills and a positive attitude

23. What is the best way to show cloud skills without work experience?

(A) List cloud vendor names
(B) Claim experience you do not have, then learn the details quickly after being hired
(C) Only read blog posts
(D) Build and document projects in a free cloud tier and home lab

24. An interviewer asks something you do not know. What should you do?

(A) Make up a confident answer, because admitting gaps always loses the job offer
(B) Change the subject
(C) Refuse to answer
(D) Admit it, explain how you would find out and link it to what you know

25. Which behaviour reflects professional ethics?

(A) Reading users' emails when bored
(B) Installing pirated tools to save the organisation money on licences
(C) Hiding your own mistakes
(D) Accessing only the data a task requires

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MODULE 8 · AUTOMATE

Answer key

Knowledge checks

1. C $_ is the current pipeline object.

2. C -WhatIf shows what would happen.

3. B (6 - 1) x 100 = 500 minutes.

4. A Execution policy prevents accidents but is not a security boundary.

5. C Parameters make scripts reusable.

6. C Minute 0, hour 2, any day of month, any month, day of week 0 (Sunday).

7. B Zero means success.

8. D set -u prevents silent use of empty variables.

9. A grep -c counts matching lines.

10. B Listing first prevents deleting the wrong files.

11. C Idempotent scripts check state before acting, so repeats are safe.

12. A Secrets belong in secure stores, never in code.

13. C git commit saves a snapshot with a message.

14. B Limiting rights limits impact.

15. B Error handling makes failures visible and safe.

16. B Runbooks contain step-by-step procedures.

17. D (100,000 - 10,000) / 3 = 30,000.

18. D Wiping and recording protect data and keep records accurate.

19. A Ownership and review prevent outdated documentation.

20. C Patterns across tickets and assets reveal problem devices.

21. C Action is what you did.

22. B Specific, measurable achievements are most persuasive.

23. D Documented hands-on projects provide evidence.

24. D Honesty plus a method shows professional judgement.

25. D Least access and honesty are core ethical behaviours.

Scenarios & case studies

Scenario · A script from a forum
B is best. Correct. You protect the organisation while still addressing the underlying goal safely.

Scenario · A clean-up script deleted too much
B is best. Correct. You recover the data and remove the root causes: unset variables, missing checks and no review.

Scenario · Automating new-starter accounts
B is best. Correct. Structured input, preview, logging and role-based groups make onboarding fast and consistent.

Case study · The undocumented network
First: regain administrative access securely (reset device admin credentials through official recovery procedures and store them in a password manager), then map the network: ISP links, firewall, switches, access points and addressing. Without this, the slow portal cannot be diagnosed. **Assets and licences:** run an inventory script and a physical audit, tag every device, record serials, users and warranty dates, and list every licence with expiry dates and renewal owners; set reminders 60 days before expiry. **Keep current:** add 'update documentation' to the change and ticket closure checklists, give each document an owner and review date, and review the asset register each term.

Scenario · A laptop returned by a leaver
C is best. Correct. This protects data, gives the new user a clean standard build and keeps records accurate.

Scenario · Your first job offer decision
A is best. Correct. Mentoring and exposure to mature processes accelerate early careers, though the right choice depends on your goals and circumstances.

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REFERENCE

Glossary

-

-WhatIf A PowerShell parameter that shows what a command would do without making changes. (Automate)

3

3-2-1 rule A backup principle: three copies of data, on two types of media, with one copy off site; often extended to 3-2-1-1-0 with one offline or immutable copy and zero restore errors. (Manage)

8

802.1X A port and Wi-Fi authentication framework where each user or device signs in individually, usually via a RADIUS server. (Connect)

A

A record A DNS record that maps a hostname to an IPv4 address. (Connect)

Account lockout A security control that disables sign-in temporarily after too many failed password attempts. (Manage)

Active listening Fully concentrating on the speaker, then summarising back what was said to confirm understanding. (Resolve)

Alert threshold The value or condition at which monitoring raises an alert, such as disk space below 10%. (Host)

APFS Apple File System: the macOS file system with snapshots, encryption and space sharing. (Boot)

APIPA Automatic Private IP Addressing: a 169.254.x.x address a host gives itself when DHCP does not respond. (Connect)

Application allow-listing A control that lets only approved applications run on a device. (Protect)

Asset register A record of an organisation's IT assets with details such as owner, location, cost and status. (Automate)

Asset tag A unique label attached to a device and used to identify it in the asset register. (Automate)

Asymmetric encryption Encryption that uses a public and private key pair, enabling secure key exchange and digital signatures. (Protect)

Availability The percentage of time a service is usable as agreed. (Host)

Availability zone A physically separate data centre location within a cloud region, used to design for failure. (Host)

awk A text-processing tool that splits lines into fields and prints or calculates on them. (Automate)

B

Bastion host A hardened server that acts as the controlled entry point for administrative remote access. (Host)

Behavioural question An interview question about how you handled real situations in the past. (Automate)

Bit The smallest unit of digital data, holding a value of 0 or 1; network speeds are measured in bits per second. (Build)

Broadcast address The last address in an IPv4 subnet, used to reach every host on that subnet. (Connect)

BYOD Bring your own device: a policy that lets employees use personal devices for work under agreed controls. (Boot)

Byte A group of eight bits; storage and file sizes are measured in bytes. (Build)

C

Cable tester A device that checks network cable pairs for continuity, wiring faults and sometimes length. (Connect)

Cache Very fast memory built into the CPU that holds recently used data to avoid slower trips to RAM. (Build)

Career path A planned sequence of roles and skills leading towards a long-term professional goal. (Automate)

Channel A specific frequency range within a Wi-Fi band; overlapping channels interfere with each other. (Connect)

Checkpoint (VM snapshot) A saved point-in-time state of a VM that can be reverted to quickly, intended for short-term use. (Host)

Checksum (hash) A value calculated from a file's contents, used to confirm the file is genuine and undamaged. (Boot)

Chipset Motherboard logic that connects the CPU to slower peripherals and determines features such as USB, SATA and PCIe lanes. (Build)

chmod The command that changes read, write and execute permissions on files and directories. (Boot)

CIA triad The three core goals of information security: confidentiality, integrity and availability. (Protect)

CIDR prefix Notation such as /24 that states how many leading bits form the network portion of an address. (Connect)

Clean boot Starting an OS with non-essential start-up programs and services disabled to isolate software conflicts. (Resolve)

Clean install Installing an OS on a wiped disk so no previous settings, apps or problems remain. (Boot)

CMDB Configuration management database: a record of IT components and the relationships between them. (Automate)

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Cmdlet A PowerShell command written in Verb-Noun form, such as Get-Service. (Automate)

CMOS battery A coin cell on the motherboard that keeps firmware settings and the real-time clock when the PC is unplugged. (Build)

Command substitution Capturing the output of a command into a variable, written $(command) in Bash. (Automate)

Commit A saved snapshot of changes in a version control repository, with a message describing it. (Automate)

Confidentiality Keeping private any information you see or hear while supporting users. (Resolve)

Configuration drift Gradual divergence of systems from their intended configuration over time. (Manage)

Container A lightweight package of an application and its dependencies that shares the host operating system kernel. (Host)

Continuous learning The ongoing habit of updating skills as technology and practices change. (Automate)

Core An independent processing unit inside a CPU that can execute its own instruction stream. (Build)

CPU socket The mounting interface on a motherboard that accepts a specific family of processors. (Build)

cron The Linux and macOS scheduler that runs commands at defined times using crontab entries. (Automate)

CSV Comma-separated values: a simple text format for tabular data used as input and output for scripts. (Automate)

D

Data classification Labelling data by sensitivity so it receives appropriate protection. (Protect)

Default gateway The router address a host uses to reach destinations outside its local subnet. (Connect)

Defence in depth Using multiple layers of security controls so that one failure does not lead to compromise. (Protect)

Deployment ring A group of devices that receives updates at a defined stage of a phased rollout. (Manage)

Depreciation The reduction in an asset's book value over its useful life. (Automate)

Desired state A declared target configuration that management tools enforce automatically. (Manage)

Device driver Software that lets the operating system communicate with a specific piece of hardware. (Boot)

DHCP Dynamic Host Configuration Protocol: automatically assigns IP addresses and network settings to clients. (Connect)

DHCP relay A function on a router that forwards DHCP broadcasts from one subnet to a DHCP server on another. (Connect)

DHCP reservation A setting that always gives the same IP address to a specific device, identified by its MAC address. (Connect)

Differential backup A backup of all data changed since the last full backup. (Manage)

Digital certificate A file that binds a public key to an identity, signed by a certificate authority. (Protect)

Direct thermal printing Printing by heating special paper so it darkens, without ink or ribbon. (Build)

Directory service A central database of users, groups, computers and other objects used for authentication, authorisation and management. (Manage)

Disk image A complete copy of a configured system used to deploy identical builds to many machines. (Boot)

DISM Deployment Image Servicing and Management: repairs the Windows component store and images. (Boot)

DisplayPort A digital display interface for monitors that carries video and audio and can drive multiple screens. (Build)

Distribution A packaged Linux operating system combining the kernel, tools, package manager and defaults, such as Debian or Rocky Linux. (Boot)

DNS Domain Name System: the distributed service that translates names into IP addresses and other records. (Connect)

Domain controller A server that holds a copy of the directory and authenticates domain sign-ins. (Manage)

Dual channel A memory configuration where matched modules in paired slots are accessed in parallel to increase bandwidth. (Build)

E

ECC memory Error-correcting code RAM that detects and fixes single-bit errors, used in servers and workstations. (Build)

EDR Endpoint detection and response: tools that monitor endpoint behaviour, detect threats and support investigation and isolation. (Protect)

Effective permissions The access a user actually has after combining all user, group, share and deny entries. (Manage)

EFI system partition A small FAT32 partition containing boot loaders used by UEFI firmware. (Boot)

Elasticity The ability to add or remove resources automatically as demand changes. (Host)

Emergency change A change, such as an urgent patch, that must be implemented faster than the normal process allows. (Manage)

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Empathy Recognising and acknowledging how a problem affects the other person. (Resolve)

End of support The date after which a vendor no longer releases security updates or fixes for a product version. (Boot)

Escalation Passing a ticket to a more specialised person or team when it exceeds your skill, access or authority. (Build)

ESD Electrostatic discharge: a sudden flow of static electricity that can damage components without visible signs. (Build)

Event Viewer The Windows console that displays system, security and application logs with event IDs and timestamps. (Boot)

Execution policy A PowerShell setting that controls which scripts are allowed to run on a system. (Automate)

Exit code A number a command returns when it finishes; 0 means success and other values signal errors. (Automate)

Expectation setting Telling users clearly what will happen next and when, so they can plan. (Resolve)

F

File server A server that stores shared files and controls access through share and file-system permissions. (Host)

File system The structure an OS uses to store, name and protect files on a volume, such as NTFS, ext4 or APFS. (Boot)

FileVault macOS full-disk encryption for the startup volume, unlocked with the user's password or a recovery key. (Boot)

Firmware update An update to the low-level software stored on hardware such as motherboards, SSDs, printers and network devices. (Manage)

First contact resolution The share of tickets resolved during the user's first contact with the service desk. (Resolve)

Fishbone diagram A cause-and-effect diagram that groups possible causes into categories such as people, process and technology. (Resolve)

Five whys A root-cause technique that asks 'why?' repeatedly until a fixable underlying cause is found. (Resolve)

Follow-up Contacting the user after a fix or at an agreed time to update them or confirm the fix held. (Resolve)

Full-disk encryption Encryption of an entire storage volume so data is unreadable without the key. (Protect)

Fuser The heated unit in a laser printer that bonds toner to paper; failure causes smudging. (Build)

G

Gibibyte (GiB) 2^30 bytes (1,073,741,824 bytes); operating systems often show this value while labelling it GB. (Build)

gpresult A Windows command that reports which group policies have been applied to a user and computer. (Manage)

GPT GUID partition table: the modern disk partitioning scheme used with UEFI, supporting very large disks. (Boot)

grep A tool that searches text for lines matching a pattern. (Automate)

Group A collection of accounts that receives permissions as a unit, usually matching a role or team. (Manage)

Group policy object (GPO) A collection of settings linked to sites, domains or OUs to configure users and computers centrally. (Manage)

Guest access Permission for an external person to access specific internal resources after signing in. (Host)

H

Half-split A diagnostic technique that tests the midpoint of a system or path to eliminate half of the possible fault locations. (Resolve)

Hash A fixed-length, one-way fingerprint of data used to check integrity or store passwords securely. (Protect)

HDD Hard disk drive: magnetic storage with spinning platters and moving heads; cheap per terabyte but slower and shock-sensitive. (Build)

Home lab A personal practice environment of physical or virtual machines used to build and test IT skills safely. (Automate)

Hot-swap Replacing a component, such as a drive or PSU, while the system keeps running. (Host)

Hypervisor Software that creates and runs virtual machines by sharing physical hardware between them. (Host)

I

IaaS Infrastructure as a service: renting virtual machines, storage and networks while managing the OS and above yourself. (Host)

Idempotent Describes an operation that produces the same result whether it runs once or many times. (Automate)

Immutable backup A backup copy that cannot be changed or deleted until its retention period ends. (Manage)

In-place upgrade Upgrading an OS to a newer version while keeping installed apps, settings and data. (Boot)

Incident An unplanned interruption to a service or a reduction in its quality. (Resolve)

Incremental backup A backup of only the data changed since the previous backup of any type. (Manage)

Inheritance The way policies linked at higher levels apply to objects in lower containers unless blocked. (Manage)

Inventory script A script that collects hardware and software details from devices for asset records. (Automate)

IP address A logical address that identifies a host interface on an IP network. (Connect)

IPv6 The 128-bit successor to IPv4 with a vastly larger address space, written in hexadecimal. (Connect)

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IT support specialist A professional who installs, maintains and troubleshoots devices, software, networks and accounts so that people can work without interruption. (Build)

J

Jargon Technical language that the listener may not understand; it should be translated into plain terms. (Resolve)

K

Kerberos A ticket-based authentication protocol used for secure sign-in within domains. (Manage)

Kernel The core of the OS that runs with full privileges and controls CPU, memory and devices. (Boot)

Keychain The macOS store for passwords, certificates and keys, unlocked by the user's login password. (Boot)

Knowledge base A searchable library of articles that explain fixes and procedures for users and support staff. (Resolve)

Known-good component A part or device verified to work, used to isolate faults by substitution. (Resolve)

L

LAN Local area network: a network covering a limited area such as an office, owned and managed by the organisation. (Connect)

Latency The delay for data to travel across a network, usually measured as round-trip time in milliseconds. (Connect)

LDAP Lightweight Directory Access Protocol: the standard protocol for querying and updating directory data. (Manage)

Least privilege The principle of granting only the minimum access needed, for the minimum time. (Protect)

Licence assignment Allocating a paid service plan to a user so they can use specific cloud applications. (Host)

Loop A structure that repeats commands, for example for each item in a list. (Automate)

M

Maintenance window An agreed time when changes such as updates and restarts may be made with minimal business impact. (Manage)

Malware Malicious software designed to damage, disrupt, steal from or gain unauthorised access to systems. (Protect)

MDM Mobile device management: a service that enrols, configures, monitors and secures phones, tablets and computers. (Boot)

Message trace An admin tool that shows the path and status of an email through the mail service. (Host)

MFA fatigue An attack that floods a user with approval prompts hoping they will accept one. (Protect)

Minimum configuration Testing with only the essential parts installed to isolate a hardware fault. (Build)

Monitoring Continuously collecting metrics and checks from systems to detect problems and trends. (Host)

Motherboard The main circuit board that connects the CPU, memory, storage, expansion cards and power, and holds the firmware. (Build)

Multi-factor authentication Sign-in that requires two or more types of evidence: something you know, have or are. (Protect)

MX record A DNS record that names the mail servers responsible for a domain. (Connect)

N

NAS Network-attached storage: a device that provides file-level storage over the network. (Host)

NAT Network address translation: rewriting private source addresses to a public address so many devices share one connection. (Connect)

Network diagram A drawing that shows network devices, connections, addressing and links. (Automate)

NTFS permissions File-system permissions on Windows volumes that control read, write, modify and full control access. (Manage)

NTP Network Time Protocol: keeps device clocks synchronised with a reliable time source. (Host)

NVMe A storage protocol that runs SSDs over PCIe lanes for very high speed and low latency. (Build)

O

Operating system Software that manages hardware and provides services such as processes, memory, files and security for applications. (Boot)

Organisational unit (OU) A container in a directory used to organise objects, delegate administration and apply policies. (Manage)

OSI model A seven-layer reference model that describes network functions from physical signals to applications. (Connect)

Out-of-band management A dedicated controller that lets administrators manage a server remotely even when its OS is not running. (Host)

P

PaaS Platform as a service: a managed platform for running applications and databases without managing servers. (Host)

Package manager A tool such as apt or dnf that installs, updates and removes software from trusted repositories. (Boot)

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Packet loss The percentage of packets that fail to reach their destination; small amounts damage voice and video. (Connect)

Passkey A cryptographic credential stored on a device that replaces passwords and is bound to the genuine website. (Protect)

Passphrase A long password made of several words, easier to remember and harder to crack than a short complex one. (Protect)

Password manager Software that generates, stores and fills unique strong passwords securely. (Protect)

Patch An update that fixes security vulnerabilities, bugs or adds improvements to software or firmware. (Manage)

Patch compliance The percentage of devices in scope that have all required updates installed. (Manage)

Phishing Fraudulent messages that trick people into revealing information, clicking malicious links or opening attachments. (Protect)

Phishing-resistant MFA Authentication methods such as FIDO2 security keys and passkeys that cannot be replayed on a fake site. (Protect)

Pilot group A small set of users or devices that receives a change first to catch problems before wide rollout. (Manage)

Ping A tool that sends ICMP echo requests to test reachability and round-trip time. (Connect)

Pipeline Passing the output of one command as input to the next, using the | symbol. (Automate)

Port forwarding A router rule that sends inbound traffic on a public port to a specific internal device. (Connect)

Port number A number that identifies a specific service or application on a host, such as 443 for HTTPS. (Connect)

Portfolio A collection of work samples, such as lab write-ups, scripts and articles, that proves practical skills. (Automate)

POST Power-on self-test: firmware checks of core hardware that run before the operating system loads. (Build)

Print server A server that hosts shared print queues and drivers for network printers. (Host)

Print spooler The operating system service that queues print jobs and sends them to printers. (Build)

Printer driver Software that translates documents into a language a specific printer understands. (Build)

Priority The relative importance of a ticket, usually derived from impact and urgency. (Resolve)

Privilege creep The gradual build-up of unnecessary access as people change roles without old access being removed. (Manage)

Privileged account An account with elevated rights, such as administrator or root, that needs extra protection. (Protect)

Problem The underlying cause, often unknown at first, of one or more incidents. (Resolve)

Problem record A record that tracks the investigation of the underlying cause of one or more incidents. (Resolve)

Professional ethics Principles of honest, confidential and responsible conduct expected of trusted IT staff. (Automate)

PSU Power supply unit: converts mains AC into the DC voltages a computer needs. (Build)

Q

Quarantine A holding area for messages blocked as spam, phishing or malware, which admins can review and release. (Host)

R

RAID Redundant array of independent disks: combining disks for redundancy, performance or both. (Host)

Ransomware Malware that encrypts data or systems and demands payment for their release. (Protect)

Redundancy Extra components that take over when one fails, removing single points of failure. (Host)

Remote support tool Approved software that lets a technician view or control a user's device over the network with consent and logging. (Resolve)

Remote wipe A management action that erases a lost or stolen device to protect organisational data. (Build)

Reproduce Make a problem happen again on purpose to observe it and test fixes. (Resolve)

Resource tag A key-value label attached to a cloud resource to record owner, cost centre, environment or purpose. (Host)

Retention policy A rule that keeps or deletes data for a defined period to meet legal or business needs. (Host)

Risk The combination of the likelihood of a harmful event and its impact. (Protect)

Role-based access control Granting permissions according to job roles, implemented through groups. (Manage)

Rollback Reversing a change to return a system to its previous working state. (Resolve)

root The Linux superuser account with unrestricted access to the system. (Boot)

Root cause The underlying reason a problem occurred, which, if fixed, prevents it recurring. (Resolve)

Router A Layer 3 device that forwards packets between different IP networks. (Connect)

RPO Recovery point objective: the maximum amount of data loss, measured in time, that is acceptable. (Manage)

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RSSI Received signal strength indicator, usually in dBm; values closer to zero mean a stronger signal. (Connect)

RTO Recovery time objective: the maximum acceptable time to restore a service after a failure. (Manage)

Runbook A documented, step-by-step procedure for carrying out a routine or emergency operational task. (Automate)

S

S.M.A.R.T. Self-monitoring data reported by drives, such as reallocated sectors, wear and temperature, used to predict failure. (Build)

SaaS Software as a service: complete applications delivered over the internet and managed by the provider. (Host)

Safe mode A diagnostic start-up mode that loads only essential drivers and services. (Boot)

SAN Storage area network: a dedicated network that provides block-level storage to servers. (Host)

Scheduled task A Windows job that runs a program or script automatically at set times or events. (Automate)

Scope (of an incident) How many users, devices, locations or services are affected by a problem. (Resolve)

Script A text file of commands that runs as a program to automate a task. (Automate)

Security advisory A vendor notice describing a vulnerability, the affected versions, its severity and the fix or workaround. (Manage)

Security filtering Limiting a policy so it applies only to specified users, computers or groups. (Manage)

Security incident An event that compromises, or threatens, the confidentiality, integrity or availability of information or systems. (Protect)

Server room A secured, cooled and powered space that houses servers and network equipment. (Host)

Service account A non-human account used by an application or service, with a documented owner and strong credentials. (Manage)

Service desk The single point of contact between users and IT, where requests and incidents are received, logged and managed. (Build)

Service request A formal request from a user for something routine, such as access, information or equipment. (Resolve)

set -euo pipefail Bash options that stop a script on errors, unset variables and failed pipeline commands. (Automate)

SFC System File Checker: scans and restores protected Windows system files. (Boot)

Shared mailbox A mailbox that several users can access, such as accounts@ or support@, without a separate sign-in. (Host)

Shared responsibility model The division of security and operational duties between a cloud provider and its customer. (Host)

Shebang The first line of a script, such as #!/usr/bin/env bash, which tells the system which interpreter to use. (Automate)

Shell A command-line interpreter, such as Bash or zsh, that runs commands and scripts. (Boot)

Single point of failure A component whose failure stops the whole service. (Host)

Single sign-on (SSO) Signing in once to access many applications without re-entering credentials. (Manage)

SLA Service level agreement: agreed targets for responding to and resolving tickets. (Resolve)

SMB Server Message Block: the protocol used for Windows file and printer sharing, on TCP port 445. (Host)

Snapshot A saved point-in-time state of a VM or volume that can be restored quickly. (Boot)

Social engineering Manipulating people into revealing information or performing actions that weaken security. (Protect)

Software licence A legal agreement setting how software may be installed and used, such as per user, per device or volume. (Boot)

Spear phishing Phishing tailored to a specific person or organisation using researched details. (Protect)

ss A Linux utility that lists open sockets and listening ports, replacing older netstat. (Connect)

SSD Solid-state drive: flash storage with no moving parts, much faster and more durable than an HDD. (Build)

SSID Service set identifier: the network name broadcast by Wi-Fi access points. (Connect)

Standoff A small spacer that holds the motherboard away from the metal case to prevent short circuits. (Build)

STAR method A structure for answering behavioural interview questions: situation, task, action, result. (Automate)

Subnet mask A value that separates the network part of an IPv4 address from the host part, such as 255.255.255.0. (Connect)

sudo A command that lets an authorised user run a single command with elevated privileges, with logging. (Boot)

Switch A Layer 2 device that connects devices in a LAN and forwards frames using MAC addresses. (Connect)

Symmetric encryption Encryption that uses the same secret key to encrypt and decrypt data. (Protect)

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systemd The init and service manager on most modern Linux distributions, controlled with systemctl and logged with journalctl. (Boot)

T

Task Manager A Windows tool to view and end processes, monitor performance and manage startup apps. (Boot)

TCP Transmission Control Protocol: connection-oriented, reliable transport that resends lost data. (Connect)

Tenant An organisation's dedicated instance of a cloud service, with its own users, settings and data. (Host)

Test restore A planned recovery of backed-up data to prove backups work and measure restore time. (Manage)

Test-NetConnection A PowerShell cmdlet that tests ping, TCP port reachability and routes to a target. (Connect)

Theory of probable cause A testable explanation of a problem that fits the symptoms, scope and timing. (Resolve)

Thermal paste A compound that fills microscopic gaps between a CPU and its cooler to improve heat transfer. (Build)

Thermal throttling Automatic reduction of processor speed to protect it when it gets too hot. (Build)

Threat Any circumstance or actor with the potential to cause harm to an asset. (Protect)

Tier (support level) A layer of support with increasing skill and access; work moves up tiers through escalation. (Build)

Time Machine The built-in macOS backup feature that keeps hourly, daily and weekly versions of files. (Boot)

Total cost of ownership The full cost of an asset over its life, including purchase, support, licences, energy and disposal. (Automate)

Traceroute A tool that lists each router hop on the path to a destination, showing where a path breaks or slows. (Connect)

Trojan Malware disguised as legitimate software to trick users into installing it. (Protect)

Troubleshooting methodology A repeatable sequence of steps for finding and fixing problems based on evidence rather than guesswork. (Resolve)

try/catch A structure that runs code and handles any errors in a controlled way. (Automate)

TTL Time to live: how long, in seconds, a DNS answer may be cached. (Connect)

Type 1 hypervisor A hypervisor installed directly on hardware, used for production servers. (Host)

U

UDP User Datagram Protocol: connectionless, low-overhead transport used for DNS, voice and video. (Connect)

UEFI Unified Extensible Firmware Interface: modern firmware that initialises hardware, offers setup screens and starts the boot loader. (Build)

UPS Uninterruptible power supply: a battery unit that keeps equipment running briefly during outages and can trigger safe shutdown. (Build)

USB-C A reversible connector whose capabilities (data speed, video, charging power) depend on the port and cable behind it. (Build)

User account An identity that lets one person sign in and be held accountable for actions on a system. (Manage)

User Account Control A Windows feature that requires consent or credentials before tasks run with administrative rights. (Boot)

V

Variable A named storage location for a value, written with a $ prefix in PowerShell and Bash. (Automate)

Verification Confirming with the user and through testing that the fix works and nothing else broke. (Resolve)

Version control A system, such as git, that records changes to files over time with author and reason. (Automate)

Virtual machine A software-based computer running on a hypervisor, isolated from the host and other VMs. (Boot)

VPN Virtual private network: an encrypted tunnel that gives remote users access to internal resources. (Host)

Vulnerability A weakness in a system, process or person that a threat could exploit. (Protect)

W

Windows service A background program that starts with the system and runs without user interaction, such as the print spooler. (Boot)

Work profile A separated, managed container on a mobile device for company apps and data. (Boot)

Workaround A temporary way for the user to keep working while the real fix is in progress. (Resolve)

WPA3 The current Wi-Fi security standard, using SAE for stronger protection of passphrase-based networks. (Connect)

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CAPSTONE

Capstone: Run IT support for a small organisation for one simulated month

Take on the role of the only IT support specialist for a realistic 40-person organisation (for example a garment buying house in Dhaka, a private clinic in Sylhet or a university department in Chattogram). Using virtual machines, a free cloud tenant or a home lab, set up, support, secure and document its IT, then hand it over as if you were leaving for a new job.

Deliverables

☐ Hardware and software inventory with lifecycle and OS end-of-support dates for at least 15 assets

☐ Build sheet for a standard workstation image (Windows or Linux) with installed software and settings

☐ Network diagram with addressing, DHCP scope, DNS records and Wi-Fi settings

☐ Ten worked support tickets: categorised, prioritised, troubleshot, communicated and closed

☐ User and group design with folder permissions based on least privilege

☐ Backup plan using the 3-2-1-1-0 rule (one immutable or offline copy), plus evidence of one successful test restore

☐ Patch and update schedule with a pilot group and rollback plan

☐ Security baseline: password, MFA and passkey policy, endpoint protection, phishing awareness note

☐ Two scripts (one PowerShell, one Bash) that automate a real support task, with comments

☐ Knowledge-base articles for three common problems and a one-page handover document

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Hardware & OS	○ Incomplete inventory or image	○ Accurate inventory and working image	○ Standardised, documented and lifecycle-aware
Networking	○ Diagram missing or wrong	○ Correct addressing and services	○ Clear, verified and easy for others to support
Troubleshooting	○ Guesswork, no method	○ Method followed in most tickets	○ Systematic, evidence-based and root cause recorded
Customer service	○ Technical but unclear to users	○ Clear and polite updates	○ Empathetic, proactive and SLA-aware communication
Administration	○ Shared or excessive access	○ Groups and permissions correct	○ Least privilege with reviewed, documented access
Backup & patching	○ No tested backup	○ Backups and patches scheduled	○ Tested restore, pilot rings and rollback evidence
Security	○ Defaults left in place	○ Baseline applied	○ Layered controls with user awareness built in
Automation & documentation	○ None or unreadable	○ Scripts run and articles exist	○ Safe, reusable scripts and professional handover

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CAPSTONE

Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 40 questions at random from a separate, secret bank that tests the same skills, with a 60-minute limit and a 80% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Build — Hardware and PC Repair

1. A finance user's desktop reboots without warning every afternoon when large reports run. Temperatures are normal, the PSU is a 250 W unit and the user recently had a graphics card fitted for a second monitor. What is the most likely cause?

   (A) The power supply cannot handle the extra load
   (B) A corrupted user profile
   (C) The monitor cable is too long
   (D) The RAM is running in single-channel mode, which limits report performance

2. A technician is asked why a new laptop with '256 GB' printed on the box shows 238 GB in the operating system. What is the best explanation to give the user?

   (A) Part of the drive is reserved for malware protection
   (B) The drive is a cheaper, lower-capacity model than the one advertised on the box
   (C) The OS shows binary units; the maker uses decimal units
   (D) The SSD has already worn out by 7%

3. A desktop powers on, its fans spin and the motherboard's CPU LED stays lit. Nothing appears on screen. The CPU was just replaced. What should the technician check first?

   (A) That the monitor's input source is set to the correct HDMI port
   (B) That the keyboard is plugged in
   (C) That the network cable is connected
   (D) That the 8-pin CPU power cable is connected

4. Students report that a lab PC's clock is wrong every morning and firmware settings revert to defaults. What is the most efficient fix?

   (A) Reinstall the operating system
   (B) Replace the motherboard
   (C) Connect the PC to a UPS so it never loses mains power
   (D) Replace the CMOS coin-cell battery

5. A label printer on a pharmacy counter prints labels that are very faint after the media supplier changed. The printer has no ribbon installed. What should be checked first?

   (A) Whether the new labels need a ribbon (thermal transfer media)
   (B) Whether the network switch port has errors
   (C) Whether the print driver is the latest version from the manufacturer's website
   (D) Whether the label printer needs a new fuser

6. A manager wants all twenty office PCs to boot faster. Each has an HDD, 16 GB RAM and a supported CPU. Which upgrade gives the biggest improvement for the money?

   (A) An SSD in place of the HDD
   (B) A larger power supply
   (C) A second monitor
   (D) A faster CPU with more cores and cache

7. After a storm, a reception PC in a clinic will not power on at all: no lights, no fans. The wall socket works for a phone charger. What should the technician try next?

   (A) Reinstall the operating system from a USB recovery drive and restore files
   (B) Check the PSU switch and test with a known-good power cable or PSU
   (C) Replace the RAM
   (D) Reset the BIOS password

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8. A technician needs to upgrade RAM in a small form factor PC but is unsure of the type supported. What is the best way to confirm before buying?

   (A) Check the specification and query the current modules
   (B) Buy DDR5 because it is backwards compatible
   (C) Ask the user which RAM they prefer
   (D) Buy the largest module available and test whether the PC accepts it

Boot — Operating Systems: Windows, Linux and macOS

9. A user's Windows laptop crashes with a blue screen every morning since a printer driver was installed yesterday. The user needs the laptop urgently. What should the technician do first?

   (A) Reset the PC, reinstall all apps and restore the user's files
   (B) Replace the system SSD
   (C) Disable automatic updates
   (D) Boot into safe mode and roll back or remove the new driver

10. A Linux server's application log says 'Permission denied' when writing to /var/app/data. The service runs as appuser, and the folder is owned by root with mode 755. What is the most appropriate fix?

    (A) chmod -R 777 /var/app
    (B) Give appuser (or its group) write access to that folder only
    (C) Run the service as root
    (D) Delete /var/app/data and recreate it while logged in as a regular user

11. An analyst must copy a 12 GB database export to a USB drive that will be read on both a Windows PC and a Mac. The copy fails although there is space. Which action solves it?

    (A) Split the export into two smaller ZIP files and keep FAT32
    (B) Back up the drive and reformat it as exFAT
    (C) Format it as ext4
    (D) Rename the file with a shorter name

12. A technician must install the same OS and 12 standard apps on 40 new laptops this week. Which approach is best?

    (A) Install each laptop manually from the vendor's website
    (B) Ask each user to install what they need
    (C) Deploy a tested standard image or automated provisioning
    (D) Copy the hard drive of one existing user's laptop to all others

13. Event Viewer shows repeated disk warnings on a PC that has started freezing for a few seconds at a time. What should the technician do next?

    (A) Ignore the warnings because the PC still boots and works most of the time
    (B) Run a registry cleaner
    (C) Increase the screen resolution
    (D) Check drive health, back up data and plan a replacement

14. A company-owned iPhone used by a field engineer is reported lost in a taxi. It is enrolled in MDM. What should happen first?

    (A) Wait a week before acting
    (B) Order a replacement phone and restore the engineer's apps on it
    (C) Remote-lock it with a contact message; wipe if not recovered
    (D) Change the office Wi-Fi password

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15. In late 2026 a company still has 60 Windows 10 PCs and bought no Extended Security Updates. Twenty of them fail the Windows 11 check only because TPM and Secure Boot are switched off in firmware. What should the support team do first for those twenty?

    (A) Enable firmware TPM and Secure Boot, recheck, then pilot the upgrade

    (B) Replace all 60 PCs at once, since none of them can be trusted any more

    (C) Keep them on Windows 10 because they still boot and work normally

    (D) Turn off the update service so the old version stays stable

Connect — Networking Fundamentals

16. A PC shows IPv4 address 192.168.40.130/25 and default gateway 192.168.40.1. The user can reach a printer at 192.168.40.140 but nothing else. What is wrong?

    (A) The DNS server is unreachable

    (B) The printer is blocking traffic

    (C) The gateway is outside the PC's subnet (192.168.40.128-255)

    (D) The subnet mask should be /16 so that all addresses are local

17. After a power cut, several new laptops in a training room get 169.254.x.x addresses, while existing PCs keep working. The DHCP scope is 192.168.5.100-192.168.5.150 with 8-day leases. What is the most likely cause?

    (A) The DNS server is down

    (B) The DHCP scope is exhausted

    (C) The switches are broken

    (D) The laptops need IPv6 disabled

18. A user can open websites by IP address but not by name. Other users are fine. `ipconfig /all` shows a manually entered DNS server address from a previous job. What should be done?

    (A) Set DNS to automatic and flush the DNS cache

    (B) Reset the internet router

    (C) Increase the DHCP lease time so the PC keeps its address

    (D) Replace the network card

19. Staff in a corner office complain that Wi-Fi calls break up. Their laptops connect on 2.4 GHz channel 3 with a signal of -79 dBm. What is the best improvement?

    (A) Rename the SSID

    (B) Switch the network to WEP encryption to reduce the protocol overhead

    (C) Increase DHCP lease time

    (D) Add or relocate an access point and use 5 GHz on a clean channel

20. A web application on 10.10.5.20 does not load for users. Ping to the server succeeds. What is the most useful next test?

    (A) ipconfig /release

    (B) arp -d *

    (C) Restart every user's PC and clear their browser caches

    (D) Test-NetConnection 10.10.5.20 -Port 443

21. A small office owner wants to view the office CCTV recorder from home and asks you to forward its web port on the router. What should you recommend?

    (A) Use a VPN or secure remote-access service instead

    (B) Forward the port and keep the default admin password

    (C) Forward the port but change it to a high number

    (D) Disable the router firewall when he needs access

22. A website moved to a new IP address an hour ago. Most staff see the new site, but some still reach the old server. What explains this, and what is the quick fix for affected PCs?

    (A) Cached DNS answers persist; flush DNS on those PCs

    (B) The subnet mask is wrong; change it

    (C) The DHCP scope is full; extend the scope and renew all leases

    (D) The website uses UDP; open the port

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23. Traceroute from a branch PC to a head-office server stops after the branch router, while internet sites work normally. What should the support specialist do?

(A) Reinstall the PC's network driver and then the operating system
(B) Escalate to the network team with the test evidence
(C) Change the PC's DNS servers
(D) Replace the user's patch cable

Resolve — Troubleshooting, Customer Service and Ticketing

24. A user reports that a spreadsheet macro stopped working today. Before changing anything, what should the support specialist do first?

(A) Reinstall the office suite
(B) Escalate to the application vendor
(C) Ask what changed, who else is affected and capture the error
(D) Restore the spreadsheet from last month's backup to undo any damage

25. A technician changes the DNS server, the proxy setting and the network driver on a PC at the same time, and the problem disappears. What is the main weakness of this approach?

(A) It took too long
(B) The cause stays unknown and extra changes may add new faults
(C) It needed manager approval
(D) It should have been done remotely so the user was not disturbed

26. At 09:00 three tickets arrive: the payroll system is down for the finance team on payday; a manager wants a new keyboard; a user's wallpaper reset after an update. Which order is best?

(A) Keyboard, wallpaper, payroll
(B) Wallpaper, payroll, keyboard
(C) All three in the order received
(D) Payroll, keyboard, wallpaper

27. An employee gets a phone call from someone claiming to be IT support, asking them to install a remote-access program to 'fix a virus'. The employee did not log a ticket. What should the employee do?

(A) Install it and watch carefully
(B) Decline, end the call and report it to IT
(C) Ask for the caller's name and then install it
(D) Give the caller their password instead

28. The VPN drops for about 40 users every weekday at 14:00. The desk has been reconnecting users individually for two weeks. What should happen next?

(A) Raise a problem record to investigate the root cause
(B) Keep resolving each incident individually as users report it
(C) Tell users to avoid the VPN after lunch
(D) Close the tickets as 'user error'

29. A frustrated user shouts that IT 'never fixes anything'. Their laptop cannot connect to the projector before a client pitch in ten minutes. What is the best response?

(A) Ask them to calm down before you continue
(B) Explain that projectors are the facilities team's responsibility, not IT's
(C) Acknowledge the deadline, try quick fixes and offer a backup device
(D) Promise a full investigation next week

30. Which ticket note best supports a colleague who takes over a ticket overnight?

(A) 'Checked network. Still broken.'
(B) 'User says it is slow. Will look later.'
(C) 'Gateway ping OK; DNS resolves; TCP 443 to erp.company.com fails since 08:40; firewall team notified at 09:10, awaiting reply.'
(D) 'Escalated.'

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Manage — System Administration

31. A sales manager moves to the marketing department. Six months later an audit shows she can still open confidential sales pricing files. Which process failed?

    (A) The joiner process
    (B) The backup process
    (C) The patching process
    (D) The mover process

32. A user is a member of two groups on a folder: one grants Modify, the other has an explicit Deny on Write. What is the user's effective permission?

    (A) Modify, because allows are combined
    (B) Full control
    (C) No access at all to the folder or its contents
    (D) Read access only; the deny blocks writing

33. A help desk receives a call: 'This is the CFO, I am travelling and locked out. Reset my password and read the new one to me now.' The number is unknown. What should the technician do?

    (A) Verify identity first, for example by calling back on the directory number
    (B) Reset the password and read it out, because the CFO is senior and the request is urgent
    (C) Refuse to help at all
    (D) Email the new password to the CFO's personal address

34. After moving computer objects into a new 'Kiosks' OU, a screen-lock policy no longer applies to those PCs. What should the technician check first?

    (A) Whether the PCs have enough RAM
    (B) Whether DNS has an MX record
    (C) Whether the screen-lock GPO is linked to (or inherited by) the new OU
    (D) Whether users have local admin rights

35. Management says the order system can lose at most 30 minutes of data and must be back within 4 hours. Which backup design meets this?

    (A) Nightly full backups only
    (B) Weekly full backups with monthly tests
    (C) Backups or replication at least every 30 minutes, with restores proven under 4 hours
    (D) A RAID 1 mirror on the order server, relying on the mirror instead of separate backups

36. A clinic's backup job reports 'success' every night, but no one has ever restored from it. What is the most important next step?

    (A) Buy a larger backup disk
    (B) Change the job to run twice a night
    (C) Switch from incremental to full backups
    (D) Perform and document a test restore, timed against the RTO

37. A critical vulnerability in VPN gateways is being actively exploited worldwide. The normal patch cycle is monthly. What should the IT team do?

    (A) Wait for the next monthly cycle
    (B) Disable all remote user accounts until the next monthly patch cycle is complete
    (C) Patch only desktop PCs
    (D) Patch the VPN gateways first through the emergency change process

38. A Linux administrator runs `sudo usermod -G finance rafiq` to add Rafiq to the finance group. Afterwards Rafiq loses access to the developers' files. Why?

    (A) The finance group has a deny rule
    (B) The developers' folder was deleted
    (C) Without -a, usermod replaced Rafiq's supplementary groups with only finance
    (D) Linux allows only one group per user

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Host — IT Infrastructure Services and Cloud

39. Users across one office suddenly fail to sign in, and some see certificate errors saying 'not yet valid'. Network and DNS checks pass. What should the technician check next?

(A) The print server's queues
(B) The size of users' mailboxes
(C) Clock synchronisation on domain controllers and clients
(D) The RAID level and free disk space on the main file server

40. A small office server uses RAID 5 with four disks. One disk has failed and the array is rebuilding. What is the most important immediate action?

(A) Add a fifth disk to increase capacity
(B) Confirm a current backup and avoid heavy load until the rebuild ends
(C) Convert the array to RAID 0 for speed
(D) Delete old files from the array so that the rebuild finishes more quickly

41. A company moves its accounting application to IaaS virtual machines in the cloud. Who is responsible for applying operating system security patches to those VMs?

(A) The cloud provider
(B) The accounting software vendor
(C) The internet service provider
(D) The company's own IT team

42. An employee resigns and leaves today. Her manager needs her emails and files for the next three months. What is the best offboarding approach in the cloud suite?

(A) Delete her account immediately to free the licence
(B) Leave the account active and give the manager her password so he can sign in as her
(C) Block sign-in, revoke sessions, delegate the mailbox and transfer her files
(D) Forward all her mail to her personal address

43. During an investigation of a compromised mailbox, what should the administrator check early because attackers commonly use it to steal mail silently?

(A) Inbox and forwarding rules that send mail to external addresses
(B) The mailbox's display picture
(C) The user's calendar colour
(D) The licence renewal date

44. A web service had 2 hours of downtime in a 30-day month (720 hours). The SLA promises 99.9% availability. Was the SLA met?

(A) No, availability was about 99.72%, below 99.9%
(B) Yes, availability was about 99.97%
(C) Yes, because downtime under a day always meets SLAs
(D) It cannot be calculated without knowing user numbers

45. A vendor asks for Remote Desktop to be exposed directly to the internet so they can support a server. What is the most appropriate alternative?

(A) Change the RDP port to a high, random number and then expose it to the internet
(B) Use a VPN or gateway with MFA and a named, time-limited account
(C) Share an administrator password by email
(D) Disable the firewall during support sessions

Protect — Security Basics for IT Support

46. A caller says he is the new regional manager, is about to present to the board and needs MFA moved to his new phone immediately. He cannot answer the verification questions. What should the help desk do?

(A) Follow the verification procedure and decline until identity is confirmed
(B) Move MFA to the new phone because he is senior, in a hurry and sounds convincing
(C) Disable MFA for him until tomorrow
(D) Ask a colleague to vouch for his voice

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47. A user received 25 sign-in approval prompts overnight and approved none. What is the best interpretation and response?

(A) The MFA app is faulty; reinstall it on the user's phone and re-register it
(B) It is normal background activity
(C) Someone likely has the password; reset it and revoke sessions
(D) Disable MFA to stop the prompts

48. Which change would most reduce the risk of account takeover by phishing for administrators?

(A) Phishing-resistant MFA such as FIDO2 security keys or passkeys
(B) Forcing password changes every 30 days
(C) Adding a security question
(D) Requiring 8-character passwords with symbols

49. A user reports that files on the shared drive are being renamed with an unknown extension while she watches. What should the support specialist instruct first?

(A) Keep the file open so it is not lost
(B) Run disk defragmentation
(C) Disconnect it from the network now and report an incident
(D) Restore the files one by one from backup while still connected

50. An accounts officer receives an email from a known supplier's lookalike domain asking to change bank details before today's payment. What is the correct action?

(A) Update the bank details because the invoice, logo and signature all look genuine
(B) Phone the supplier on the number on file to verify, and report the email
(C) Reply to the email to confirm
(D) Forward it to all colleagues as a warning

51. A laptop with full-disk encryption and an escrowed recovery key is stolen while powered off. What is the main remaining action for IT?

(A) Revoke its sessions and certificates, send a remote wipe and log it
(B) No action is needed at all
(C) Change the office Wi-Fi password only
(D) Reset every user's password company-wide in case the laptop is unlocked

52. A risk assessment rates an unpatched internet-facing server at likelihood 4 and impact 5, and a lost-badge risk at likelihood 3 and impact 2. Which should be addressed first and why?

(A) The badge risk, because badges are physical
(B) Both equally, because all risks are equal
(C) Neither until the annual review
(D) The server risk, scoring 20 against 6

53. Users keep clicking through a certificate warning on the internal HR portal. What should IT do?

(A) Tell users that clicking through warnings is fine for internal sites
(B) Disable certificate checks in browsers
(C) Move the HR portal to HTTP
(D) Fix the cause, such as an expired or misnamed certificate

Automate — Scripting, Documentation and Career Growth

54. A technician wants a PowerShell command that deletes log files older than 30 days from a shared folder. What should be done before running it for real?

(A) Run it as a domain administrator to avoid errors
(B) Run it with -WhatIf and check the list of files it would remove
(C) Change execution policy to Unrestricted
(D) Run it twice to make sure it worked

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55. A Bash clean-up script uses `rm -rf "$TARGET"/*`. On one server TARGET was empty and the script began deleting from the root directory. Which addition would most directly have prevented this?

    (A) Running the script with sudo
    (B) Adding `set -u` and a check that TARGET is an existing directory
    (C) Using a faster disk
    (D) Scheduling it with cron so that it never runs interactively by hand

56. A cron entry reads `15 18 * * 1-5 /opt/scripts/report.sh`. When does it run?

    (A) At 15:18 every day
    (B) At 18:15 Monday to Friday
    (C) On the 15th and 18th of each month
    (D) Every 15 minutes between 18:00 and 19:00

57. An onboarding script is accidentally run twice with the same CSV and creates duplicate accounts. Which design principle was missing?

    (A) Least privilege
    (B) Idempotence: checking whether each account exists before creating it
    (C) Execution policy
    (D) Version control

58. A team's scripts are stored on individual laptops, and nobody knows which version is current. What is the best improvement?

    (A) Email the latest scripts to each other after every change is made
    (B) Print the scripts and keep them in a binder
    (C) Keep them in a shared version-control repository with reviews
    (D) Allow only one technician to write scripts

59. A school bought 50 laptops at BDT 90,000 each with an expected residual value of BDT 10,000 per laptop after 4 years. What is the annual straight-line depreciation for the whole fleet?

    (A) BDT 1,125,000
    (B) BDT 1,000,000
    (C) BDT 4,000,000
    (D) BDT 20,000

60. In an interview you are asked, 'Tell me about a time you resolved a difficult support issue.' Which answer structure is most effective?

    (A) List every certification you hold
    (B) Explain why the user was wrong
    (C) Say that you have never had a difficult issue in any support role
    (D) Situation, task, your actions and the measurable result

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EXAM PREPARATION

Practice exam answers & explanations

1. A Adding a graphics card raised peak demand beyond a small 250 W PSU, so heavy work trips it. Profiles, cable length and channel mode do not cause power-related restarts.

2. C 256 x 10^9 bytes equals about 238 GiB. There is no fraud, wear or hidden malware partition behind the difference.

3. D The CPU LED points to the processor or its power; a missing 8-pin cable after CPU work is common. Monitor, keyboard and network checks do not explain a CPU POST failure.

4. D A flat CMOS battery loses settings and time when mains power is off overnight. Reinstalling, replacing the board or adding a UPS do not address the cause cheaply.

5. A Direct thermal printers need heat-sensitive labels; thermal transfer media needs a ribbon. The change of media is the obvious variable. Thermal printers have no fuser.

6. A Replacing an HDD with an SSD transforms boot and app load times. A new CPU costs more for less gain, and a PSU or monitor does not affect boot speed.

7. B No lights or fans means the system is not receiving power; the PSU and its cable are the next link to test. RAM, OS and passwords matter only once the system powers up.

8. A The manual and an inventory query show the exact type, speed and slots. DDR5 is not backwards compatible, users rarely know, and trial purchases waste money.

9. D The change that coincides with the fault is the new driver, and safe mode lets you remove it with minimal disruption. Resetting is excessive, hardware is not implicated and disabling updates creates risk.

10. B Granting the service account just the access it needs follows least privilege. 777 and running as root expose the system, and recreating the folder randomly does not define correct ownership.

11. B FAT32 cannot hold files over 4 GB; exFAT can and is read-write on both systems. ext4 is not natively supported, renaming does nothing and splitting adds needless complexity.

12. C A tested standard build deployed automatically is fast and consistent. Manual or user installs drift, and cloning a user's disk spreads their personal data and problems.

13. D Disk warnings plus freezes suggest a failing drive, so protect data and verify health. Ignoring it risks data loss, and registry cleaners and display settings are irrelevant.

14. C Locking first protects data while allowing recovery; wiping follows if it stays lost. Waiting is risky and the other options do not protect the device's data.

15. A These twenty only need a firmware setting before a normal, piloted upgrade, which ends their unpatched state quickly. Replacing everything wastes money, staying on Windows 10 leaves known holes open and blocking updates makes it worse.

16. C A /25 starting at .128 covers .128-.255, so .1 is not on the local subnet and cannot be used as the gateway. DNS, the printer and a /16 mask are not the cause.

17. B Existing PCs keep their leases, while new devices cannot get one because the small scope with long leases has run out. Broken switches or DNS would affect everyone.

18. A The user's static, outdated DNS setting is the fault; returning to automatic DNS from DHCP and clearing the cache fixes it. The router, lease time and hardware are working.

19. D Weak signal on an overlapping 2.4 GHz channel causes drops; better coverage and a cleaner band address the cause. SSID names and leases do not affect radio quality, and WEP is insecure.

20. D Ping proves IP reachability; a port test shows whether the HTTPS service is reachable and listening. Releasing addresses, clearing ARP or restarting PCs do not test the service.

21. A Exposing a recorder to the internet invites attacks; a VPN or secure remote service avoids it. Changing the port number is not real protection and disabling the firewall is worse.

22. A Cached records last until their TTL expires; flushing the client cache forces a fresh lookup. Masks, DHCP and transport protocols are unrelated.

23. B The evidence shows the path breaks beyond the branch router, which is network-team territory. PC reinstall, DNS changes and cabling are disproven by working internet access.

24. C Identifying the problem through facts, scope and recent changes comes first. Reinstalling, escalating or restoring old data are actions taken before you know the cause.

25. B Changing one variable at a time shows which change fixed the issue; multiple changes hide the cause and can introduce new faults. Time, approval and location are not the core issue.

26. D Payroll on payday has the highest impact and urgency; the routine requests follow. Ordering by arrival or ease ignores business impact.

27. B Unsolicited remote-support calls are a common scam; staff should refuse and report. Watching, asking for a name or sharing a password still hands control to the caller.

28. A Recurring incidents with a pattern call for problem management to find and remove the cause. Repeating fixes, workarounds or blaming users leave the cause in place.

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29. C Empathy plus immediate action and a workaround protects the meeting. Telling people to calm down, deflecting or deferring helps no one.

30. C It gives tests, results, timing and next step, so no one repeats work. The others lack evidence and status.

31. D When people change roles, old group memberships must be removed on the day of the move. Joiner covers new starters, and backup and patching do not control access.

32. D Explicit deny entries override allows, so write-related actions are blocked while read remains. Allows do not cancel a deny, and nothing grants full control or removes read.

33. A Urgency and seniority are classic social-engineering pressure; the approved verification protects the account. Refusing outright is unhelpful, and sending passwords to personal email is insecure.

34. C Policy follows OU links and inheritance; the move is the change that matches the timing. RAM, MX records and admin rights do not determine policy scope.

35. C The RPO needs copies at least every 30 minutes and the RTO needs a proven restore time. Nightly or weekly backups lose too much, and RAID is not a backup.

36. D Only a test restore proves the backup is usable and fast enough. More capacity, frequency or type changes do not prove recoverability.

37. D Actively exploited flaws on internet-facing systems justify an emergency change, prioritising exposed devices. Waiting leaves the door open, and the other actions miss the target.

38. C -G alone sets the full list of supplementary groups, removing developers; -aG appends. Linux supports many groups per user, and nothing suggests deletion or deny rules.

39. C Clock differences break ticket-based authentication and certificate validity checks. Print queues, mailbox size and RAID level do not cause sign-in and certificate errors together.

40. B A second failure during a RAID 5 rebuild loses the array, so a current backup and a careful rebuild are vital. Expanding, converting or deleting files add risk without protecting data.

41. D In IaaS the customer manages the guest OS and everything above it. The provider handles the physical and virtualisation layers, and the vendor and ISP do not patch your VMs.

42. C This removes her access while preserving data for the business. Deleting loses data, sharing her sign-in removes accountability and forwarding to a personal address leaks data.

43. A Malicious forwarding and inbox rules are a common way to exfiltrate mail. Pictures, colours and licence dates are unrelated.

44. A (720 - 2) / 720 = 99.72%, which misses 99.9% (about 43 minutes allowed per month). User numbers are not needed.

45. B A gateway or VPN with MFA, logging and limited time controls the risk. Changing ports, emailing passwords or disabling the firewall all leave the server exposed.

46. A Urgency and seniority are typical pretexting tactics; MFA changes require verified identity. Disabling MFA or relying on voice recognition creates the opening attackers want.

47. C Unexpected prompts mean the password is likely known to an attacker, so reset and investigate. Reinstalling, ignoring or disabling MFA leaves the account at risk.

48. A Phishing-resistant MFA cannot be replayed on fake sites. Frequent changes, security questions and short complex passwords do not stop credential phishing.

49. C Active ransomware must be isolated at once to stop further encryption. Keeping files open, defragmenting or restoring while still infected spreads the damage.

50. B Out-of-band verification through known contacts defeats payment diversion fraud, and reporting lets IT block it. Replying reaches the attacker, and mass forwarding spreads the malicious message.

51. A Encrypted data is protected, but the device's tokens and certificates should be revoked and the loss recorded. Ignoring it, changing Wi-Fi only or mass resets are inappropriate.

52. D Using likelihood x impact, the server scores 20 and the badge 6, so the server takes priority. Risks are not equal, and waiting leaves a high risk open.

53. D Fixing the certificate removes the warning properly and keeps users wary of real attacks. Training users to ignore warnings, disabling checks or dropping encryption weakens security.

54. B -WhatIf previews exactly which files would be deleted so mistakes are caught. Extra privilege, weaker policy or repeated runs do nothing to prevent deleting the wrong files.

55. B Strict unset-variable handling plus an explicit directory check stops the script before damage. sudo increases the damage, and disk speed or scheduling are irrelevant.

56. B The fields are minute 15, hour 18, any day of month, any month, days 1-5 (Monday to Friday). The other options misread field order.

57. B An idempotent script checks state first, so repeats are harmless. Privilege, execution policy and version control matter, but none prevents duplicates by itself.

58. C Version control gives one current source, history, review and rollback. Email, paper and single-person ownership create confusion and risk.

59. B Per laptop (90,000 - 10,000) / 4 = 20,000 a year; for 50 laptops that is BDT 1,000,000. The other values ignore the residual value, the fleet size or the useful life.

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60. D STAR gives a clear, evidence-based story. Listing certificates, blaming users or denying difficulty do not answer the question.

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ANSWER KEY

Subnetting worksheet answers

ADDRESS / PREFIX	NETWORK	MASK	FIRST HOST	LAST HOST	BROADCAST	HOSTS
`192.168.10.77/26`	192.168.10.64	255.255.255.192	192.168.10.65	192.168.10.126	192.168.10.127	62
`10.45.130.9/20`	10.45.128.0	255.255.240.0	10.45.128.1	10.45.143.254	10.45.143.255	4094
`172.20.5.200/27`	172.20.5.192	255.255.255.224	172.20.5.193	172.20.5.222	172.20.5.223	30
`10.0.0.129/25`	10.0.0.128	255.255.255.128	10.0.0.129	10.0.0.254	10.0.0.255	126
`192.168.1.5/30`	192.168.1.4	255.255.255.252	192.168.1.5	192.168.1.6	192.168.1.7	2
`172.16.99.1/22`	172.16.96.0	255.255.252.0	172.16.96.1	172.16.99.254	172.16.99.255	1022

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NOTES

My notes

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